Representable but not found: what path-integrated rotary attention learns, and how it fails

Main paper, draft 1 (2026-10-18), from `docs/paper/OUTLINE_main_v2.md`. *Appendix:* `docs/paper/MAIN_APPENDIX1.md`; *full theory appendix with check scripts:* `docs/paper/APPENDIX_theory_v3.md`. *Word counts, the source of every number and the deviations from the outline:* `docs/paper/MAIN_DRAFT1_notes.md`. *Delete this line before submission.*

Abstract

Path-integrated rotary attention (MapFormer) replaces RoPE's token index with a learned phase that sums a content-dependent step per token. We ask what such models learn and how they fail. On the standard short-walk benchmark a map-free rule that retraces the last run of moves already scores 0.934, so we train and test on 1024-move torus walks, where that rule scores 0.843: one path-integrated layer reaches 0.998 (8 of 8 seeds solved) against 0.731 for index RoPE (0 of 8). At matched depth one path layer is worth about three index layers on Dyck-2 and a biased 1D walk, and more than four on the torus, where index depth helps only short-gap revisits. Told in English, the walk is learned from words (0.979 vs 0.505), with opposite direction words cancelling. Where these models fail, they fail by search, not capacity. With rank 2 per head the content-to-angle map solves the torus on at most 2 of 8 seeds, against 8 of 8 at rank 4, although a rank-2 solution exists and training keeps it. A non-negative step represents the map exactly and keeps or re-finds a constructed solution, yet from scratch solves 0 of 8 against 8 of 8 for signed steps, a gap that persists, on a ring, at each step's best of four learning rates. Both failures are optimisation, not expressivity.

1. Introduction

Rotary position encodings are integrators. Index RoPE (Su et al. 2024) adds a constant angle per token; content-dependent variants let each token set its own step (CARoPE, Veisi et al. 2025; CoPE, Golovneva et al. 2024; Selective RoPE, Movahedi et al. 2026; Mamba-3, Lahoti et al. 2026; MapFormer, Rambaud et al. 2026). When the step is driven by actions, the summed phase is a position on a map, and attention compares net displacements instead of token distances. MapFormer built this into a transformer and showed that it helps navigation. We ask what such a model actually learns, how much it buys over index attention, and why training sometimes misses a map the model can represent.

MapFormer shows that path-integrated attention beats index RoPE, TAPE and CoPE baselines with up to four layers on 1D and 2D grid navigation with 128-step walks (and PathAtt in 2D), generalises out of distribution to sparser and denser walks, extrapolates to deeper Dyck-2 strings with one layer, and learns opposite latents for opposite actions (their Table 2, Figs. 3 and 9). Three questions remain open. How much of the short-walk benchmark needs a map at all? What is one path layer worth in index layers when both are trained and tested at the same length and depth? And why does the model sometimes fail on its home task, for which it can represent the solution?

We answer each in turn. Measured against map-free floors, short walks are mostly solvable by retracing; at 1024 moves the floor drops and a single path layer still solves the map. Trained and tested at matched depth, one path layer is worth about three index layers on Dyck-2 and on a biased 1D walk, and more than four on the 2D torus. The same mechanism learns a map from navigation told in words. And its failures share one signature: a solution in the constrained class exists and training keeps it, but training from scratch does not find it.

Contributions.

1. *A sharper measurement of the advantage.* A map-free retrace rule scores 0.934 on MapFormer's short walks, so we measure on 1024-move torus walks against measured floors (blank 0.506, n-gram 0.575, retrace 0.843). There one path layer scores 0.998 against index RoPE's 0.731 (8/8 vs 0/8 seeds solved, p 0.0002), and index depth or a looped block lifts only short-gap revisits (to about 0.94), while long-gap and wrap-only revisits stay at the floor (0.46-0.51).
2. *An exchange rate at matched depth.* One path layer is worth about three index layers on Dyck-2 trained and tested at nesting depth 12 (position effect +0.353 / +0.130 / +0.045 / +0.024 at 1-4 layers) and on a biased 1D walk (three index layers reach 0.998-0.999 against path 1.000), and more than four on the 2D torus (four index layers 0.860 vs 1.000).
3. *Failure by search, not capacity, along two axes.* With per-head rank 2 the content-to-angle map solves the torus on 0-2 of 8 seeds (rank 3: 6/8; rank 4: 8/8); twice the budget leaves it at 0/8 (rank 4: 7/8); a rank-2 solution exists (0.9955) and is held under training. A non-negative (|.|) step represents the map exactly and keeps or re-finds a constructed solution, yet from scratch solves 0/8 against 8/8 for signed steps on the torus (-0.177), a gap that holds without weight decay, in a 2-layer d 256 model, and at each step's best of four learning rates (2/8 vs 8/8, p 0.007).
4. *Maps from words.* Navigation told in English is learned by one path layer (0.979 vs index 0.505, 6/6 vs 0/6 seeds, p 0.0022), and every model encodes opposite direction words as cancelling steps.

2. Background

2.1 Rotary position as an integrator. Write the phase of rotary channel c (a head and a 2D block) at token t as theta_{c,t} = omega_c Phi_{c,t}, with Phi_{c,t} = sum_{u <= t} Delta_{c,u} the channel's summed step. The channel's contribution to the score of query t and key t' then depends on position only through exp(-i omega_c (Phi_{c,t} - Phi_{c,t'})), the identity behind Mamba-3's Proposition 3 (App. E). Howard et al.'s (2014) rate-driven integrators give the vocabulary. A constant rate codes elapsed time (index RoPE); a non-negative rate codes path length (CARoPE, CoPE, and GRAPE's additive phase in the real slot, Zhang et al. 2026); a signed rate codes position on a map (MapFormer, Selective RoPE). TEM and TEM-t (Whittington et al. 2020, 2022) frame such a map as a "where" learned apart from the "what".

2.2 MapFormer. MapFormer's working-memory model (MapWM) rotates content queries and keys by theta_t = omega * cumsum(W_out W_in x_t), a low-rank content-to-angle map; observations learn near-zero steps and opposite actions opposite steps (their Fig. 3d). In their Table 2, one-layer MapWM with rank 2 solves 2D navigation at 128

steps, rank 1 does not, and RoPE, TAPE and CoPE up to four layers fall short; they attribute CoPE's failure to the sign of its gate, which cannot represent an action and its inverse. We build on exactly this model. Our implementation reproduces their Table 2 2D result (0.989 for MapWM and 0.987 for MapEM, against 1.00 reported in v4 and 0.99 / 1.0 in v1; App. A).

2.3 Representability. Real non-negative eigenvalues cannot track parity (Sarrof et al. 2024, Thm. 2; Grazzi et al. 2025, Thm. 1). Rotations escape this, and rotations by non-negative angles generate cyclic groups (Grazzi et al., Thm. 3; Mamba-3, Sec. 3.2). In the rotary slot, then, the sign of a channel's step does not limit what that channel can represent; Proposition 1 below extends this to our rank-limited step maps, so any difference between signed and non-negative steps is a difference in what training finds.

2.4 Optimisation, not expressivity. Shakerinava et al. (2026) find that input-dependent rotations in unit-modulus SSMs can represent non-Abelian state tracking but do not learn it from scratch, while near-solution starts recover it. Orientation-fixing parameterisations have a history: scoRNN fixes the +/-1 diagonal of its Cayley transform during training (Helfrich et al. 2018); a reparameterised weight w = |u| cannot change sign under gradient flow (Amid and Warmuth 2020, Prop. 2; Woodworth et al. 2020); the LRU initialises its phases near zero (Orvieto et al. 2023). Depth can substitute for recurrence through shortcuts (Liu et al. 2023), though shallow transformers learn sensitive functions unreliably (Hahn and Rofin 2024), while Selective RoPE and PaTH (Yang et al. 2025) solve parity or Flip-Flop with one layer. We bring these lines to path-integrated attention, measuring the depth exchange rate and the learnability gap in one model family.

2.5 Two results we use (proofs in App. E). *Proposition 1 (representability by wrapping).* If at most r tokens step, a |.| model of rank r reproduces any signed angle table exactly mod 2 pi. Annihilating the observation embeddings in W_in leaves only the moves stepping, so every |.| model in this paper can represent the map. Softplus can too: with the observations annihilated each steps softplus(0) = ln 2, and omega' = 2 pi / ln 2 makes that step a full turn. *Lemma 1 (relative-sign gauge).* Negating a channel's steps (through omega_c) together with the second coordinate of that channel's query and key projections leaves every attention score unchanged, since J R(phi) J = R(-phi) for J = diag(1, -1). A non-negative step therefore fixes only the relative signs of the tokens that step in a channel. It excludes no map function, only parameter points: for an inverse pair (two tokens that undo each other, such as opposite moves) it removes cancellation through zero, angles (a, -a), while the wrapped code (a, 2 pi - a), the same function, remains. With fixed frequencies the parameter symmetries of rotary attention are per-plane complex scalars (Wang and Wang 2025; Tran et al. 2026); a learnable step adds this reflection.

3. Setup

Models. MapWM with one layer (up to four where stated), 2 heads, d_model 128, and step Delta = W_out W_in x through a rank-r latent. Unless stated, r = 4 and the latent is shared across heads; MapFormer's latent is per head, and per-head variants are named wherever rank is compared. *Per-head rank* is the rank of the map from token to the angles of one head. Three step parameterisations: *signed* (Delta = s = W_out W_in x), |.| (Delta = |s|, applied per channel after W_out), and *softplus* (Delta = softplus(s), so every

step starts at ln 2). *Observation-step-free* variants give steps to the move tokens only. Each head has 32 two-dimensional rotary blocks with frequencies omega_i = 2 pi (1/ Delta_max)^(i/31), Delta_max the grid size, as in MapFormer; omega is a free parameter whose sign is not constrained, so a channel's direction is never fixed by the parameterisation, only the relative signs of its tokens. The layer is otherwise a standard pre-norm transformer block (feed-forward width 4 d_model, dropout 0.1; about 204k parameters at one layer). The index baseline is RoPE with the canonical ladder base^(-2c/d), at 1-4 layers or as one block looped four times. On Dyck-2 we also pair PoPE (Gopalakrishnan et al. 2025), a rotary variant with a polar score, with its path-integrated version MapPoPE (App. F, L). Faithful CARoPE and CoPE serve as the published monotone comparison.

Training and evaluation. AdamW (learning rate 1e-3, weight decay 0.05), 5% linear warm-up then cosine decay to 10% of the peak, 98 batches per epoch (16 walks per batch on the 1024-move torus, 128 on the ring and the 128-move torus), cross-entropy on revisited observations only (Dyck-2: learning rate 1e-4, next-token loss; App. A). Torus models are scored on 100 walks over one held-out map; ring models on 400 walks with freshly drawn maps. A run that is not SOLVED (defined below) is *STALLED* if its final-10% mean loss is within 5% of the preceding 10%, and *DESCENDING* otherwise.

Tasks. (i) *2D torus*: a 64 x 64 torus walked by runs of 1-10 repeated moves, interleaved with observations drawn from 16 objects; the map is redrawn at evaluation, so every score is transfer. Walks have T = 1024 moves (2,048 tokens; MapFormer's T = 128 for the floor analysis), 900 epochs. (ii) *1D ring* of 32 cells, T = 128, forward-step probability p_plus (0.5 balanced; 1.0 never reverses), maps redrawn per walk, 300 epochs. (iii) *Dyck-2* at length 32 and nesting depth 12, scored by Hewitt et al.'s (2020) distance-averaged closing accuracy (chance 0.5; best map-free floor 0.594). (iv) *Text world*: the torus walk rendered as English (58 words: three synonyms per direction, movement-free verbs, fillers, asides), T = 1024 words. (v) 2D and 3D tori of other sizes for the test of rank against dimension. The loss and the accuracy are on *revisits*: observations at cells visited before, the only predictable targets.

Floors. For each task we measure the best map-free predictor: always-blank, the best n-gram over orders 0-5, and *retrace* (while the current run of moves undoes the previous run, copy the observation seen when the walk was last at the cell it is returning to). Revisits are split into *strata*: short-gap (fewer than 128 moves since the last visit), long-gap, and *wrap-only* (the cell was seen before only from a different unwrapped position, a full lap of the torus away). Floors, strata floors and cancelling shares were computed after the batches on the stored streams and checkpoints; we mark such numbers "(analysis)" where they first appear. App. B lists every floor.

Protocol. Every verdict was pre-registered with its decision rule against the measured noise floor, with n = 8 seeds per arm unless stated, every model trained and tested at the same length (and nesting depth), *SOLVED* defined as a mean training loss below 0.05 over the final 5% of epochs, and exact permutation (accuracy) and Fisher (SOLVED) tests. Before any result was read, 12 of the 21 main-text batches were checked by an independent, results-blind code review, and the analyses of 20 were re-run from stored checkpoints by an independent reproduction audit that reproduced every one; App. A records both per batch.

4. A sharper measurement of the advantage

Short walks mostly do not need a map (analysis). On MapFormer's T = 128 torus walks retrace scores 0.934 (blank 0.507, n-gram 0.598), and no revisit is reachable only by wrapping around the torus (wrap-only share 0.000). At T = 1024 retrace falls to 0.843 (blank 0.506, n-gram 0.575) and 7.9% of revisits are wrap-only (Table 1). The long-walk regime is where map use becomes measurable. At T = 128 the path model still beats index attention (+0.243 position effect at rank 2; path 0.971 at rank 2 and 1.000 at rank 4, index 0.805; n = 8), but index RoPE falls below retrace there and the rank-2 path model clears it by only 0.04.

Retrace is strong on short walks for a simple reason. The walk moves in runs of repeated moves, and a run that reverses the previous one returns, step by step, to cells seen moments earlier. Over 128 moves nearly every revisit is of this local kind, so a model can score well above 0.9 without integrating anything. Over 1024 moves the walk drifts far from where it started, revisits arrive after hundreds of moves or across the wrap of the torus, and a rule that only undoes the last run reaches 0.843. A score counts as evidence of a map when it clears this floor, and most directly when it holds on the long-gap and wrap-only revisits that no local rule can reach.

Table 1. Torus floors and accuracies at MapFormer's walk length and at ours (held-out map; floors (analysis) on the same evaluation walks as the models; n = 8 per model).

T (moves)	wrap-only revisits	blank	n-gram	retrace	index RoPE, 1 layer	path, rank 2	path, rank 4
128	0.000	0.507	0.598	0.934	0.805	0.971	1.000
1024	0.079	0.506	0.575	0.843	0.731	0.894	0.998

One path layer solves the long-walk map; index attention does not. On the torus at T = 1024, one path layer scores 0.998 +/- 0.004 (8/8 SOLVED) against 0.731 +/- 0.008 for index RoPE (0/8), a difference of +0.267 (permutation and Fisher p 0.0002). Index RoPE sits below retrace on every run (0.72-0.74 against 0.843; analysis): it does not learn even the map-free rule. The same contrast appears in two further registered batches: 1.000 against 0.722 (n = 8) and 1.000 against 0.719 (n = 4; drawn with its own floors in Fig. 1c).

Where the difference lives. By stratum (short gap / long gap / wrap-only), the path model scores 1.000 / 0.995 / 0.982 and one-layer index RoPE 0.784 / 0.462 / 0.471, against stratum floors of 0.506 / 0.500 / 0.507 and retrace 0.911 on short gaps (analysis; retrace equals the constant on the other two strata) (Fig. 2c). Index attention gains its margin over the constant on short gaps, where retrace does better still; long-gap and wrap-only revisits require the map, and only the path model has it.

5. An exchange rate at matched depth

Depth is the obvious way to give index attention more computation, so we ask how many index layers one path layer is worth, with both trained and tested at the same length and nesting depth, on three tasks that need integration.

Dyck-2: about three index layers. A bracket string is a walk on a stack: an opening bracket steps forward, its closing partner steps back, and MapFormer's models encode the two as opposite directions (their Fig. 3f). Trained and tested at length 32 and nesting depth 12, one path layer scores 0.996 (1.000 at 2-4 layers), and index attention 0.648 / 0.889 / 0.964 / 0.982 at 1-4 layers, against a floor of 0.594 (Fig. 1a). The position main effect, averaged over two encodings (MapWM vs RoPE and MapPoPE vs PoPE; App. F), is +0.353 / +0.130 / +0.045 / +0.024 at 1-4 layers, positive on 8/8 seeds at each depth (sign-flip p 0.0078). With three times the budget, four index layers reach the ceiling (0.997-0.998), and training on a mixture of depths 4-12 keeps an effect of +0.110 at depth 12 (8/8). The index arms are not held back by their frequency ladder (base 32 and base 10000 agree to 0.001 at four layers with three times the budget), so the rate measures what a layer of attention computes, not a mis-set encoding.

A biased 1D walk: the same rate. The ring lets us dial how often a step is undone: at p_plus 0.5 the walk is balanced, at 0.9 one segment in ten runs backwards, and at 1.0 the walk never reverses and position is a count. On the ring (fresh seeds, n = 6), one path layer scores 1.000 at every bias, and index attention with 1 / 2 / 3 layers 0.710 / 0.949 / 0.998 at p_plus 0.5, 0.677 / 0.962 / 0.998 at 0.75, and 0.824 / 0.987 / 0.999 at 0.9 (Fig. 1b). Wherever a step can be undone, two index layers stay 0.013-0.051 short of the path layer and three come within 0.002. At p_plus 1.0, where no step is undone, a single index layer reaches 1.000. One path layer solves every run in every cell.

The 2D torus: more than four. At T = 1024 (n = 4) one path layer scores 1.000 (4/4 SOLVED) against index attention at one layer 0.719, at four real layers 0.860 and as one block looped four times 0.864 (each 0/4, every run stalled; complete separations, p 0.029) (Fig. 1c). Depth and the loop each help index attention (+0.140 and +0.145), and the loop matches four real layers with a quarter of the parameters (203,925 against 798,741). The four-layer index model scores 0.897 on its own training map against 0.860 on the held-out map, so part of what depth adds is map-specific; the path model scores 1.000 on both.

What depth buys index attention. Four layers or a loop of four lift short-gap revisits from 0.769 to 0.943 / 0.938, just above retrace on this stream (0.912), and leave long-gap (0.458 / 0.507) and wrap-only (0.457 / 0.509) revisits at or below the constant floor (0.515 / 0.517; analysis) (Fig. 2a; App. F). Depth buys index attention the local part of the map; only the path code integrates over hundreds of moves.

Depth substitution is known qualitatively, from shortcuts to automata (Liu et al. 2023) and one-layer parity with Selective RoPE. We give the rate at matched depth on three tasks and show where it breaks: on a 2D map with long gaps and wrap-around, four index layers recover only the short-gap revisits.

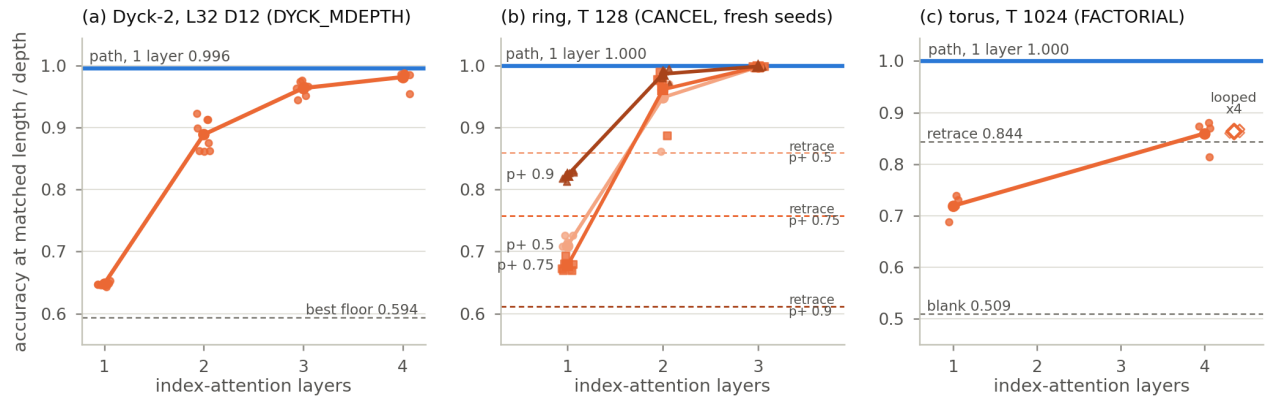


Figure 1 (`fig_exchange_rate`). Accuracy of index-RoPE attention against its number of layers (orange; dots: runs, line: means) beside one layer of path integration (blue line), each trained and tested at the same length and depth. (a) Dyck-2 at length 32 and depth 12 (Hewitt closing accuracy; n = 8): index attention needs about three layers to approach one path layer (0.964 at 3, 0.982 at 4, against 0.996). (b) Ring of 32 cells at T = 128 for three move biases (n = 6 fresh seeds): three index layers reach one path layer (0.998-0.999 against 1.000). (c) Torus at T = 1024 (n = 4): four index layers (0.860) or one layer looped four times (0.864) stay far below one path layer (1.000), just above the retrace floor (0.844). Dashed lines: map-free floors (analysis; Dyck: the best stack-free floor; ring: retrace per bias, on another stream of the same task; torus: always-blank and retrace on this batch's own walks).

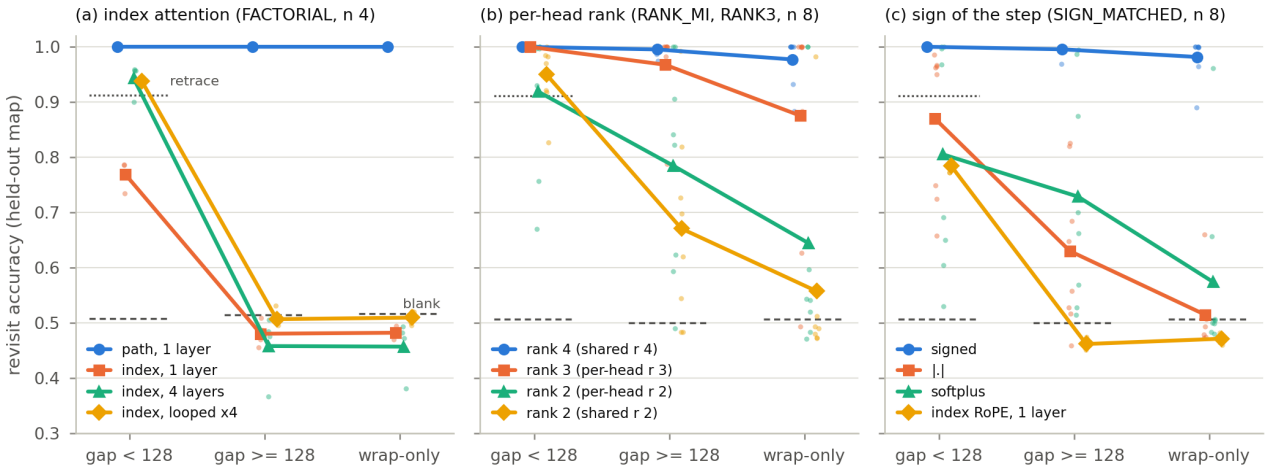


Figure 2 (`fig_strata_modes`). Revisit accuracy by stratum on the torus at T = 1024 for the three ways a model fails to learn the map; lines: arm means, dots: runs. (a) Index attention (n = 4): depth and a loop help short-gap revisits only. (b) Per-head rank of the content-to-angle map (n = 8, same seeds and initial weights): rank 4 is near 1.000 in every stratum; rank 2 loses accuracy with the gap and most at wrap-only revisits. (c) Sign of the step (n = 8): signed near 1.000 everywhere; |.| and softplus fall from short gaps to wrap-only revisits; one-layer index RoPE for reference. Dashed: always-blank within each stratum; dotted: retrace within the short-gap stratum (0.91); floors (analysis) on each batch's own walks.

6. Failure by search, not capacity

Two design choices of the content-to-angle map decide whether the torus map is learned: its per-head rank and the sign parameterisation of its step. For each we show the same three facts. The constrained class fails from scratch; a solution in the class exists, constructed explicitly; and that solution, installed and trained, is kept or re-found.

6.1 Per-head rank of the content-to-angle map

Per-head rank decides it. On the torus at T = 1024 we compare six bottlenecks, all built from the same initial weights at each seed (Table 2, Fig. 3a). Every arm whose heads each read a rank-2 map solves 0-2 of 8; rank 3 solves 6/8 and rank 4 8/8, whether the latent is shared or per head. At matched sharing (per-head rank 4 against the block-diagonal rank-4 latent, in which each head reads rank 2) the split is 8/8 against 2/8 (Fisher and permutation p 0.0070). Per-head rank 3 improves on per-head rank 2 by +0.102 in accuracy (permutation p 0.027). Total latent size does not track success: the per-head rank-2 arm and the shared rank-4 arm both have four latent dimensions and land on opposite sides. Nor is it the initial angle scale: the block-diagonal arm starts with smaller angle increments, but the two other rank-2 arms start at the same scale as rank 4 and fail the same way. The training curves show the split early: most rank-2 runs flatten at losses of 0.2-0.5, while the rank-3 and rank-4 medians cross the SOLVED threshold at epochs 478-532 (App. G). Read with MapFormer's Table 2: rank 1 cannot represent a 2D map; rank 2 can, and learns 128-step walks, but at 1024 moves reliable learning needs rank above the map's dimension.

Table 2. SOLVED (of 8) and held-out accuracy at T = 1024 (mean over 8 seeds) by per-head rank of the content-to-angle map; torus, 900 epochs, every arm built from the same initial weights at each seed.

per-head rank	arm	SOLVED	accuracy
2	one rank-2 latent shared by the heads	0/8	0.894
2	per-head rank-2 latents	1/6 fresh seeds (2/8 all seeds)	0.885
2	block-diagonal rank 4 (each head reads rank 2)	2/8	0.948
3	per-head rank 3	6/8	0.987
4	shared rank 4	8/8	0.998
4	per-head rank 4	8/8	0.999

Twice the budget does not rescue rank 2. Trained for 1800 epochs from scratch, rank 2 solves 0/8 (0.908) against rank 4's 7/8 (Fisher p 0.0014; accuracy +0.086, permutation p 0.0003).

The rank-2 solution exists and is held. The rank-2 projection of each solved rank-4 model scores 0.9955 frozen (at least 0.981 on every seed). Trained from there it solves 7/8, as many as the rank-4 control trained from its own solution (7/8), against 1/8 for rank 2 trained further from its own stalled weights (accuracy p 0.0009) (Fig. 4a). The restart at learning rate 1e-3 lifts every projected run's training loss to 0.37-0.46 (control

0.26-0.30), and seven return below 0.05 at epochs 553-654, inside the control's window (497-669). A rank-2 map that solves the torus exists, is kept by training, and is not reached from scratch: the failure is search.

Rank against the map's dimension. Rank equal to the map's dimension D is hard in 2D and 3D and not in 1D (Fig. 3b). On a 2D torus of 32 cells per side rank 2 solves 1/8 against rank 3's 8/8 (Fisher p 0.0014). On a 3D torus of 10 cells per side rank 3 solves 1/8 (rank 4, 4/8), and rank 4 solves an 18-per-side 3D torus 8/8 (1.000). On the 1D ring rank 1 solves 7/8 against rank 2's 8/8. Per-head rank D + 1 learns reliably in 2D and on the larger 3D torus.

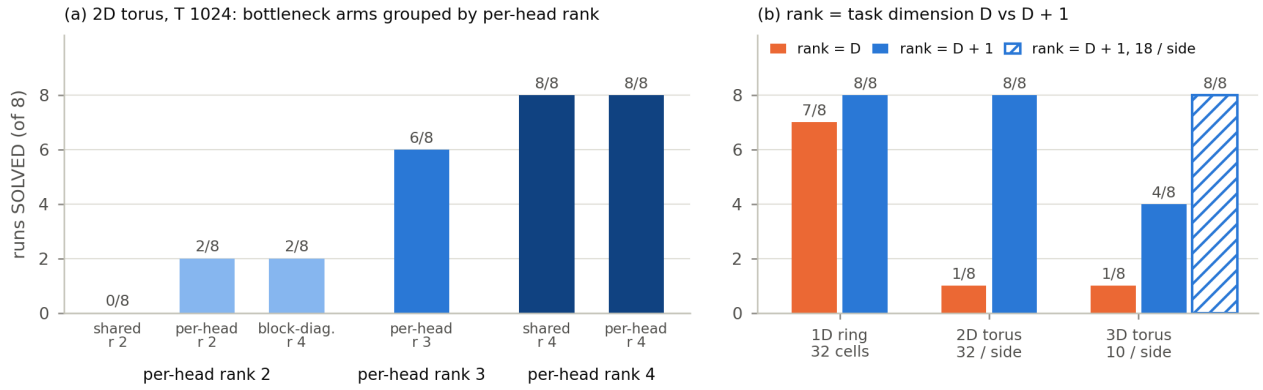


Figure 3 (`fig_rank_solved`). Runs (of 8) that solve the map, by the per-head rank of the content-to-angle map. (a) Torus, T = 1024, six bottlenecks built from the same initial weights: every arm whose heads each read a rank-2 map solves 0-2/8, rank 3 solves 6/8 and rank 4 8/8, whether the latent is shared or per head. (b) Rank equal to the task's dimension D against D + 1 (separate batches and recipes): on the 1D ring rank 1 solves 7/8 (rank 2, 8/8); on a 2D torus rank 2 solves 1/8 (rank 3, 8/8); on a small 3D torus (10 per side) rank 3 solves 1/8 and rank 4 4/8, and rank 4 solves 8/8 on an 18-per-side torus with few wrap-only revisits (hatched). Torus budget 900 epochs; at 1800 epochs rank 2 still solves 0/8.

6.2 The sign of the phase step

Non-negative parameterisations rarely learn the map from scratch. On the torus at T = 1024 the |.| step scores 0.821 against 0.998 for the signed step, -0.177 (permutation p 0.0002), and solves 0/8 against 8/8; softplus scores 0.781 and solves 1/8 (-0.218, p 0.0003). On the ring |.| solves 2/8 and 1/8 against 8/8 at p_plus 0.5 and 0.75 (Fisher p 0.007 and 0.0014). Observation-step-free models show the same: on the ring 0/8 against 8/8, on the torus 0/8 against 8/8 (0.880 against 1.000). The training curves separate (Fig. 5): the signed median crosses the SOLVED threshold at epoch 480 / 57 / 309 / 183 in four batches and keeps descending, while most |.| runs flatten at a loss of 0.3-0.5.

The cost holds at each step's best learning rate, without weight decay and at 2 layers d 256 (Table 3). With each step at its own best of four learning rates (ring, fresh seeds; the |.| optimum, 3e-3, is bracketed by the grid), |.| solves 2/8 against signed 8/8 (selection-aware p 0.0070 on SOLVED and 0.0057 on accuracy). Without weight decay the ring gives 0/8 against 8/8 (-0.127); at a third of the learning rate, 0/8 against 6/8 (Fisher p 0.007). A 2-layer, d_model 256 torus model with eight times the parameters

gives 2/8 against 8/8 (Fisher p 0.007; 0.967 against 0.999, p 0.0005). Across the learning-rate grid, higher rates hurt the signed step first (8, 8, 4 and 1 of 8 solved at 1e-3, 3e-3, 1e-2 and 3e-2) and do not help |.| past 3e-3 (1, 2, 1 and 0 of 8). In every recipe the |.| models learn little cancelling code: opposite moves cancel mod 2 pi in 0.04-0.13 of their channels (arm means), against 0.43-1.00 for signed steps, including the recipes in which some |.| runs solve.

Table 3. The sign cost by task, model and recipe (observation-step-free models unless marked "full"; n = 8 per arm).

task and model	recipe	|.| accuracy, SOLVED	signed accuracy, SOLVED	test
torus, 1 layer, full	default (lr 1e-3, weight decay 0.05)	0.821, 0/8	0.998, 8/8	perm p 0.0002; Fisher 0.0002
torus, 1 layer	default	0.880, 0/8	1.000, 8/8	perm p 0.0002; Fisher 0.0002
torus, 2 layers, d_model 256	default	0.967, 2/8	0.999, 8/8	perm p 0.0005; Fisher 0.0070
ring, full, p_plus 0.5	default	0.973, 2/8	1.000, 8/8	Fisher 0.0070
ring	default	0.911, 1/8	1.000, 8/8	perm p 0.0002; Fisher 0.0014
ring	weight decay 0	0.873, 0/8	1.000, 8/8	perm p 0.0002; Fisher 0.0002
ring	lr / 3 (3.3e-4)	0.841, 0/8	0.950, 6/8	Fisher 0.0070
ring	each step at its best of four rates (|.| 3e-3, signed 1e-3)	0.963, 2/8	1.000, 8/8	selection-aware p 0.0070 (SOLVED), 0.0057 (accuracy)

Per-rate results, the threefold learning rate and the SOLVED-threshold sensitivity are in App. H.

The non-negative solution exists, and training keeps or re-finds it. Proposition 1 constructs it: each solved signed observation-step-free model is converted exactly into |.| form: the move latents are re-solved to orthogonal directions, each channel's frequency is set to max(|omega|, 1), and each step to the signed angle reduced mod 2 pi. Angles are unchanged mod 2 pi (error at most 1.4e-6 rad), frozen accuracy equals the source's on 8/8 seeds, and the weights stay within about 5x of the signed model's, so the test is posed at the scale training uses. Softplus is constructed the same way, with logits equal to the source's to 2.8e-11 (App. D). On the ring the constructed |.| model, trained for 300 more epochs, stays solved on 8/8 seeds and never leaves the solved region, keeping its cancelling code in the channels attention uses (98%, usage-weighted). On the torus, trained at learning rate 1e-4, it is perturbed early (peak loss 0.11-0.21), re-enters the solved region for good between epochs 119 and 204, and ends solved on 8/8 seeds at accuracy 1.000, equal to its signed control. From scratch the same class solves 0/8 on both tasks (Fig. 4b-c).

How signed steps reach the map (ring). Started from the |.| start, with both moves rotating the same way, signed steps still solve 7/8 (|.| from the same start: 0/8; Fisher p

0.0014). They end with 98-99% of channels opposite-signed (analysis): they cancel opposite moves through zero, which Lemma 1 shows the non-negative step excludes for an inverse pair (Fig. 6). The escape is collective. Freezing the two move latents, with W_out and omega still free, blocks it: 1/8 against 7/8 (Fisher p 0.010): with the latents free the model moves one move's latent toward the negative of the other's and cancels every channel together (82% of channels end opposite-signed, against 15% with the latents frozen). MapFormer described the antipodal end state (their Figs. 3d and 9b); this is the causal test of how it is reached. A |.| step has only the wrapped branch, and on the ring it does learn the map when its opposite-move angles start with sums past pi (8/8, cancelling by wrapping; Fig. 6d); on the torus that start does not help within the budget (App. H).

Published monotone content phases also fail these maps. Faithful CARoPE cannot represent cancellation (Proposition 2, App. E) and scores 0.667 on the torus (0/8), below index RoPE's 0.722 (p 0.040). Faithful CoPE can represent ring cancellation (we construct the solution; App. I) yet solves 0/8 on the ring (0.887 against signed 1.000, p 0.0002), while beating index RoPE by +0.157. It also fails a 256-cell line that the walk never wraps (1/16 solved against signed 8/8), so wrap-around is not its obstacle either. MapFormer's Table 2 reported CoPE failing 1D and 2D navigation and attributed it to the gate's sign; our construction shows that the obstacle is not what CoPE can represent.

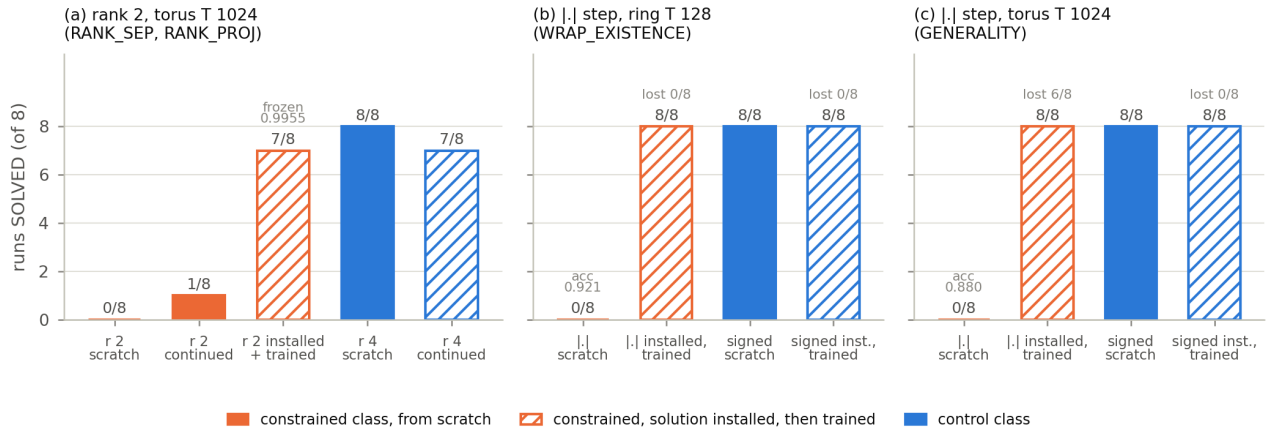


Figure 4 (`fig_existence`). A solution exists in each constrained class and training keeps or re-finds it, but training from scratch does not find it. Bars: runs SOLVED of 8 (hatched: the solution was installed and then trained). (a) Rank 2 on the torus: a rank-2 projection of a solved rank-4 model scores 0.9955 frozen and is held when trained (7/8), while rank 2 from scratch solves 0/8, and 1/8 when the stalled runs are trained further. (b) The |.| step on the ring: an installed exact non-negative solution is held under training (8/8 SOLVED, lost on 0/8), while |.| from scratch solves 0/8 against signed 8/8. (c) The |.| step on the torus: the installed solution is perturbed (loss above 0.05 on 6/8) and re-found (8/8 SOLVED); from scratch 0/8 against 8/8. "Lost": the maximum per-epoch training loss after the start exceeds 0.3, or exceeds 10x the signed control's maximum at that seed and 0.05.

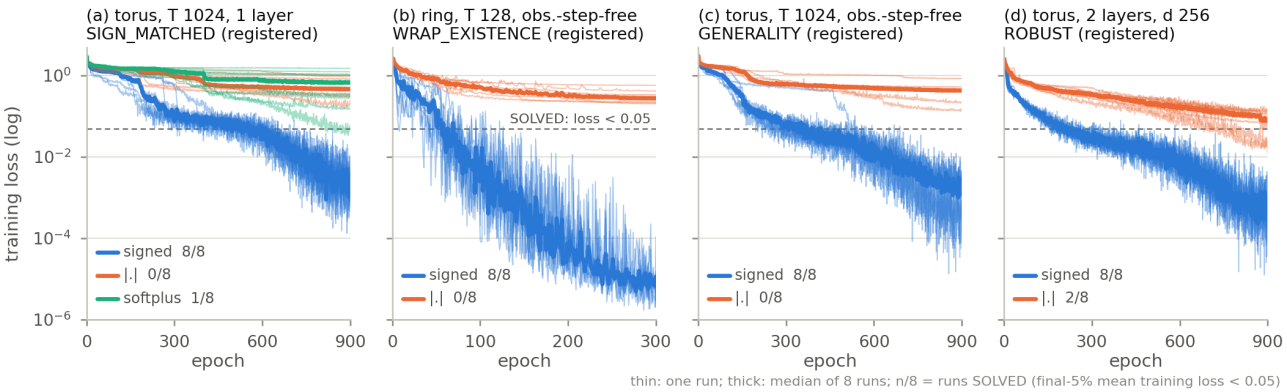


Figure 5 (`fig_loss_curves`). Training loss per epoch for the signed step (blue) and the non-negative steps |.| (orange) and softplus (green), every run of four batches (thin lines; thick: median of 8 runs; log scale; training loss with dropout on). Dashed line: the SOLVED threshold (final-5% mean training loss below 0.05); n/8 = runs SOLVED. (a) Torus, T = 1024, one layer: signed 8/8, |.| 0/8, softplus 1/8. (b) Ring of 32 cells, observation-step-free: 8/8 against 0/8. (c) Torus, observation-step-free: 8/8 against 0/8. (d) As (c) at two layers and d_model 256: 8/8 against 2/8. The signed median crosses the threshold at epoch 480 / 57 / 309 / 183 of 900 / 300 / 900 / 900 and keeps descending; the |.| median never does.

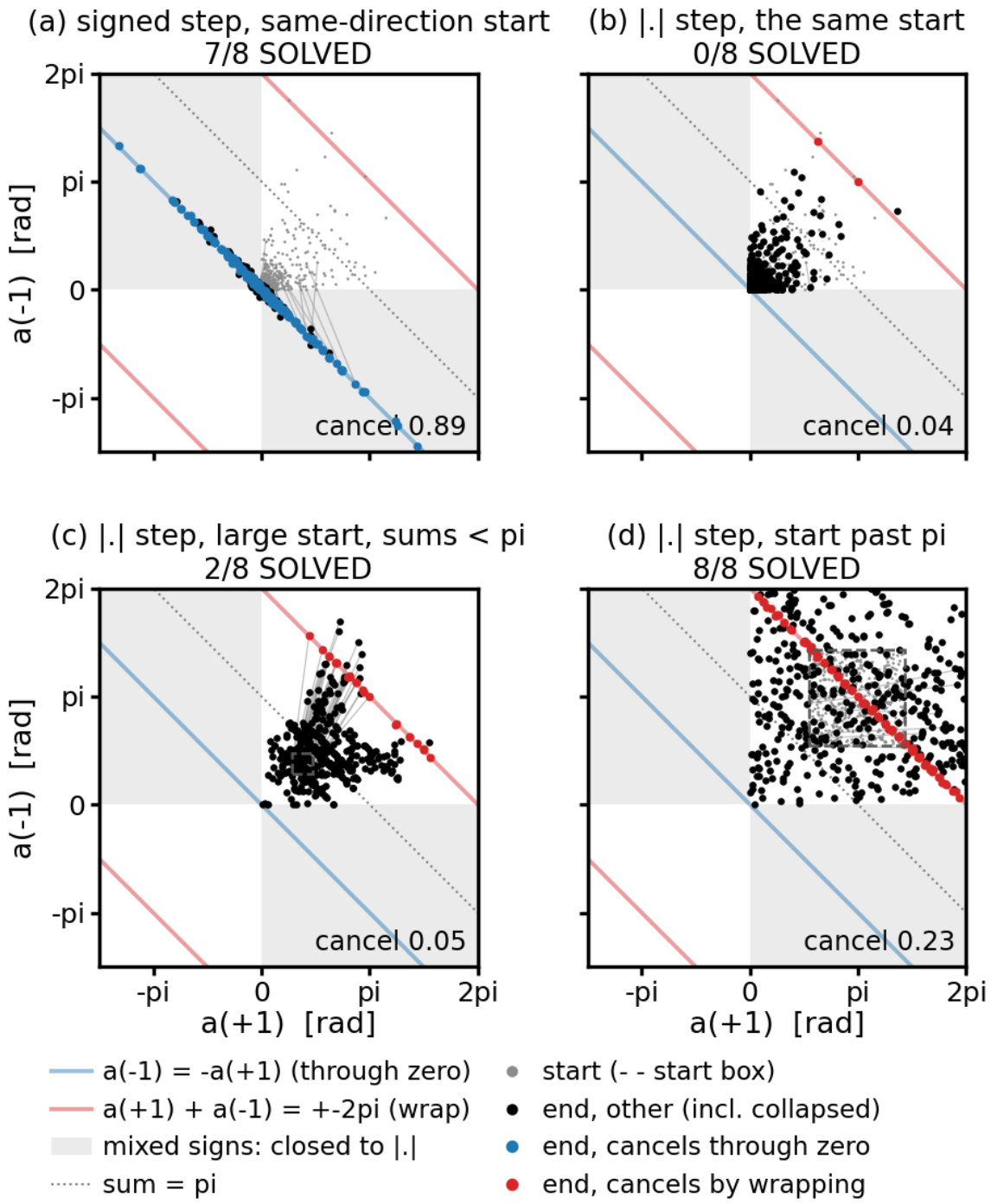


Figure 6 (`fig_mechanism`; ring, 8 runs per panel, analysis of a registered batch). Move angles (a(+1), a(-1)) of every rotary channel at the start (grey) and after training (dots). A channel cancels opposite moves when a(+1) + a(-1) = 0 mod 2 pi: on the blue line (through zero, reached only with opposite signs) or the red line (by wrapping). Grey shading marks the two mixed-sign quadrants, the only region the |.| step cannot reach (omega's sign is free, so both angles may be negative); dotted: a(+1) + a(-1) = pi; dashed boxes: start ranges. (a) Signed step from the same-sign start: 7/8 runs solve; one move's angle crosses zero, and 89% of channels cancel at the end (6% at the start). (b) The |.| step from the same start: 0/8 solve, 4% cancelling. (c) The |.| step from a large start whose sums stay below pi: 2/8 solve, 5% cancelling, all by wrapping. (d) The |.| step from

a large start whose sums exceed pi: 8/8 solve; cancelling rises from 7% to 23% of channels, all by wrapping (App. H).

6.3 One failure signature

Every failing model keeps short-gap revisits and loses long-gap and wrap-only ones (short / long / wrap-only): one-layer index RoPE 0.784 / 0.462 / 0.471; shared rank 2 0.950 / 0.671 / 0.557; |.| 0.870 / 0.630 / 0.515. The solving models are at 0.977-1.000 in every stratum: signed 1.000 / 0.995 / 0.982 and rank 4 1.000 / 0.995 / 0.977 (Fig. 2b-c). The non-negative codes also do not cancel opposite moves: weighted by how much attention uses each channel, signed steps cancel in 70-93% of channels and |.| steps in 2-10% (analysis; App. H). Index attention, a rank-2 map and a non-negative step all end in the same place: a code that serves local revisits and does not carry position over long gaps or around the torus.

7. Maps from words

The text world asks whether the path model can find its actions when nobody marks them. Each step of the torus walk is written as a sentence: a direction word chosen from three synonyms, verbs that do not move the walker, fillers, and asides that name objects without visiting them. Nothing in the input says which words move the walker; the model must learn from the prediction task which words should rotate the phase and by how much (vocabulary and per-seed step tables: App. J).

One path layer learns navigation told in English. With the torus walk rendered as 1024 words (fresh seeds, n = 6), one path layer scores 0.979 +/- 0.048 (6/6 SOLVED) against 0.505 for one-layer index RoPE (0/6), +0.474 (permutation and Fisher p 0.0022); two index layers reach 0.793 (0/6). The floors are 0.505 for the best constant and 0.597 for a reversal-copy rule. Over all eight seeds one-layer index RoPE sits exactly at the constant floor and below the reversal-copy rule on every seed, and two index layers clear both floors but stall on every seed.

It learns which words move it, and opposite directions cancel. On every seed (declared secondaries, n = 8), once the component common to the four direction steps is removed, north + south and west + east cancel to 0.002-0.012 (Fig. 7a). Zeroing the direction words' steps drops every seed to or below the constant floor (0.39-0.51), and synonyms collapse to one step (cosine 1.000) (Fig. 7c). Half the seeds add a per-step clock alongside the map, in 31-38 of 64 channels, and solve as well (Fig. 7b). On those seeds the common component points along the verbs' step (cosine +0.999 to +1.000): the model counts steps in some channels and integrates directions in the others, and zeroing the non-direction words' steps costs it more (0.33-0.69 against 0.05-0.30). The cancellation that signed steps build on the ring (Sec. 6.2) appears here over a vocabulary, learned from the prediction task alone.

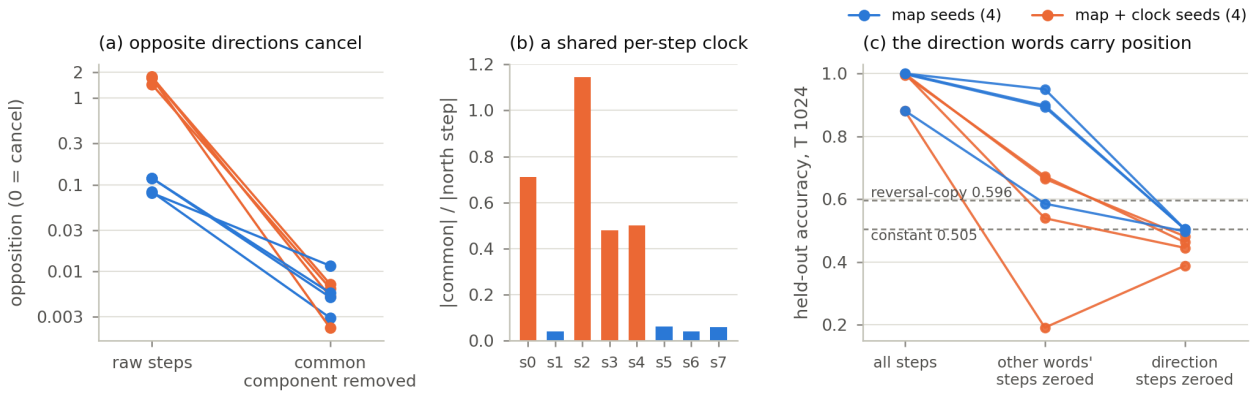


Figure 7 (`fig_textworld`). What a one-layer path model learns from a torus walk told in English (n = 8 seeds). (a) Opposition of the learned direction-word steps, |A + B| / |A| averaged over north/south and west/east (0 = opposites cancel): after removing the component common to all four directions, opposites cancel to 0.002-0.012 on every seed. (b) That common component is small on four seeds (blue, 4-6% of a direction step) and large on the other four (orange, 0.48-1.14), where it points along the verbs' step: a per-step clock layered on the map (31-38 of 64 phase channels drift between visits to the same cell). (c) Held-out accuracy with every step, with the steps of all non-direction words zeroed, and with the direction words' steps zeroed: removing the direction steps drops every seed to the constant floor (0.505, dashed) or below; the clock seeds also depend on the other words. Eval mode without the dropout re-scale (App. C).

8. Discussion

What the two failure axes share. For per-head rank 2 and for non-negative steps, a solution in the class exists (the rank-2 projection scores 0.9955; the |.| construction is exact), training keeps it (7/8 held at rank 2; held 8/8 on the ring and re-found 8/8 on the torus for |.|), and training from scratch does not reach it (0-2/8; 0/8). The obstacle is optimisation, not expressivity. Shakerinava et al. report the same pattern for non-Abelian rotations; we find it in single-layer Abelian tasks, produced by one rank or parameterisation choice. Both failures leave the same trace in the strata: a code that serves short-gap revisits while long-gap and wrap-only revisits fall toward the floor; for the non-negative step we also see directly that opposite moves do not cancel. On the ring we also see the route that the successful parameterisation takes: signed steps move one move's latent toward the negative of the other, cancelling every channel at once, a route Lemma 1 shows is closed to the non-negative step. In the rotary slot, then, the interesting design variables are not those that change what a model can represent, but those that change which representations gradient descent reaches from a small random start.

Design guidance. Keep the phase step signed. Give each head rank above the map's dimension (D + 1). Measure map use at walk lengths where retracing no longer suffices, against measured map-free floors and, where possible, by stratum: the long-gap and wrap-only revisits separate a map from a local rule, and an aggregate accuracy near 0.9 can be either.

What and where. The maps these models learn keep position apart from content in the sense TEM proposed. In converged path models on the short-walk torus, removing object

identity from the score leaves accuracy at 0.974-0.989, and replacing the score by one shared position kernel times a content gain gives 0.988-1.000, while index models fall below the floor (analysis; App. K). MapFormer showed the separation in the learned embeddings (their Fig. 9); these checks show that the trained models rely on it.

The integrator view. In Howard et al.'s terms signed steps make a map and non-negative steps a path-length clock. In the rotary slot these coincide in what they can represent (Proposition 1), so the difference we measure lies in what training finds. The text world shows the other side of the same coin: given a vocabulary in which only some words move, the signed step learns which ones, and that their opposites cancel.

Text. On ordinary text, at small scale (analysis, n = 3), the learned phase behaves as a token clock: it grows linearly with token index (R^2 0.994-0.999 on TinyStories, Eldan and Li 2023, and on code), and MapWM matches index RoPE (1.5053 against 1.5050 nats per token on TinyStories), consistent with there being no map to build.

9. Limitations

- Small models (1-4 layers, 2 heads, d_model 128-256) on synthetic map tasks; natural text is touched only at small scale.
- "Not found" is relative to the stated budgets (300-1800 epochs). Learning rates were swept on the ring, while the torus was run at one rate and two model sizes; in the larger torus model the accuracy gap is a quarter of the one-layer gap and most $|.|$ runs were still descending, so at scale part of the sign cost may be speed.
- Rank results use two heads, and rank equal to the map's dimension is hard in 2D and 3D but not in 1D. Our latent is shared across heads where MapFormer's is per head; per-head arms are reported alongside.
- We show how signed steps reach the map; why the non-negative parameterisations stall from small starts is not identified.
- No recurrent or SSM baselines. The torus exchange rate is a lower bound (more than four layers), measured at n = 4.

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Appendix to "Representable but not found: what path-integrated rotary attention learns, and how it fails"

Draft 1 (2026-10-18), for `docs/paper/MAIN_DRAFT1.md`. The full theory appendix, with every check script and the results not used here (P1(c)-(e), P2, P5, P7, Remark R), is `docs/paper/APPENDIX_theory_v3.md`; App. E below states and proves what the main text uses and gives the v3 label of each statement. Every number is copied from the file named beside it; a CORRECTED or AUDIT block at the top of a results file supersedes its body. "Registered" = a pre-registered decision or declared secondary; "(analysis)" = computed after the batch on stored streams or checkpoints, never changing a verdict. Delete this paragraph before submission.

Contents. A: models, recipes, batches. B: floors. C: per-seed tables, SOLVED-threshold sensitivity, evaluation scale. D: constructions. E: theory. F: exchange-rate details. G: rank details. H: sign details. I: CARoPE and CoPE. J: text world. K: what and where. L: PoPE's score.

A. Models, recipes, batches

A1. Architecture

component	setting
layers	1 (Sec. 5: 1-4 real layers and one block looped x4; ROBUST torus: 2 layers, d_model 256)
heads, width	2 heads, d_model 128, d_head 64; feed-forward 4 x d_model with GELU; LayerNorm before attention and before the feed-forward; dropout 0.1 on attention weights, attention output and feed-forward output
rotary channels	32 two-dimensional blocks per head, 64 channels; every query and key coordinate is rotated
score	channel c contributes q_c^T R(theta_{t',c} - theta_{t,c}) k_c (content Q and K rotated by the path angle; not additive)
step map	Delta = W_out W_in x, W_in: 128 -> r, W_out: r -> 64, no bias; r = 4 shared across heads unless stated (MapFormer's latent is per head); per-head variants: one W_in / W_out block per head; block-diagonal rank 4: the shared rank-4 latent with the off-diagonal blocks of W_out held at zero (each head reads rank 2)
frequencies	omega_c = 2 pi (1 / Delta_max)^(i / 31) for block i = 0..31, the same ladder in each head; Delta_max = grid size (torus 64, ring 32, line 256); omega is a free parameter, its sign unconstrained
vocabulary	torus: 4 moves + 17 observation tokens (16 types + blank); ring and line: 2 moves + 17
parameters	about 204k for the one-layer torus MapWM; 798,741 for four index layers; 203,925 for the looped block (`FACTORIAL_RESULTS.md`); 1.59M for the ROBUST torus model

Step parameterisations (`model.py`, `model_sign.py`, `model_wrap_exist.py`): signed (`Signed_r4` = `Vanilla_r4`), |.| (`Abs_r4`; the absolute value is applied after W_out on each of the 64 channel steps, adding no parameters), softplus (`Pos_r4`; softplus(0) = 0.693, so every token starts at about the same step), observation-step-free (`SignedAO_r4`, `AbsAO_r4` on the ring,

`SignedAO4_r4`, `AbsAO4_r4` on the torus: every non-move token's step is 0 by construction). Index RoPE (`RoPE`): canonical ladder base^(-2c / d_head), base 10000.

A2. Recipes

	torus (Secs. 4-6, 7)	**ring and line (Secs. 5, 6)**	**Dyck-2 (Sec. 5)**
optimiser	AdamW, weight decay 0.05	AdamW, weight decay 0.05	AdamW, weight decay 0.01
peak learning rate	1e-3	1e-3 (ROBUST and LRSWEEP vary it)	1e-4
schedule	5% linear warm-up, cosine to 10% of the peak	same	same
batch	16 walks of T = 1024 moves (2,048 tokens); T = 128 batches of 128	128 walks of T = 128	128 sequences
budget	98 batches per epoch, 900 epochs (1800 in Sec. 6.1; 300 at T = 128)	98 batches per epoch, 300 epochs	560,000 sequences (1x) or 1,680,000 (3x)
loss	cross-entropy on revisited observations	same	next-token cross-entropy on every position
training map	one per run	redrawn for every walk	--
evaluation	100 walks on one held-out map, eval mode	400 walks, maps redrawn	L32 D12 test set

Run classes (`stats_core.classify_run`, per-epoch training loss): SOLVED if the mean of the final 5% of epochs is below 0.05; otherwise STALLED if the final-10% mean is within 5% of the preceding 10% mean, else DESCENDING. Continuations of constructed models (App. D): ring 300 epochs at lr 1e-3 (schedule re-warmed, fresh optimiser state, fresh walk stream); torus 300 epochs at lr 1e-4, fixed before launch (about the rate at which the source solution was reached).

A3. Batches

"Code check" = an independent, results-blind code review, whose findings were amended into the pre-registration before any result was read (before launch, except TEXTWORLD and LEAK, checked while the batch trained): a separately launched read-only AI code-review session given the pre-registration and the code but no results. "Reproduced" = the analysis was re-run from stored checkpoints by an independent reproduction audit (`docs/paper/review_draft5/REPRO_AUDIT.md`, `REPRO_AUDIT_nonsign.md`, 2026-10-16) and matched byte for byte (spot re-evaluations within 7e-4). "Attention": standard, or `--fast-attn` (SDPA; agrees to ~1e-6 in logits, not bit-identical in training). Pilots ran on seeds outside the batch except where noted.

batch	**main text**	**task**	**arms**	**seeds**	**epochs**	**attention**	**code check**	**reproduced**
PAPER2X2	4	torus T = 128	RoPE, PoPE, MapWM r2 / r4, MapPoPE r2 / r4	0-7	300	standard	no	yes

batch	main text	task	arms	seeds	epochs	attention	code check	reproduced
SIGN_MATCHED	4, 6.2	torus T = 1024	signed, |.|, softplus, RoPE	0-7	900	standard	no (authors' pre-launch gate)	yes
GENERALITY	4, 6.2	torus; ring p_plus 0.5	obs.-step-free signed / |.| and constructions; signed, CARoPE, CoPE, RoPE	700-707	900 (ring 300); +300	fast-attn (torus)	yes	yes
FACTORIAL	5	torus T = 1024	path 1 layer; index 1, 4 layers, looped x4	500-503 (n = 4)	900	fast-attn	yes	yes
DYCK_MDEPTH	5	Dyck-2 L32 D12	MapWM, MapPoPE (r2), RoPE, PoPE at 1-4 layers; 3x; mixture	0-7	1x / 3x	standard	no	yes
CANCEL	5	ring p_plus 0.5-1.0	path 1 layer; index 1-3 layers	0-7; main text 2-7 (0-1 were the pilot)	300	standard	no	yes
RANK_MI	6.1	torus	shared r2, per-head r2, shared r4	0-7 (per-head r2 s0-1 were a pilot)	900	standard	no	yes
RANK_SEP	6.1	torus	block-diagonal r4, per-head r4	0-7	900	standard	no	yes
RANK3	6.1	torus	per-head r3	0-7	900	standard	no	yes
LOOP_RANK_E1800 (part 1)	6.1	torus	shared r2, shared r4	0-7	1800	standard	no	yes
RANK_PROJ	6.1	torus	projected r2 (frozen, trained), r4 control, r2 continued	0-7	+900	standard	no	yes
RANK_ND	6.1	2D (32 per side), 3D (10 per side) tori	per-head rank D, D + 1	0-7	900	standard	yes	yes
RANK_WRAP	6.1	2D and 3D tori	per-head rank D + 1	0-7	900	standard	yes	yes

batch	main text	task	arms	seeds	epochs	attention	code check	reproduced
		of two sizes						
RANK_TESTS	6.1	ring; torus escape test	rank 1, 2 (+ variants); lifted / fresh rank 4	20-27; 0-7	300; 600	standard	yes	yes
RING_DIAL	6.2	ring p_plus 0.5 / 0.75 / 0.9	signed, |.|	30-37	300	standard	yes	yes
WRAP_EXISTENCE	6.2	ring p_plus 0.5	obs.-step-free signed / |.|, constructions	600-607	300 + 300	standard	yes	yes
LRSWEEP	6.2	ring	obs.-step-free signed / |.| at 1e-3 / 3e-3 / 1e-2 / 3e-2	1500-1507	300	standard	yes	no (windowed run classes cross-checked)
ROBUST	6.2	ring (four recipes); torus 2 layers d 256	obs.-step-free signed / |.|	1400-1407	300; 900	fast-attn (torus)	yes	yes
ACCOUNTS	6.2	ring; line	T1 continuations; T2 frozen latents; T4 CoPE on a line	600-607; 1000-1007; 1100-1115	300	standard	yes	yes
MECHANISM	6.2	ring p_plus 0.5	eight start / step arms	800-807	300	standard	yes	yes
TEXTWORLD	7	torus in words	path 1 layer; index 1, 2 layers	0-7; main text 2-7 (0-1 were the pilot)	900	standard	yes	yes
RING_BUDGET (H)	--	ring p_plus 0.5	|.| at 300 and 900 epochs	40-47	300 / 900	standard	yes	yes
TORUS_START (H)	--	torus	start arms	1200-1207	900	fast-attn	yes	yes
GAUGE (H)	--	ring, moves +1/-1 or +1/+2	obs.-step-free signed / |.|, RoPE	1300-1307	300	standard	yes	yes
WRAP_COPE (I)	--	ring	CoPE table starts	900-907	300	standard	yes	yes
LEAK (K)	--			0-7	900	standard	yes	yes

batch	main text	task	arms	seeds	epochs	attention	code check	reproduced
		new-object torus	MapWM, ActOnly, NormStep					
GAIN_PHASE (K)	--	new-object torus	step x score 2x2	8-15	900	standard	yes	no
MAPPOPE_PAIR (L)	--	torus T = 128	MapWM r2, MapPoPE r2 (32 / 64 angles)	10-25	300	standard	yes	no

ACCOUNTS T1's weight-decay-0.05 arms are bit-identical retrains of WRAP_EXISTENCE's constructed arms (a second readout of the same runs, not a replication). The shared rank-2 arm of every rank contrast is one stored set of eight runs.

A4. Reproduction of MapFormer v4 Table 2 (2D)

With the paper's recipe (16 epochs x 98 batches, linear decay, lr 3e-4; `python3 -m mapformer.main --device cuda --epochs 16 --n-batches 98`), our MapWM scores 0.989 +/- 0.010 and our MapEM (single p_0) 0.987 +/- 0.012 on fresh maps at T = 128 (`PAPER_TASK_ACCURACY.md`), against 1.00 / 1.00 in v4 and 0.99 / 1.0 in v1 (MapWM-r2, MapEM-os, 2D IID).

B. Floors

A floor is the accuracy of the best map-free predictor on the same kind of evaluation walk. Rows marked (analysis) were computed after the batches.

task	predictor	value	stream	source
torus T = 128 (analysis)	blank / n-gram (orders 0-5) / retrace	0.507 / 0.598 / 0.934 (retrace per seed 0.928-0.946); wrap-only share 0.000	PAPER2X2 eval walks, seeds 0-7	`docs/audits/2026-10-16/paper2x2_T128_floors_out.txt`
torus T = 1024 (analysis)	blank / n-gram / retrace	0.506 / 0.575 / 0.843 (retrace per seed 0.837-0.849); wrap-only share 0.079	SIGN_MATCHED eval walks, seeds 0-7 (shared by RANK_MI, RANK_SEP, RANK3)	`docs/audits/2026-10-16/sign_floors_out.txt`
torus T = 1024 (analysis)	same	0.508 / 0.578 / 0.843; wrap-only 0.081	GENERALITY, seeds 700-707	`generality_floors_out.txt` (same directory)
torus T = 1024, 2 layers d 256 (analysis)	same	0.509 / 0.578 / 0.844; every run of both arms above retrace (lowest |.| 0.910)	ROBUST, seeds 1400-1407	`robust_floors_out.txt`

task	predictor	value	stream	source
torus T = 1024 (analysis)	blank / retrace	0.509 / 0.844	FACTORIAL, seeds 500-503	`docs/paper/figs/fig_factorial_strata_out.txt`
ring p_plus 0.5 / 0.75 / 0.9 (analysis)	retrace	0.860 / 0.758 / 0.612 (blank 0.497-0.499)	2000 walks per bias	`docs/audits/2026-10-16/ring_floors_out.txt`
ring p_plus 0.5 / 0.75 / 0.9 / 1.0	better of best constant and n-gram (orders 1-5)	0.554 / 0.544 / 0.520 / 0.498	1000 held-out walks	`CANCEL_PREREG.md`, `CANCEL_RESULTS.md`
line (256 cells)	blank	0.503	ACCOUNTS T4	`ACCOUNTS_RESULTS.md`
ring, moves +1/+2	blank (retrace reduces to it)	~0.50	GAUGE	`GAUGE_RESULTS.md`
torus in words	best constant / reversal-copy	0.505 / 0.597	TEXTWORLD eval set	`TEXTWORLD_RESULTS.md`
Dyck-2 L32 D12	best stack-free A2f floor (n-grams of orders 1-6)	0.594	gate G4	`DYCK_MDEPTH_RESULTS.md`

Per-stratum floors (analysis). Torus, seeds 0-7 (`docs/paper/figs/torus_strata_floors_s0-7.json`): best constant 0.506 / 0.500 / 0.507 (short gap / long gap / wrap-only), retrace 0.911 on short gaps and equal to the constant on the others (walk runs are at most 10 moves, so a retrace never reaches a gap of 128 moves or a wrap-only revisit). FACTORIAL's stream: 0.508 / 0.515 / 0.517, retrace 0.912. Ring (`ring_strata_floors_out.txt`): retrace 0.9998 at gaps 2-7, 0.943 at 8-31, 0.570 at 32-127, 0.493 wrap-only; constant 0.49-0.50 in every stratum.

Against its own seed's retrace floor (analysis, torus T = 1024): signed falls below on 0/8 runs, |.| on 3/8 (0.637, 0.690, 0.717), softplus on 4/8, index RoPE on 8/8.

C. Per-seed tables

Accuracy at the training length on held-out maps, eval mode. SOLV / STAL / DESC = SOLVED / STALLED / DESCENDING; a number in brackets is the final-5% training loss.

C1. Torus sign batch (Secs. 4, 6.2; `SIGN_MATCHED_ANALYSIS.txt`)

seed	signed	|.|	softplus	index RoPE
0	1.000 SOLV (0.003)	0.931 DESC (0.201)	0.590 STAL (1.250)	0.730 STAL (0.983)
1	0.997 SOLV (0.003)	0.917 DESC (0.314)	0.527 STAL (1.510)	0.729 STAL (0.939)

seed	signed	|.|	softplus	index RoPE
2	1.000 SOLV (0.005)	0.868 STAL (0.492)	0.630 STAL (1.084)	0.720 STAL (0.931)
3	1.000 SOLV (0.008)	0.637 STAL (1.043)	0.972 DESC (0.158)	0.739 STAL (0.910)
4	1.000 SOLV (0.004)	0.690 DESC (0.962)	0.674 STAL (1.009)	0.737 STAL (0.928)
5	0.990 SOLV (0.003)	0.888 STAL (0.446)	0.935 DESC (0.295)	0.744 STAL (0.885)
6	1.000 SOLV (0.004)	0.922 STAL (0.349)	0.922 DESC (0.345)	0.722 STAL (0.956)
7	1.000 SOLV (0.004)	0.717 STAL (0.834)	0.995 SOLV (0.049)	0.728 STAL (0.976)
mean	0.998	0.821	0.781	0.731

Registered: |.| - signed -0.177 (permutation p 0.0002; 0/8 vs 8/8, Fisher p 0.0002); softplus - signed -0.218 (p 0.0003; 1/8 vs 8/8, p 0.0014); signed - RoPE +0.267 (p 0.0002; 8/8 vs 0/8, p 0.0002). r(final loss, accuracy) -0.992 over 32 runs.

C2. Torus, index depth (Sec. 5; `FACTORIAL_{L1,L4,LOOP}.json`)

seed	path, 1 layer	index, 1 layer	index, 4 layers	index, looped x4
500	1.000	0.719	0.814	0.868
501	1.000	0.731	0.870	0.861
502	1.000	0.688	0.874	0.860
503	1.000	0.739	0.881	0.868

Every path run SOLVED, every index run STALLED. Registered C0-C3: path beats index at one layer (+0.281), four layers (+0.141) and loop x4 (+0.136); depth and loop each help index (+0.140, +0.145); every firing a complete separation (p 0.029).

C3. Ring exchange rate, fresh seeds 2-7 (Sec. 5; `docs/audits/2026-09-27/fresh_seeds.txt`)

p_plus	path 1 layer	index 1 layer	gap to path at 1 / 2 / 3 index layers
0.5	1.000	0.710 +/- 0.020	+0.290 / +0.051 / +0.002
0.75	1.000	0.677 +/- 0.009	+0.323 / +0.038 / +0.002
0.9	1.000	0.824 +/- 0.007	+0.176 / +0.013 / +0.001
1.0	1.000	1.000	0 / 0 / 0

All eight seeds (`CANCEL_RESULTS.md`): index 1 / 2 / 3 layers 0.717 / 0.954 / 0.998 (p_plus 0.5), 0.676 / 0.969 / 0.998 (0.75), 0.825 / 0.988 / 0.999 (0.9), 1.000 (1.0); path 1.000 in every cell (32/32 SOLVED); registered exchange rate k = 3, 3, 3, 1 index layers.

C4. Rank (Sec. 6.1; `RANK_MI_ANALYSIS.txt`, `RANK_SEP_ANALYSIS.txt`, `RANK3_ANALYSIS.txt`, `RANK_SEP.json`, `RANK3.json`)

seed	shared r2	per-head r2	block-diag r4	per-head r3	shared r4	per-head r4
0	0.849 STAL (0.587)	0.656 STAL (1.049)	0.879 STAL (0.482)	0.942 DESC (0.243)	1.000 SOLV (0.004)	0.996 SOLV (0.004)
1	0.998 DESC (0.051)	1.000 SOLV (0.007)	0.936 DESC (0.302)	0.990 SOLV (0.003)	1.000 SOLV (0.003)	1.000 SOLV (0.010)

seed	shared r2	per-head r2	block-diag r4	per-head r3	shared r4	per-head r4
2	0.917 DESC (0.308)	1.000 SOLV (0.003)	1.000 SOLV (0.010)	1.000 SOLV (0.003)	1.000 SOLV (0.003)	1.000 SOLV (0.004)
3	0.917 DESC (0.321)	0.723 STAL (0.910)	0.945 DESC (0.234)	0.967 DESC (0.134)	1.000 SOLV (0.003)	1.000 SOLV (0.004)
4	0.947 DESC (0.177)	0.959 DESC (0.194)	0.964 DESC (0.202)	1.000 SOLV (0.003)	1.000 SOLV (0.002)	0.998 SOLV (0.002)
5	0.907 STAL (0.404)	0.949 DESC (0.211)	0.917 DESC (0.310)	1.000 SOLV (0.006)	0.994 SOLV (0.004)	1.000 SOLV (0.008)
6	0.773 STAL (0.675)	0.943 DESC (0.248)	1.000 SOLV (0.010)	1.000 SOLV (0.003)	1.000 SOLV (0.004)	1.000 SOLV (0.005)
7	0.842 STAL (0.605)	0.851 STAL (0.539)	0.947 DESC (0.166)	1.000 SOLV (0.005)	0.987 SOLV (0.010)	1.000 SOLV (0.003)
mean	0.894, 0/8	0.885, 2/8	0.948, 2/8	0.987, 6/8	0.998, 8/8	0.999, 8/8

Per-head r2 seeds 0-1 were a pilot run before registration (1/6 SOLVED on seeds 2-7). Registered: per-head rank 4 vs block-diagonal 8/8 vs 2/8 (Fisher and permutation p 0.0070); shared r4 vs block-diagonal the same (it also changes per-head rank); sharing (per-head r4 vs shared r4, +0.001) and W_out scale (block-diagonal vs per-head r2, +0.063) not detected; per-head r3 vs r2 +0.102 (permutation p 0.027; SOLVED 6/8 vs 2/8, Fisher p 0.13); r4 vs r3 +0.012, not detected. Shared r4 vs shared r2: 8/8 vs 0/8 (Fisher p 0.0002), +0.104 (p 0.0005).

Twice the budget (`LOOP_RANK_E1800_P1_ANALYSIS.txt`, 1800 epochs from scratch): shared r2 0/8 (final-5% losses 0.489, 0.383, 0.390, 0.340, 0.141, 0.231, 0.305, 0.571; 3 STALLED, 5 DESCENDING), accuracy 0.908; shared r4 7/8 (s5 STALLED at 0.180), 0.994.

C5. Sign, ring and torus (Sec. 6.2)

Ring dial (`RING_DIAL_VERDICTS.json`; full models, S = SOLVED):

seed	0.5 signed	0.5 |.|	0.75 signed	0.75 |.|	0.9 signed	0.9 |.|
30	1.000 S	1.000 S	1.000 S	0.994 S	1.000 S	1.000 S
31	1.000 S	0.977	1.000 S	0.958	1.000 S	0.999 S
32	1.000 S	0.987	1.000 S	0.957	1.000 S	0.885
33	1.000 S	0.964	1.000 S	0.954	1.000 S	1.000 S
34	1.000 S	1.000 S	1.000 S	0.964	1.000 S	1.000 S
35	1.000 S	0.944	1.000 S	0.967	1.000 S	0.825
36	1.000 S	0.977	1.000 S	0.993	1.000 S	0.960
37	1.000 S	0.931	1.000 S	0.958	1.000 S	0.964

Registered: p_plus 0.5 -0.028 (permutation p 0.007), 2/8 vs 8/8 (Fisher p 0.0070); 0.75 -0.032 (p 0.0002), 1/8 vs 8/8 (p 0.0014); at 0.9 the contrast did not reach the registered threshold (-0.046, below the 0.05 materiality; 4/8 vs 8/8, p 0.077).

Observation-step-free ring from scratch and constructed (`WRAP_EXISTENCE_ANALYSIS.txt`):

seed	signed	|.|	constructed |.|: frozen -> trained (max continuation loss)	constructed signed: frozen -> trained
600	1.000 S	0.905 STAL	1.000 -> 1.000 S (0.041)	1.000 -> 1.000 S (0.019)
601	1.000 S	0.949 STAL	1.000 -> 1.000 S (0.030)	1.000 -> 1.000 S (0.042)
602	1.000 S	0.853 STAL	1.000 -> 1.000 S (0.154)	1.000 -> 1.000 S (0.072)
603	1.000 S	0.951 STAL	1.000 -> 1.000 S (0.046)	1.000 -> 1.000 S (0.017)
604	1.000 S	0.907 STAL	1.000 -> 1.000 S (0.024)	1.000 -> 1.000 S (0.039)
605	1.000 S	0.930 STAL	1.000 -> 1.000 S (0.032)	1.000 -> 1.000 S (0.028)
606	1.000 S	0.950 STAL	1.000 -> 1.000 S (0.031)	1.000 -> 1.000 S (0.112)
607	1.000 S	0.922 STAL	1.000 -> 1.000 S (0.036)	1.000 -> 1.000 S (0.015)

Registered: from scratch 0.921 vs 1.000, -0.079 (permutation p 0.0002), 0/8 vs 8/8 (Fisher p 0.0002); constructed |.| HELD (8/8 SOLVED, 0/8 LOST).

Observation-step-free torus (`GENERALITY_ANALYSIS.txt`):

seed	signed	|.|	constructed signed (lr 1e-4)	constructed |.| (lr 1e-4): peak loss, epoch of peak / of re-entry
700	1.000 S	0.874 STAL	1.000 S	1.000 S; 0.168, 29 / 153
701	1.000 S	0.707 STAL	1.000 S	1.000 S; 0.110, 40 / 121
702	1.000 S	0.946 DESC	1.000 S	1.000 S; 0.173, 19 / 204
703	1.000 S	0.879 STAL	1.000 S	1.000 S; 0.143, 52 / 149
704	1.000 S	0.892 STAL	1.000 S	1.000 S; 0.131, 13 / 119
705	1.000 S	0.907 STAL	1.000 S	1.000 S; 0.155, 13 / 148
706	1.000 S	0.877 STAL	1.000 S	1.000 S; 0.208, 36 / 164
707	1.000 S	0.960 DESC	1.000 S	1.000 S; 0.117, 23 / 139

Registered: from scratch 0.880 vs 1.000, -0.120 (p 0.0002), 0/8 vs 8/8; the constructed |.| model leaves the solved region on 6/8 seeds (peak loss 0.11-0.21; signed control 0.004-0.017) and ends SOLVED on 8/8 at 1.000: by the reading fixed before launch, perturbed and re-found.

Recipe and scale (`ROBUST_ANALYSIS.txt`, seeds 1400-1407; accuracy at the training length, class):

```
ring  seed | def:SO | def:AO | wd0:SO | wd0:AO | lrx3:SO | lrx3:AO | lrd3:SO | lrd3:AO
1400 | 1.0000 SOLV | 0.8230 STAL | 1.0000 SOLV | 0.9078 DESC | 1.0000 SOLV | 1.0000 DESC |
1.0000 SOLV | 0.8893 STAL
1401 | 1.0000 SOLV | 0.9155 STAL | 1.0000 SOLV | 0.6239 STAL | 1.0000 SOLV | 0.9596 STAL |
0.6194 STAL | 0.6221 STAL
1402 | 1.0000 SOLV | 0.9237 STAL | 1.0000 SOLV | 0.8878 STAL | 1.0000 SOLV | 0.9702 STAL |
1.0000 SOLV | 0.8319 STAL
1403 | 1.0000 SOLV | 0.9136 STAL | 1.0000 SOLV | 0.8800 STAL | 1.0000 SOLV | 0.9995 SOLV |
1.0000 SOLV | 0.9167 STAL
1404 | 1.0000 SOLV | 0.9118 STAL | 1.0000 SOLV | 0.9119 STAL | 1.0000 SOLV | 1.0000 SOLV |
1.0000 SOLV | 0.8313 STAL
1405 | 1.0000 SOLV | 0.8791 STAL | 1.0000 SOLV | 0.8858 STAL | 1.0000 SOLV | 0.9650 STAL |
1.0000 SOLV | 0.8526 STAL
1406 | 1.0000 SOLV | 0.9211 DESC | 1.0000 SOLV | 0.9110 STAL | 1.0000 SOLV | 1.0000 SOLV |
0.9794 STAL | 0.9062 STAL
1407 | 1.0000 SOLV | 0.9997 SOLV | 1.0000 SOLV | 0.9735 DESC | 1.0000 SOLV | 0.9998 SOLV |
1.0000 SOLV | 0.8801 STAL
torus 1400 SO 1.0000 SOLV | AO 0.9941 DESC     1404 SO 1.0000 SOLV | AO 0.9935 SOLV
      1401 SO 1.0000 SOLV | AO 0.9981 SOLV     1405 SO 1.0000 SOLV | AO 0.9706 DESC
      1402 SO 0.9941 SOLV | AO 0.9705 DESC     1406 SO 1.0000 SOLV | AO 0.9098 DESC
      1403 SO 1.0000 SOLV | AO 0.9454 DESC     1407 SO 1.0000 SOLV | AO 0.9544 DESC
```

Learning-rate sweep (`LRSWEEP_RESULTS.md`, seeds 1500-1507): per-seed table in H2.

C6. Text world (Sec. 7; `TEXTWORLD_ANALYSIS.txt`)

seed	path, 1 layer	index, 1 layer	index, 2 layers
0 (pilot)	0.881	0.504	0.695
1 (pilot)	0.998	0.504	0.723
2	0.999	0.503	0.927
3	0.995	0.506	0.722
4	0.999	0.505	0.926
5	1.000	0.506	0.706
6	1.000	0.506	0.760
7	0.882	0.504	0.720

Fresh seeds 2-7 (`docs/audits/2026-09-27/fresh_seeds.txt`): path 0.979 +/- 0.048 (6/6 SOLVED), index 1 layer 0.505 (0/6), index 2 layers 0.793 (0/6); path - index one layer +0.474, permutation and Fisher p 0.0022. All eight seeds: path 0.969 +/- 0.054 (7/8; seed 0 still descending), +0.464 (p 0.0002; Fisher p 0.0014); path - index two layers +0.197 (p 0.0009).

C7. Dyck-2

Per-seed values are in `DYCK_MDEPTH_RESULTS.json` (cells `T12|{RoPE,PoPE,MapWM,MapPoPE}|{1-4}L|L32D12`); every one of the eight seeds has a positive position effect at every depth (sign-flip p 0.0078 per depth).

C8. SOLVED-threshold sensitivity

Sign batches (`docs/audits/2026-10-15/reanalyses/solved_thresholds_out.md`; predictions and decision rules written before the re-analysis; two-sided Fisher p in brackets):

batch	contrast	loss < 0.02	loss < 0.05 (registered)	loss < 0.1	accuracy >= 0.99
SIGN_MATCHED torus	|.| vs signed	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	0/8 vs 7/8 (0.0014)
SIGN_MATCHED torus	softplus vs signed	0/8 vs 8/8 (0.0002)	1/8 vs 8/8 (0.0014)	1/8 vs 8/8 (0.0014)	1/8 vs 7/8 (0.0101)
RING_DIAL p 0.5	|.| vs signed	2/8 vs 8/8 (0.0070)	2/8 vs 8/8 (0.0070)	4/8 vs 8/8 (0.0769)	2/8 vs 8/8 (0.0070)
RING_DIAL p 0.75	|.| vs signed	1/8 vs 8/8 (0.0014)	1/8 vs 8/8 (0.0014)	2/8 vs 8/8 (0.0070)	2/8 vs 8/8 (0.0070)
WRAP_EXISTENCE ring	|.| vs signed	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)
MECHANISM ring	|.| vs signed, same-direction start	0/8 vs 7/8 (0.0014)	0/8 vs 7/8 (0.0014)	0/8 vs 7/8 (0.0014)	0/8 vs 7/8 (0.0014)
ACCOUNTS T2 ring	frozen vs free move latents	1/8 vs 7/8 (0.0101)	1/8 vs 7/8 (0.0101)	1/8 vs 7/8 (0.0101)	1/8 vs 7/8 (0.0101)
GENERALITY torus	|.| vs signed	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)
GENERALITY ring	CoPE vs signed	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	1/8 vs 8/8 (0.0014)	0/8 vs 8/8 (0.0002)

ROBUST (declared secondary, loss < 0.1): ring default 1/8 vs 8/8 (p 0.0014), weight decay 0 1/8 vs 8/8 (0.0014), lr / 3 0/8 vs 7/8 (0.0014), lr x3 4/8 vs 8/8 (0.077), torus 2 layers d 256 5/8 vs 8/8 (0.20); on that torus cell the accuracy contrast (0.967 vs 0.999, p 0.0005) carries the result. LRSWEEP (loss < 0.1, selection redone): 2/8 vs 8/8, same label.

Rank batches (analysis; read from the per-seed final-5% losses in C4): at loss < 0.02 the counts are 0 / 2 / 2 / 6 / 8 / 8 of 8 (shared r2, per-head r2, block-diagonal r4, per-head r3, shared r4, per-head r4), identical to the registered 0.05; at loss < 0.1 they are 1 / 2 / 2 / 6 / 8 / 8 (shared r2 seed 1 at 0.051 enters).

C9. Attention-dropout evaluation scale

Models trained with attention dropout expect the 1/(1 - p) attention scale; evaluated with it, below-ceiling one-layer runs score higher. The registered text-world evaluation re-run with the scale (`docs/audits/2026-10-04/DROPOUT_RESCORE.md`, all 24 runs) gives path 0.969 -> 0.993, index one layer 0.505 -> 0.506, index two layers 0.772 -> 0.772; path - index one layer +0.464 -> +0.488 (p 0.0002). The re-score was a declared secondary of RING_DIAL, RING_BUDGET, WRAP_EXISTENCE, GENERALITY, MECHANISM and WRAP_COPE and changed no label there; it was applied to SIGN_MATCHED by the same audit with no verdict changed. ACCOUNTS, TORUS_START, GAUGE and ROBUST were not re-scored (the correction does not apply at two layers). All main-text accuracies are eval mode without the scale.

D. Constructions

D1. The rank-2 projection (Sec. 6.1; `RANK_PROJ_RESULTS.md`)

Each solved shared rank-4 model (seeds 0-7, a stored batch) is projected onto the top two directions of its action latent over the vocabulary (top-2 spectral energy 0.99952-0.99994), every other weight kept, and installed in the shared rank-2 architecture. Frozen, the projections score 0.9955 at T = 1024 against their sources' 0.9966 (at least 0.981 on every seed; per seed 0.9993, 1.0000, 0.9988, 1.0000, 0.9850, 0.9806, 1.0000, 1.0000). Trained 900 epochs from a learning-rate restart at 1e-3, 7/8 are SOLVED (one DESCENDING), as for the rank-4 control trained from its own solution (7/8); rank 2 continued from its own 900-epoch weights solves 1/8. Registered verdict: STABLE.

D2. Per channel (Lemma P1.0 in v3)

For a signed step Delta and omega != 0, Delta' = Delta mod (2 pi / |omega|) lies in [0, 2 pi / |omega|) and omega Delta' = omega Delta (mod 2 pi) for either sign of omega. The model reads each channel's angle only through cos and sin, so the model with Delta' is the same function.

D3. The balanced |.| construction (Proposition 1(a); the registered existence test)

Given a solved signed observation-step-free model with m stepping move tokens (m = 2 ring, 4 torus; r = 4):

1. Re-solve W_in so that the move latents are c e_1, ..., c e_m (c = the source's mean move-latent norm). W_in acts only on the move embeddings, so this is a free gauge.
2. Per channel set omega' = max(|omega|, 1) and the steps to remainder(omega Delta, 2 pi) / omega' (angles in [0, 2 pi) for either sign of omega).
3. Set W_out[:, j] = step_j / c for the m move columns and 0 elsewhere; copy every other weight.

The signed control is built by the same function with the unwrapped signed angle (steps omega Delta / omega'), so the two differ only in the wrap. On the torus m = 4 = r, exactly at Proposition 1(a)'s limit.

Exactness. Ring: frozen accuracy = source on 8/8; maximum angle error 1.2e-6 rad (|.|), 6.1e-7 (signed); maximum logit difference 5.1e-4 at T = 128. Torus: frozen = source on 8/8; maximum angle error 1.4e-6 rad (|.|), 5.8e-7 (signed). *Weight scale.* Ring: maximum |W_out| 0.31-0.38 in the signed sources, 1.07-1.70 built; |W_in| 0.15-0.21 -> 0.19-0.33. Torus: maximum |W_out| 0.48-0.58 -> 1.23-1.55, maximum |omega| 0.68-0.96 -> 1.00. Within about 5x of the trained scale.

Why the scale matters. A construction that keeps the source's W_in and solves W_out at minimum norm is also exact, but in a solved signed model the two move latents are nearly antiparallel (cosine -0.987 to -0.998), so W_out must be huge (up to 16,320 against 0.31-0.38). Adam's per-entry steps then lose that solution in the warm-up (8/8 LOST)

before re-finding it (7/8 SOLVED). An existence test is informative only near a trained scale, which the balanced construction provides.

D4. The full |.| arms

To show that the task, not only a given signed table, is representable by a full |.| model (observations stepping), W_in is chosen to project onto two directions orthogonal to all 17 observation embeddings and W_out is solved per channel (`docs/audits/2026-10-12/ring_dial_audit.md` M1). In the real modules: |.| steps >= 0, observation-token |Delta| <= 1.7e-6, cos / sin equal to the signed code's to 6.3e-4 over a 512-step walk, circular mismatch <= 4.6e-4 on 2,000 same-cell revisit pairs.

D5. Softplus (Proposition 1(f))

Softplus cannot reach 0, but observation steps need only be 0 mod 2 pi. With W_in annihilating every observation embedding, each observation steps softplus(0) = ln 2 in every channel, and omega'_c = 2 pi / ln 2 makes that exactly one turn; the move columns are set to softplus^{-1}((remainder(Theta, 2 pi) + 2 pi) / omega'). Built in the real softplus model from a solved ring model (`docs/paper/review_draft5/theory_check_extra.py` (2), CPU): minimum step 0.693 > 0, angle error mod 2 pi 8.4e-15 rad over all 19 tokens, logits equal to the source's to 2.8e-11 (logit scale 21.6). On the torus m = 4 = r, so the same construction applies.

D6. Continuations and the LOST rule

A constructed model is trained further on a fresh walk stream and compared with its signed control at the same seed. *LOST* (registered): the run's maximum per-epoch training loss during the continuation exceeds 0.3, or exceeds 10x its control's maximum at that seed and 0.05. *HELD* = SOLVED on at least n - 1 seeds, mean accuracy at least control - 0.01, and LOST on at most one seed. Ring (lr 1e-3): HELD, 8/8 SOLVED, 0/8 LOST (maximum per-epoch loss 0.024-0.154 against the control's 0.015-0.112). Torus (lr 1e-4): LOST on 6/8 and SOLVED on 8/8 at the end, read by the rule fixed before launch as perturbed and re-found; at lr 1e-3 (descriptive) 6/8 SOLVED (0.988).

D7. What the held model keeps (ACCOUNTS T1, registered)

Weighted by usage (mean query norm at move tokens times mean key norm at observation tokens, per channel), the held ring |.| model keeps its cancelling code from 0.99 at construction to 0.98 after training; unweighted (every channel equal) the share falls from 0.87 to 0.18, because channels attention does not read leave the code. Weight decay 0 vs 0.05 changes nothing detected (-0.003).

E. Theory

Labels: main-text Proposition 1 = P1(a) and P1(f) of `APPENDIX_theory_v3.md`; Lemma 1 = T1; Proposition 2 = P6; Propositions 3 and 4 below = P3 and P4. Every check script is in `docs/paper/appendix_checks/` (CPU, seconds each).

Notation. A rotary channel is c = (h, b), head h and block b, acting on coordinates (2b, 2b + 1) of head h; R(phi) is the 2D rotation. The phase is theta_{t,c} = omega_c sum_{u <= t} S_{c, x_u}, with S = W_out W_in E the C x V step table (rank at most r) and Theta = diag(omega) S the angle table. omega is free in sign. For an inverse pair of tokens (x, y), channel c *cancels through zero* if Theta_{c,x} + Theta_{c,y} = 0 with Theta_{c,x} != 0, *cancels by wrapping* if the sum is a non-zero multiple of 2 pi, and is *blind* if both angles are 0 mod 2 pi.

E1. Score identity (A0)

With alpha piecewise constant, alpha = Delta_u on (u - 1, u], the homogeneous part of Howard et al.'s integrator, dF/dt = -s alpha(t) F, has propagator F(t) = exp(-s (Phi_t - Phi_{t'})) F(t'). With s = i omega, channel c of the rotary score is q_c^T R(theta_s - theta_t) k_c = Re(conj(q~*c) k~_c exp(-i omega_c (Phi_t - Phi_s))), with q~ = q*{2b} + i q_{2b+1}. *Proof.* R(theta) acts on q~ as multiplication by e^{i theta}; flows over tokens compose by adding exponents. The identity for complex SSMs is Mamba-3's Proposition 3; ours is only its mapping onto Howard's three rate cases. Checked (`a0_propagator.py`): 3e-14 (propagator), 4e-15 (score).

E2. Proposition 1 (representability by wrapping; P1(a), P1(f))

(a) Let S be a signed table with at most m <= r non-zero columns (stepping tokens x_1, ..., x_m), whose embeddings are linearly independent modulo the span of the other embeddings (generic when V <= d_model, here 19 or 21 <= 128; automatic in observation-step-free models). Then a |.| model with the same embeddings and content weights has an angle table equal to Theta mod 2 pi, entrywise.

Proof. Pick W_in' with W_in' e_{x_j} = u_j (the j-th unit vector of R^r) and W_in' e_y = 0 for every other token. Set omega' = max(|omega|, 1) per channel and W_out'[:, j] = remainder(Theta_{., x_j}, 2 pi) / omega' >= 0, other columns zero. Then W_out' W_in' E is entrywise non-negative, so |.| leaves it unchanged, and it reproduces each stepping column mod 2 pi. [] *Full arms:* choosing W_in' to annihilate every observation embedding leaves only the moves stepping, so every |.| model in the paper can represent any code in which only the moves step, including the map codes of the trained signed observation-step-free models. Checked (`p1_wrap.py`): angle error <= 7e-14, logits <= 8e-14.

(f) Softplus. Under the same assumptions, send the stepping embeddings to u_1, ..., u_m and every other embedding to 0, set omega'c = 2 pi / ln 2 for every c, and W_out'[c, j] = softplus^{-1}((remainder(Theta{c,x_j}, 2 pi) + 2 pi) / omega'_c). Every non-stepping token then steps softplus(0) = ln 2, an angle of exactly 2 pi, and every stepping token's angle is Theta mod 2 pi.

Proof. softplus is a bijection from R onto (0, inf); the targets lie in [ln 2, 2 ln 2); omega' ln 2 = 2 pi. [] Checked (`theory_check_extra.py` (2)): D5.

E3. Lemma 1 (relative-sign gauge; T1)

Take a MapWM model with any number of layers and a set F of channels. Form a second model by (a) negating every token's step in each channel of F (by negating omega_c; in the signed class equivalently row c of W_out, which is not available in the |.| class since |.|

is applied after W_out) and (b) negating output coordinate h d_head + 2b + 1 of the query and key projections (weight row and bias entry, every layer) for each c = (h, b) in F. The two models have identical attention scores at every layer, hence identical logits.

Proof. With J = diag(1, -1), J R(phi) J = R(-phi) and J is orthogonal. (a) replaces theta_{t,c} by -theta_{t,c}; (b) replaces q_c, k_c by J q_c, J k_c, exactly including biases, since nothing between the projection and the rotation mixes coordinates. The channel term becomes (J q_c)^T R(-(theta_s - theta_t)) (J k_c) = q_c^T R(theta_s - theta_t) k_c. All other components are untouched, and every layer uses the same cos / sin, so the scores agree layer by layer. [] Checked (`t1_gauge.py`): 35 channels flipped and reflected give bitwise-identical logits (control without the reflection: 6.6).

Corollaries. (i) The |.| class realises a signed angle table Theta exactly as real numbers, with the same latent, iff every row of Theta is sign-coherent (all stepping tokens in a channel step the same way); omega carries each channel's direction. So the |.| constraint removes only the relative signs of the tokens that step in a channel. (ii) It therefore excludes no map *function* for which a code with sign-coherent rows exists, and that code needs no wrapping: one stepping token (parity, Z_m counting), or two moves that step the same way (+1 and +2: a(+2) = 2 a(+1)). Other codes of the same function can still be excluded as parameter points: on +1/+2 the code a(+1) = 1, a(+2) = 2 - 2 pi has an opposite-sign row. (iii) Up to the gauge, the excluded tables are exactly those with a row containing entries of strictly opposite sign. For an inverse pair, every through-zero cancelling channel is such a row; cancellation by wrapping, (a, 2 pi - a), is allowed and is representable whenever Proposition 1 applies. The constraint removes the through-zero branch of the inverse-pair problem, not cancellation.

E4. One channel's geometry (P3, P4)

Take one channel and an inverse pair (x, y) with angles (a_x, a_y), everything else fixed; the loss is a function on the angle torus. Write sigma = a_x + a_y.

Proposition 3 (cut circle). (a) The closed quadrant [0, inf)^2 maps onto the torus, so the signed and |.| classes attain the same loss values in this channel. (b) The non-blind cancelling set in the quadrant consists of the segments sigma = 2 pi k, k >= 1; the through-zero line meets the quadrant only at the blind origin. (c) From a start in the open quadrant with sigma < 2 pi, the infimum distance (in angle coordinates) to a non-blind cancelling point is min(sigma, 2 pi - sigma) / sqrt 2 in the signed class and (2 pi - sigma) / sqrt 2 in the |.| class; they coincide iff sigma >= pi. (d) Any continuous |.| path from |sigma| < pi to a non-blind cancelling point passes through |sigma| = pi.

Proof. (a), (b) from the quotient map. (c) The distance to sigma = 2 pi k is |sigma - 2 pi k| / sqrt 2; the |.| class has only k >= 1 (with omega < 0 also the negative quadrant, which is farther). (d) |sigma| is continuous along a |.| path and a non-blind cancelling point has |sigma| >= 2 pi; intermediate value theorem. [] Checked (`p3_p5_geometry.py`): a brute-force grid matches both formulas to 2e-4.

Proposition 4 (sticky wall). Let s be the pre-|.| step of one token in one channel and a = omega |s|, with L(s) = l(|s|) and l continuously differentiable near 0. (a) For s != 0, dL/ds = omega sign(s) dL/da. (b) If l'(0) > 0, s = 0 is a strict local minimum of L along s with every other parameter fixed: L(s) - L(0) = l'(0) |s| + o(|s|). (c) Gradient descent on s with

step eta, l' in [mu, M] on [0, rho], mu > 0, eta M <= rho and |s_0| <= rho, gives |s_{k+1}| <= max(|s_k| - eta mu, eta M): the iterate chatters within eta M of the wall.

Proof. (a) chain rule for |.|; (b) first-order expansion; (c) case split on |s_k| >= eta l'. [] Proposition 4 is a statement along one coordinate; in parameter space s = w_c . z_x is shared across tokens and channels.

E5. Proposition 2 (CARoPE cannot cancel; P6)

CARoPE (faithful, `model_mono_baselines.MapFormerWM_CARoPEF`) advances channel i of head h by f_h(x)^i per token, with f in (0, 1), so every per-token advance lies in (0, 1]; channel 0 advances by exactly 1, an index clock.

(a) The phase of any interval of k >= 1 tokens lies in (0, k]. The shortest return (out and back) spans 3 tokens between the move query and the observation key, and 4 between two observation tokens, so its phase lies in (0, 4] subset (0, 2 pi): no channel, at any parameter value, reads every return at phase 0. (b) Fix eps in (0, pi - 2). If a channel reads each return of an excursion of J moves out and J back within eps of 0 mod 2 pi, then the move-driven phase of any interval with at most 2J moves lies in [0, 2 eps]: every channel is either a clock (it misses some short return) or blind over that range.

Proof. (a) Sum k numbers in (0, 1]. (b) The depth-j return phase P_j satisfies P_1 in (0, 3] and P_{j+1} - P_j in (0, 4]; the arc (eps, 2 pi - eps) is longer than 4 when eps < pi - 2, so a sequence starting in (0, eps] with increments at most 4 cannot jump it; induction gives P_j in (0, eps], hence J (c_+ + c_-) <= eps. [] Checked (`p6_carope.py`): eight trained ring CARoPE models; 10,331 random draws satisfy the bound.

F. Exchange-rate details

Dyck-2 (`DYCK_MDEPTH_RESULTS.md`). Task and metric as MapFormer v4 App. B.4, trained and scored at length L = 32 and nesting depth D = 12; primary metric A2f, Hewitt's distance-averaged closing accuracy over prefixes where a closer can be emitted (CE-optimal predictor 1.000, chance 0.500, best stack-free floor 0.594). 2 heads, d_model 128, 1-4 layers; path-integrated arms MapWM and MapPoPE at rank 2; index arms RoPE and PoPE at base 10000 (and base 32 in the 4-layer 3x cell); n = 8 paired seeds. Position main effect = ((MapWM - RoPE) + (MapPoPE - PoPE)) / 2.

condition	layers	RoPE	PoPE	MapWM	MapPoPE	position main effect
1x budget	1	0.648	0.640	0.996	0.998	+0.353
1x budget	2	0.889	0.850	1.000	1.000	+0.130
1x budget	3	0.964	0.946	1.000	1.000	+0.045
1x budget	4	0.982	0.971	1.000	1.000	+0.024
3x budget	4	0.998	0.997	1.000	1.000	+0.002
mixture of depths 4-12	4	0.898	0.881	0.999	0.999	+0.110

Every effect is positive on 8/8 seeds (sign-flip p 0.0078). Index base 32 minus base 10000 at four layers, 3x budget: +0.001 (RoPE), -0.001 (PoPE), within the MDE. Three times the budget raises the index arms by +0.021 at four layers (8/8).

Ring (C3). The index-one-layer accuracies are not monotone in p_plus (0.710, 0.677, 0.824, 1.000 on fresh seeds; every pair differs, p 0.0022-0.0108), so the exchange rate, not the one-layer accuracy, is the quantity that tracks whether steps are undone.

Torus (`FACTORIAL_RESULTS.md`, Fig. F1). Strata (short gap / long gap / wrap-only; mean and run range, n = 4): path one layer 1.000 / 1.000 / 1.000; index one layer 0.769 (0.735-0.787) / 0.480 (0.456-0.508) / 0.482 (0.470-0.495); four layers 0.943 (0.900-0.958) / 0.458 (0.367-0.506) / 0.457 (0.381-0.494); looped x4 0.938 (0.937-0.940) / 0.507 (0.496-0.531) / 0.509 (0.496-0.521). Floors on this stream (analysis): constant 0.508 / 0.515 / 0.517, retrace 0.912 on short gaps. Memorisation check: four index layers 0.897 on their own training map against 0.860 held out; looped 0.862 against 0.864; path 1.000 on both.

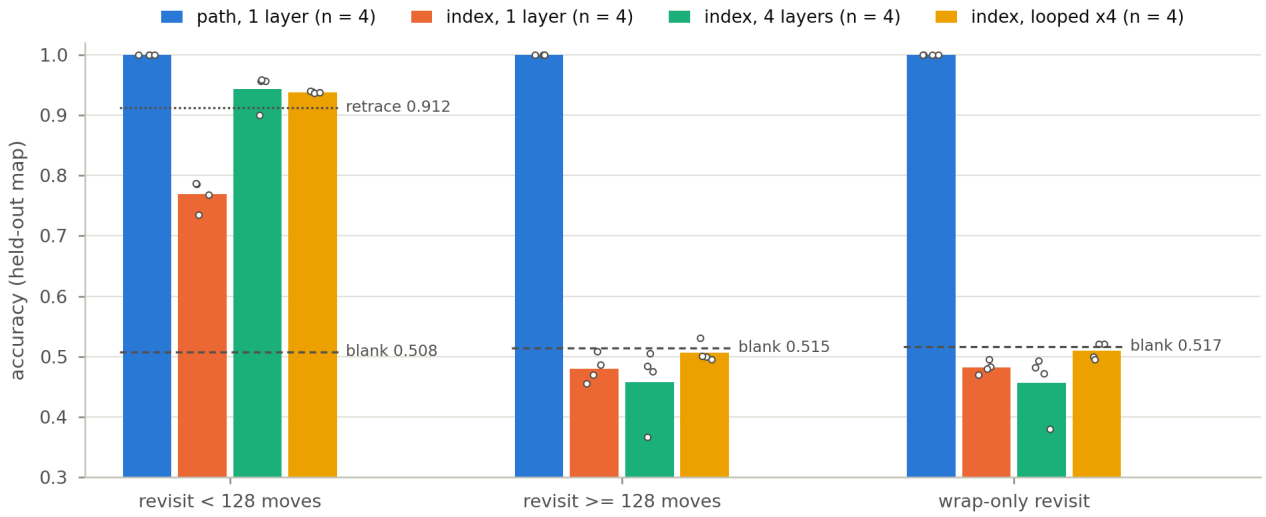


Figure F1 (`fig_factorial_strata`). Revisit accuracy by stratum on the torus at T = 1024 (trained and tested at T = 1024; bars: means of n = 4 runs, dots: runs). Dashed: always-blank within each stratum, 0.508 / 0.515 / 0.517; dotted: retrace within the short-gap stratum, 0.912 (elsewhere it equals the constant); floors on this batch's own evaluation walks. Path integration with one layer is at 1.000 in every stratum. Four index layers or a loop of four lift short-gap revisits to 0.94, just above retrace; at long gaps and wrap-only revisits every index model stays at or below the constant (0.46-0.51).

G. Rank details

Arms (A1). Shared r2 (`Vanilla`), per-head r2 (`Vanilla_r2ph`), block-diagonal r4 (`Vanilla_r4mibd`: the shared rank-4 latent with W_out's off-diagonal blocks held at zero), per-head r3 (`Vanilla_r3ph`), shared r4 (`Vanilla_r4mi`), per-head r4 (`Vanilla_r4ph`), all built from the shared rank-2 model's initial weights at each seed. Initial angle-increment scale (std): 0.208 for block-diagonal r4 against 0.33-0.34 for the other arms, which fail (rank 2) or solve (rank 4) from the same scale. Per-seed results: C4.

Training curves (Fig. G1; `fig_loss_curves_rank_out.txt`). Shared r2 0/8 SOLVED (4 STALLED, 4 DESCENDING; median final loss 0.35) against shared r4 8/8 (2.8e-3); per-head r2 2/8 (median 0.22), r3 6/8 (3.5e-3), r4 8/8 (3.1e-3). The rank-3 and rank-4 medians cross the threshold at epochs 478-532; the rank-2 medians never do.

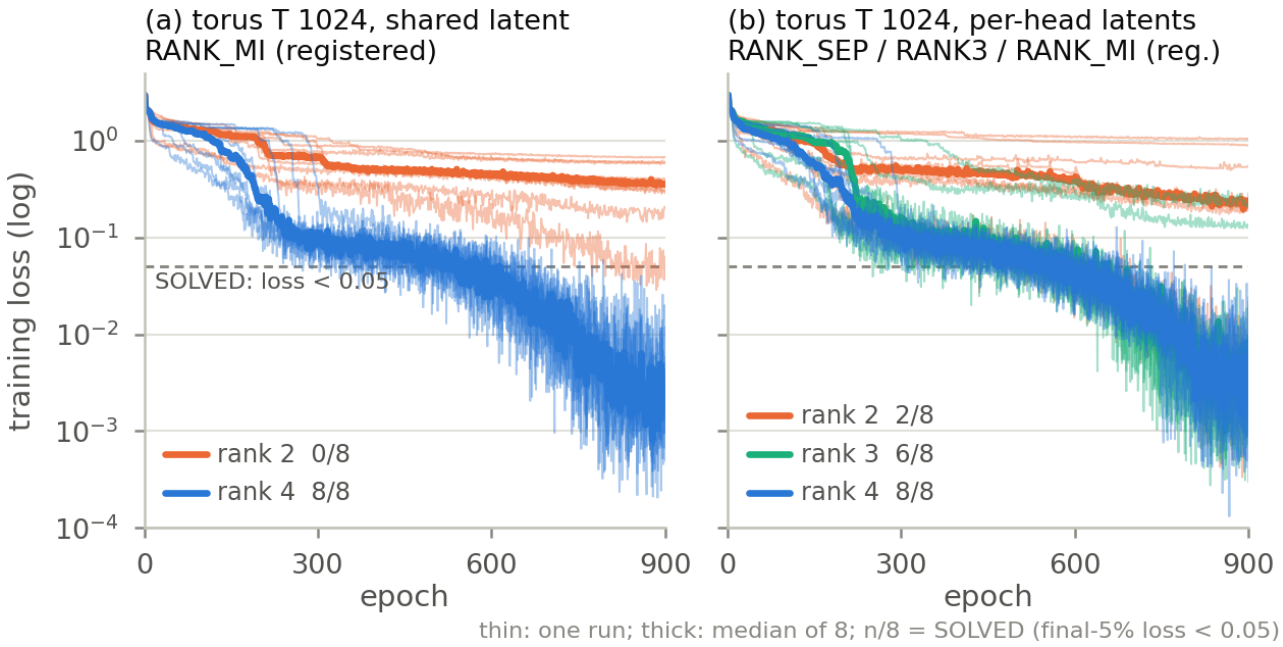


Figure G1 (`fig_loss_curves_rank`). Training loss per epoch on the torus (T = 1024) by the per-head rank of the content-to-angle map; thin lines are runs, thick lines medians of 8; dashed: the SOLVED threshold. (a) One latent shared by the heads: rank 2 solves 0/8, rank 4 8/8. (b) Per-head latents: rank 2 2/8, rank 3 6/8, rank 4 8/8.

Restart curves (`RANK_PROJ_ANALYSIS.txt`). Projected rank 2: peak training loss 0.37-0.46 after the restart, back below 0.05 at epochs 553-654 on 7/8 seeds. Rank-4 control: peak 0.26-0.30, back at 497-669 on 7/8. Rank 2 continued: peak 0.55-0.82, back on 1/8.

Rank against dimension (`RANK_ND_RESULTS.md`, `RANK_WRAP_RESULTS.md`, `RANK_TESTS_RESULTS.md`). Per-head rank, every arm from the same base, trained and tested at T = 1024, 900 epochs:

torus	D	per-head rank	wrap-only share of revisits	SOLVED	held-out accuracy
32 per side	2	2	0.37	1/8	0.769
32 per side	2	3	0.37	8/8	0.999
10 per side	3	3	0.71	1/8	0.800
10 per side	3	4	0.71	4/8	0.847
18 per side	3	4	0.35	8/8	1.000

2D rank 3 vs rank 2: 8/8 vs 1/8 (Fisher p 0.0014), +0.230 (permutation p 0.0014). 3D, 18 vs 10 per side at rank 4: +0.153 (p 0.0085). On a 100-cell 2D torus (10 per side) rank 3 memorises its training map (0.986 on it, 0.273 on an unseen map), so that cell does not measure map learning. 1D ring (32 cells, T = 128, 300 epochs): rank 1 7/8 (0.993), rank 2 8/8 (1.000); with the component common to the move steps removed by construction, rank 1 solves 8/8 in a median of 24 epochs (58 without).

Escape test (`RANK_TESTS_RESULTS.md`). Lifting the eight stalled shared-rank-2 runs to rank 4 (function-preserving) and training 600 epochs solves 3/8, against 1/8 for rank 2 continued

and 2/8 for a fresh rank-4 step map on the same networks; at this budget the contrasts are not resolved.

H. Sign details

H1. Learning rates (`LRSWEEP_RESULTS.md`; ring, observation-step-free, seeds 1500-1507)

learning rate	signed SOLVED, accuracy	|.| SOLVED, accuracy
1e-3	8/8, 1.000	1/8, 0.917
3e-3	8/8, 1.000	2/8, 0.963
1e-2	4/8, 0.972	1/8, 0.872
3e-2	1/8, 0.559	0/8, 0.584

Registered B1 (each arm at its best rate, selection redone in the null over all 12,870 relabellings): |.| 3e-3 2/8 vs signed 1e-3 8/8, -0.037, decision p 0.0070 (SOLVED) / 0.0057 (accuracy). The |.| optimum is interior to the grid. Three |.| runs at 3e-3 were still descending at 300 epochs. At 3e-2 both arms sit near the 0.50 blank floor. Per seed:

```
seed | 1e-3:SO | 1e-3:AO | 3e-3:SO | 3e-3:AO | 1e-2:SO | 1e-2:AO | 3e-2:SO | 3e-2:AO
1500 | 1.0000 SOLV | 0.9362 STAL | 1.0000 SOLV | 0.9111 STAL | 0.9460 DESC | 0.9936 DESC |
0.4960 STAL | 0.4958 STAL
1501 | 1.0000 SOLV | 0.8759 STAL | 1.0000 SOLV | 0.9662 DESC | 0.9460 STAL | 0.4960 STAL |
0.4960 STAL | 0.4960 STAL
1502 | 1.0000 SOLV | 0.8995 STAL | 1.0000 SOLV | 0.9428 STAL | 0.9407 STAL | 0.9703 DESC |
0.4960 STAL | 0.4960 STAL
1503 | 1.0000 SOLV | 0.9116 STAL | 1.0000 SOLV | 1.0000 SOLV | 1.0000 SOLV | 0.6000 STAL |
0.4960 STAL | 0.4960 STAL
1504 | 1.0000 SOLV | 0.9415 STAL | 1.0000 SOLV | 0.9686 STAL | 0.9460 STAL | 1.0000 DESC |
0.4960 STAL | 0.8159 DESC
1505 | 1.0000 SOLV | 1.0000 SOLV | 1.0000 SOLV | 0.9993 SOLV | 1.0000 SOLV | 0.9950 DESC |
0.9998 SOLV | 0.4961 STAL
1506 | 1.0000 SOLV | 0.8736 STAL | 1.0000 SOLV | 0.9595 DESC | 1.0000 SOLV | 0.9996 SOLV |
0.4960 STAL | 0.4960 STAL
1507 | 1.0000 SOLV | 0.8971 STAL | 1.0000 SOLV | 0.9587 DESC | 1.0000 SOLV | 0.9184 DESC |
0.4959 DESC | 0.8799 DESC
```

H2. Recipe cells (`ROBUST_RESULTS.md`, seeds 1400-1407)

cell	recipe	|.| accuracy, SOLVED	signed accuracy, SOLVED	d [95% CI]	perm p	Fisher p
ring	lr 1e-3, weight decay 0.05	0.911, 1/8	1.000, 8/8	-0.089 [-0.125, -0.055]	0.0002	0.0014
ring	lr 1e-3, weight decay 0	0.873, 0/8	1.000, 8/8	-0.127 [-0.200, -0.075]	0.0002	0.0002
ring	lr 3e-3, weight decay 0.05	0.987, 4/8	1.000, 8/8	-0.013 [-0.025, -0.005]	0.026	0.077
ring	lr 3.3e-4, weight decay 0.05	0.841, 0/8	0.950, 6/8	-0.109 [-0.215, +0.010]	0.089	0.0070
torus		0.967, 2/8	0.999, 8/8		0.0005	0.0070

cell	recipe	|.| accuracy, SOLVED	signed accuracy, SOLVED	d [95% CI]	perm p	Fisher p
	2 layers, d_model 256			-0.032 [-0.055, -0.015]		

At lr 3e-3 the registered contrast was below its decision threshold in this batch (4/8 vs 8/8); on fresh seeds in the sweep the same rate gives 2/8 vs 8/8 (H1). Declared secondaries: cancelling share (unweighted) |.| 0.04-0.13 in every cell, signed 0.43-1.00; torus strata (short / long / wrap-only) signed 1.000 / 0.999 / 0.992, |.| 0.989 / 0.911 / 0.799; on the torus 6/8 |.| runs were still descending at 900 epochs (final losses 0.018-0.144).

H3. Starts (`MECHANISM_RESULTS.md`, ring p_plus 0.5, seeds 800-807; `TORUS_START_RESULTS.md`, torus, seeds 1200-1207)

Observation-step-free models differing only in the initial move angles (every other weight identical at a seed):

initial move angles	signed (ring)	|.| (ring)	signed (torus)	|.| (torus)
default (small, signs independent)	8/8	0/8	8/8	1/8
the |.| start: both moves rotating the same way	7/8	0/8	--	--
large, opposite-move sums below pi, U(0.9, 1.5)	--	2/8	--	0/8
large, sums above pi, U(1.7, 4.5)	8/8	8/8	2/8	0/8

Ring, registered: same start, sign matters (0/8 vs 7/8, Fisher p 0.0014; -0.092 [-0.135, -0.045]); a start past pi rescues |.| (8/8 vs 0/8, +0.109 [+0.085, +0.135], p 0.0002); a large start with sums below pi does not (2/8; vs past pi Fisher p 0.0070); from past pi the sign makes no difference (8/8 vs 8/8). The large start also moves the angles 13-22x faster per Adam step (the sub-pi control about 5x), so sum and size are only partly separated. Analysis: signed models from the small starts end with 98-99% of channels opposite-signed and cancel through zero; from past pi both signs cancel by wrapping (signed 88%, |.| 100% of cancelling channels); the rescued |.| models cancel in 23% of channels. Torus, registered: the past-pi start does not rescue |.| (0/8 vs 1/8, +0.005 [-0.060, +0.070]), and it slows the signed model (2/8; 5/8 runs per large-start arm still descending at 900 epochs); the default-start sign cost replicates (1/8 vs 8/8, Fisher p 0.0014). The start lever is a ring result.

H4. Moves +1/-1 vs +1/+2 (`GAUGE_RESULTS.md`, ring, seeds 1300-1307)

By Lemma 1 (ii) a |.| step represents, without wrapping or crossing zero, a map whose two moves step the same way. With moves +1/-1 (an inverse pair) the cost replicates: |.| 0.931, 1/8 vs signed 1.000, 8/8 (-0.069 [-0.095, -0.045], Fisher p 0.0014). With moves +1/+2, which needs integration (index RoPE 0.649, 0/8), |.| scores 0.883, 2/8 against signed 0.947, 5/8 (-0.064 [-0.200, +0.075], Fisher p 0.31): the registered comparison is not resolved, because +1/+2 (every revisit a lap of at least 16 moves) is harder for the signed step as well. Signed models on +1/+2 end with 0.92 of channels having both move angles of one sign (0.01 on +1/-1), the sign-coherent code Lemma 1 allows. The |.| step does not learn +1/+2 more reliably than +1/-1 (2/8 vs 1/8).

H5. Three times the ring budget (`RING_BUDGET_RESULTS.md`, seeds 40-47)

The |.| step at 900 epochs solves 3/8, as at 300 epochs on the same seeds (3/8); the solved sets differ.

H6. Cancelling share and ring strata

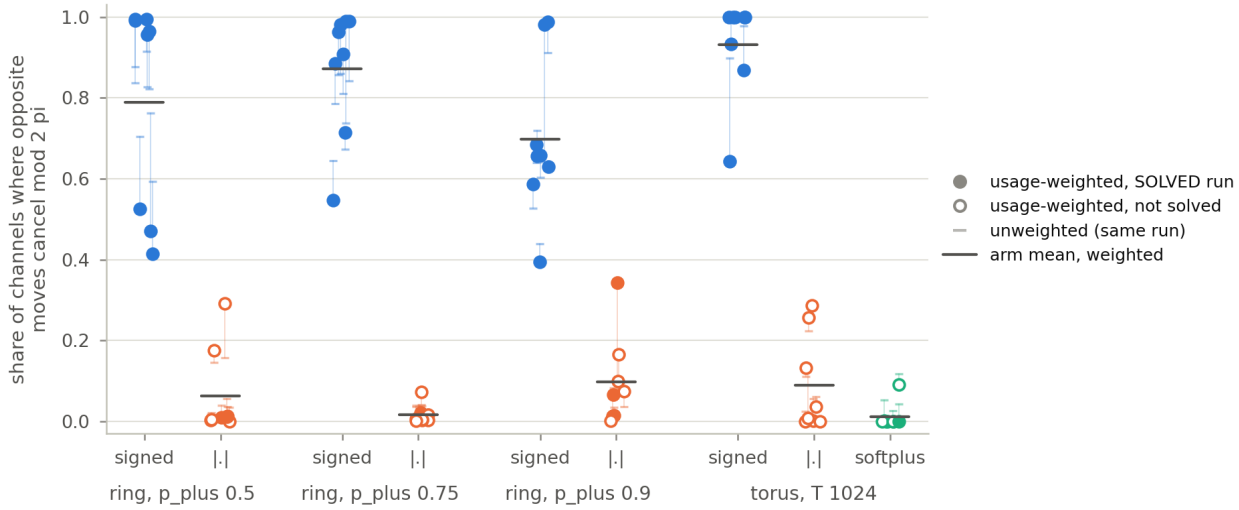


Figure H1 (`fig_wrap_share`; analysis). Share of rotary channels in which opposite moves cancel mod 2 pi, per run, weighted by how much attention uses each channel (dots; filled = SOLVED) and unweighted (faint dashes, joined to the same run). A channel cancels when |wrap(a(move) + a(opposite move))| < 0.1 rad; channels whose move angle has collapsed to within 0.05 rad of 0 mod 2 pi are excluded from the denominator. Usage weight: mean query norm of the channel at move tokens times mean key norm at observation tokens. Ring of 32 cells at three move biases (n = 8 each) and the torus at T = 1024 (n = 8). Bars: arm means of the weighted share. Signed steps 0.70-0.93; |.| steps 0.02-0.10; softplus 0.01. Seven of the eight non-negative runs that solve stay at or below 0.07.

cell	arm	SOLVED	weighted mean (range)	unweighted mean (range)
ring p_plus 0.5	signed	8/8	0.79 (0.41-0.99)	0.79 (0.59-0.92)
ring p_plus 0.5	|.|	2/8	0.06 (0.00-0.29)	0.06 (0.02-0.16)
ring p_plus 0.75	signed	8/8	0.87 (0.55-0.99)	0.78 (0.64-0.86)
ring p_plus 0.75	|.|	1/8	0.02 (0.00-0.07)	0.03 (0.02-0.04)
ring p_plus 0.9	signed	8/8	0.70 (0.39-0.99)	0.64 (0.44-0.91)
ring p_plus 0.9	|.|	4/8	0.10 (0.00-0.34)	0.06 (0.02-0.10)
torus T 1024	signed	8/8	0.93 (0.64-1.00)	0.98 (0.90-1.00)
torus T 1024	|.|	0/8	0.09 (0.00-0.29)	0.11 (0.00-0.27)
torus T 1024	softplus	1/8	0.01 (0.00-0.09)	0.03 (0.00-0.12)

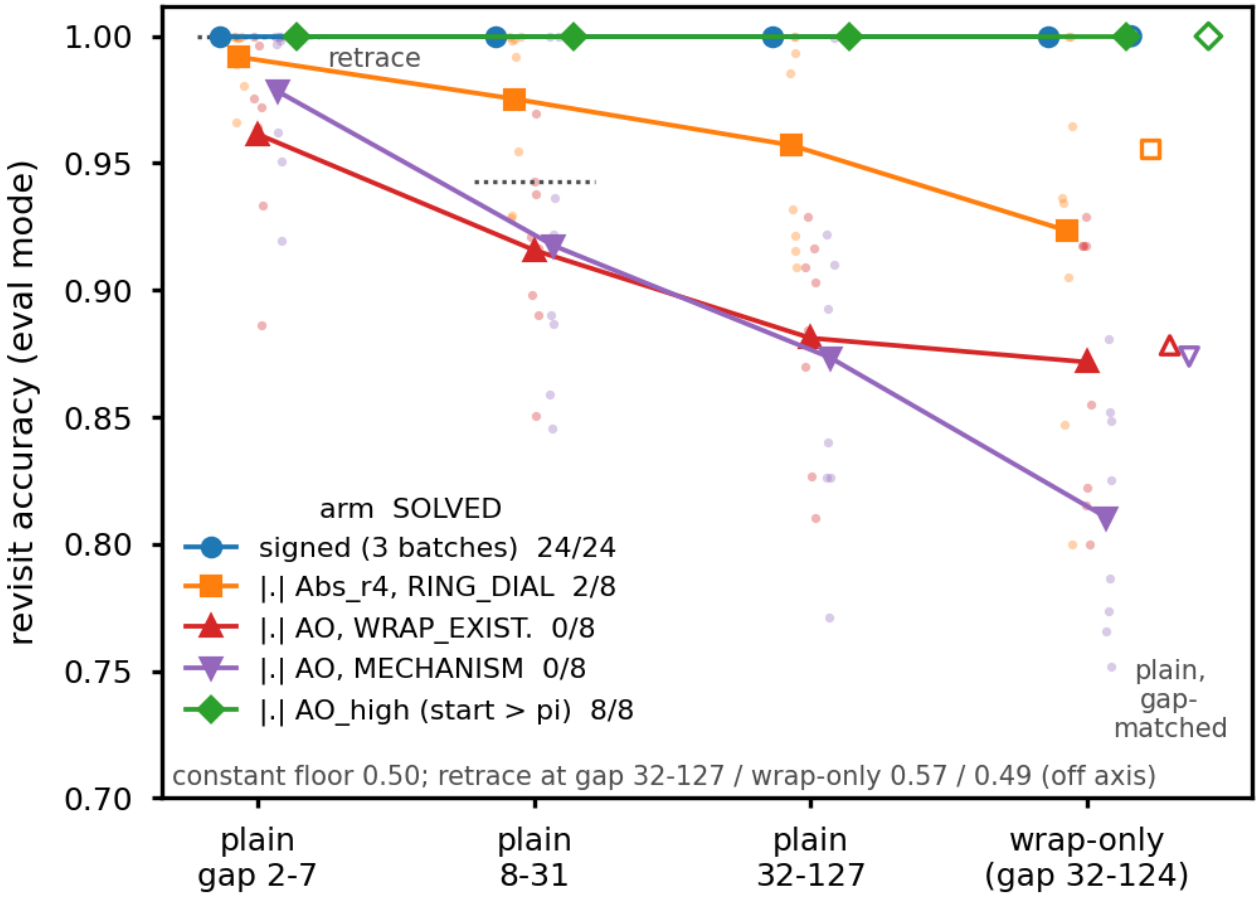


Figure H2 (`fig_ring_strata`; analysis). Revisit accuracy on the 1D ring by stratum (eval mode; 400 trajectories, T = 128; maps redrawn). "Plain" revisits have an earlier visit at the same unwrapped position; "wrap-only" revisits were seen only a lap away. Lines: arm means; dots: runs; hollow markers: plain revisits re-weighted to the wrap-only gap distribution. Signed steps (24 runs over three batches) and |.| steps started past pi are at 1.000 everywhere. |.| steps from the default start lose accuracy with the gap (0.96-0.99 at gaps 2-7, 0.87-0.96 at gaps 32-127). Dotted: retrace within each stratum, 1.000 at gaps 2-7 and 0.943 at gaps 8-31 (0.57 and 0.49 in the long-gap and wrap-only strata, off axis); constant floor 0.50.

I. CARoPE and CoPE

CARoPE (`model_mono_baselines.py`, `CARoPEF`). Per head h and token t a scalar rate f_h(x_t) = 1 / (softplus(x_t W)_h + 1) in (0, 1) replaces RoPE's base frequency, so channel i advances by f_h(x_t)^i per token; initialised as index RoPE (W = 0, a per-head bias with 1 / (softplus(b) + 1) = 10000^(-2 / d_head)). Every token, observations included, advances the phase; the layer is MapWM's, so only the phase differs. Proposition 2 (E5): no channel can read every return. Registered (GENERALITY): torus 0.667 (0/8), below index RoPE (0.722, -0.055, p 0.040); ring 0.741 against index 0.730 (+0.011, not resolved).

CoPE (`CoPEF`). Golovneva et al.'s Listing 1 in torch: per layer one module shared by the heads; gates g_ij = sigmoid(q_i . k_j / sqrt(d_h)); count p_ij = sum_{k=j}^{i} g_ik, clamped to 2047; position logit by linear interpolation of q_i . e[floor p] and q_i . e[ceil p],

table e at zeros initially; added to the attention logits; no rotary position. *Representability (constructed):* gates 17/16 and 15/16 per (move, observation) pair make p_ij mod 2 a function of net displacement, read by a period-2 table (1.000 revisit accuracy). The exact solutions form a lattice (P7 of v3: period 2 or 3).

batch	task	CoPE	SOLVED	comparison
GENERALITY (700-707)	ring	0.887	0/8	signed 1.000, 8/8 (-0.113, p 0.0002); index 0.730 (+0.157, p 0.0002)
WRAP_COPE (900-907)	ring, table at zeros / sinusoid past pi / below pi / slope-matched	0.840 / 0.913 / 0.867 / 0.943	1/8, 2/8, 1/8, 0/8	contrasts not resolved (bimodal seeds)
ACCOUNTS T4 (1100-1115)	256-cell line (never wraps) / ring	0.914 / 0.873	1/16 / 0/16	line: signed 1.000, 8/8 (Fisher p < 1e-4); index 0.772, 0/8

On both line and ring several CoPE seeds reach 0.99-1.00 held-out accuracy without meeting the loss criterion (at a held-out accuracy cut of 0.999 the line reads 5/16 and the ring 2/16; analysis). MapFormer v4 Table 2 reports CoPE failing 1D and 2D navigation (one layer 0.63 +/- 0.21 in 1D and 0.65 in 2D; four layers 0.88 and 0.76).

J. Text world

Task (`TEXTWORLD_RESULTS.md`, `train_textworld.py`). The 64 x 64 torus walk rendered as English: 58 words, three synonyms per direction, movement-free verbs, fillers and asides; T = 1024 words (about 132 steps); held-out map; predict the observation word at revisits. A direction word appears only inside a movement clause (context-free steps). Floors: best constant 0.505, reversal-copy 0.597. Recipe: the torus recipe (A2), path MapWM r4 at one layer against index RoPE at one and two layers; 200 evaluation walks.

Step tables per seed (declared secondaries, `fig_textworld_out.txt`; "minus common" = opposition after removing the component shared by the four direction steps; ablation columns: held-out accuracy, eval mode without the dropout scale):

seed	type	raw opposition	minus common	common / north	accuracy	non-direction steps zeroed	direction steps zeroed
0	clock	1.730	0.0071	0.711	0.881	0.191	0.388
1	map	0.083	0.0029	0.041	0.998	0.893	0.505
2	clock	1.434	0.0064	1.144	0.999	0.665	0.482
3	clock	1.707	0.0022	0.480	0.995	0.673	0.463
4	clock	1.768	0.0052	0.501	0.999	0.539	0.445
5	map	0.120	0.0050	0.060	1.000	0.898	0.505
6	map	0.080	0.0117	0.040	1.000	0.950	0.505
7	map	0.119	0.0057	0.059	0.882	0.586	0.498

On the clock seeds the common component is parallel to the verbs' step (cosine +0.999 to +1.000), and 31-38 of 64 phase channels drift between visits to the same cell (2-4 on the map seeds). Synonym cosine 0.9996-1.0000 on every seed.

K. What and where (analysis)

Do trained path models need the content x position interaction? (`docs/WHAT_WHERE_CHECKS.md`; eval-only on the 48 short-walk torus checkpoints of PAPER2X2.) Held-out accuracy when the score is rebuilt (floors: n-gram 0.598, blank 0.507):

score used	RoPE	PoPE	MapWM r2	MapWM r4	MapPoPE r2	MapPoPE r4
intact	0.805	0.679	0.971	1.000	0.999	1.000
position only (object identity removed, token type kept)	0.463	0.476	0.837	0.989	0.974	0.986
shared position kernel x content gain	0.430	0.435	0.791	1.000	0.988	1.000

Converged path models keep 0.974-0.989 with object identity removed and 0.988-1.000 with a shared kernel times a content gain; index models fall below the floor. Separation as such is MapFormer's (their Fig. 9).

Converging on new objects (`LEAK_RESULTS.md`, `GAIN_PHASE_RESULTS.md`, registered). On a task with a fresh map and 16 fresh objects per sequence (32 x 32 torus, T = 1024, rank 4), object tokens leak a small step into the phase. Removing the step from object tokens (ActOnly) or normalising the step input (NormStep, the step reads LayerNorm(embedding)) lets the model converge: 16/16 SOLVED (unseen-object accuracy 0.9997 / 0.9998) against 0/8 for MapWM (0.9890, still descending). On fresh seeds NormStep again converges 8/8 against MapWM's 0/8 (unseen-object accuracy 0.9940 against 0.9895).

L. PoPE's score

On the short-walk torus (T = 128, rank 2, n = 16, fresh seeds), MapPoPE's gain over MapWM comes with PoPE's score rule (+0.0243, permutation p 0.012; SOLVED 16/16 vs 10/16), and doubling the angle count adds +0.0002 at a ceiling where both arms sit at 0.9995-0.9997 (`MAPPOPE_PAIR_RESULTS.md`).

Appendix: theory (v3)

v3, 2026-10-16, for draft 6 (`docs/paper/DRAFT6_workshop.md`, appendix E). Replaces `APPENDIX_theory_v2.md` (kept unchanged). What changed in v3 (from the referee check `docs/paper/review_draft5/THEORY_CHECK.md`; list "Corrections in v3" at the end): A0's inhomogeneous term corrected (it was exact only to first order); a new P1(f) proves softplus representability exactly; P4 states the omega factor and "along s"; P7's precision bound separates the proved sufficient condition from a heuristic; wording fixes in A0 notes, B0, T1, P1, P3, P5 and Remark R; the "three integrators" paragraph moved here from the main text (A0, end). v2 (2026-10-15, for draft 4) changed from v1: a new section B0 (non-negative values vs barrier parameterisations); T1 is presented as a lemma with its corollaries qualified; the drift / weight-decay account under P2 is removed (ACCOUNTS T1: no drift in the channels attention reads; weight decay changes nothing detected); P4 gains the trained-model data; Remark R is updated with ACCOUNTS T2 (the signed collective route, SUPPORTED) and the loose fit on the |.| side; P7 is updated with ACCOUNTS T4 (CoPE fails a non-wrapping line too); a pointer to the GAUGE test (result: `GAUGE_RESULTS.md`); draft section references updated. Every check pointer of v1 is kept. Sources: the theory review (`docs/audits/2026-10-14/theory_review/THEORY_REVIEW.md`), the CoPE diagnosis (`docs/audits/2026-10-14/cope_diagnosis/COPE_DIAGNOSIS.md`), `ACCOUNTS_RESULTS.md`, `MECHANISM_RESULTS.md`, `docs/paper/figs/FIGS_NOTES.md`, `docs/audits/2026-10-12/ring_dial_audit.md` (M1). Every statement is numbered and labelled proved, proved (special case), sketch, conjecture or account. After each one, a "Checked" line names a script in `docs/paper/appendix_checks/`; its output sits beside it as `*_out.txt`. All checks are CPU-only and eval-only, and none trains a model. The checkpoints used are `runs/wrap_mech/p0/{SO,AO}_s800` (MECHANISM, ring) and `runs/generality/ring/CARoPEF_s70x`. Nothing here is registered, and nothing changes a registered verdict. Statements that cite registered batches (ACCOUNTS, MECHANISM, WRAP_EXISTENCE, GAUGE) say so; post hoc readouts are marked.

Notation and conventions

- **Channel.** A rotary channel is a pair c = (h, b), meaning head h and block b. It acts on query and key coordinates (2b, 2b+1) of head h (`model._apply_rope`). Write q_c, k_c in R^2 for those coordinates and R(phi) = [[cos phi, -sin phi], [sin phi, cos phi]]. The code rotates Q by R(theta_t) and K by R(theta_s), so channel c contributes q_c^T R(theta_s - theta_t) k_c to the score of query t and key s. Every coordinate of a head is rotated, and the score is the sum over the head's channels.
- **Step table and angle table.** The phase is theta_{t,c} = omega_c * sum_{u <= t} S_{c, x_u}. Here S = W_out W_in E is the C x V step table, with C = H * n_b channels, V tokens and E the embedding matrix; its rank is at most r. The angle table is Theta = diag(omega) S. In MapWM the row of W_out for channel (h, b) is h * n_b + b. omega is a free real parameter: nothing constrains its sign (`model.PathIntegrator`). Write M_r for the set of C x V tables of rank at most r.
- **Classes.** The *signed* class takes steps S. The |.| class (`model_sign`, mode "abs") takes steps |S| entrywise, so its angle table is diag(omega) |X| for some X in M_r. The *softplus* class (mode "pos") takes softplus(S). The observation-step-free versions (AO, `model_wrap_exist`) give non-zero steps only to the move tokens.

- **Angle sum.** For an inverse pair in one channel, sigma = Theta_{c,x} + Theta_{c,y} (v1 wrote S for this; S is the step table).
- **Ring intervals.** Each move token a_t at position 2t is a query that predicts the observation o_t. A revisit key is the observation token o_s at position 2s+1 from an earlier visit to the same cell. The tokens in (2s+1, 2t] are g = t - s moves (n+ of them +1 and n- of them -1) and g - 1 observations. Write d = n+ - n-. On a ring of N cells, the pair is a same-cell pair iff d = 0 mod N.
- For an inverse pair of tokens (x, y), such as the +1 and -1 moves, channel c *cancels through zero* if Theta_{c,x} + Theta_{c,y} = 0 with Theta_{c,x} != 0. It *cancels by wrapping* if Theta_{c,x} + Theta_{c,y} is in 2 pi Z \ {0}. It is *blind* if both angles are 0 mod 2 pi.

A0. The propagator identity (proved; prior art for the identity itself)

Proposition A0. Let alpha(t) be piecewise constant, with alpha = Delta_u on (u-1, u], and let Phi_t = sum_{u <= t} Delta_u. The homogeneous part of Howard et al.'s integrator, dF/dt = -s alpha(t) F, has the propagator

F(t) = exp(-s (Phi_t - Phi_{t'})) F(t').

The solution with input is F(t) = e^{-s Phi_t} F(0) + sum_{u <= t} e^{-s (Phi_t - Phi_u)} ((1 - e^{-s Delta_u}) / s) f_u, exactly, for piecewise-constant f (zero-order hold); the weight (1 - e^{-s Delta_u})/s is Delta_u + O(s Delta_u^2), and a zero-rate token writes nothing. For an impulse input f_u delta(t - u) the weight is exactly Delta_u. (v2 wrote Delta_u f_u as exact; it is the first-order term. Not used in the main text.) Take s = i omega. Channel c of the rotary score is then

q_c^T R(theta_s - theta_t) k_c = Re(conj(q~_c) k~_c G_c(t, s)), G_c(t, s) = exp(-i omega_c (Phi_t - Phi_s)),

with q~ = q_{2b} + i q_{2b+1}. So the position factor of a rotary score is exactly Howard's propagator with an imaginary spectrum, driven by the rate Delta.

Proof. On each interval the ODE is linear with constant coefficient -s Delta_u, so the flow over one token is exp(-s Delta_u). Flows compose by multiplication, and the exponents add to Phi_t - Phi_{t'}. Variation of constants gives the inhomogeneous term. For the score, R(theta) acts on q~ as multiplication by e^{i theta}. So (R(theta_t) q)^T (R(theta_s) k) = Re(conj(q~ e^{i theta_t}) k~ e^{i theta_s}) = Re(conj(q~) k~ e^{-i (theta_t - theta_s)}), and theta_t - theta_s = omega (Phi_t - Phi_s). The zero-order-hold discretisation is the exact flow, not an approximation. []

What the identification does not carry. 1. *Input gain.* In Howard's equation alpha also multiplies the input f, so a token with zero rate writes nothing. In attention, a zero-step token still writes its key and value at full strength. Softmax normalisation has no counterpart either. 2. *Real s.* Real s > 0 with a non-negative rate gives a decay exp(-s (Phi_t - Phi_s)) in (0, 1] (with a signed rate it can exceed 1). Its logarithm, -s (Phi_t - Phi_s), is the additive logit bias of GRAPE-AP / FoX / ALiBi (`papers/txt/grape.txt:682-686`). In that slot, non-negativity of the rate is what makes the bias a forgetting term, and

wrapping does not exist. GRAPE-AP is Case II with real s, not a rotary phase. 3. *Mamba-3.* The per-channel angle is Delta_t theta_t[i], with Delta_t > 0 (`papers/txt/mamba3.txt:154`; softplus in the implementation) and theta_t a signed, data-dependent projection (`papers/txt/mamba3.txt:637`). This is the propagator of dF/dt = -i Omega_t F with a content-dependent spectrum Omega_t = Delta_t theta_t. Howard's scalar rate times a fixed spectrum is the special case where theta_t is constant. The sign therefore sits in theta_t, not in Delta_t. The identity "complex SSM = data-dependent RoPE" is Mamba-3's Proposition 3 (`papers/txt/mamba3.txt:662`). Ours is only the mapping onto Howard's three rate cases. 4. *Vector rates.* MapWM's step is an r-dimensional latent read per channel, which is the D-dimensional (wave-vector) generalisation, not one scalar rate driving a bank. 5. *The sign of s.* s = +i omega and s = -i omega give the same model up to the reflection of Theorem T1. The convention is a gauge.

Checked (`a0_propagator.py`): the RK4 propagator of the ODE matches exp(-i omega (Phi_t - Phi_s)) to 3e-14 for signed, non-negative and constant rates. The `_apply_rope` score matches sum_c Re(conj(q~) k~ G) to 4e-15. The script tests only the homogeneous part; the corrected input term is checked by `docs/paper/review_draft5/theory_check_extra.py` (1): RK4 vs the corrected formula 1.1e-14, vs v2's Delta_u f_u 0.857 (|F| = 2.19; s = 0.7i, T = 12, F(0) = 0).

Three integrators (moved from draft 5 Sec. 3; the main text keeps one sentence in Related work). Howard et al. (2014) drive integrators by a rate: constant gives elapsed time, a speed path length (a clock), a signed velocity position (a map). Per token the rate is Delta_t. In MapWM the step is a vector, so with Phi_{c,t} = sum_{u <= t} Delta_{c,u}, channel c of the score of query t and key t' depends on position only through exp(-i omega_c (Phi_{c,t} - Phi_{c,t'})): Howard's propagator with an imaginary rate constant (for complex SSMs, Mamba-3's Prop. 3). Index RoPE is the constant case; CARoPE / CoPE and MapFormer / Selective RoPE / Mamba-3 are the non-negative and signed cases. In the real (decay) slot nothing wraps; in the rotary slot the last two cases represent the same per-channel codes (B0).

B0. Non-negative values versus barrier parameterisations (proved; elementary)

Remark B0. (a) *Non-negativity of the step value is not a constraint on the model.* For any signed step Delta and omega != 0, Delta' = Delta mod (2 pi / |omega|) lies in [0, 2 pi / |omega|), and omega Delta' = omega Delta - 2 pi k sign(omega) = omega Delta (mod 2 pi) for either sign of omega. The score reads each channel's angle only through cos and sin, so the model with Delta' is the same function. If the step is parameterised as s mod (2 pi / |omega|), its derivative in s is 1 except on a null set, so gradient descent on s is gradient descent on the signed model. *Rank caveat:* the entrywise remainder of a rank-r step table need not have rank r; it stays in the rank-r class when at most r tokens step (its non-zero columns then number at most r), which is P1(a); it covers every observation-step-free model and, with the observation embeddings annihilated, a map code for every |.| arm (P1(a), full arms). The full arms' trained signed tables, in which every token steps (V = 19 or 21 > r = 4), are not covered. (b) *What differs is the parameterisation.* |s| has a fold at s = 0 (P4); softplus(s) has no fold, but its slope sigmoid(s) vanishes as the step goes to 0, and it cannot reach 0. In both, the step is never negative, so no path moves one token's

angle in a channel through zero while another's stays put (omega flips a whole channel; T1): we call them *barrier parameterisations*. The paper's experiments compare the signed parameterisation with these two; "non-negative" in the paper means barrier-parameterised. (c) *Initial steps.* softplus(0) = 0.69, so the softplus class starts with every token at about the same step (an index clock with rescaled omega), unlike |.| and signed, which start near 0 (`model_sign.py` docstring; the pre-registration calls this an initialisation confound).

Proof. (a) remainder arithmetic and cos/sin periodicity. (b) chain rule. []

Not checked by script: (a) is the construction used, per channel, in P1 and in `model_wrap_exist.wrap_construct`.

T1. The relative-sign gauge (proved; Lemma 1 in the draft)

Status. T1 is an elementary identity, J R(phi) J = R(-phi) applied per channel. It is stated as a lemma because its corollaries say exactly which parameter points the |.| class excludes, and corollary (ii) makes a prediction (tested in GAUGE, not confirmed; see the Test paragraph under T1(ii)).

Theorem T1. Take a MapWM model with any number of layers. Fix a set F of channels and form a second model by two changes: (a) negate every token's step in each channel c in F; (b) negate output coordinate h * d_head + 2b + 1 of q_proj and k_proj, both the weight row and the bias entry, in every layer, for each c = (h, b) in F. Change (a) can be made by negating row c of W_out, which is a reparameterisation of W_out and leaves its rank and the latent unchanged. It can also be made by negating omega_c. In the |.| class only the omega_c form is available (|.| is applied after W_out, so negating a row of W_out does nothing there). The two models have identical attention scores at every layer, and hence identical logits.

Proof. Let J = diag(1, -1). Then J R(phi) J = R(-phi), and J is orthogonal. Change (a) replaces theta_{t,c} by -theta_{t,c} for every t, because the phase is linear in the steps and in omega_c. Change (b) replaces q_c by J q_c and k_c by J k_c at every position. This is exact including biases, because negating one output coordinate of an affine map negates its weight row and its bias entry, and nothing between the projection and the rotation mixes coordinates (`norm1` acts before the projection). The new channel term is

(J q_c)^T R(-(theta_s - theta_t)) (J k_c) = q_c^T J R(-(theta_s - theta_t)) J k_c = q_c^T R(theta_s - theta_t) k_c.

Other channels, V, O, FFN, the norms and the output head are untouched. Every layer uses the same cos/sin, so applying (b) in each layer makes every layer's scores identical. Inputs to the next layer are therefore identical, and so are the logits, by induction. In MapEM the same argument applies with the reflection on the c-th pair of q0_pos and k0_pos. []

Remarks. - *The gauge group per channel is O(2).* A common rotation U in SO(2) of (q_c, k_c) commutes with R(phi) and leaves the step unchanged. A reflection reverses the step. The sign of a channel's step is the determinant of its O(2) gauge element. - *Floating point.* IEEE negation is exact, and cos(-x) = cos x and sin(-x) = -sin x hold bitwise in the CPU

kernels used (checked; this is a property of the math library, not of IEEE 754). The two models therefore agree bitwise there, not only to rounding.

Corollary T1(i): sign-coherence characterisation (proved). Let S be in M_r, with angle table Theta = diag(omega) S. The |.| class realises Theta exactly *as real numbers* (no wrapping), with the same omega up to sign and the same content weights (the same latent), if and only if every row of Theta is *sign-coherent*: Theta_{c,x} Theta_{c,y} >= 0 for all tokens x, y. (Equivalently every row of S, when omega_c != 0; a row with omega_c = 0 is zero in Theta and realised whatever the signs of S's row.)

Scope note. This does not contradict P1(a). P1(a) reproduces opposite-sign rows exactly *mod 2 pi* with a *different* latent (W_in', W_out'), when at most r tokens step. T1(i) is about the same table as real numbers.

When it holds, take D = diag(sign of row c) and a |.| model with W_out unchanged and omega' = D omega. Then omega'$_c$ $|S\{c,x\}|$ = omega_c S_{c,x}. No reflection is needed, because omega carries the channel's sign. If omega were constrained positive, T1 would absorb D into the q/k reflection instead.

Conversely, every row of a |.| angle table, omega'$_c$ $|X\{c,.\}|$, is sign-coherent. So, modulo T1, the |.| constraint does not fix the direction of any channel. It removes only the *relative* signs of the tokens that step in a channel: the |.| class is "in each channel, every stepping token rotates the same way". Two side remarks. The |.| class also contains tables |X| of rank greater than r, when X has mixed-sign rows; all of these are sign-coherent. On the torus the four moves form two inverse pairs, and the constraint makes all four rotate the same way in each channel.

Corollary T1(ii): one stepping token, no representational sign cost (proved; the learnability prediction is untested for one stepping token and not confirmed for +1/+2, see the Test below). Suppose a single token x steps, so S = s e_x^T. Then the signed and |.| classes realise the same angle tables: {omega_c |s_c|} = {omega_c s_c} = R^C. There is also no cut, because omega_c is a continuous parameter that crosses zero, so any channel's angle moves continuously through 0 in either class. Parity, written as R(pi x_t) (Mamba-3 Sec. 3.2, `papers/txt/mamba3.txt:576-578`), and Z_m counting by rotations through 2 pi/m (Grazzi et al. 2025, Thm. 3 and App. C.1) are cases of this kind. The prediction is no sign cost on single-stepping-token tasks. It is not tested here. The |.| reduced loss still has a kink at s = 0 (P4), so the corollary says nothing about learning speed.

Extension (by T1(i)). The same holds, for map *functions*, whenever the task has a solution code whose rows are sign-coherent, for any number of stepping tokens. (Other codes of the same function can still be excluded as parameter points: on +1/+2 the code a(+1) = 1, a(+2) = 2 - 2 pi = 2 a(+1) mod 2 pi has an opposite-sign row and is not in the |.| class. What is not excluded is the function, which has the sign-coherent representative a(+2) = 2 a(+1).) Example: a ring walked by moves +1 and +2. The map code a(+2) = 2 a(+1) has both angles of one sign in every channel, so it lies in the |.| class without wrapping and can be reached from a small same-sign start without crossing zero. The prediction is no sign cost there, against a cost for +1/-1 (an inverse pair).

Test (registered, `GAUGE_RESULTS.md`). Ring 32 cells, observation-step-free SignedAO_r4 / AbsAO_r4, moves +1/-1 vs +1/+2, n = 8, seeds 1300-1307. +1/-1: |.| 1/8 vs signed 8/8 (Fisher p 0.0014). +1/+2 (primary G1): |.| 2/8 vs signed 5/8, d -0.064 [-0.200, +0.075]: UNMEASURED; |.| on +1/+2 vs +1/-1: 2/8 vs 1/8. The signed model learns a sign-coherent code on +1/+2 (0.92 of channels same-sign vs 0.01 on +1/-1), so T1(i)'s representational statement is visible in what signed steps learn; the learnability prediction is NOT confirmed (the task is harder for both arms), and nothing measured suggests the |.| cost needs an inverse pair.

Corollary T1(iii): the excluded codes (proved). Up to the gauge, the signed tables that the |.| class excludes (without wrapping) are exactly the tables with a row containing entries of strictly opposite sign. For an inverse pair (x, y), every through-zero cancelling channel (Theta_{c,x} = -Theta_{c,y} != 0) is such a row. The only unwrapped cancellation the |.| class allows is a blind channel. Cancellation by wrapping, Theta_{c,x} + Theta_{c,y} in 2 pi Z \ {0}, is allowed: it is representable whenever P1 applies. So the constraint removes the through-zero branch of the inverse-pair problem, not cancellation itself. The sign question is about inverse pairs, not about wrapping.

Functions versus parameter points. A through-zero code (a, -a) and the wrapped code (a, 2 pi - a) are the same function (B0(a)). What T1(iii) excludes are parameter points; when P1 applies, every excluded function has a representative in the |.| class. The cost the paper measures is therefore a cost of reaching that representative from where training starts (P3), not of representing it. "Exactly the through-zero codes" holds within the set of cancelling codes of an inverse pair; the full excluded set is every row with strictly opposite-signed entries.

Checked (`t1_gauge.py`). - SO_s800, with 35 random channels flipped (70 move angles change sign) and reflected: logits are bitwise identical in float32 and float64. The control, a flip without the reflection, moves the logits by 6.6. - AO_s800 with the flip carried by omega: identical (control: 22). - A random 2-layer Signed_r4 in which every token steps and the q/k biases are N(0, 0.5): identical (control: 0.23). - MapEM with q0_pos and k0_pos reflected: identical, but this sub-check is uninformative as run: its control (a flip without the reflection) moves the logits by only 4.7e-5, because q0_pos / k0_pos sit at their 0.02 initialisation. It would need q0_pos / k0_pos scaled up (e.g. x 50) to test anything. - (i), constructive: AO_s800 is rewritten as a signed model with steps D|S|. Logits agree to 1e-11 with the reflection, and bitwise with the sign in omega instead. - (iii): SO_s800 has 53/64 opposite-sign channels, and its weights loaded into the |.| class change the logits by up to 35.

P1. Representability by wrapping

Lemma P1.0 (per channel; proved). For any target angle A and any omega' != 0 there is a step Delta' >= 0 with omega' Delta' = A (mod 2 pi). For example, Delta' = remainder(A, 2 pi) / |omega'| with omega' > 0. []

Proposition P1(a) (at most r stepping tokens; proved). Let S be a signed table with at most m <= r non-zero columns (the stepping tokens x_1..x_m). Assume the stepping tokens' embeddings are linearly independent modulo the span of the other embeddings. This holds generically when V <= d_model (here 19 <= 128). In the AO models it is

automatic, because only the moves' embeddings enter the step path. Then a |.| model with the same embeddings and content weights has an angle table equal to Theta mod 2 pi, entrywise.

Proof. Pick W_in' with W_in' e_{x_j} = u_j, the j-th unit vector of R^r, and W_in' e_y = 0 for every other token. This is possible by the independence assumption, since m <= r. Set omega' = max(|omega|, 1) per channel and W_out'[:, j] = remainder(Theta_{., x_j}, 2 pi) / omega' >= 0 (Theta = diag(omega) S), with all other columns zero. Then W_out' W_in' E has non-negative entries, so |.| leaves it unchanged. It reproduces each stepping column mod 2 pi and leaves every other column at zero. []

This is the observation-step-free construction of draft 6 Sec. 4.2 (`model_wrap_exist.wrap_construct`, mode `balanced`). On the ring m = 2 and r = 4. On the torus m = 4 = r, exactly at the limit. P1(a) is sufficient but not necessary: by T1(i), any table with sign-coherent rows is representable for any m, with no wrapping at all.

The full (observation-stepping) |.| arms. Representing the *task* does not require reproducing a given signed table in which observations step: choose W_in' to annihilate every observation embedding, which P1(a)'s independence assumption allows when V <= d_model, so that only the moves step. Every |.| arm in the paper can therefore represent any code in which only the moves step, in particular the map codes of the trained signed observation-step-free models. The ring_dial audit built this in the real modules on the ring (`docs/audits/2026-10-12/ring_dial_audit.md` M1; `wrap_construction_t2.py`): Abs steps >= 0 (min 5e-11), observation-token |Delta| <= 1.7e-6, cos/sin equal to the signed code's to 6.3e-4 over a 512-step walk, circular mismatch <= 4.6e-4 on 2,000 same-cell pairs. P1(c) and P1(e) below concern reproducing a *given* table with more than r stepping tokens; they are not needed for the paper's claims.

Proposition P1(f) (softplus, exact; proved; new in v3). Let at most r tokens step in a signed table Theta, with embeddings as in P1(a). Choose W_in' sending the stepping embeddings to u_1..u_m and every other embedding to 0, set omega'*c = 2 pi / ln 2 for every c, and W_out'[c, j] = softplus^{-1}((remainder(Theta*{c,x_j}, 2 pi) + 2 pi) / omega'_c). Every non-stepping token then steps softplus(0) = ln 2 in every channel, an angle of exactly 2 pi, and every stepping token's angle is Theta mod 2 pi. (Any omega'_c = 2 pi k / ln 2, k >= 1, works. The + 2 pi keeps every target step strictly positive: the target (remainder + 2 pi) / omega' lies in [ln 2, 2 ln 2), inside softplus's range (0, inf), so softplus^{-1} is defined even where the remainder is 0.)

Proof. softplus is a bijection from R onto (0, inf); the targets are positive; a non-stepping token's pre-softplus step is 0. Angles: omega' ln 2 = 2 pi, and omega' (remainder + 2 pi)/ omega' = remainder + 2 pi = Theta mod 2 pi. []

So softplus, like |.|, can represent the map on every task here: it cannot set a step to zero, but it need not, since the observations' common step can be a full turn. Its existence under training (frozen and trainable continuations, as for |.| in draft 6 Sec. 4.2) was not tested.

Checked (`docs/paper/review_draft5/theory_check_extra.py` (2), CPU): a full `MapFormerWM_Pos_r4` built this way from SO_s800 has min step 0.693 > 0, angle error mod 2 pi 8.4e-15 rad

over all 19 tokens, and logits equal to SO_s800's to 2.8e-11 (logit scale 21.6). On the torus m = 4 = r, so the construction applies (not built).

P1(b) (the general condition; definition-level). A signed angle table Theta has an exact |.| rank-r representation mod 2 pi iff there exist an integer matrix K and a sign matrix Sigma with two properties: every row of Theta' = (Theta mod 2 pi) + 2 pi K is sign-coherent, and rank(Sigma o Theta') <= r. Here o is the entrywise product, and the rows of Theta' may be rescaled freely by omega'. Realisability through the embeddings is as in P1(a). This merely restates the definition.

Proposition P1(c) (generic failure, rank 1; proved). Let r = 1 and C >= 2. Let the signed table be Theta = alpha v^T with alpha in R^C, and suppose two tokens step with v_1 > 0 > v_2. Then the set of alpha for which the |.| rank-1 class represents Theta exactly mod 2 pi has Lebesgue measure zero. The same holds for any m >= 2 stepping tokens whose v has entries of strictly opposite sign: restrict to such a pair.

Proof. A rank-1 |.| angle table is diag(omega') |w z^T| = a b^T, with a = omega' o |w| arbitrary and b = |z| >= 0. So we need a_c b_j = alpha_c v_j + 2 pi k_{cj} for j = 1, 2, with integers k_{cj}.

If b_1 = 0, then alpha_c v_1 is in 2 pi Z for every c, a null set. Otherwise scale b_1 = 1. Then a_c = alpha_c v_1 + 2 pi k_{c1}. This is non-zero except on a null set (if a_c = 0, alpha_c v_2 is in 2 pi Z). So b_2 = (alpha_c v_2 + 2 pi k_{c2}) / (alpha_c v_1 + 2 pi k_{c1}), and this must be the same for c = 1 and c = 2.

Clearing denominators gives, for each integer quadruple (k_1, k_2, l_1, l_2), an equation that is *affine* in (alpha_1, alpha_2). The alpha_1 alpha_2 terms cancel:

2 pi alpha_1 (l_1 v_2 - l_2 v_1) + 2 pi alpha_2 (k_2 v_1 - k_1 v_2) + 4 pi^2 (k_2 l_1 - k_1 l_2) = 0.

Suppose it holds identically. Then k_2 v_1 = k_1 v_2 and l_2 v_1 = l_1 v_2, which forces b_2 = v_2 / v_1 < 0 in each channel, contradicting b_2 >= 0. Otherwise its zero set is a line in the (alpha_1, alpha_2) plane. There are countably many quadruples, so the admissible set is a countable union of lines, a null set in R^2, and its product with R^{C-2} is null in R^C. []

Proposition P1(d) (approximation at rank 1, any m; proved; the scale is heuristic). Let b = (1, b_2, ..., b_m) >= 0 be linearly independent over Q. By Kronecker's theorem, t -> (t b_j mod 2 pi)*j is dense in the m-torus. So for every channel there is a_c with |a_c b_j - Theta*{c,j}|_{mod 2 pi} < eps for all j. A rank-1 non-negative code therefore approximates *any* angle table mod 2 pi to any precision. The cost is the angle size. Heuristically, |a_c| ~ eps^{-(m-1)}: this is the return time of a linear flow on T^m, measured roughly, not proved.

So "cannot represent" in P1(c) and in the conjecture below means *exactly*. Approximately, representation is never the obstruction. Its price is weight scale, which is also the rescaling sensitivity of P2.

Conjecture P1(e) (r >= 2; sketch only). For r >= 2 and m > r, consider signed rank-r tables that have at least one row with strictly opposite-sign entries. Among these, those with an exact |.| rank-r representation mod 2 pi form a null set in the rank-r variety.

Sketch (from the review). For fixed (Sigma, K), the condition is the vanishing of the (r+1)-minors of Sigma o Theta', which is polynomial in Theta on each cell where Theta mod 2 pi is affine. There are countably many (Sigma, K). The missing step is to show that no minor condition holds identically on the mixed-sign set. For r = 1 that is the computation above. For r >= 2, |X| can have rank greater than r, so the r = 1 argument does not transfer. The restriction to mixed-sign rows is needed: the sign-coherent set and the m <= r set have positive measure and are representable (T1(i), P1(a)).

Checked (`p1_wrap.py`). - (a) A random rank-4 signed model whose W_in annihilates all but m stepping embeddings (m = 1..4; for m = 2..4, 39-56 of 64 channels have opposite-signed steps; m = 1 has none) is rebuilt in the |.| class. The built table is >= 0, the angle error mod 2 pi is <= 7e-14 and the logit difference is <= 8e-14 (float64). - (c) (labelled "(b)" in the script) Sympy confirms that the compatibility numerator is affine in alpha with the stated coefficients, and that the identically vanishing case forces b_2 = v_2 / v_1. A search over integer wraps |k| <= 5 / 20 / 80 never finds an exact solution, but the b_2 mismatch shrinks (1e-3 -> 2e-8), which is the approximation of (d). - (d) (labelled "(c)" in the script) Max-over-64-channels error vs allowed wrap count K = 10..10^4: m = 2: 0.17 -> 2e-4 rad. m = 3: 1.15 -> 0.065. m = 4: 1.53 -> 0.15. This is roughly the eps^{-(m-1)} scaling, and fitted to nothing.

P2. Rescaling sensitivity (proved; mathematics only)

Proposition P2. Fix a channel with move angles a+ and a-. Over an interval with n+ forward and n- backward moves, the raw phase is phi = a+ n+ + a- n-. Multiply the channel's angles by (1 + eps), whether through omega, through W_out's row, or through any factor of the step. The phase changes by eps * phi exactly, and hence by eps * phi mod 2 pi.

- Signed cancelling code, a- = -a+: phi = a+ d. The error is eps a+ d. It is zero at every d = 0 revisit for every eps: the set {a+ + a- = 0} is a linear subspace, invariant under rescaling.
- Wrapped code, a- = 2 pi - a+: phi = a+ d + 2 pi n-. The error is eps (a+ d + 2 pi n-), which is 2 pi eps n- at d = 0. It grows with the number of backward moves, a path length, not with the displacement. The set {a+ + a- = 2 pi} is affine and off the origin, so it is not invariant.
- Wrap revisits (d = +-N) need the lattice condition a+ N in 2 pi Z in both codes. That condition is not rescaling-invariant in either code. The signed code's error there is eps a+ d = 2 pi k eps.

Proof. phi is linear in (a+, a-). Substitute each code. []

Remark P2.1 (AdamW decoupled decay is a rescaling; proved). AdamW applies p <- (1 - lr_t wd) p to every parameter tensor (`train.py:57`), separately from the gradient step. A move angle is the product of four decayed tensors (omega_c, the row of W_out, W_in,

the move embedding), so the decay term alone is a uniform eps < 0 rescaling of every angle, to which P2 applies.

Status (v2). v1 offered P2 and P2.1 as an account of a "drift" of the held non-negative ring model off its cancelling code (87% -> 18% of channels) and of the torus knock-out. Both uses are withdrawn. ACCOUNTS T1 (registered; its wd 0.05 runs are bit-identical retrains of WRAP_EXISTENCE's, so this is a second readout of the same runs, not a replication): weighted by how strongly attention reads each channel, the held model keeps 98% of its cancelling code (0.99 -> 0.98); the unweighted drop was in channels attention barely reads; and removing weight decay changes nothing detected (D1, -0.003). P2 remains a statement about sensitivity: the wrapped code is not invariant under rescaling and the signed cancelling code is. Its frozen-model consequence is measured (WRAP_EXISTENCE secondary, a probe of frozen models, not a basin width): 1% Gaussian noise on omega leaves the signed solution at 0.998 and the wrapped one at 0.802 (W_out: 0.975 vs 0.815). Nothing here says how training moves either code.

Checked (`p2_rescale.py`). - 106,602 same-cell pairs from 400 registered ring walks (0.787 at d = 0). The measured error matches the formula to <= 1e-13 rad for both codes at eps = 1e-3 and 1e-2. rms(n-) = 26.6, so the predicted rms wrapped-code error at eps = 0.01 is 1.67 rad. - On AO_bal / SO_bal built from SO_s800 (exact to 7e-7), omega x (1 + eps) in every channel gives the following revisit accuracies:

eps	wrapped	signed
1e-3	1.000	1.000
3e-3	0.941	1.000
1e-2	0.700	1.000
3e-2	0.485	0.866 (the lattice term)

- v1's integrated-decay figures (wd * sum(lr) = 0.805 ring, 0.080 torus) are still printed by the script; they are no longer used.

P3. The cut circle (proved)

Take one channel and an inverse pair (x, y) with angles (a_x, a_y), and hold everything else fixed. The model depends on the angles only through cos and sin, so the loss is a function on the angle torus (R / 2 pi Z)^2.

Proposition P3. (a) The closed quadrant [0, inf)^2 maps onto the torus, so the signed class and the |.| class attain the same set of loss values in this channel. The point (a, 2 pi - a) is the point (a, -a). (b) The non-blind cancelling set is {a_x + a_y in 2 pi Z}, excluding both angles = 0 mod 2 pi. Its part in the quadrant consists of the segments a_x + a_y = 2 pi k with k >= 1. The through-zero line k = 0 meets the quadrant only at the blind origin. (c) Take a start in the open quadrant with sigma = a_x + a_y < 2 pi. The infimum Euclidean distance to a non-blind cancelling point is min(sigma, 2 pi - sigma) / sqrt 2 in the signed class and (2 pi - sigma) / sqrt 2 in the |.| class. They coincide iff sigma >= pi. (d) Any continuous path in the |.| class, with omega's sign free, from |sigma(0)| < pi to a non-blind cancelling point passes through |sigma| = pi. A signed path to the through-zero

branch exists that keeps |sigma| <= |sigma(0)|: the straight segment to the projection onto sigma = 0, perturbed off a_x = a_y if needed.

Proof. (a) and (b) are immediate from the quotient map. (c): the distance from a point to the line a_x + a_y = 2 pi k is |sigma - 2 pi k| / sqrt 2. The signed class may use k = 0 or k = 1. In the |.| class, k = 0 is only the origin, which is blind, leaving k >= 1, and k = 1 is nearest. With omega < 0 the class also contains the negative quadrant, which is farther. (d): along a path in the |.| class, |sigma| is continuous and starts below pi. A non-blind cancelling point has |sigma| >= 2 pi. Apply the intermediate value theorem. []

So the constraint removes no solution. It cuts each channel's circle at zero. From a same-direction start with sigma < pi, the nearest non-blind cancelling point (in angle coordinates, an infimum; not a distance in parameter space) lies across the cut, and the only branch on the near side is the wrap, which is reachable, but only through |sigma| = pi. From sigma >= pi the two classes have the same nearest point. P3 is a statement about distances; it does not say that a channel started past pi flows to the wrap (Remark R).

Checked (`p3_p5_geometry.py`, P3): a brute-force grid (step 0.025) from 60 random starts reproduces both distance formulas to 2e-4.

P4. The sticky wall (proved for the reduced coordinate)

Proposition P4. Let s be the pre-|.| step of one token in one channel, as a function of the parameters (W_out row, latent), and let a = omega |s|. Write L(s) = l(|s|), with l C^1 near 0 and everything else fixed. (a) For s != 0, dL/ds = sign(s) l'(|s|); in terms of the angle, dL/ds = omega sign(s) dL/da. Through the chain rule, the gradient with respect to every parameter that moves s is sign(s) l'(|s|) grad s. (b) If l'(0) > 0, meaning the loss wants that angle smaller at the wall, then s = 0 is a strict local minimum of L *along s, with every other parameter fixed*: L(s) - L(0) = l'(0) |s| + o(|s|). In parameter space s = w_c . z_x is shared (w_c moves every token's step in channel c, z_x every channel's step for token x), so this is not a local minimum of the loss. The signed model's loss l, extended to negative arguments, decreases into a < 0 at the same point. So the stationary points of |.| pinned at the wall (with l'(0) != 0) are exactly the points where the signed loss decreases into s < 0. (c) Gradient descent on s itself with step eta, with l' in [mu, M] on [0, rho], mu > 0, eta M <= rho and |s_0| <= rho, gives |s_{k+1}| <= max(|s_k| - eta mu, eta M). The iterate enters |s| <= eta M in finitely many steps and stays there: a chatter of amplitude eta M, not a bounce. (Gradient descent on the factors (w, z) moves s by eta l' sign(s) (|w|^2 + |z|^2) + O(eta^2).)

Proof. (a) is the chain rule for |.|. (b) is a first-order expansion. (c): s_{k+1} = s_k - eta sign(s_k) l'(|s_k|), so |s_{k+1}| = | |s_k| - eta l'(|s_k|) |. Split into the cases |s_k| >= eta l' and |s_k| < eta l'. []

Under Adam the normalised step makes the chatter amplitude ~lr. This is a remark, not proved. Weight decay (P2.1) adds a deterministic pull toward 0.

Trained models (post hoc, eval-only; `docs/paper/figs/fig_mechanism_out.txt`, MECHANISM seeds 800-807). Share of channels whose forward-move angle ends

within 0.05 rad of 0 (mod 2 pi): |.| from the same-direction (|default|) start (AO_absmag) 0.441; signed from the same start (SO_absmag) 0.166; |.| from a large start with sums below pi (AO_mid) 0.002; |.| from a start past pi (AO_high) 0.014. Of AO_absmag's registered 0.04 cancelling share, 0.035 is same-sign channels near the origin, not cancellation. This is what P4 predicts if the loss pushes those steps toward zero: they settle at the wall. It is a competing (or additional) account of the |.| failure beside P3's wrap, which lies past |sigma| = pi. *Checked since (post hoc, eval-only, rules fixed in advance in `docs/audits/2026-10-15/reanalyses/PREDICTIONS.md; collapse_wall_out.md`):* counted per move step, collapse is common to every |.| arm from a small start (24-42% on the ring, 60-70% on the torus, softplus included) and smaller in signed controls from the default start (10-16%, 34-40%); it is partly inherited (77-81% of |.| steps starting within 0.05 rad of zero end there; signed 29-35%); attention reads the collapsed channels (usage ratio 0.39-1.10 of the others'); and at the end of training the loss gradient points into the wall (l'(0) > 0) for only 51-57% of the collapsed steps (eval mode, ring), which the pre-stated rule reads as MIXED, not as P4. Torus gradients were not computed.

Checked (`p3_p5_geometry.py`, P4): on AO_s800 (float64, revisit cross-entropy), autograd gives dL/ds = sign(s) dL/d|s| exactly (difference 0; the script's a is the pre-omega table | s|). 69 of 128 pre-|.| entries are negative.

P5. The pi-mirror (proved, with the scope stated)

Lemma P5.0. On a ring of even size N, every same-cell pair has an even move count g.

Proof. d = 0 mod N with N even gives d even, and g = n+ + n- = d + 2 n- = d mod 2. []

Proposition P5. Let Pm act on every channel as (a+, a-) -> (pi - a+, pi - a-). This sends sigma -> 2 pi - sigma and delta -> -delta, where delta = (a+ - a-) / 2. (a) In the observation-step-free model (no observation steps), Pm maps the raw phase of every interval to pi g - phi. So it negates every even-g phase mod 2 pi, and in particular every same-cell phase, and it sends every odd-g phase to pi - phi. (b) Consequently, any functional of the same-cell phases through cos is exactly invariant under Pm. This includes the attractive term V_s(a) = E_q[mean over same-cell keys of cos phi] of the review's small-beta potential, and it needs no assumption on the walk. (c) *Model level.* In MapWM (AO), Pm applied to the move angles of every channel, together with the T1 reflection of every channel, leaves every even-g attention score unchanged and negates every odd-g score. Odd-g pairs are never same-cell pairs. (d) Pm maps the signed through-zero cancelling line {a- = -a+, 0 < |a+| < pi} onto the wrap segment {a+ + a- = 2 pi, a+ and a- in (0, 2 pi)} minus the point (pi, pi) (whose preimage is the blind a+ = 0), which lies in the non-negative quadrant. The fixed set of the mirror in sigma is sigma = pi. (e) The mirror sigma -> 2 pi - sigma *at fixed delta*, i.e. (a+, a-) -> (pi - a-, pi - a+), leaves V_s invariant under either of two conditions: (1) the law of (g, d) over same-cell pairs is symmetric under d -> -d, which holds in expectation at p_plus = 0.5 by the walk's reflection symmetry; or (2) delta N is in pi Z, including the wrap lattice. It is not a symmetry of the all-keys term V_a, nor of the transformer's loss.

Proof. (a): (pi - a+) n+ + (pi - a-) n- = pi g - phi. For g even, pi g is in 2 pi Z. (b): cos is even, and same-cell pairs have g even (P5.0). (c): an even-g phase becomes -phi, and with T1's J the channel term returns to its original value. An odd-g phase becomes pi - phi, and

q^T J R(pi - phi) J k = q^T R(phi - pi) k = -q^T R(phi) k. Every channel is transformed, and every coordinate of a head is a rotary channel, so the whole score is negated. (d): substitution. (e): the phase is sigma g/2 + delta d. Under the map it becomes pi g - sigma g/2 + delta d = -(sigma g/2 - delta d) mod 2 pi. This equals the original after d -> -d, or when 2 delta d is in 2 pi Z for d in N Z. []

Scope. P5 needs moves +-1 (it fails on the +1/+2 task, where it is not used). It is a symmetry of the same-cell (attractive) part of the objective and of the even-gap scores. It is not a symmetry of the loss, whose odd-gap competitors are negated, nor of the dynamics. "A start with sigma > pi is on the wrap's side" means that it lies on the wrap's side of this mirror and is nearer the wrap than the through-zero branch (P3(c)). It does not mean that the flow from such a start reaches the wrap.

Checked (`p3_p5_geometry.py`, P5; `p5_pm_accuracy.py`). - Same-cell move counts are all even in 58k revisit queries at p_plus 0.5 and 0.75. - |V_s(Pm a) - V_s(a)| <= 2e-15 on 20 random points per sample, at both p_plus values. - The fixed-delta mirror is off by 7e-3 on a plain p_plus 0.5 sample, exact (9e-16) once the sample is closed under walk reflection, off by 0.46 at p_plus 0.75, and exact (2e-15) on the lattice. - The all-keys term changes by 0.06-0.18. - Model level on SO_s800 (float64): even-g scores are unchanged to 1e-10 and odd-g scores negated to 1e-10, on a score scale of 121. The control without the reflection moves even-g scores by 45. The image has non-negative angles on 128/128 entries, and its opposite-sign channels' sums are 2 pi - sigma in [5.69, 6.67]. - Descriptive: the mirror image, untrained, scores 0.850 revisit accuracy against the source's 1.000. The odd-gap negation costs accuracy, as (c) allows.

Remark R. The collective route (account; signed side tested)

The mechanism in the draft is: "a signed model escapes the same-direction start by moving one move's angle through zero, collectively through the move latents; a |.| model cannot, and reaches the map reliably only from a start past pi". P3-P5 give the geometry. The dynamics are not derived.

Per channel the flow does not reproduce the run-level pattern. The review computed the exact one-channel small-beta potential (the retrieval objective at a fresh model's small attention scale, q.k / sqrt(d) std ~0.3, where softmax is close to linear in the scores) and ran gradient flow on it (`theory_review/potential.py`, `mech_capture.py`). |.| starts reach the wrap only from within ~0.5 rad of 2 pi (28% of channels from U(1.7, 4.5)), not from everywhere past pi. Signed starts from the |.| start cross zero in only 12% of channels.

The collective route. In latent space the set of latents at which every channel cancels through zero is {W_out (z(+) + z(-)) = 0}; for W_out of full column rank (C = 64 >= r = 4, generically) it is {z(-) = -z(+)}, otherwise a larger linear subspace. Either way it is linear in the latents at fixed W_out (bilinear jointly), so a single latent direction cancels every channel at once. Under |.| that route yields a(-) = a(+), a clock. In the review's toy the route starts on a plateau (`path_cut_out.txt`).

Prior observation. Trained signed MapFormers end in this set: opposite actions map to antipodal latents, cos(Delta_left, Delta_right) = -1, and observations to near-zero steps

(Rambaud et al., Fig. 3d and Fig. 9b, `papers/txt/mapformer.txt:650-655, 2120-2155`). That is the end state, described; ACCOUNTS T2 tests whether it is the route.

Signed side: tested (ACCOUNTS T2, registered, seeds 1000-1007). From the |.| start, freezing the two move latents (W_out and omega free per channel) blocks the escape: 1/8 vs 7/8 SOLVED (Fisher p 0.010; -0.075 [-0.130, -0.020]); 15% vs 82% of channels end opposite-signed. From the default start freezing has no detected effect (7/8 vs 8/8). SUPPORTED.

|.| *side: open, and it fits loosely (post hoc).* - The |.| (|default|) start has per-channel sums 0.01-6.58 rad (`fig_mechanism_out.txt`); 5% of channels start past pi (review `mech_channels_out.txt`). Among the AO channels whose forward step does not end collapsed (|wrap(a(+1))| > 0.05; 321 channels over 8 runs), the 20 that start with sums in [pi, 3 pi / 2) end with none cancelling, and fewer than 5 start beyond 3 pi / 2 (review `mech_bins_out.txt`). Past pi is not sufficient per channel. - From past pi (AO_high, 8/8 SOLVED) only 23% of channels end cancelling (SO_high 34%; 7% at the start by chance), and the final cancel share falls gently with the initial distance from 2 pi (0.34 within 0.25 rad, 0.19-0.25 at 0.5-2 rad, 0.08 at 2-3 rad; SO_high 0.43-0.52 -> 0.11). The per-channel flow's 28% is close to the observed 23% in aggregate, but not channel by channel (post hoc, `docs/audits/2026-10-15/reanalyses/capture_by_channel_out.txt`): run from each AO_high channel's own start the flow predicts 0.33 cancelling against 0.23 observed, agrees with the trained models barely above chance (Cohen's kappa +0.09; SO_high +0.20), captures 95% of the channels starting within 0.5 rad of 2 pi where the models cancel 28%, and does not capture 59% of the models' cancelling channels. So per-channel capture does not explain the rescue; whether it is collective is untested. What the other 77% of channels do is not identified. - From the small start, the wall (P4, trained-model data above) is a competing account of the failure.

So the run-level result (8/8 vs 2/8 vs 0/8) needs enough channels to reach a cancelling code, and the start shifts how many can. What is proved is the geometry (P3-P5); what is tested is the signed collective route; the |.| dynamics are open.

P6. CARoPE cannot cancel (proved)

CARoPE (`model_mono_baselines.MapFormerWM_CARoPEF`): every token, observations included, advances channel i of head h by c(x) = f_h(x)^i, with f in (0, 1), so c is in (0, 1]. Channel i = 0 has c = 1 for every token, which is an index clock.

Proposition P6. (a) The phase of any interval containing k >= 1 tokens lies in (0, k]. The shortest return is out and back, g = 2. Between the move-token query and the observation key it contains 3 tokens (2 moves and 1 observation), so its phase is in (0, 3]. Between two observation tokens it contains 4 tokens and its phase is in (0, 4]. Either way the phase is in (0, 2 pi), never 0 mod 2 pi. No channel, at any parameter value, reads every return at phase 0. (b) *Clock or blind.* Fix eps in (0, pi - 2). Consider the excursion J moves forward and then J back, with nested returns j = 1..J. If a channel reads each of its returns within eps of 0 mod 2 pi, then J (c+ + c-) <= eps. The move-driven phase of any interval with at most 2J moves then lies in [0, 2 eps]. Over displacements up to J, the channel separates cells by at most 2 eps of phase from moves. So every channel is either a clock (it misses some short return by more than eps) or blind over that range.

Proof. (a): sum k numbers in (0, 1]. (b): the return at depth j spans 2j moves and 2j - 1 observations. Its phase P_j satisfies P_1 in (0, 3] and P_{j+1} - P_j = c+ + c- + c(o) + c(o') in (0, 4], where the two observations are at the same cell. The arc (eps, 2 pi - eps) has length 2 pi - 2 eps > 4 when eps < pi - 2, so a sequence that starts in (0, eps] with increments of at most 4 cannot jump over it. Since every P_j must stay within eps of 0 mod 2 pi, induction gives P_j in (0, eps] for all j. Hence J (c+ + c-) <= P_J <= eps, and 2J max(c+, c-) <= 2 eps. []

This replaces the draft's earlier reason, "steps are at most 1 rad", with the review's corrected argument. The (0, 4] in the review is the observation-observation pair; the move-query pair gives (0, 3]. The conclusion is the same.

Checked (`p6_carope.py`). - Eight trained ring CARoPEF checkpoints: the g = 2 advance lies in [0.000, 3.000] (channel 0 is exactly 3). - 0-15 channels per run read every return within 0.1 rad. All of them are blind: their per-token step is <= 1e-3 rad, and their phase over *all* query-key pairs, any cells, is <= 0.105 rad. - 10,331 random (rates, excursion) draws satisfying the premise of (b): the bound held in all.

P7. CoPE on the ring: the lattice family

Setup (exact for this model; `COPE_DIAGNOSIS.md` Sec. 1). - The model is one layer with no other position signal, so each gate is a table G[h, query type, key type]. - Only move queries carry loss. - The count from the key o_s to the query a_t sums one move gate and one observation gate per step: p = n+ alpha + n- beta, with alpha = G[a, +] + cbar and beta = G[a, -] + cbar, when the observation gates are constant (tau = 0). - The position logit is a table z read by linear interpolation between integer knots.

Lemma P7.0 (proved). A periodic, non-constant, piecewise-linear function with knots at the integers has an integer period P >= 2.

Proof. Its kink set K is a non-empty subset of Z, since a function with no kinks is affine, and a periodic affine function is constant. K is invariant under the period, so the period is a difference of integers. Period 1 makes all knot values equal, which is constant. []

Proposition P7 (lattice; proved under the setup). Let z have period P and a unique maximum per period at p = 0 mod P, as the triangle-type tables do. Suppose the walk produces out-and-back returns (D = 0), wrap returns (D = +-32) and every D with 1 <= |D| <= 31; the ring walks do. Then z(p) is maximal exactly at the returns, the counts with D = 0 mod 32, iff all three of the following hold: - (i) alpha + beta = P, and since all four gates lie in (0, 1), alpha + beta < 4, so P is 2 or 3; - (ii) alpha = k P / 32; - (iii) k is odd.

Proof. Write p = n- (alpha + beta) + D alpha. (i): D = 0, n- = 1 requires alpha + beta in P Z, and 0 < alpha + beta < 4 with P >= 2 leaves alpha + beta = P in {2, 3}. Then p = D alpha mod P. (ii): D = 32 requires 32 alpha / P in Z. (iii): no non-return may read as a return. If k = 2k', then D = 16 gives 16 k P / 32 = k' P = 0 mod P, an alias. If k is odd, D k / 32 is in Z iff 32 | D. Every non-return count then sits at a non-zero multiple of P/32 mod P, strictly below the peak. []

The family is: P = 2, alpha - beta = (k - 16) / 8 with k odd, i.e. g+ - g- an odd multiple of 1/8; or P = 3 with k odd. At initialisation, gates of ~1/2 already give alpha + beta = 2.

Precision (the first two terms are a proved sufficient condition; the last is a heuristic estimate). Returns sit at 0 mod P, and the nearest non-returns sit at +-P/32. For a table strictly decreasing in the distance to P Z, the readout is therefore unchanged if every count stays within P/64 of its lattice value:

N |d sigma| + |D| |d k| P/32 + |sum of observation-gate deviations| < P/64

for all (N, D) in context. Here sigma = (alpha + beta)/2, CoPE's mean count per step (in P7 only; not the angle sum of P3), and N = g. So |d k| < 1/(2 D_max), not 1/64. The diagnosis's 1/64 is the |D| = 32 term alone, and with D_max = 97 in T = 128 walks it is not enough. The other terms give |d sigma| < P/(64 N_max), and for the observation gates |sum of observation-gate deviations| <= N max_o |dg_o| < P/64 is sufficient; for independent deviations of size tau the typical size of that sum is tau sqrt(N) (a heuristic, not a bound).

Why a non-wrapping line removes the lattice (proved; the margin caveat is proved too). On a line the only returns are D = 0. Condition (i) is unchanged. Conditions (ii)-(iii) become "D alpha / P is not in Z for 1 <= |D| <= D_max". That excludes a finite set of rationals, so the exact solution set in alpha is open, dense and of full measure, and there is no target point to hit.

The ring-lattice points are *not* line solutions: they alias D = 32. By Dirichlet's theorem, every alpha has some 1 <= |D| <= D_max with dist(D alpha / P, Z) <= 1/(D_max + 1). So the line's worst-case margin is at most P/(D_max + 1), below the ring lattice's P/32 once D_max > 31. The line trades a measure-zero target for a smaller margin.

ACCOUNTS T4 (registered, seeds 1100-1115). CoPE fails the non-wrapping line too: 1/16 SOLVED (0.914) vs 0/16 on the ring (0.873); L1 NO EFFECT DETECTED. Signed steps solve the line 8/8 and index RoPE does not (0.772; floor 0.503). So the wrap (and the lattice it creates) is not CoPE's main obstacle. The precision margin above remains a plausible account, not a tested one.

Checked (`p7_cope.py`, with the paper's CoPE module from `model_mono_baselines` and constructed gate logits). - The module's count equals n+ alpha + n- beta exactly. - Over 11,600 ring revisit queries (1,452 wrap-only) and 10,148 line returns: - Lattice points (P = 2: k = 13, 15, 17; P = 3: k = 11, 21) read every ring return (1.000, wrap-only 1.000). On the line the same gates read only 0.83-0.98 (the D = 32 aliases). - Even k: k = 18 reads 0.289 (wrap-only 0.000). k = 16 (parity only) reads 0.001. - Off-lattice k = 17.3 reads ring returns 0.723 (wrap-only 0.000) and line returns 1.000. - k = 17 + 0.95/(2 D_max) reads 1.000. k = 17 + 1/64 reads 0.926 (wrap-only 0.679). - sigma + 0.95 P/(64 N_max) reads 1.000. sigma + 4 P/(64 N_max) reads 0.369.

What P7 does not cover (from the diagnosis): - The NONE regime (20/40 runs: no locked count, identity-dependent observation gates). - The sigma = 0.5 / 0.75 / 1.25 locks. - Why training stops off the lattice. The barrier on the straight path is measured (11/11 runs) but not derived. - Non-periodic tables are not excluded by the proposition.

Summary of status

#	statement	status	check
A0	rotary position factor = Howard propagator with s = i omega (exact; ZOH discretisation exact); input weight (1 - e^{-s Delta_u})/s (v2's Delta_u is first order)	proved (identity is Mamba-3 Prop. 3); input term corrected in v3	`a0_propagator.py`; `theory_check_extra.py` (1)
B0	Delta mod (2 pi / |omega|) is non-negative and the same model; |.| and softplus are barrier parameterisations	proved (elementary)	--
T1	channel step flip + q/k reflection (rows and biases, every layer; MapEM: q0/k0) leaves all scores unchanged	proved (elementary identity)	`t1_gauge.py`
T1(i)	the |.| class realises a signed table exactly as real numbers (same latent) iff its rows are sign-coherent; only relative signs are removed	proved	`t1_gauge.py` (d, d2)
T1(ii)	one stepping token, or any task with a sign-coherent solution code (e.g. moves +1/+2): no representational sign cost	proved; "no learning cost" is a prediction, NOT confirmed: registered GAUGE G1 UNMEASURED (+1/+2 harder for both arms; |.| 2/8 vs signed 5/8)	`GAUGE_RESULTS.md`
T1(iii)	excluded codes = opposite-sign rows, which include every through-zero cancelling channel	proved	`t1_gauge.py` (e)
P1(a)	<= r stepping tokens: exact non-negative representation mod 2 pi	proved	`p1_wrap.py`
P1(f)	softplus: same, with the observations' common step ln 2 made a full turn by omega' = 2 pi / ln 2	proved; built (logits to 2.8e-11); existence under training untested	`theory_check_extra.py` (2)
P1(c)	r = 1, opposite-signed pair, C >= 2: exact representation fails for a.e. alpha	proved (special case)	`p1_wrap.py`
P1(d)	rank-1 non-negative codes approximate any table mod 2 pi (Kronecker)	proved; the eps^-(m-1) scale is a heuristic	`p1_wrap.py`
P1(e)	r >= 2, m > r, mixed-sign rows: exact representation generically fails	conjecture (sketch)	--
P2	rescaling error eps phi_raw: 0 (signed, d = 0) vs 2 pi eps n- (wrapped); AdamW decay = rescaling	proved (mathematics only; the drift account is withdrawn, ACCOUNTS T1)	`p2_rescale.py`

#	statement	status	check
P3	cut circle: same loss on the torus; distances min(sigma, 2pi - sigma)/sqrt2 vs (2pi - sigma)/ sqrt2; paths cross sigma = pi	proved	`p3_p5_geometry.py`
P4	sticky wall: attracting kink along s (other parameters fixed); dL/ds = omega sign(s) dL/da; GD on s chatters <= eta M	proved (reduced coordinate); trained-model collapse 44% (post hoc); end-of-training gradient into the wall for 51-57% of collapsed steps, MIXED by the pre-stated rule (post hoc), so its causal role is not established	`p3_p5_geometry.py`, `fig_mechanism_out.txt`
P5	even gaps; Pm negates even-g phases; model-level even-g scores invariant, odd-g negated; sigma-mirror conditions	proved, scope stated	`p3_p5_geometry.py`, `p5_pm_accuracy.py`
R	signed escapes the |.| start through zero, collectively; |.| reaches the map reliably only from past pi	account; signed collective route tested (ACCOUNTS T2, SUPPORTED); |.| side open (loose fit; the per-channel flow does not predict which channels cancel, kappa +0.09, post hoc)	review's `potential.py`, `mech_capture.py`; `ACCOUNTS_RESULTS.md`
P6	CARoPE: no channel cancels; every channel a clock or blind	proved	`p6_carope.py`
P7	CoPE lattice family, precision, line removes the lattice target (smaller margin)	proved under the stated setup (the tau sqrt(N) precision term is a heuristic); the line also fails (ACCOUNTS T4), so the lattice is not the main obstacle	`p7_cope.py`

Corrections found while writing. 1. P3: from a start with sigma >= pi, the signed class's nearest cancelling point is also the wrap. The through-zero branch is nearer only for sigma < pi. (My first numerical check assumed sigma/sqrt 2 and failed; the statement above is the corrected one.) 2. P1: "a generic signed code has no exact rank-r non-negative counterpart" is true only in the exact sense. It is proved for r = 1 and conjectural for r >= 2. Approximately, a rank-1 non-negative code represents anything, at a large angle scale. 3. P7: the precision on k is 1/(2 D_max), not 1/64. The ring-lattice gates are not line solutions, and the line's margin is at most P/(D_max + 1). 4. P6: the shortest-return bound is (0, 3] for the move-query / observation-key pair. (0, 4] is the observation-observation pair. 5. T1: with omega free in sign, the |.| class already contains both directions of every channel. T1 shows that the direction is a gauge even if omega were constrained positive. The |.| class also contains tables |X| of rank greater than r, all of them sign-coherent. 6. P2.1: with the exact schedule, wd * sum(lr) is 0.805 on the ring and 0.080 on the torus (review: ~0.7 and ~0.07). An angle is a product of four decayed tensors.

Corrections in v2. 7. P2: the drift / weight-decay account is withdrawn (ACCOUNTS T1). P2 is kept as mathematics with its frozen-model probe. 8. Remark R: the signed collective route has been tested (ACCOUNTS T2, SUPPORTED); the |.| side is stated as open, with the numbers that fit it loosely. 9. P7: the line was tested (ACCOUNTS T4);

CoPE fails it too. 10. T1(i): qualified "as real numbers, same latent", which removes an apparent contradiction with P1(a). 11. P1(a): stated to cover the full |.| arms (observation steps annihilated), with the ring_dial M1 construction. 12. P3: the angle sum is written sigma (v1 used S, which is also the step table). 13. P4 and Remark R (2026-10-15 end-to-end read): the post hoc re-analyses (`docs/audits/2026-10-15/reanalyses/`) replace the "not established / not done" statements: the wall gradient check reads MIXED; the per-channel flow does not predict, channel by channel, which channels cancel. T1(ii)'s heading now records the GAUGE result.

Corrections in v3 (2026-10-16; referee check `docs/paper/review_draft5/THEORY_CHECK.md`). 14. A0: the input term's weight is (1 - e^{-s Delta_u})/s, not Delta_u (exact only to first order; counterexample 0.857 vs |F| = 2.19). Note 2 needs a non-negative rate; note 3 writes Delta_t > 0. 15. P1(f) (new): softplus represents the map exactly (observations annihilated, omega' = 2 pi / ln 2); v2 called its representability untested. 16. P4: the omega factor in dL/ds; "strict local minimum along s, other parameters fixed"; (c) needs mu > 0, |s_0| <= rho and is gradient descent on s. 17. T1: in the |.| class the flip is carried by omega only; the bitwise claim is a property of the CPU kernels; the MapEM sub-check's control is uninformative; T1(i) is stated on Theta (the omega_c = 0 edge). 18. T1(ii) extension and the main text's Lemma 1(ii): "no map *function*" (the +1/+2 code (1, 2 - 2 pi) is excluded as a parameter point). 19. B0: (a)'s rank caveat no longer claims to cover every model; (b) states what is excluded (relative sign within a channel). 20. P3 / P5 / Remark R: "in angle coordinates", |sigma|, "reachable through |sigma| = pi" (not "unreachable"); P5 scope (moves +-1, AO) and the punctured image; R's cancelling set needs W_out of full column rank. 21. P7: the tau sqrt(N) term is a heuristic; app. H's "move-gate difference an odd multiple of P/32" is P/16 (alpha is the odd multiple of P/32). 22. Check-text slips: P1 (a) m = 1 has no opposite-signed channels; script labels "(b)" / "(c)" are P1(c) / P1(d); runtimes are seconds, not minutes; header pointers updated to draft 6.

Reproduce

From `docs/paper/appendix_checks/` (CPU, OMP_NUM_THREADS=4; each step takes 1.5-20 s):

```
python3 a0_propagator.py
python3 t1_gauge.py
python3 p1_wrap.py
python3 p2_rescale.py
python3 p3_p5_geometry.py
python3 p5_pm_accuracy.py
python3 p6_carope.py
python3 p7_cope.py
```

and, for A0's input term and P1(f), `python3 docs/paper/review_draft5/theory_check_extra.py` (CPU, ~5 s; reads `runs/wrap_mech/p0/SO_s800`).

`common.py` holds the ring-walk and reflection helpers. The scripts read only the committed `runs/wrap_mech/p0` and `runs/generality/ring` checkpoints.

Does PoPE's pinned phase offset explain the MapPoPE interaction? -- results (registered, `DELTA_FREE_PREREG.md` + Amendment 1)

Batch 2026-10-20 10:26-14:00 on GPU 1 (shared with another user, <= 4 runs), 40 runs, seeds 1600-1607 (fresh). Paper torus task, T = 128, rank 2, PAPER2X2's recipe (300 epochs, cosine, lr 1e-3, wd 0.05). Arms: PoPE-Flat (index phase, delta pinned: zero init held at 0 by the clamp's dead zone), PoPE-FreeDelta (delta init U(-2 pi, 0), clamp removed), MapPoPE-Flat (path phase, pinned), MapPoPE-FreeDelta (path, free), MapPoPE-FrozenDelta (path, delta frozen at the free arms' draw). At each seed twins differ only in delta. Independent check before launch (`docs/audits/2026-10-20/delta_free_audit.md`, Amendment 1); pilot on seed 1690 (never pooled). Analysis `analyze_delta_free.py` -> `DELTA_FREE_ANALYSIS.txt`, `DELTA_FREE_VERDICTS.json`; recipe asserted on all 40 checkpoints. `experiment_audit`: all arms converged, windowed classes agree. Own-stream retrace floor 0.938. Account under test (`docs/paper/expansion/MAPPOPE.md`): PoPE's content-free offset delta, pinned at 0, fixes every kernel's peak at zero phase difference; with a path phase that is the revisit target (helps), with an index phase it is the query's own token (hurts: no multi-step retrace kernel).

PRIMARY (registered): PEAK-PINNING SUPPORTED (path unchanged within noise). Freeing delta raises the index model by +0.122 [90% CI +0.076, +0.167] (8/8 seeds, sign-flip p 0.0078), concentrated in multi-step reversals (R3+ +0.180, Holm p 0.031), and leaves the path model at ceiling (-0.018, p 0.31; SOLVED 6/8 vs 8/8, Fisher p 0.47).

arm	T = 128	T = 512	T = 1024	SOLVED
PoPE-Flat (index, pinned)	0.649	0.615	0.599	0/8
PoPE-FreeDelta (index, free)	**0.771**	0.536	0.442	0/8
MapPoPE-Flat (path, pinned)	0.999	0.980	0.933	8/8
MapPoPE-FreeDelta (path, free)	0.981	0.901	0.831	6/8
MapPoPE-FrozenDelta (path, random fixed)	0.999	0.929	0.832	8/8

Interaction I = dP - dI = -0.140 [-0.186, -0.092] (0/8 seeds positive, p 0.0078); PAPER2X2's pinned score x phase interaction was +0.154 (cross-batch, descriptive). The licensed sentence (Amendment 1 MAJOR 5) is the one above, not "freeing delta reverses the interaction".

Mechanism readouts (declared secondaries; eval-only probes, rebuilt logits exact)

- **Index kernel:** with delta pinned, 0% of content pairs prefer offset 3 over offset 1 (kernel argmax at offset 1 in 1074/1088 pairs); freed, 50-100% per seed, with argmax spread over offsets 3, 4, 7, 8 (the retrace comb). Attention on the newest observation (never the target) 0.287 -> 0.003.
- **Where the index gain comes from:** R1 +0.039 (n.s.), R2 +0.117, R3+ +0.180 (Holm 0.031 each), unvisited-run +0.014; R3+ carries 0.093 of the 0.122.

- **Path target alignment:** C (same-cell key alignment) 0.987 pinned vs 0.616 free and 0.614 frozen; same-cell attention mass 0.909 vs 0.809 / 0.876. Accuracy at T = 128 stays at ceiling because the code itself (C0 0.993 / 0.798 / 0.884) still carries the map.
- **Delta (S4):** pinned arms 96-100% in the dead zone. Free arms moved mostly by weight decay (x0.447); the learned part beyond decay is largest in the fastest frequency quartile (index 0.447, path 0.371 rad).
- **S7 (diagnostic):** resetting delta to 0 on either half of the free index model's channels drops it below the pinned model (0.43 / 0.47 vs 0.649): the trained network relies on its learned offsets across the spectrum, so the reset is not a clean ablation of one mechanism.

Beyond the training length (robustness, rule 10)

- **P1 (registered):** path frozen / free minus pinned at T = 1024: -0.101 (p 0.047, Holm 0.094) and -0.102 (Holm 0.195): both UNMEASURED by the registered rule; descriptively both lose robustness (T = 512: -0.052, -0.079).
- Index free minus pinned at T = 1024: -0.158 (p 0.016, Holm 0.094): the retrace kernel the free offset buys is length-specific, while the pinned recency kernel extrapolates (descriptive).
- Training (S6): the free index arm learns faster (loss AUC -0.131, p 0.0078); the free and frozen path arms slower (+0.122, +0.040).

Reading

PoPE's polar score helps a path-integrated phase and hurts an index phase because its pinned offset fixes the attention kernel's peak at zero phase difference. Freeing the offset lets the index model build the multi-step retrace kernel RoPE uses (+0.122 at T = 128, mostly on multi-step reversals) and leaves the path model's accuracy at ceiling while its target alignment falls (0.987 -> 0.62). The account is causal at T = 128; the costs beyond the training length are descriptive. Scope: one layer, 2 heads, rank 2, the paper torus at T = 128.

What/where at T = 1024 (post hoc, eval-only; 2026-10-20)

Re-run of the causal what/where test (`docs/audits/2026-10-03/causal_whatwhere.py`) in the regime where a map is needed: the SIGN_MATCHED checkpoints (torus, trained and tested at T = 1024, 1 layer, rank 4, n = 8 per arm; floors on these walks blank 0.506, n-gram 0.575, retrace 0.843). Script `causal_whatwhere_T1024.py`, output `causal_whatwhere_T1024.txt` / `.json`, log `.log`. Rebuilt forward = model logits exactly (max | dlogit| 0.0) on all 32 checkpoints; intact accuracy matches the stored registered eval to <= 0.0007. Changes from the T = 128 script (memory): evaluation in chunks of 10 trajectories, calibration on 20 trajectories, graded phase-lesion and lam_add grids dropped.

condition	signed (path, learns the map)	abs	softplus	index RoPE
intact	0.998	0.821	0.780	0.731
object identity removed from the score (type kept)	0.966 (-0.032, p 0.012)	0.544 (-0.277)	0.476 (-0.305)	0.479 (-0.252)
one shared kernel x content gain	**0.998 (-0.000)**	0.584 (-0.237)	0.675 (-0.106)	0.439 (-0.293)
additive position + content	0.895 (-0.104)	0.350	0.512	0.492
type removed too (one global mean)	0.113	0.295	0.302	0.399
content-only (no rotation)	0.505	0.399	0.385	0.490
share of score variance explained by position alone	0.925	0.644	0.577	0.618
content x position interaction share	0.061	0.187	0.174	0.353

Reading (analysis): - The model that learns the map factorises what and where at T = 1024: a single position kernel per head, scaled by a content gain, reproduces it exactly (0.998); removing object identity costs 0.032 (p 0.012, 8/8 seeds lower; at T = 128 the cost was 0.011-0.025); position alone explains 92.5% of the score's variance; token TYPE (move vs observation) is essential (0.113 without it). - The models that fail do not: removing object identity costs 0.25-0.31 and a shared kernel x gain does not restore them. - Within the non-negative arms, separation tracks success: across 16 runs, accuracy correlates with the share of it that survives removing object identity (Spearman 0.84, p 5e-5); the two non-negative runs that reach > 0.95 keep 0.97 / 0.99 of their accuracy, the failing ones collapse to 0.03-0.07. So in the map regime what/where separation is a property of the solution training finds, not of the architecture.

Draft 2 of the main paper: integrated plan from the seven expansion reports (2026-10-20)

Inputs, read in full: `PAPER_LEVEL.md`, `MAPPOPE.md`, `RECURRENCE.md`, `RANK_THEORY.md`, `SEARCH_THEORY.md`, `DEPTH_THEORY.md`, `LANGUAGE.md` (this directory), against `docs/paper/MAIN_DRAFT1.md`, `MAIN_APPENDIX1.md`, `APPENDIX_theory_v3.md`, `OUTLINE_main_v2.md` and CLAUDE.md. Spot checks made for this plan (CPU, read-only): the SEARCH / RANK clock outputs (`scripts/sign/rank_clock_out.txt`, `scripts/rank/rank_geom.py`), the running what/where re-analysis at T = 1024 (`docs/audits/2026-10-20/causal_whatwhere_T1024.{log,json}`; no `.txt` yet), the T = 1024 floors (`docs/audits/2026-10-16/sign_floors_out.txt`), `figs/fig_exchange_rate_out.txt`. Nothing trained, nothing committed, no existing file edited.

Labels used throughout: **registered** (a registered verdict or declared secondary; the results file is authoritative), **analysis** (post hoc, eval-only on stored checkpoints or CPU pilots; in the paper every such number carries "(analysis)"), **theory-proved** (a proof is given in a report or the theory appendix), **theory-conjecture** (a derivation whose dynamics or assumptions are not checked), **speculation** (untested). Nothing withdrawn in CLAUDE.md is used; the guards are listed in Sec. 1.3.

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1. The thesis after the expansion

1.1 One paragraph

A map in the phase is a code without a clock. Path integration writes the prefix sum of the moves into the rotation, so one attention layer compares two positions at any gap and across the wrap (theory-proved; registered: 0.998, 8/8 at T = 1024 against retrace 0.843). Index attention sees only token offsets and must build that sum in its residual stream. Two layers suffice in principle given a normaliser and enough width (Liu et al. 2023, Lemma 6), so the exchange rates (about 3 on Dyck-2 and the biased ring, more than 4 on the torus) are what training finds, not representational bounds. On the torus the trained index models learn local undo rules whose reach grows segment by segment with depth and never find the global count (analysis); on the small ring three layers do find it (analysis). Whether a path model finds its map depends on whether training removes the clock: the phase that an out-and-back adds at zero displacement. Any clock serves short-gap revisits and loses long-gap and wrap-only ones (theory-proved). That single signature is shared by every failing model, index or path, at the level of what one attention pass can read. A signed step at per-head rank above the map's dimension D has two map-free ways to remove the clock. One is collective: driving the latents to z_x + z_y = 0 zeroes every channel at once (theory-proved; registered causal test on the ring: frozen move latents 1/8 vs 7/8). The other is per channel: each channel can stop reading the part of the clock that lies off the map plane ("parking"; theory-proved; solved heads show it, analysis). A non-negative step has neither route from a small start. Under |.| the collective move makes a clock, at a generic readout no latent choice cancels more than 2r channels, and each channel's clock-free points lie past pi (theory-proved). At rank D the collective route remains, but no channel can be both clock-free and a full map (theory-proved). The free alternative, "hiding" the drift by no longer resolving its direction, costs nothing to first order in D >= 2 and is conjectured to starve the collective route. In 1D

there is nothing to hide along, which fits rank 1 learning the ring (theory-conjecture; decisive test G-1). Downstream computation reads a code but does not repair it. A looped block or extra layers read the same rank-2 code with context (code quality unchanged, analysis). PoPE's polar score fixes every kernel's peak at zero phase difference, which aims attention at the target when the phase is a place and away from it when the phase is an index (theory-proved given delta = 0; analysis). In language the same rules hold word by word. A word gets a step if and only if its count per move is fixed. Asides keep a small offset so that what is mentioned is not filed at the place. A per-move tick survives only in channels that retrieval does not use (tick rule, theory-proved; analysis). The used-channel clock ratio tells map data from clock data where the pooled growth exponent cannot (analysis).

1.2 How the pieces fit (one line each, with the report that carries it)

piece	statement	label	report
why one path layer	the integrator supplies P_t exactly; the score is a function of omega (P_t - P_s)	theory-proved	DEPTH 3.2
why index depth is not enough on the torus	route A (global count) is depth 2 in principle but needs a normaliser, 1/n precision and width ~ N x T; route B (local return sets) is what is learned, reach measured in direction runs	theory (cited construction) + analysis	DEPTH 3.3-3.4, 4.1-4.5
the failure signature	a channel with clock k serves revisits with gap < pi / k; a channel serves place-keyed retrieval only if its tick is 0 mod 2 pi / g	theory-proved	SEARCH S5; LANGUAGE T1 (the same statement, generalised)
what removes a clock	collective route R1 (signed, any rank); per-channel route R2 (signed, rank > D: parking)	theory-proved	SEARCH S2, S6(b); RANK Prop. 1(a)
why sign matters		.	has no R1 for an inverse pair and its R2 targets lie past pi; clock >= map below pi
why rank D matters (D >= 2)	no clock-free full-map channel; hiding is free to first order and lowers the resolution that drives R1	proved (static, first-order); dynamics conjecture	SEARCH S6(a); RANK Props. 1(b), 2, 3
why 1D is exempt	no hiding direction; the readout cannot change the drift-to-map ratio	theory-proved	RANK Prop. 1(c)
"exists, kept, not found"	the start distribution lies outside the basin of a non-empty, stable solution set	theory (definitional) + analysis	SEARCH S8, 3.3-3.4
readers vs code	one-layer models are position-only readers of their own code; extra passes widen the reader, never the code; a score rule acts on the readout term only	theory-proved (architecture) + analysis	RECURRENCE T1-T2; MAPPOPE Theory 2
the polar score	peak pinned at dphi = 0: helps a path phase, hurts an index phase	theory-proved (delta = 0, measured) + analysis	MAPPOPE Prop. 1, Cor. 1a-1c

piece	statement	label	report
words	role, not identity, sets a token's offset; tick rule; used-channel rho	theory (light) + analysis	LANGUAGE 2-3, 5

1.3 Contradictions between the reports, and their resolution

#	contradiction	resolution	status
X1	SEARCH: the strong unification ("the failing class lacks a linear route to cancellation") is **false for rank D**, which has R1.	RANK Props. 2-3 supply the missing piece (hiding starves R1 at D >= 2; none in 1D). The first-order statement is proved. The feedback is **not shown**: the one failing pilot seed inherited its weak axis (RANK C6), and the readout gradient at initialisation is uninformative (C8, drift too large for assumption A1). The paper states the route inventory as proved and the hiding dynamics as a conjecture. G-1 decides it.	flagged: conjecture
X2	RANK: solved rank > D heads **park** the clock (off-plane 0.96-1.00, leftover lam 0.003-0.004). SEARCH: at rank 3/4 it is **removed in the latent (R1), not parked** (latent clock 0.045-0.065, in-span 0.999-1.000).	The two use different definitions. RANK's c is the clock common to all four moves plus the mean observation step, measured in the native latent against U. SEARCH's are per inverse pair (c_x = z_-x + z_+x, c_y), and its "in-span" is the share of the *read* channel clock inside the map span; a parked clock is by definition the unread part, so the two agree that what is still read lies in-plane and is small. The magnitudes (0.004 vs 0.065) are not comparable. Unresolved: whether the final shrinkage at rank > D is gradient-driven (R1) or weight decay on a parked, unread direction (RANK C2: "consistent, not a test"). Before drafting: **one definition** (per inverse pair, SEARCH's) and a recomputation of RANK's off-plane share and readgain with it (CPU-3). Add a rank-4 wd 0 arm to G-1 (parking predicts a solved model with a large parked clock; R1 predicts the clock is removed anyway).	resolved on end states; mechanism of final removal open
X3	SEARCH: solved rank-2 heads "keep a moderate clock, 0.26-0.40". RANK: in solved rank-D heads the clock "is simply gone (lam <= 0.001)".	SEARCH's numbers are means over **both heads** of solved runs, and solved rank-2 runs have one clean head and one drifting head (per-head r2 s1: h0 lat-clock 2.10, h1 0.028; block-diagonal s6: h0 0.082, h1 3.50; `rank_clock_out.txt`). This matches "SOLVED iff a clean head" (39/40, close to circular). Wording fix: report the clean head and the other head separately.	resolved
X4	RECURRENCE calls the loop's 1D cancelling code "a third end state". RANK classes are CLOCK (A) and COLLAPSE (B), "points on one curve" (R2).	They are the same object: the 1D code is RANK's collapse corner with the common mode removed (hiding complete). One taxonomy for the paper: **map** (2D, clock-free), **collapsed** (clock-free, one map dimension lost), **clock** (drifting). RECURRENCE's "cancel" is not wrap-aware (its reader columns are), so use the reader-based classification.	resolved (naming)
X5	PAPER_LEVEL 3.1: LOOP_RANK shows "more search (depth) partly recovers rank 2, as a search account	Follow RECURRENCE, as CLAUDE.md already does ("NOT search at constant capacity"). The registered UNMEASURED verdict stands. The	resolved

#	contradiction	resolution	status
	predicts". RECURRENCE: the gain is a wider **reader** on an unchanged code, not search (code quality L - A +0.027, p 0.28; L4 - A +0.031, p 0.27; 2D codes 1/16 vs rank 4 8/8).	paper's sentence: "more computation reads a rank-2 code; it does not repair it" (analysis).	
X6	PAPER_LEVEL abstract and contribution: "every failing model, index or path, ends with the same code: short-gap revisits served, long-gap and wrap-only at the floor". RECURRENCE: unsolved L / L4 rank-2 runs reach 0.93-0.99 on long gaps.	The signature is a property of the **code read by one pass** (SEARCH S5, RECURRENCE T1). Deep readers recover accuracy, but not the code. State it at the code level. At the accuracy level it holds for one-layer models only.	resolved (wording)
X7	MAPPOPE's proposed main text: "the rank-2 phase does not place a wrapped cell at zero". This is close to the **withdrawn** "the rank defect is the periodic code" (CLAUDE.md, 2026-10-11).	MAPPOPE's own A2 shows that the code term C0 is also low at gap >= 128 (0.81-0.84 vs 0.99 at rank 4), and RANK R1 explains wrap-only revisits as long-gap revisits read through a drift. Rephrase as "does not place long-gap or wrapped revisits at zero phase (C0 0.81-0.84 at gap >= 128, 0.41-0.46 wrap-only; analysis)". Wrap and gap are still not separated (RANK_NOWRAP would separate them).	resolved (guard)
X8	RECURRENCE: readers never repair the code. MAPPOPE: at T = 128, MapPoPE's code term is also higher (C0 0.996 vs 0.905, p 0.0002).	At T = 1024 the score leaves C0 unchanged (short gaps 0.900 vs 0.960, p 0.16; wrap 0.457 vs 0.411). At T = 128 a reader change co-occurs with a better code, plausibly through easier finding (MAPPOPE: speculation; fits GAIN_GRAIN's faster convergence, which did not replicate in GAIN_PHASE). Scope the claim: "at T = 1024, neither the loop, extra depth nor the polar score changes the code".	scoped
X9	LANGUAGE: on the text world's own eval stream, retrace scores **0.934** (0.9337; TW_NORMSTEP stream 0.9365). The draft and PAPER_LEVEL's abstract cite only the 0.505 / 0.597 floors.	Correction C-1 (Sec. 4). The text-world claim must rest on (i) the 179 / 3,055 NEEDS-MAP revisits: path 0.961 on fresh seeds 2-7 and 0.920 on all 8 re-scored, against RoPE 2L 0.138 and best constant 0.112 (analysis); (ii) the mechanism (cancelling steps; direction steps zeroed -> floor, registered). PAPER_LEVEL's CCN abstract must carry the same caveat.	resolved: correction
X10	LANGUAGE counts clock channels as 26-38 (drift between **consecutive** visits). The registered CORRECTED block counts 31-38 (drift from the **first** visit).	Keep the registered 31-38 with its definition. Add the consecutive-visit count as analysis only where the tick-predicts-drift result is quoted.	resolved
X11	PAPER_LEVEL: EM's recency deficit is a **contrast**, not a third axis ("kept" only after restricting the model; 4x budget closes most of it). No other report disagrees.	Adopt: two sentences in the Discussion plus one appendix table. Keep the 1800-epoch caveat (5/8 rank-2 runs still descending) on "stays".	adopted

#	contradiction	resolution	status
	SEARCH S8's "held" leg fits rank and sign, not EM.		
X12	RANK, PAPER_LEVEL and RECURRENCE each propose a new "Appendix M" (or "X").	Re-letter as in Sec. 3.3.	resolved
X13	RECURRENCE E4 proposes a registered loop vs Level 1.5 replication under action noise. CLAUDE.md lists Level 1.5 / InEKF and "filter x loop" as **live negatives: do not re-run**.	Drop E4. The n = 3 loop-under-noise observation stays out of the paper.	dropped
X14	SEARCH S6(b): at rank 3 only **one** clock direction can be parked (two independent pair clocks need rank 4), which would explain rank 3 6/8 vs rank 4 8/8. RANK P9: the expected off-plane share of a random clock is 0.58 vs 0.71.	Two competing accounts of rank 3's shortfall. CPU-4 checks whether the unsolved rank-3 heads carry two independent pair clocks. Neither account goes in the main text until checked.	flagged
X15	RANK C2 explains the parked remainder by AdamW decay. The theory appendix v2 **withdrew** "the drift / weight-decay account" (ACCOUNTS T1) for the **sign** axis.	Different claims: the residual size of a solved rank > D head versus the cause of the	.
X16	DEPTH: the rates are not representational bounds. The draft reads them as "what a layer of attention computes".	This strengthens the paper's thesis: it is a third learnability statement, for index attention. The construction's MLP width (Lemma 6: 4nT, here 64 x 2048 per axis) exceeds our 512, so state it as "in principle, with a normaliser and enough width"; we do not claim our index models could represent it.	adopted with caveat
X17	LANGUAGE 4.2 lists the Case I/II/III identification for rotary attention as "ours". CLAUDE.md: the taxonomy is published (GRAPE, Mamba-3, Selective RoPE).	LANGUAGE itself says "present it as an organising remark, not a result". The draft already uses it as background. Claim only the measurements (which case a used channel realises is set per channel by the readout; Cases II and III in one model).	guarded

Not contradictions, but joint support worth one sentence each: MAPPOPE Cor. 1c (RoPE's one-layer kernel is a comb at the retrace offsets 4j - 1) and DEPTH 4.5 (one index layer learns exactly retrace and the clock) describe the same one-layer solution from the score side and the accuracy side. RECURRENCE 5.7 and DEPTH 4.2 agree that the index loop and index depth add only short-gap retrieval, pass by pass. LANGUAGE's pi-tick parity channels are the even-gap lemma of v3 P5.0 at work.

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2. Title, abstract, contributions

2.1 Title

Recommended: **A map is a phase without a clock: what path-integrated attention learns, and what decides whether training finds it**

Alternatives: - *Representable, kept, not found: what decides whether path-integrated attention learns a map* (PAPER_LEVEL 2.1). Best if the theory section is cut for an 8-page venue. - *Clocks and maps in rotary attention: depth, rank and sign as learnability variables*. For a main track, once the SSM arm (G-3) lands; then generalise to "content-dependent phases".

2.2 Abstract (about 240 words; registered numbers only, theory statements only where proved)

> Path-integrated rotary attention replaces RoPE's token index with a learned phase that sums a content-dependent step per token. We give an account of what such models learn and of what decides whether training finds a map. On 1024-move torus walks, where a map-free rule that retraces the last run of moves scores 0.843, one path-integrated layer scores 0.998 (8 of 8 seeds solved) against 0.731 for index RoPE. The rotation carries the prefix sum of the moves, so one layer compares positions at any gap. Index attention must build that sum in its residual stream: at matched depth one path layer is worth about three index layers on Dyck-2 and on a biased 1D walk, and more than four on the torus, where depth helps only short-gap revisits. We show that a map code is one whose phase carries no clock, no phase gained over an out-and-back, and that any clock serves only short-gap revisits. Two choices decide whether training removes it. A signed step at per-head rank above the map's dimension has two clock-removing routes that leave the map intact. A non-negative step has neither from a small start, and at rank equal to the dimension no channel can be clock-free and a full map at once. In line with this, rank 2 solves the torus on at most 2 of 8 seeds against 8 of 8 at rank 4, and non-negative steps on 0 of 8 against 8 of 8 for signed steps. Yet both constrained classes represent the map exactly and keep or re-find an installed solution, and the sign gap holds on a ring at each step's best learning rate. Told in English, the walk is learned by one path layer (0.979 vs 0.505), which learns which words move it and cancels opposite directions.

Notes: - "on a ring" stays: the best-rate test is ring-only until G-4. - Do not add "every failing model ends with the same code" (X6). - Swap the last clause for the SSM result if G-3 is positive.

2.3 Contributions

1. *Measuring map use.* Map-free floors (blank, n-gram, retrace) and revisit strata (short-gap, long-gap, wrap-only) defined once and used throughout. At T = 128 retrace scores 0.934, so we measure at T = 1024 (retrace 0.843). There one path layer scores 0.998 against index RoPE's 0.731 (8/8 vs 0/8, p 0.0002), and only the path model holds long-gap and wrap-only revisits.

2. *Why one path layer, and what depth buys index attention.* - The prefix sum is in the rotation (proved). - Index attention reaches the map only through a global count. That count is depth 2 in principle but needs a normaliser, 1/n precision and large width. - Trained index models instead learn local return rules, whose reach grows with depth in direction runs (analysis). - Registered exchange rates: about 3 on Dyck-2 and the ring, more than 4 on the torus.
3. *A theory of when the map is found.* - The clock / map decomposition of a learned phase, and the failure signature it implies (proved). - A route inventory of each step class (proved): signed above rank D, two routes; signed at rank D, one; |.|, none from a small start. - Registered support on both axes: the solution exists, is kept or re-found, and is not found from scratch. Rank 2: at most 2/8; |.|: 0/8 against 8/8; 2x budget; best of four rates on the ring. - The causal test of the collective route: freezing the move latents gives 1/8 vs 7/8. - End-state checks on every stored model (analysis): failures end with a clock about as large as the map, successes clock-free.
4. *Readers and codes.* A score rule, a loop or extra layers change how a code is read, never the code (analysis at T = 1024). - Loops and depth raise rank-2 accuracy (+0.079 / +0.096, registered) without improving the code (analysis). - PoPE's polar score pins the kernel peak at zero phase difference (proved given delta = 0). This explains the registered +0.154 interaction: about 80% of it is the index side's loss on revisits at fixed far offsets (analysis).
5. *Maps from words.* - One path layer learns navigation told in English (registered). - Above retrace, the evidence is the 5.9% of revisits that retrace gets wrong: path 0.961 vs two index layers 0.138 (analysis). - The step table follows one rule: a token's offset from its place depends on its role, not its identity (analysis; the leak, registered, is the same rule). - Clock channels are tolerated only where retrieval does not look (tick rule, proved; analysis).

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3. Section-by-section plan for draft 2

3.1 Structure and page budget (TMLR, main text about 13-14 pages; 8-page cut in 3.5)

#	section	pages	built from
--	title, abstract	0.3	Sec. 2 here
1	Introduction	1.0	rewrite around the thesis; PAPER_LEVEL's bridge sentence ("depth does not buy the map, so the step map must find it")
2	Background	0.6	draft 2.1 replaced by LANGUAGE 6.2; 2.2-2.3 kept; 2.4 moved to Related Work in Sec. 9; 2.5 (Prop. 1, Lemma 1) moved into Sec. 4
3	Setup and how map use is measured	0.8	draft Sec. 3, plus the strata defined here (PAPER_LEVEL 1), the scoped protocol sentence (C-3), and the statement that the theory and its checks are post hoc
4	Theory: maps and clocks in the phase	1.6	Sec. 3.2 here
5	One path layer solves the long-walk map	0.8	draft Sec. 4 (Table 1), shortened
6	What index depth buys	1.3	

#	section	pages	built from
			draft Sec. 5 + DEPTH 5 text + RECURRENCE 6.2 sentence + segments figure
7	What decides whether training finds the map	3.6	draft Sec. 6 + SEARCH 5.1 + RANK 5.1 + RECURRENCE 6.1 + MAPPOPE main paragraph
8	Maps from words	1.2	LANGUAGE 6.1 (replaces draft Sec. 7)
9	Discussion and related work	1.0	draft Sec. 8 + LANGUAGE 6.3 + PAPER_LEVEL 3.2 (EM) + what/where at T = 1024 + draft 2.4
10	Limitations (the one place scope is stated)	0.4	Sec. 3.4 here
	total	about 12.6 + figures	

3.2 Section 4, the theory section (one place, every statement with its status)

Put it before the results. Its opening sentence says: "This account was developed after the registered batches; every check in Secs. 5-8 marked (analysis) was computed afterwards on stored models." Each proposition is stated in two to four lines in the main text, with the proof in App. E. The status column goes into a table at the end of the section (Table 4).

#	statement (short)	status	proof / source
4.1	**Score identity.** The rotary position factor is the Howard propagator with s = i omega. Position enters a channel only through omega_c (Phi_t - Phi_s).	theory-proved (identity is Mamba-3 Prop. 3)	v3 A0
4.2	**The prefix sum is in the phase.** With move steps fixed per move and observation steps zero, theta_t = omega P_t: one layer, any gap, any T, wrap free when omega x step is a multiple of 2 pi / N.	theory-proved	DEPTH 3.2
4.3	**Index attention needs a count.** Route A (global count) is depth 2 in principle (Liu et al. Lemma 6; Yao et al. 2021). Its costs are a normaliser, precision 1/n_t, MLP width ~4nT and sensitivity growing with the gap (Hahn and Rofin, used for direction only). Route B: bounded return rules (retrace, constant offset) extended by depth.	theory (cited upper bound); route B's reach is measured, not derived	DEPTH 3.3-3.4
4.4	**Clock and map.** For an inverse pair, clock k_c = a_c(x) + a_c(y) and map m_c = a_c(y). Phase over an interval = m_c d + k_c n_x. **Signature:** a clock serves revisits with gap < pi / k and loses the rest. **Tick rule:** a channel serves place-keyed retrieval only if its tick is 0 mod 2 pi / g (g = 2 on the even torus).	theory-proved	SEARCH S5; LANGUAGE T1 (merge into one proposition)
4.5	**Representability** (Prop. 1) and **relative-sign gauge** (Lemma 1), unchanged from draft 1.	theory-proved	v3 P1(a), P1(f), T1
4.6	**Route inventory.** (a) Signed: the clock is diag(omega) W u with u = z_x + z_y. The set	.	: the same move ends at a(y) = a(x), a clock. At a

#	statement (short)	status	proof / source
	{u = 0} cancels every channel and leaves the map free. Gradient flow on the clock surrogate converges to it from every start, and it is generically the only such set. (b)		generic readout at most 2r channels cancel by any latent choice. Each channel has clock >= map below pi, and its non-blind zero-clock points are at
4.7	**Hiding.** At rank D >= 2 with isotropic resolution, rotating a channel's resolution off the drift direction lowers the drift loss at zero first-order map cost, and that lowers the rate at which the global route removes the drift.	first-order gradient statement theory-proved under A1-A3; feedback and plateau theory-conjecture (P2 NOT SHOWN)	RANK Props. 2-3
4.8	**Exists, kept, not found** = the initial distribution lies outside the basin of a non-empty, stable solution set.	theory (definitional)	SEARCH S8
4.9	**Readers and codes.** In one layer, Q and K at a scored query depend on token identity only, so the model is a position-only reader of its own code. Theta is computed once (`model.py`, `model_looped.py`), so loops and layers widen the reader and never change the code class. Any score of the form sum A_c cos(dphi_c + beta_c) splits target coherence into a code term C0 and a readout term C - C0; a score rule acts only on the second.	theory-proved (architecture + algebra)	RECURRENCE T1-T2; MAPPOPE Theory 2
4.10	**Peak pinning.** With beta = 0 (PoPE, delta in its dead zone), every head's kernel peaks at dphi = 0 for every content pair (Bochner). Path phase: dphi = 0 is the same place, which is the target. Index phase: it is the current token, so a fixed far offset is unreachable in one layer.	theory-proved given delta = 0 (measured: 98-100% of entries in the dead zone)	MAPPOPE Prop. 1, Cor. 1a-1c
4.11	**Used-channel ratio.** Steps split into the frequency-weighted mean (clock) and a content part. Place-keyed readouts push used channels to rho << 1, count-keyed readouts to rho >> 1.	theory (light)	LANGUAGE 3.1-3.3

Speculations, which stay out of Sec. 4 or are flagged in App. E only: RECURRENCE T4 (deep readers make the collapsed code a usable "compensation basin"); RANK's reduced plateau law; DEPTH's per-layer hop model (linear vs logarithmic reach); LANGUAGE's "the clock is used early and abandoned later"; MAPPOPE's "a pinned peak makes a good code easier to find on short walks".

Figure T1 (new, schematic, no data): left, one channel's phase over an out-and-back for a map (returns to 0), a clock (advances by k per pair) and a |.| channel (cone: clock >= map). Right, the route diagram: collective (latent u -> 0), parking (off-plane), hiding (rotate off the drift direction). Five minutes of readers' time saved; costs about 0.3 pages.

3.3 Section by section: add, move, cut

1 Introduction. - Add: the thesis paragraph (1.1 here, compressed to about 150 words) and the five contributions (2.3). - Add PAPER_LEVEL's positioning sentence, early: MapFormer already shows path integration helps; ours are the floors, the matched-depth rate and its mechanism, the theory of the two axes, and the words. - Cut: the three-questions paragraph (it is replaced by the thesis).

2 Background. - Replace 2.1 by LANGUAGE 6.2. Howard's three cases become an organising remark (X17), and the new citations go in: Alipour et al. 2025, GridPE (arXiv: 2406.07049), Mochizuki-Freeman et al. 2023, Kabir et al. 2025; optionally Sarkar and Howard 2026 and RoMem. - Keep 2.2 (MapFormer) and 2.3 (representability). - Move 2.4 (orientation-fixing parameterisations, Shakerinava, Liu, Hahn and Rofin) to Related Work at the end of Sec. 9. - Move 2.5 (Prop. 1, Lemma 1) into Sec. 4.

3 Setup and measurement. - Define the strata here and make Fig. 2 the figure the paper returns to (PAPER_LEVEL 1). - Fix the protocol sentence (C-3). - Add one sentence on the analyses: post hoc, eval-only, scripts in `docs/paper/expansion/scripts/`, with independent-audit status per script in App. O (CPU-9).

4 Theory. As in 3.2.

5 One path layer solves the long-walk map (draft Sec. 4). - Keep Table 1 and the headline paragraphs. - Add one sentence tying it to 4.2: "the score depends on P_t - P_s directly". - Shorten "Retrace is strong on short walks" to three sentences.

6 What index depth buys (draft Sec. 5). - Keep the three registered rate paragraphs and Fig. 1. - Replace "What depth buys index attention" and the positioning paragraph by DEPTH 5 (the "Why the rate differs by task" text), each analysis number labelled. - Add RECURRENCE 6.2's sentence (the same four passes lift long-gap revisits on a rank-2 path code, where the code has something to read). - Add the exact accounting of "just above retrace" (DEPTH 4.1: (0.680 x 0.996 + 0.148 x 0.699) / 0.828 = 0.943, analysis). - Add the one-sentence statement that the rates are not representational bounds (X16). - New Fig. 8 (segments; Sec. 3.4). - The ring's 0.75 dip decomposition goes to App. F only, labelled descriptive (DEPTH 4.5; the registered primary attached no mechanism sentence).

7 What decides whether training finds the map (draft Sec. 6, reorganised). - *7.1 The clock and the signature.* - Move draft 6.3 here, now derived from 4.4. - Add SEARCH's end-state table (S5) as Table 5: usage-weighted clock / map per arm, analysis. - Signed solutions: 0.001-0.018. - |.| from small starts: 1.10-1.68. - Frozen-latent signed: 1.39. - Unsolved rank-2 heads: 0.5-1.9. - Solved rank-3/4 heads: 0.043-0.051. - Use the clean-head / other-head split (X3) and SEARCH's per-pair definition (X2). - *7.2 Rank.* - Keep the draft's registered paragraphs and Table 2; add Holm (C-11). - Add RANK 5.1 ("Why the extra rank helps"), trimmed to about 180 words. Keep its wording constraints: "parks the clock outside the map plane" only with the off-plane share and its random baseline beside it, and nothing causal until G-1. - Add the SCORE_RANK sentence (PAPER_LEVEL 3.1): the polar score does not rescue rank 2 (3/12 vs 3/12). The mechanism is in 7.4. - *7.3 Sign.* - Keep the draft's registered paragraphs, Table 3, Figs. 4-5. - Add SEARCH 5.1, cut to about 220 words: the two routes and their absence under |.|, the path-profile result and the frozen-content qualification. - Keep "How signed steps reach the map" (the registered

causal test) and the CoPE / CARoPE paragraph. - *7.4 Readers cannot repair a code.* - RECURRENCE 6.1 (about 190 words) on loops and depth. - MAPPOPE's main-text paragraph (about 200 words), with the X7 rephrasing. - Draft 2 should leave out "loop on Match-Query +0.346" (PAPER_LEVEL keep-out list).

8 Maps from words (draft Sec. 7). - Replace with LANGUAGE 6.1. It reports retrace 0.934, the NEEDS-MAP stratum, the step table by word class, the aside collapse / lift test, the idle clock, and "what must and must not move" (the leak with its failed accuracy replication). - Fig. 7 gains panels (d), the step table by word class, and (e), aside collapse / lift. - Clock-seed per-seed detail goes to App. J.

9 Discussion and related work. - Rewrite "What the two failure axes share" in route terms (one paragraph, from 1.1), with RECURRENCE 6.3's sentence. - Keep design guidance, and add: "measure the used-channel clock ratio, not the pooled growth exponent". - *What and where:* replace with the T = 1024 result once complete (CPU-1). Current partial numbers (analysis, from `causal_whatwhere_T1024.json` at 2026-10-20 ~02:00; .txt pending): - Signed rank-4 models keep 0.925-0.999 (mean 0.966) with object identity removed. That is above retrace (0.843) on every seed, but the cost (mean 0.032, up to 0.064) is larger than at T = 128 (0.011-0.025). - The shared position kernel times a content gain keeps 0.990-1.000 (cost <= 0.001). - |.| models lose 0.02-0.61 (mean 0.28) with identity removed and score 0.48-0.90 under the gain-only score. - Pos (softplus) has n = 5 so far; RoPE is pending. - If it holds, the new reading is: in the long-walk regime the solving models separate where from what, and the failing ones lean on content-position interactions. - *Integrator view* and *Text:* LANGUAGE 6.3. The pooled growth exponent is replaced by the used-channel ratio (C-2). - *EM contrast:* PAPER_LEVEL's two-sentence version with the 1800-epoch caveat (X11). - *Egocentric actions:* two sentences as a boundary only. Rotation actions break a cumsum phase and allocentric recoding restores it (+0.050 -> +0.488, 8/8; 16-epoch recipe, index arm on the floor, not registered). - *Related work:* draft 2.4 plus PAPER_LEVEL's positioning (Shakerinava owns "optimisation not expressivity" for non-Abelian; Grazzi / Mamba-3 own representability; ours is the learnability cost in the rotary slot and its route account).

10 Limitations. Sec. 3.4.

Appendix, re-lettered.

letter	content	source
A	models, recipes, batches, reproduction of MapFormer v4 Table 2	draft A
B	floors (add the text-world retrace 0.934 / 0.937 and the NEEDS-MAP share; T = 1024 floors per seed)	draft B + LANGUAGE 2.6
C	per-seed tables	draft C
D	constructions (add the exp-step construction, CPU-7)	draft D
E	theory: v3 (A0, B0, T1, P1-P7, Remark R updated) + SEARCH S1-S8 + RANK Props. 1-3 with A1-A3 + DEPTH 3.3 + RECURRENCE T1-T2 + MAPPOPE Prop. 1 / Theory 2 + LANGUAGE T1-T2; one status table	all reports
F	exchange rate: draft F + DEPTH 4.1-4.6 (class splits, attention, probes, run-length eval, ring rules and count, Dyck blocks, the 0.75 decomposition labelled descriptive)	DEPTH
G		RANK

letter	content	source
	rank: draft G + RANK C1-C9 (C6 and C9 as CPU pilots; the 5D non-reproduction stated as a recipe difference, rule 16) + Fig. G2 (pilot dynamics)	
H	sign: draft H + SEARCH 3.1-3.4 (latent geometry, reachable share, path profiles, basin); H4 rewritten (C-7)	SEARCH
I	CARoPE and CoPE	draft I
J	text world: draft J + LANGUAGE 2.1-2.6	LANGUAGE
K	what / where at T = 128 and T = 1024; the leak with its fresh-seed accuracy non-replication (+0.0045, p 0.46)	draft K + CPU-1
L	PoPE's score: MAPPOPE's one-page appendix (Tables L1, L2; CORRECTED qualifiers; "what NOT to claim")	MAPPOPE
M	loops and depth on a rank-2 code	RECURRENCE 6.4
N	EM recency as a contrast case (WARM, NOLEAK, SEARCH, SPREAD rows)	PAPER_LEVEL 3.2
O	analysis scripts: path, what each computes, verification check (logit-exact / reproduces stored accuracy), audit status	all reports' reproduce sections

3.4 Limitations: the single scope statement (replaces draft Sec. 9)

> Scope. - The models are small (1-4 layers, 2 heads, d_model 128-256) and the tasks synthetic; natural text appears only as a three-seed pilot. - "Not found" is relative to the stated budgets (300-1800 epochs). Learning rates were swept on the ring only; on the torus, in the larger model, the sign gap is a quarter of the one-layer gap with most |.| runs still descending. - Rank results use two heads; our latent is shared where MapFormer's is per head (per-head arms are reported). - The torus exchange rate is a lower bound measured at n = 4. - The theory was developed after the registered batches. Its proved statements concern the step parameterisation and quadratic surrogates. The hiding feedback at rank D is a conjecture. Every check marked (analysis) is post hoc, on one batch per arm, without fresh-seed replication [and without an independent code audit, until CPU-9]. - The text world is scripted, its direction words are context-free, and its walks are short enough that retrace scores 0.934. - No recurrent or SSM baseline [until G-3].

3.5 For an 8-page cut

Keep Secs. 1, 3, 5, 6, 7.1-7.3 and 8. Shrink Sec. 4 to Figure T1 plus 4.4, 4.6 and 4.9 (half a page). Move Fig. 6 (mechanism), 7.4 and the language theory to the appendix.

3.6 Figure plan

fig	file	status	section	content / data source
1	`fig_exchange_rate`	exists	6	caption: name the stream for retrace (0.844 is the FACTORIAL stream; Table 1's 0.843 is the sign batch's)
2	`fig_strata_modes`	exists	3, 5-7	recurring strata figure; add "one-pass reader" ticks for rank 2 if 7.4 keeps the reader (optional)
3	`fig_rank_solved`	exists	7.2	unchanged

fig	file	status	section	content / data source
4	`fig_existence`	exists	7.2-7.3	unchanged
5	`fig_loss_curves`	exists	7.3	unchanged
6	`fig_mechanism`	exists	7.3 (App. H in 8 pages)	unchanged
7	`fig_textworld`	extend	8	add (d) step table by word class (`scripts/language/tw_step_account.json`), (e) aside collapse / lift (`tw_attention_asides_out.txt`, `tw_attention_asides_dironly.json`); caption: eval mode for (c), re-scored for (e)
T1	`fig_theory_routes`	new, schematic	4	clock / map / cone for one channel; route diagram (collective, parking, hiding)
8	`fig_segments`	new	6	accuracy vs direction runs spanned (2, 3-4, 5-8, 9-16, 17+): torus index 1L / 4L / loop / path (`scripts/depth/out/torus_depth_*.json`) beside ring index 1L / 2L / 3L (`ring_depth.json`); optional inset: fixed-gap 32-127 rows
9	`fig_clock_endstate`	new	7.1	x: usage-weighted clock / map (log), y: arm; dots = runs, colour = SOLVED; both axes (SEARCH `sign_clock_rows.json`, `rank_clock_out.txt` after CPU-3 harmonisation); side panel: off-plane share of solved rank > D heads vs the random-direction baseline (RANK `sweep_final.json`)
10	`fig_loop_reader`	new	7.4	per stratum, bars for A / L / L4 / rank 4 split into "read by one pass" (soft reader) and "added by later passes"; index 1L / loop as a cross-batch group (RECURRENCE 6.4, `out/SUMMARY.txt`)
11	`fig_kernel_peak`	new	7.4 (or App. L)	(a) content-pair kernel vs token offset: PoPE peaks at D = 1, RoPE a comb at 3, 7, 10-11; (b) target coherence C vs C0 for MapWM vs MapPoPE at T = 128; (c) C0 by stratum at T = 1024, r2 vs r4 (`scripts/mappope/kernel_probe_*.json`)
G2	`fig_rank_pilot_dynamics`	new	App. G	kappa and off-plane share vs step for r2 / r4 / R2Clock (pilot n = 3; `scripts/rank/pilot_t128/`)
H7	`fig_path_profiles`	new	App. H	loss along straight paths, signed vs
J2	`fig_used_channel_rho`	new	App. J (or Sec. 9)	used-channel rho across torus / text world / code / TinyStories / recency (`crossover_usage.json`, `usage_tick.json`)

4. Corrections to draft 1 found by the reports

id	where in draft 1	problem	fix	source
C-1	Sec. 7, App. B, App. J, abstract	Text-world floors given as 0.505 / 0.597 only; retrace	Report retrace beside the others. Rest the map claim on the NEEDS-	LANGUAGE 2.6

id	where in draft 1	problem	fix	source
		scores 0.934 on that stream	MAP revisits (path 0.961 fresh seeds 2-7, 0.920 all eight re-scored; RoPE 2L 0.138; best constant 0.112; analysis) and the registered mechanism. Note that DirOnly (0.970-0.974) is barely above retrace. Run CPU-2 first	
C-2	Sec. 8 "Text"	"the phase grows linearly with token index ... consistent with there being no map to build": text-world models with a perfect map show the same alpha (0.98-1.02)	Replace with LANGUAGE 6.3 "Text". Linear growth is not diagnostic; the used-channel ratio (0.06-0.23 vs 9.9-12.3) separates the two	LANGUAGE 3.4
C-3	Sec. 3 "Protocol"	"Every verdict was pre-registered" sits beside the ring exchange rate (a declared secondary; CANCEL's registered primary landed in no branch) and beside analysis floors	"Every contrast in Tables 1-3 and Figs. 1, 3-5 was pre-registered with its decision rule; declared secondaries and analyses are marked". Check against App. A	PAPER_LEVEL 1
C-4	Sec. 8 "What and where", App. K	The checks ran on T = 128 checkpoints, where retrace scores 0.934 and a map is not needed	Re-run at T = 1024 (running). Replace the paragraph with the T = 1024 numbers and keep T = 128 in App. K	PAPER_LEVEL 1, 3.1
C-5	Sec. 7 clock seeds	"the model counts steps"; "solve as well"; "zeroing the non-direction words' steps costs it more" read as clock use	The clock counts **moves** (an aside adds <= 0.06 of a tick, a filler <= 0.007). It sits in channels retrieval does not use, at a cost <= 0.009 to remove on the 7 clock models at ceiling. Seed 0 (a clock seed, 0.881 eval mode) is below ceiling and still uses it, so drop "solve as well" for all four. The zeroing cost tracks the direction words' common component in the used channels (Spearman +0.905), not clock use. All analysis	LANGUAGE 2.2-2.3, 6.5

id	where in draft 1	problem	fix	source
C-6	Sec. 9 Limitations	"why the non-negative parameterisations stall from small starts is not identified"	Replace with the route account (theory-proved inventory; analysis end states). Keep "the per-channel	.
C-7	App. H4 / theory T1(ii)	"+1/+2 is harder for both; why is open"	"On +1/+2 the	.
C-8	Sec. 5	The rates read as what a layer "computes"	Add: two layers suffice in principle (cited construction, normaliser and width needed); the rates are what training finds at these budgets	DEPTH 3.3
C-9	Sec. 5 "What depth buys"	"just above retrace" stated without an account	The exact class accounting 0.943 (analysis), and the reach in direction runs	DEPTH 4.1
C-10	any mention of LOOP_RANK	Risk of "the loop is a search aid at constant capacity"	"Same parameters and initialisation, larger reader function class; the code is unchanged (analysis)"	RECURRENCE 4
C-11	Sec. 6.1	"rank 3 improves on rank 2 by +0.102 (permutation p 0.027)" without the family correction	Add Holm 0.054 (CLAUDE.md, `RANK3_RESULTS.md`)	CLAUDE.md
C-12	Fig. 1 caption vs Table 1	Retrace 0.844 vs 0.843	Say which stream each is (FACTORIAL s500-503 vs SIGN_MATCHED s0-7)	`fig_exchange_rate_out.txt`, `sign_floors_out.txt`
C-13	Sec. 6.3	Signature stated as an observation over accuracies	Derive it from 4.4. State it at the code level (X6)	SEARCH S5
C-14	Sec. 8 "Integrator view"	"signed steps make a map and non-negative steps a path-length clock"	Which case a used channel realises is set by what the readout keys on. The same signed class gives rho 0.064-0.079 on the torus and 21.8-22.1 on recency (analysis)	LANGUAGE 3.3, 6.3
C-15	App. L	Two lines	MAPPOPE's appendix. The path-side gain is small (+0.0243, shrinking to +0.009 on fresh seeds, plausibly convergence speed); the interaction is about 80% index side	MAPPOPE

id	where in draft 1	problem	fix	source
C-16	App. K leak paragraph	Gives the fresh-seed accuracies 0.9940 vs 0.9895 without saying the accuracy gain did not replicate	Add "+0.0045, p 0.46: the convergence gain replicated, the accuracy gain did not"	LANGUAGE 5; CLAUDE.md
C-17	Clock-channel counts	31-38 vs 26-38	Registered 31-38 (first visit); analysis 26-38 (consecutive visits), each with its definition	X10
C-18	Abstract (PAPER_LEVEL's proposed version)	"at each parameterisation's best learning rate" drops "on a ring"	Keep "on a ring"	PAPER_LEVEL 2.2 vs draft

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5. GPU experiments, merged and ranked; CPU checks before drafting

Every GPU batch needs a pre-registration, a CPU task gate, pilots on seeds outside the batch, a results-blind verification agent with its amendment before reading (rules 7, 11, 29), a check of `nvidia-smi` owners (shared server), and the user's approval of the cost. Costs are the reports' estimates, scaled from measured timings (torus T = 1024, 1 layer: about 2,700 run-epochs per hour on the two 4090s; 4-layer or loop runs about 3.5x).

5.1 Ranked list

"Decisive" means the theory predicts opposite outcomes under its rival accounts.

rank	id	experiment (merged from)	design	cost	claim it unlocks	decisive?
1	**G-1 Clock-route batch (rank)**	RANK G1 + G1+, SEARCH 6.1, SEARCH 6.3, RECURRENCE E2 (snapshot part)	Torus T = 1024, 900 ep, n = 8 fresh seeds. Arms: shared r2; **r2 + clock slot** (identical to r2 at init, +64 params); **r2 drift-forbidden** (inverse-pair latents tied, observation steps 0); r2 ActOnly (observation steps 0, moves free); shared r4; optional r4 wd 0. Geometry snapshots every 10-50 epochs	5 arms x 8 = 40 runs, about 13-14 h wall; +r4 wd 0 about +2.7 h	Rank theory (4.6c, 4.7). Clock slot >= 6/8 supports "extra rank helps by giving the clock an exit"; <= 2/8 supports over-parameterisation (Burer-Monteiro). Drift-forbidden r2 solving like r4 shows the rank-2 failure is clock removal; failing shows a second obstacle. r4 wd 0 separates parking from removal (X2). Snapshots test P2 / P4 at scale	**yes** (the central rank test)

rank	id	experiment (merged from)	design	cost	claim it unlocks	decisive?
			(check snapshots do not change the RNG stream, rule 13)			
2	**G-2 Sign-route batch (ring + torus)**	SEARCH 6.4, 6.2; PAPER_LEVEL #3	Ring, observation-step-free, n = 8: signed;	.	; **hybrid a = omega abs(W) z** (non-negative readout, signed latent: collective route restored); **	.
3	**G-3 Sign beyond attention: diagonal complex SSM**	PAPER_LEVEL #1	One-layer h_t = exp(i omega Delta(x_t)) h_{t-1} + B x_t on Z_m (m 8, 32; T 128-512) with +1 / -1 / stay; signed,	.	, softplus, exp; n = 8; existence construction per Prop. 1	5-8 GPU-h; 1-2 days code and checks
4	**G-4 Torus best-rate sign sweep (+ rank)**	PAPER_LEVEL #2	Observation-step-free torus, signed vs	.	at 1e-3 / 3e-3 / 1e-2, n = 8, selection-aware test as LRSWEEP; optional r2 vs r4 at 3e-3, 1e-2	sign only 40-48 runs, about 12.5-15 h wall; + rank about 25 h
5	**G-5 Depth routes**	DEPTH E1, E3 (ring step)	(a) Ring T = 128, index 2 layers with and without BOS + absolute channel (does the count appear at 2 instead of 3?), n = 4-8. (b) Torus walks with run length fixed at 2 / 5 / 10, index 4L and path 1L, trained and tested on each; T = 512 version	(a) about 3 GPU-h; (b) about 24 GPU-h at T = 512 (48 at T = 1024)	Route B predicts accuracy is a function of runs spanned in every condition; route A predicts dependence on T and the gap. (a) tests whether a missing normaliser blocks route A	**yes** (between the two routes)
6	**G-6 Delta-freedom 2x2 (polar score)**	MAPPOPE #1	Paper torus T = 128, r2, 300 ep: {index, path} x delta {pinned, free U(-2 pi, 0)}; optional MapPoPE with delta frozen random; n = 8	about 2.5-3.5 h	Peak pinning (4.10) predicts a **sign reversal**: free delta helps the index arm (concentrated in R3+) and hurts or slows MapPoPE	**yes** (cheap)
7	**G-7 Long-walk text world +**	LANGUAGE E4, E3	(a) Text world at 4096 words (~530 moves): path vs RoPE 1-2	(a) about 20-35 GPU-h;	(a) Removes the short-walk caveat that C-1 creates for contribution 5. (b)	(b) yes; (a) strengthens

rank	id	experiment (merged from)	design	cost	claim it unlocks	decisive?
	double dissociation		layers, n = 8. (b) A text world where some queries are count-keyed ("what did she see k moves ago") and the rest place-keyed: used channels should split into rho << 1 and >> 1, and each set's ablation should remove only its query type	(b) about 3 GPU-h	Turns the tick rule and the used-channel ratio into a within-model test	
8	G-8 Rank x sign x passes factorial	RECURRENCE E1	Torus T = 1024: {r2, r4} x {signed,	.	} x {1 pass, loop x4}, n = 8; primary = geometry class and one-pass reader quality	64 runs, about 74 GPU-h (about 19 h wall); trimmed 48 runs about 55 GPU-h
9	G-9 Rank follow-ups	RANK G2, G4, G3	Seeded escape (lift with clock off-plane share ~0.7 vs default), n = 8, 3.6 h; ring rank 1 vs 2 at T = 1024, 2 h; 3D grid 10 rank 5 vs 4, 5.3 h	about 11 h total	Escape mechanism; whether the 1D exemption is the short task; P9 (D + 2)	G2, G4 yes
10	G-10 Reviewer-level breadth	PAPER_LEVEL #4, #5, #6, #8 + DEPTH E2	LSTM / GRU torus baseline (6-10 GPU-h); egocentric actions at T = 1024 (about 10 h); sign and rank at scale (about 27 h); index depth to 8 layers (DEPTH E2 / PAPER_LEVEL #8, about 35-44 h)	80-90 h	Main-track objections: "why attention", scale, "more than four" turned into a number	no
11	G-11 Smaller items	LANGUAGE E1; SEARCH 6.5; PAPER_LEVEL #7 + RECURRENCE E5; MAPPOPE #2;	Clock emergence snapshots in the text world (1.5 GPU-h); +1/+2 at 900 epochs (3 h); registered Match-Query	--	appendix-level	mostly no

rank	id	experiment (merged from)	design	cost	claim it unlocks	decisive?
		RECURRENCE E3; DEPTH E4, E5	rerun with destruction (5-10 h); pinned-peak MapWM at T = 1024 (about 1 day); pass count x8 (30-50 GPU-h); ring vs T (15-60 GPU-h); Dyck across D (about 20 GPU-h)			
--	dropped	RECURRENCE E4 (loop vs Level 1.5 under noise: CLAUDE.md live negatives, X13); TW_AMBIG (stopped 2026-10-11; low for this paper); TW_LANDMARK (live negative)	--	--	--	--

Cheapest set that makes the theory registered: G-2 ring part (about 5 h) + G-6 (about 3 h) + G-1 (about 14 h), about 22 h wall. Add G-3 and G-4 for a main track.

5.2 CPU checks before drafting (no GPU)

id	check	why	cost
CPU-1	Finish the what/where re-analysis at T = 1024 (running: Pos_r4 5/8 done, RoPE pending) and write its `.txt`	C-4; Discussion and App. K	running
CPU-2	Retrace floor and NEEDS-MAP strata on the TW_NORMSTEP and TW_STATECHANGE batches (LANGUAGE E5)	C-1: no text-world number goes into the paper until done	about 1 h
CPU-3	Harmonise the clock metric: recompute RANK's off-plane share and readgain with SEARCH's per-inverse-pair clocks on the same checkpoints; report clean and drifting heads separately	X2, X3; Fig. 9	about 1 h
CPU-4	Rank-3 heads: are the two pair clocks independent in the unsolved runs (S6(b): one parkable at r = 3)?	X14	< 1 h
CPU-5	Polarise MapWM r2 / r4 at eval (beta -> 0) at T = 1024, by stratum (MAPPOPE #3)	7.4: a cheap check of Theory 2	< 1 h
CPU-6	Loss by stratum for the LOOP_RANK arms (RECURRENCE T3) and strata for the aids (PAPER_LEVEL CPU (c), mostly in RECURRENCE 5)	7.4, App. M	< 1 h
CPU-7	Exp-step existence construction (as softplus, P1(f))	needed by G-2, G-3	< 1 h
CPU-8	The evidence table (PAPER_LEVEL 2.4) from existing files; MAPPOPE's T = 1024 probe at 100 walks instead of 40	Sec. 7 table; App. L numbers	about 1 h
CPU-9	Independent, results-blind code audit of the analysis scripts whose numbers enter the main text: `mappope/`	None had one (the reports say so). Rule 9:	an agent,

id	check	why	cost
	`kernel_probe.py`, `recurrence/phase_reader_soft.py`, `geometry.py`, `pass_strata.py`, `depth/torus_depth.py`, `ring_depth.py`, `torus_probe.py`, `language/tw_retrace_floor.py`, `tw_strata.py`, `tw_step_account.py`, `tw_attention_asides*.py`, `usage_tick.py`, `crossover_usage.py`, `sign/sign_clock.py`, `rank_clock.py`, `path_profiles.py`, `rank/rank_geom.py`, `c1_runlevel.py`, `escape_start.py`	probes lie confidently. The reports' self-checks (logit-exact forwards, stored accuracies reproduced) are necessary, not sufficient	half a day
CPU-10	Build Figs. T1, 8-11, 7d-e, G2, H7, J2 from stored outputs	Sec. 3.6	half a day
CPU-11	End-to-end read of draft 2 after assembly (rule 28); grep the generators for any withdrawn number (rule 18)	--	--

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6. Status table: every new claim draft 2 would carry

"New" = not in draft 1. R = registered, A = analysis, P = theory-proved, C = theory-conjecture, S = speculation.

#	claim (as the paper would state it)	label	source
1	The integrator puts P_t in the rotation; one layer compares P_t - P_s at any gap	P	DEPTH 3.2
2	Index attention can represent the count in two layers in principle (normaliser, 1/n precision, large width)	P (cited construction)	DEPTH 3.3; Liu et al. 2023 Lemma 6
3	Exchange rates are learned rates, not representational bounds	P (from 2) + R (rates)	DEPTH 3.3
4	Deep torus index models solve retraces (0.996 / 1.000) and decay with direction runs spanned (0.93 / 0.98 at 3-4, 0.74 / 0.65 at 5-8, 0.56 / 0.50 at 9-16)	A	DEPTH 4.1
5	At a fixed gap of 32-127 moves the same decay appears (0.87 / 1.00 at 2 runs to 0.49 / 0.50 at 17+)	A	DEPTH 4.1
6	0.943 short-gap accuracy is exactly retrace plus partial 3-8-run returns	A	DEPTH 4.1
7	No global count in torus index models (P_t / t R^2 <= 0.21); a recent-displacement code degrading with the window	A	DEPTH 4.3
8	Three ring index layers are flat in runs spanned and carry a count (P_t / t R^2 0.69)	A	DEPTH 4.5
9	One ring index layer learns exactly retrace and the one-lap clock	A	DEPTH 4.5
10	Deep index models' last layer attends to the last visit on retraces (0.55-0.58) and not at long gaps (0.00-0.01; path 0.81)	A	DEPTH 4.2
11			DEPTH 4.4

#	claim (as the paper would state it)	label	source
	Run-length-shifted eval: two-run returns gap-invariant; runs set the level, the gap trims it	A (distribution-shifted)	
12	Clock / map decomposition; a clock serves gaps < pi / k only (signature)	P	SEARCH S5
13	Tick rule: usable for place-keyed retrieval only if tick = 0 mod 2 pi / g	P	LANGUAGE T1
14	Signed collective route {z_x + z_y = 0}: cancels every channel, map free, surrogate flow converges, generically unique	P	SEARCH S2
15	Freezing move latents blocks the escape (1/8 vs 7/8)	R	ACCOUNTS T2
16		.	cone: clock >= map below pi; zero-clock non-blind points at 2 pi n
17		.	has no collective route for an inverse pair; <= 2r latent-cancellable channels at a generic readout
18	+1/+2:	.	has two mirror collective zeros; frustration contracts to the blind origin when both coherence classes are populated
19	Latent-reachable share of the needed change at init: signed 1.000,	.	0.34 / 0.56 (unweighted / usage)
20	Straight-path loss to the solution:	.	rises 0.6-1.9 nats, signed 0.04-0.48 (complete separation); signed latent route alone 0.30 -> 0.72,
21	With the content frozen, the start decides the step map's search; the signed escape from the same-direction start is not reproduced by the step map alone	A	SEARCH 3.4
22	Failing arms end clock about the size of the map (1.1-1.9	.	; 0.5-1.9 unsolved rank-2 heads); successes clock-free (<= 0.065)
23		.	trained ring models: 0.30-0.36 of sub-pi channels at the cone boundary (0.056 at init)
24	On +1/+2 the	.	step code is learned (8/8 in used channels); unsolved runs fail outside the step code
25	Rank D: no channel is clock-free and a full map while the clock is non-zero; rank > D: per-channel parking at zero map cost	P	SEARCH S6; RANK Prop. 1
26	1D exemption: no readout change alters the drift-to-map ratio	P	RANK Prop. 1(c)
27	Hiding is free to first order at D >= 2 (isotropic P)	P (under A1-A3)	RANK Prop. 2

#	claim (as the paper would state it)	label	source
28	Hiding starves the global route (feedback, plateau)	C (P2 NOT SHOWN)	RANK Prop. 3
29	Drifting rank-D heads drift along their least-resolved axis (57/62 runs vs 28.7 expected)	A (not discriminating)	RANK C1
30	Solved rank > D heads hold the small leftover clock off the map plane (0.96-1.00 in 9 of 10 sets vs random 0.50-0.71) and barely read it	A	RANK C2 (re-check CPU-3)
31	A fresh rank-4 step map removes the drift in 16/16 heads, by parking; lifted runs solve in order of initial off-plane share (rank-sum p 0.018)	A	RANK C5
32	Clock slot rescues a failing rank-2 start (one matched pair, T = 128)	A (pilot, n = 1 pair)	RANK C6
33	Rank 2 + clock slot learns the T = 1024 torus like rank 3-4	S (prediction; G-1)	RANK P7
34	One-layer models are position-only readers of their own code (model - reader +0.028 rank 2, -0.002 rank 4)	P (structure) + A	RECURRENCE T1, 5.3
35	Loop and depth leave the code unchanged (reader quality L - A +0.027, p 0.28; L4 - A +0.031, p 0.27) and read it better (+0.32-0.44 on long / wrap)	A	RECURRENCE 5.3
36	The loop is a 4-step computation: one pass of its weights scores 0.381; retrieval forms at passes 2-4	A	RECURRENCE 5.2
37	Aids end in 1D clock-free (8/16) or drifting codes, never rank 4's 2D code (1/16 vs 8/8)	A	RECURRENCE 5.4
38	The loop gives no larger or more coherent step-map gradient at init (Adam-invariant x2.9)	A	RECURRENCE 5.6
39	Deep readers make the collapsed code a usable basin	S	RECURRENCE T4
40	Peak pinning under beta = 0	P (delta = 0 measured)	MAPPOPE Prop. 1
41	Target coherence C 0.70 (MapWM) vs 0.995 (MapPoPE), T = 128; action-key attention 0.179 vs 0.041	A	MAPPOPE Cor. 1a-1b
42	PoPE's index kernel cannot prefer offset 3 over 1 (K(3) > K(1) in 0% of pairs); 0.110 of the -0.126 gap is on R3+ revisits	P (Cor. 1c) + A	MAPPOPE Cor. 1c
43	About 80% of the registered +0.154 interaction is the index side	A	MAPPOPE
44	At T = 1024 the score repairs the readout term on short gaps (C +0.34, p 0.0002; C0 unchanged) and leaves the code term on long-gap and wrap revisits	A (40 walks; CPU-8)	MAPPOPE Theory 2 check

#	claim (as the paper would state it)	label	source
45	Free delta reverses the interaction	S (prediction; G-6)	MAPPOPE #1
46	Text world: retrace 0.934; NEEDS-MAP revisits path 0.961 (fresh) vs RoPE 2L 0.138	A	LANGUAGE 2.6
47	Step table by word class: optional words 0.004-0.03 of a move; aside words large but cancelling to a 0.04-0.16 offset	A	LANGUAGE 2.1
48	Collapsing asides onto the place drops learned models to 0.952-0.975 with 71-100% aside errors; offsetting DirOnly's asides lifts it to 0.999-1.000 (8/8)	A	LANGUAGE 2.4
49	The text-world clock counts moves (Case II) in channels retrieval does not use; removal costs <= 0.009 at ceiling	A	LANGUAGE 2.2-2.3
50	The tick alone predicts which channels drift (agreement 0.98-1.00, 32 models)	A	LANGUAGE 2.2
51	Used-channel rho: torus 0.064-0.079, text world 0.06-0.23, code 4.5-6.6, TinyStories 9.9-12.3, recency 21.8-22.1	A (code, TinyStories n = 3 pilots)	LANGUAGE 3.3
52	The pooled growth exponent does not separate map from clock data	A	LANGUAGE 3.4
53	"A token's offset depends on its role, not its identity" (leak, DirOnly and asides as instances)	C (light) + R (leak) + A	LANGUAGE 5
54	What/where at T = 1024: signed r4 keeps 0.925-0.999 with identity removed, 0.990-1.000 with a shared kernel x gain;	.	models depend on content-position interactions
55	EM's recency deficit is a different kind of search failure (closed mostly by 4x budget)	R pieces + A contrast (not registered)	PAPER_LEVEL 3.2
56	The exact start-to-solution basin does not reach halfway along the straight	.	segment within 2000 step-map steps

Appendix to this plan: what was checked here, and what was not

- The X3 resolution was checked on `rank_clock_out.txt` (per-head rows of the solved per-head r2 and block-diagonal runs).
- X2's definitional difference was checked against the docstrings of `sign/rank_clock.py` and `rank/rank_geom.py`. The 0.004-vs-0.065 gap was not recomputed (CPU-3).
- The what/where numbers were read from the JSON at about 02:00 on 2026-10-20 (Signed_r4 8/8, Abs_r4 8/8, Pos_r4 5/8, RoPE 0/8). The `.txt` does not exist yet.
- Every other number in this plan is copied from the named report or results file. None was re-derived.

MapPoPE: what PoPE's polar score adds to a path-integrated phase (expansion note, 2026-10-20)

Labels: **ESTABLISHED** = registered result (results file named); **ANALYSIS** = new post hoc CPU analysis run for this note, eval-only on stored checkpoints (script and output named); **THEORY** = derivation, with its assumptions and a check against data; **SPECULATION** = untested.

Summary

1. **THEORY.** Both rotary scores have the form `s_ij = sum_c A_c cos(dphi_c + beta_c)`. RoPE and MapWM let content set the amplitude A and the phase beta. PoPE lets content set only A (softplus) and fixes beta = delta_c, a content-free bias. In our runs delta_c is effectively 0. So for every content pair, every PoPE head's kernel peaks at zero phase difference: s_ij <= sum_c A_c, with equality iff dphi_c = 0 mod 2 pi on every weighted channel.
2. With a **path phase**, "zero phase difference" means the same place, which is exactly the revisit target. PoPE's score removes MapWM's content phase, and MapWM's content phase misaims its kernel at the target. ANALYSIS: the kernel's alignment on the target key (C, defined in the theory section) is 0.70 for MapWM and 0.995 for MapPoPE (p 0.0002, 8/8). Removing the content phase alone would lift MapWM by +0.20. MapPoPE also spends 4x less attention on action keys.
3. With an **index phase**, "zero phase difference" means the query's own token. The 1-layer torus solution needs kernel peaks at fixed past offsets (4j-1). RoPE builds those peaks with its content phase; PoPE cannot. ANALYSIS: delta sits in the clamp's dead zone (98-100% of entries), K(3) > K(1) holds in 0% of PoPE content pairs, and 88% of the PoPE - RoPE gap (-0.110 of -0.126) is on multi-step reversals. This sign flip gives the +0.154 interaction, and about 80% of the interaction is the index side.
4. The score acts only on the readout term (C - C0). It never acts on the code term C0, which is how well the phase alone puts the same cell at zero. At T = 1024 and rank 2 (ANALYSIS), the score repairs short-gap revisits (C +0.34, p 0.0002; C0 unchanged). On wrap-only revisits C0 is 0.41-0.46 under both scores, and only rank 4 raises it. So no score rule can rescue rank 2 there.
5. Paper text: a short paragraph (Sec. 6 or App. L) and a 1-page appendix are proposed below. The single new GPU experiment is a delta-freedom 2x2 (index/path x delta pinned/free), about 2.5-3 GPU-hours. It predicts that the interaction reverses.

Established facts (registered unless marked)

fact	numbers	file
2x2 at T = 128, r2	RoPE 0.805, PoPE 0.679, MapWM 0.971, MapPoPE 0.999; interaction **+0.154** (MDE 0.076, 8/8). Encoding main effect -0.049. At r4 the interaction is +0.125, with both path arms at 1.000	`PAPER2X2_RESULTS.md`
		`PAPER2X2_RESULTS.md`

fact	numbers	file
Beyond training length the encoding helps the index arm	PoPE 0.639 / 0.607 vs RoPE 0.470 / 0.374 at T = 512 / 1024 (+0.169 / +0.233); r2 encoding main effect +0.114 / +0.189. At r4 the interaction is -0.172 / -0.221 (OOD; robustness, rule 10)	
T = 128 floors	blank 0.507, n-gram 0.598, **retrace 0.934** (wrap-only share 0). Both index arms sit below retrace	`docs/audits/2026-10-16/paper2x2_T128_floors_out.txt`
The gain is the score rule, not the frequency count	SCORE +0.0243 (p 0.012, 16/16 vs 10/16 SOLVED); COUNT +0.0002 (at ceiling, uninformative). CORRECTED: plausibly convergence speed; it shrank to +0.009 on fresh seeds	`MAPPOPE_PAIR_RESULTS.md`, `GAIN_GRAIN_RESULTS.md`
A peak-at-zero gain kernel is enough on a task with no headroom	GainScalar (one gain per token, delta 0) 20/20, the same as MapPoPE (descriptive: loss < 0.05 at epoch 23 vs 74 vs MapWM 132; confounded)	`GAIN_GRAIN_RESULTS.md`
No rank-2 rescue at T = 1024	3/12 vs 3/12 SOLVED. Secondaries: short gap +0.115 (p 0.036), gap >= 128 +0.043, wrap-only -0.015	`SCORE_RANK_RESULTS.md`
The score does not touch the what-to-where leak	GainRaw keeps MapWM's leak. ONE defect shown (CORRECTED)	`GAIN_PHASE_RESULTS.md`
Prior probes (post hoc)	MapWM r2 observation-key fields peak at d = 0 for 7% of content pairs (mean shift 13.7 cells); MapPoPE for 100%. Same-cell coherence C vs C0 at T = 128	`docs/audits/2026-10-05/remap_probe_out.txt`, `neuro_samecell_phase_out.txt`
Language / other	Bach MapPoPE - MapWM -0.0165 NLL (5/5); PoPE - RoPE -0.032. Code and Dyck MapPoPE rows are UNMEASURED or depth-OOD. PoPE faithful (1.7e-6); 80.7% of `pope_delta` in its clamp dead zone on the language runs	`JSB_RESULTS.md`, `.claude-memory/project_mappope_asymmetry.md`, `ABLATE_RESULTS.md`

Theory

THEORY 1 (the polar decomposition of a rotary score). Write channel c's query and key as complex numbers. RoPE / MapWM score: `s_ij = sum_c Re(q_c conj(k_c) e^{i dphi_c})` with `dphi_c = phi_c(i) - phi_c(j)`, so A_c = |q_c||k_c| and beta_c = arg q_c - arg k_c, both content-dependent. PoPE / MapPoPE (code: `model_pope.py`, `cos(dphi - delta)`): `A_c = softplus(q_c) softplus(k_c)`, beta_c = -delta_c, content-free. The phase dphi is the WHERE half: path integration changes where it comes from (`omega * cumsum(W_out W_in x)` instead of `t * theta`). The score map is the HOW half: whether content may enter the phase. The two halves are orthogonal design choices. This was already noted in `.claude-memory/reference_language_and_pope.md`; what follows makes it predictive.

Proposition 1 (peak pinning). If beta_c = 0 for all c, then for any content pair `s_ij <= sum_c A_c`, with equality iff dphi_c = 0 mod 2 pi on every channel with A_c > 0. As a function of displacement, the kernel is a non-negative combination of cosines, which is

positive-definite (Bochner). Content can reweight the frequencies, which changes the kernel's width and shape, but it cannot move or invert the peak. *Assumption:* delta = 0. **Check:** delta is in the clamp's dead zone (>= 0, zero gradient) on 98-100% of entries in every PoPE-Flat run, with mean |delta| = 0.00 (ANALYSIS A1). In MapPoPE C = C0 to 3 decimals (A1, A2), so delta does not shift the peak there either.

Corollary 1a (path phase). If the code is faithful (equal phase iff equal cell, within the head's resolved range), the pinned peak is "attend to where I am", which is the revisit target. A RoPE-form score must instead learn beta_c = 0 for every (action query, observation key) pair. A residual beta_c displaces the field in space by beta_c / omega_c, and the displacement is largest on the coarse channels. **Check (A1, T = 128, r2, n = 8):** target coherence is C = 0.703 for MapWM vs 0.995 for MapPoPE (p 0.0002, 8/8 seeds above). MapWM's content phase costs C0 - C = +0.20 (7/8 seeds). Same-cell attention mass is 0.747 vs 0.925 (p 0.0008).

Corollary 1b (key suppression must be by magnitude). Under a pinned peak, an action token at the same cell has the same phase as the observation there, so the score can only down-weight it through softplus(k). **Check:** mean softplus(k) is 0.61-1.56 for action keys vs 0.91-1.80 for observation keys. Attention on action keys is 0.041 for MapPoPE vs 0.179 for MapWM (p 0.0002). MapWM's amplitude-weighted cos(beta) is nearly the same for both key types (+0.76 vs +0.75), so it barely separates them in phase.

Corollary 1c (index phase). With an index phase, dphi = 0 is the query's own token. PoPE's frequencies are theta_c = 10000^(-c/64) <= 1 rad/token. Each cos(theta_c D) decreases on D in [0, pi], so with beta = 0 **every** content pair's kernel strictly decreases over offsets 0..3. A pinned-peak index head cannot prefer the observation 3 tokens back to the one 1 token back. Every fixed nonzero target offset must come from beta, which RoPE supplies through content and PoPE through delta (stuck at 0). The 1-layer torus task needs exactly such offsets: during a run that reverses the previous run, step j's target sits at token offset 4j-1 (the retrace rule, 0.934 at T = 128). **Check (A1):** - K(3) > K(1) holds in 0% of PoPE-Flat (head, action, observation) pairs on every seed, and in 53-100% for RoPE. - The argmax over D in 1..16 of the content-pair kernel is at D = 1 for 1068/1088 PoPE pairs. For RoPE it is at D = 3 (414), 7 (277), 10-11 (147) and 2 (168), a comb over the retrace offsets. - PoPE puts 0.307 of its attention on the newest observation (never the target; every step moves), vs 0.002 for RoPE (p 0.0002). - The accuracy gap decomposes over retrace strata as R1 -0.006, R2 -0.005, R3+ **-0.110**, non-retrace -0.004, out of -0.125. R3+ accuracy is 0.566 vs 0.779 (p 0.0002, 0/8 seeds overlap). The cost is entirely where the target is far back at a fixed offset; R1/R2, where the target is among the few most recent keys, are nearly intact.

Reading of the +0.154 interaction (THEORY + ANALYSIS). (MapPoPE - MapWM) - (PoPE - RoPE) = (+0.028) - (-0.126). The same constraint (peak pinned at zero phase difference) is aligned with the target under a path phase and misaligned under an index phase. About 80% of the interaction is the index-side loss; at r4 the path side is at ceiling and the interaction is +0.125, all index side. A pinned recency kernel is also length-agnostic. That fits PoPE's better extrapolation as an index code (0.607 vs RoPE 0.374 at T = 1024) and the PoPE paper's own remark that zero-init delta is "important for length generalization". This is a consistency check, not a test.

THEORY 2 (what a score rule can and cannot fix). Any score of this family sees position only through dphi. Write the target coherence as `C = C0 + (C - C0)`: - C0 is the code term: the A-weighted cos(dphi) on the true target, i.e. whether the phase alone puts the same cell at 0. - C - C0 is the readout term, which is beta's contribution.

A pinned-peak score sets the readout term to 0 by construction. It can also reweight channels (A >= 0), so it can **select** channels whose code is consistent but cannot **create** them. Wrap-only revisits need the code to be periodic on the torus, so they live in C0. **Check (A2, T = 1024, SCORE_RANK checkpoints, r2 n = 12, r4 n = 8):** - Short-gap revisits: C rises from 0.617 (MapWM) to 0.960 (MapPoPE), p 0.0002. C0 does not move (0.900 vs 0.960, p 0.16). Accuracy is +0.114 (p 0.036, replicating the registered +0.115). The repair is entirely the readout term. - Gap >= 128: C rises from 0.524 to 0.840 (p 0.029), but accuracy (+0.042, p 0.63) and target mass (-0.035) do not. C0 stays at 0.81-0.84 vs 0.99 at r4. The code has drifted, so many wrong-cell keys score as well as the target. Aiming the kernel exactly is not enough when the code cannot discriminate. - Wrap-only: C0 is 0.457 (MapWM) vs 0.411 (MapPoPE), p 0.79, and accuracy is 0.641 vs 0.631. Rank 4 raises MapWM's C0 to 0.840 (p 0.03) and accuracy to 0.998.

The score rule cannot rescue rank 2 because the rank-2 failure is in C0 (the code's periodicity and drift), which is outside any score's reach. Note: the score does not touch the code at T = 1024, but at T = 128 MapPoPE's C0 is also higher (0.996 vs 0.905, p 0.0002). A pinned peak may make a good code easier to *find* on short walks, consistent with GAIN_GRAIN's faster convergence; the effect does not carry to T = 1024 (SPECULATION as to mechanism).

Scope and assumptions. All checks are 1-layer, 2-head models on the 64x64 torus. In a 1-layer model Q and K depend only on the current token, so content cannot encode a computed offset. Deeper models could carry position into the amplitudes; PoPE's Indirect Indexing result (`INDIRECT_RESULTS_200k.md`, 7/8) is multi-layer and outside this account. "delta = 0" is a property of the faithful zero-init plus clamp. With delta ~ U(-2 pi, 0) the peak is still content-independent but no longer at zero, which is the lever of the proposed experiment.

Analyses run (all post hoc, eval-only, CPU)

All are in `/home/prashr/mapformer/docs/paper/expansion/scripts/mappope/`. - `kernel_probe.py` - Rebuilds layer 0 by hand; every run's logits equal the model's own to 0.0e+00 (`verify`). - Uses the same evaluation stream as `eval_noise_refine` / `eval_rank_strata` (held-out map seed 10000, np seed 1234 + s). Per-seed accuracies reproduce `runs/paper2x2/eval.log` exactly, e.g. RoPE s0 0.7892 and PoPE-Flat s0 0.6244. - A1 `paper2x2` (T = 128, 100 walks per seed, seeds 0-7): `kernel_probe_paper2x2_out.txt`, `.json`, `.log`. - A2 `score_rank` (T = 1024, **40** walks per seed, a subsample of the registered 100): `kernel_probe_score_rank_out.txt`, `.json`, `.log`. - Readouts: accuracy by retrace stratum (R1/R2/R3+/none) and by SCORE_RANK stratum (gap < 128 / >= 128 / wrap-only); attention mass on same-cell observation keys, the retrace target, the offset-1 observation, action keys and other observations; target coherence C and C0 on the most recent same-cell observation key; attention by token offset; content-pair kernel checks (K(3) > K(1), argmax offset, delta dead-zone share, zero-displacement score for action vs observation keys). - `probe_stats.py` -> `probe_stats_out.txt`: unpaired permutation tests (`stats_core.perm2_p`) on per-seed readouts,

and the stratum decomposition of PoPE - RoPE. - Caveats: - The masses and C use the head with the larger mean same-cell mass. In MapPoPE r4, wrap-only C0 is 0.55 while accuracy is 0.995, so another head or the FFN may carry wrap revisits there, and C0 is not a sufficient statistic for success. - A2 uses 40 walks per seed. - All p-values are post hoc and unregistered. They describe these checkpoints; they are not verdicts.

Proposed paper text

Main text, one paragraph (end of Sec. 6 or App. L lead):

> *Where the angle comes from and how content uses it are separate choices.* Any rotary score can be written `sum_c A_c cos(dphi_c + beta_c)`: the phase difference dphi says where, and content sets an amplitude A and, in RoPE, a phase beta. Path integration changes dphi. PoPE's polar score (Gopalakrishnan et al. 2025) removes beta. Without beta, every head's kernel peaks at zero phase difference whatever the content (App. L). Under a path phase, zero phase difference is the same place, which is the revisit target. Under an index phase it is the current token. The two predictions have opposite signs, and so do the results. On the short-walk torus PoPE's score raises the path-integrated model (MapPoPE 0.999 vs MapWM 0.971). There it aligns the attention kernel with the target (alignment 0.995 vs 0.70) and removes the cost of the content phase. It lowers the index model (PoPE 0.679 vs RoPE 0.805), whose solution needs kernel peaks at fixed past offsets. Nearly the whole loss (0.110 of 0.126) is on revisits where the target lies more than ten tokens back. A score rule acts only on how the kernel reads the phase, never on the phase itself. On long walks it repairs short-gap revisits (+0.115) and leaves wrap-around revisits, and the rank-2 deficit, untouched (3/12 vs 3/12 SOLVED): the rank-2 phase does not place a wrapped cell at zero.

Appendix L (replacing the current two lines), about one page: 1. The decomposition and Proposition 1, with the proof (one line) and its assumption (delta = 0; the dead-zone measurement). 2. Table L1, T = 128 2x2, one row per arm. Columns: accuracy; R3+ accuracy; target coherence C and C0; attention on the newest observation; attention on action keys; K(3) > K(1) share. Numbers are from `kernel_probe_paper2x2_out.txt` / `probe_stats_out.txt`. 3. The stratum decomposition of PoPE - RoPE (-0.006 / -0.005 / -0.110 / -0.004). 4. Table L2, T = 1024. C, C0 and accuracy by stratum for MapWM / PoPE score at r2 and r4. The readout term repairs short gaps; wrap-only lives in C0. 5. The registered facts (MAPPOPE_PAIR SCORE vs COUNT, SCORE_RANK NO RESCUE, GAIN_GRAIN no-headroom), each with its CORRECTED qualifier. 6. Label every probe number "post hoc, eval-only". Say plainly that the interaction is ~80% index side, and that the path-side gain is small (+0.024, shrinking to +0.009 on fresh seeds) and at T = 128 plausibly convergence speed.

What NOT to claim: that the polar score fixes rank or wrap; that MapPoPE beats MapWM at long walks (unmeasured at T = 1024, 3/12 vs 3/12); "PoPE hurts index models" in general (one layer, one task; PoPE helps index models on Bach and beyond training length here); "two defects, two fixes" (withdrawn).

Proposed experiments (GPU), ranked

1. **DELTA-FREEDOM 2x2 (the causal test of peak pinning).** - Design: phase {index, path} x delta {pinned: zero-init + clamp (current PoPE-Flat / MapPoPE-Flat); free: init U(-2 pi, 0), clamp removed}. Paper torus, T = 128, r2, 300 epochs cosine lr 1e-3, n = 8 fresh seeds, one batch (rule 12). Optional 5th arm: MapPoPE with delta frozen at U(-2 pi, 0), which removes the pin with no learning route back. - Primary readout (pre-register): the delta x phase interaction on T = 128 accuracy. - Predictions:
   - Free delta helps the index arm, concentrated in R3+ (a comb at offsets 4j-1 becomes reachable).
   - Free or frozen-random delta hurts or slows MapPoPE, since target coherence C falls below C0.
   - Secondary: free delta costs the index arm at T = 1024 (loss of the recency kernel).
   - Gates: delta dead-zone share and K(3) > K(1) share, eval-only with `kernel_probe.py`.
   - Cost: 4 arms x 8 seeds = 32 runs at about 38 min each (7.5 s/epoch at 4 jobs/GPU) is about 2.5-3 h on two GPUs, plus eval. The 5-arm version is about 3.5 h. A pilot on seeds outside the batch and a code-verification agent come first (rule 29).
   - Why first: it is the only test where the account predicts a sign reversal. It also turns the interaction from a description into a mechanism.
2. **Pinned-peak MapWM at T = 1024, r2.** Score `sum_c |q_c||k_c| cos(dphi_c)` (content phase removed, signed magnitude kept), plus MapWM and MapPoPE-Pair as twins, n = 12. Separates "beta removed" from "softplus magnitudes". Predicted: it matches MapPoPE on short gaps and does nothing on wrap-only. Cost about 36 runs at T = 1024 / 900 ep, which is substantial (SCORE_RANK-scale, about 1 day). Lower priority: GAIN_GRAIN already shows a pinned scalar gain suffices at T = 128.
3. **Eval-only (no GPU needed, can run now): polarise trained MapWM r4.** Replace beta by 0 at eval on the SCORE_RANK r4 checkpoints and read the accuracy change by stratum. `WHAT_WHERE_CHECKS` did the related "shared kernel x gain" rebuild at T = 128 (MapWM r2 0.971 -> 0.791). Expect small losses at r4 (C0 - C = 0.07) and large ones at r2. Diagnostic only, because the network was trained with beta.

Why per-head rank decides whether the map is learned: hiding the clock, and parking it

Expansion note for `docs/paper/MAIN_DRAFT1.md` Sec. 6.1 (2026-10-20). Labels: **ESTABLISHED** (registered results, files named), **ANALYSIS** (computed here on stored checkpoints or CPU pilots, script named; post hoc unless said), **THEORY** (statement + assumptions + the check that could break it), **SPECULATION**. Scripts and outputs: `docs/paper/expansion/scripts/rank/`. Nothing here is registered; nothing was committed; no GPU was used.

Summary

The account that survives (THEORY, end-state checks pass, dynamics partly checked): rank above D gives every channel a map-free way to cancel the per-move clock; at rank D there is none. Every move adds a step common to all moves (a "clock" c) to the latent. A channel can stop seeing it in two ways only (Prop. 1): through the latent (shrink c: one global coordinate, removed at a rate proportional to the head's own resolution along the drift direction) or through its own readout. At per-head rank r > D the readout can turn away from the part of c that lies outside the map plane **without touching the map** ("parking"). At r = D the clock lies inside the map plane, c = U beta, and a readout that stops seeing it must stop resolving the drift direction ("hiding"), which in turn slows the global route (Props. 2-3). At D = 1 hiding kills the channel and leaves the drift-to-map ratio unchanged, so only the global route acts: rank 1 learns the ring (7/8).

Strongest evidence (ANALYSIS, post hoc; 1-3 on the stored checkpoints of registered batches, 4 a CPU pilot): 1. **Solved rank > D heads park the clock.** The off-plane share |c_perp| / |c| of their leftover latent clock has medians 0.96-1.00 in nine sets of solved rank > D heads (0.81 in the tenth, 3D grid 18); a random direction gives 0.50-0.71; in those nine sets the readout sees it at 0.09-0.27 of the map's gain. Unsolved rank > D heads hold it in the plane and read it (torus r3 off-plane 0.27, lifted r4 0.08; readgain 0.61-0.78). In solved rank-D heads the clock is simply gone (lam <= 0.001). `summarise_final_out.txt`. 2. **Escape test, re-read.** A fresh rank-4 step map on a stalled network removes the drift in 16/16 heads, from initial drift up to 5.9 cells per move, every one by parking (off-plane 0.91-1.00); rank 2 continued removes it in 2/16 heads. The lifted rank-4 runs start with an off-plane share of only 0.01-0.14, and the three that solve are exactly the three with the largest initial off-plane share (0.09-0.14 vs 0.01-0.06; exact rank-sum p 0.018, n 8). `escape_start_out.txt`. 3. **Rank-D drift sits on the least-resolved axis**: 57/62 rank-D runs with a drifting head have it at r_beta < 0.5 against 28.7 expected for a random drift direction (P < 5e-5); median r_beta 0.087 over 116 drifting heads. `c1_runlevel_out.txt`. (Also predicted by "gradient pruning after collapse": not discriminating on its own.) 4. Pilot dynamics (T = 128, CPU, n = 3 per arm; `pilot_t128/`): rank 4 parks (off-plane share 0.55-0.81 -> ~1.0 while the visible drift drops 10x); rank 2 removes c through the latent in lockstep with the visible drift, or not at all. The one rank-2 seed that failed started with a near-collinear frame (indep 0.03) and kept its drift on the unresolved axis (r_beta 0.00, loss 0.23); **the same initial weights plus a per-channel clock slot (rank 2 map + a fixed extra latent coordinate) solved (loss 0.006, frame indep 0.44)**. n = 1 matched pair: suggestive only.

Not supported or not tested: the positive feedback in which hiding *creates* the weak axis (P2) was not seen -- the pilot failure's weak axis was there at initialisation, and the readout gradient at initialisation shows no hiding direction (drift there is ~1 cell per move, outside the small-drift regime; `grad_hiding_out.txt`, NEGATIVE). Initial frame conditioning does not predict rank-D outcomes (p 0.19, as `03_formal.md` 1.5); initial drift weakly does (pooled p 0.012 with shared seeds; independent score_rank seeds p 0.073). The measure-zero account (b) fails the 1D test. MapFormer's 5D failure did not reproduce under our recipe at T = 16 (all ranks 0.999-1.000; RoPE 0.95), so it says nothing here. **Decisive test (GPU, ~8 h): rank 2 + clock slot on the T = 1024 torus** (Sec. 6, G1): the account predicts rank-3/4 behaviour; an over-parameterisation account predicts rank-2 behaviour.

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0. What a theory has to explain (ESTABLISHED; files are authoritative)

fact	numbers	file
per-head rank D = 2 fails on the T = 1024 torus	shared r2 0/8, per-head r2 2/8 (fresh seeds 1/6), block-diagonal r4 2/8; score_rank shared r2 3/12	`RANK_SEP_RESULTS.md`, `RANK_MI_RESULTS.md`, `SCORE_RANK_RESULTS.md`
rank D + 1 and D + 2 succeed	per-head r3 6/8, r4 8/8 (shared or per head)	`RANK3_RESULTS.md`, `RANK_SEP_RESULTS.md`
it is per-head rank, not latent size	D - C_bd 8/8 vs 2/8 (Fisher p 0.0070); B and C both 4 latent dims, opposite sides	`RANK_SEP_RESULTS.md`
search, not capacity	rank-2 projection of solved r4: 0.9955 frozen, held under training 7/8	`RANK_PROJ_RESULTS.md`
budget does not fix it	1800 epochs: r2 0/8 vs r4 7/8 (5/8 r2 still descending)	`LOOP_RANK_E1800_P1_RESULTS.md`
rank = D is hard in 2D and 3D, not in 1D	2D G32 r2 1/8 vs r3 8/8; 3D G10 r3 1/8, r4 4/8; 3D G18 r4 8/8; ring rank 1 7/8 vs rank 2 8/8	`RANK_ND_RESULTS.md`, `RANK_WRAP_RESULTS.md`, `RANK_TESTS_RESULTS.md`
failing runs end with a drifting head	"SOLVED iff a clean head" 39/40 (post hoc thresholds, close to circular: CORRECTED block)	`SCORE_RANK_RESULTS.md`
forbidding the drift speeds rank 1 on the ring	R1DF 8/8, median 24 epochs vs R1 58; R1AO (observation steps 0) 52	`RANK_TESTS_RESULTS.md`
escape	stalled r2 lifted to r4: 3/8 in 600 ep vs r2 continued 1/8, fresh r4 step map 2/8 (unmeasured); unsolved lifted runs keep a CLOCK head (5/5), fresh step maps end CLEAN (8/8)	`RANK_TESTS_RESULTS.md`
failure signature	rank 2 keeps short-gap revisits, loses long-gap (0.671) and wrap-only (0.557)	`MAIN_DRAFT1.md` Sec. 6.3, `RANK_MATCHED_RESULTS.md`
MapFormer's own Table 6	inner rank = D: 3D (T 256, grid 16) MapWM 1.00 IID; 5D (T 16, grid 5) 0.75, "on par with RoPE"	`papers/txt/mapformer.txt` l. 2385-2455

Withdrawn accounts that are NOT revived here: "r = 2 loses because its basis is skewed" (skew is a symptom), "the r = 4 gap is training speed", packing geometry (`DXR_RANK_THRESHOLD.md`). The end-state lemma of `docs/theory/2026-10-04/03_formal.md` (Theorem 1, "T1") is used and extended; its causal reading was explicitly left open there (toy negative, 1.9).

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1. Setup and notation (exact for a one-layer MapWM)

Per head h, every token x adds a latent step u(x) = W_in^h e(x) in R^r (r = per-head rank; for a shared latent, the head's read subspace, i.e. the row space of W_out^h). Channel i (i = 1..n_b) has readout row f_i (a row of diag(omega_h) W_out^h) and phase phi_i(t) = , z_t = sum_{s <= t} u(x_s) (`model.py:83,120`). The attention score between a query at t and a key at s is sum_i A_i cos(phi_i(t) - phi_i(s) + psi_i), A_i the content amplitudes.

On a D-torus with actions +-e_j and strictly alternating action / observation tokens, one move adds Delta z = U a + c + eta(o), U_j = (u(+e_j) - u(-e_j)) / 2, c = mean_a u(a) + E_o u(o), U in R^{r x D} the **map frame**, c the **per-move clock** (common to every move), eta the zero-mean observation leak (ignored here; `LEAK_RESULTS.md`). Channel i reads a phase per unit displacement k_i = U^T f_i in R^D and a **drift** tau_i = per move. A revisit at displacement 0 after g moves carries a phase error g tau_i in channel i.

- At r = D with U invertible, c = U beta for a unique **phantom displacement** beta = U^{-1} c (cells per move), and tau_i = in every channel (T1). Write P = sum_i A_i k_i k_i^T (the head's positional precision, D x D) and r_beta = (bhat^T P bhat) / (tr P / D): the head's resolution along its own drift direction, relative to its mean (1 = isotropic or unaligned).
- At r > D, split c = c_par + c_perp with c_par in range(U) and c_perp orthogonal to it (latent metric); c_perp is the part of the clock **outside the map plane**.

Readouts used below (`rank_geom.py`): kappa (visible drift per move in units of one move, wrap-aware, channel-weighted as `analyze_score_rank.head_stats`: verified identical on the stored checkpoints), indep (frame conditioning), beta, r_beta, lam = |c| / |U| in the native latent (the clock whether read or not), off = |c_perp| / |c|, and readgain (how strongly the readout sees the leftover clock relative to the map; ~1 = read like the map, ~0 = invisible).

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2. Theory

2.1 Proposition 1 (two routes to a drift-free channel; THEORY, proved -- linear algebra)

A change of parameters lowers channel i's drift tau_i = only by (G) the **global route**: changing c (shared by every channel of the head; for a shared latent, by every head), or (L) the **local route**: changing the channel's readout f_i by delta with = -tau_i. The change L makes to the channel's map is delta k_i = U^T delta. Among all delta that cancel tau_i: (a) if c_perp != 0 (possible only at r > D), delta = -tau_i c_perp / |c_perp|^2 cancels it with **U^T delta = 0 exactly**: the map is untouched to all orders ("parking": the clock

stays in the latent, off the map plane, and the readout stops seeing it); (b) if c in range(U) (always at r = D), = , so every cancelling delta changes k_i, by at least |tau_i| / |beta|, and the minimum-norm one sets k_i <- k_i - (k_i . bhat) bhat: **it removes the channel's resolution along the drift direction** ("hiding": the channel stops seeing the drift by no longer resolving bhat). (c) at D = 1, bhat-perp = {0}: hiding sets k_i = 0 (the channel dies). More strongly, in 1D the drift a channel sees, measured in cells, is tau_i / k_i = beta for every channel and every readout: **no readout change alters the drift-to-map ratio**. Only the global route (shrinking c, or growing U) reduces the phantom displacement. Proof: (a) take delta along c_perp; U^T c_perp = 0 by definition. (b) c = U beta gives = ; Cauchy-Schwarz; the minimiser is delta k_i parallel to beta. (c) D = 1: k_i = U f_i, tau_i = c f_i, ratio c / U. QED. Per head, because each head's channels read only that head's r-dim image (T1's per-head argument): a block-diagonal rank-4 latent gives each head rank 2 and no c_perp, which is exactly the RANK_SEP split.

2.2 Proposition 2 (at rank D >= 2 hiding is free to first order; at D = 1 it is not; THEORY)

Assumptions (A1) small drift: for the gaps that carry gradient, g |tau_i| << 1, so the drift loss is L_d = (1/2) sum_i a_i tau_i^2 with a_i = gamma A_i E[g^2; g < g_c] > 0 (gamma: curvature of the retrieval loss in the score); (A2) the map loss depends on the frame only through the spectrum of P, L_m = Phi(P) = sum_j f(lambda_j(P)) with f decreasing and convex (more resolution helps, with diminishing returns), plus a norm cost on readout rows (weight decay); (A3) U near-isometric on the readout's span, so rotating k_i preserves |f_i|. Claim. (i) D >= 2, P isotropic (P = lambda I): rotating any channel's k_i by a small angle eps from bhat towards a direction v orthogonal to bhat changes L_m by 0 at first order (dPhi = f'(lambda) tr(dP) and a rotation leaves tr P unchanged), and changes L_d by -eps a_i |k_i|^2 |beta|^2 sin(2 phi_i) / 2 < 0 for every channel not already aligned with or orthogonal to bhat. So the gradient flow on the readout moves resolution **off the drift axis** at once, for any beta != 0; the equilibrium anisotropy solves f'(lambda_perp) - f'(lambda_beta) = (a/2) |beta|^2 (D = 2, a_i = a A_i), growing with the drift. (ii) D = 1: the only readout move that lowers L_d is shrinking |k_i|, whose first-order map cost is -f'(lambda) A_i > 0; it is taken only above the threshold a_i |beta|^2 > -2 f'(lambda) A_i, and even then it does not change beta (Prop. 1c). (iii) r >= D + 1 with c_perp != 0: parking (Prop. 1a) lowers L_d at rate a_i |c_perp|^2 per channel with zero change in L_m at all orders. Derivation: (i) k_i = |k_i|(cos phi bhat + sin phi v); tau_i = |k_i||beta| cos phi; d/deps of (a_i/2) tau_i^2 = -a_i |k_i|^2 |beta|^2 cos phi sin phi. (ii)-(iii) as stated. The step is a statement about the gradient, so it also holds for sign-based (Adam) steps when the gradient component is consistent across steps.

2.3 Proposition 3 (hiding starves the global route: a positive feedback; THEORY)

Under (A1), the gradient of L_d with respect to the phantom displacement is P_a beta (P_a = sum_i a_i k_i k_i^T), so the global route removes the drift along bhat at a rate proportional to the head's own (drift-weighted) resolution along bhat, bhat^T P_a bhat. Hiding (Prop. 2) lowers exactly that quantity. Hence at rank D >= 2: drift -> hiding (resolution leaves bhat) -> slower global removal -> drift persists -> more hiding. The clean state (beta = 0, P isotropic) remains a stable fixed point (it is the projection that RANK_PROJ installs and training holds), but a run that starts with, or develops, a drift

reaches it only after a plateau. In a reduced model (f(lambda) = k0 / lambda; quasi-static hiding) the hidden equilibrium is lambda_beta ~ sqrt(2 k0 / a) / |beta| and the drift then shrinks only linearly, d|beta|/dt = -eta_G sqrt(2 k0 a): a plateau whose length grows with the initial drift and shrinks with the latent learning rate -- slow, not absorbing (SPECULATION as to the exact law; the qualitative statement follows from Props. 1-2). Two refinements. (R1) Long gaps carry the loss but not the gradient: once g |tau_i| > 1 the cosine term averages out (Riemann-Lebesgue over the gap distribution), so drift removal is driven by short gaps while the failure shows on long gaps and wrap-only revisits (which are long-gap by construction, T5 of `03_formal.md`). At T = 128 the hidden drift is nearly harmless: rank-2 heads drift or collapse on 32/32 while fitting the training task (`03_formal.md` 1.4); at T = 1024 it is the failure. (R2) With hiding the head stays near the "COLLAPSE" corner (T1's class B) when hiding wins completely, and in the "CLOCK" class (A) when it is partial: both are points on one curve, not two mechanisms.

2.4 Corollaries and predictions

P1 (rank D >= 2). Drifting heads have their drift along their least-resolved direction: r_beta << 1. P2 (rank D >= 2, dynamics). The anisotropy builds up along an early, persistent drift direction (rather than the drift turning onto an axis that was already weak). P3 (rank >= D + 1, end state). In solved heads the leftover latent clock lies outside the map plane (off -> 1, above the random-direction value sqrt((r - D) / r)) and the readout is nearly blind to it (readgain << 1): parked, not removed. Its size then decays only through weight decay, so it should shrink with training length. P4 (rank >= D + 1, dynamics). Parking precedes removal: the visible drift kappa falls well before the latent clock lam. P5 (rank >= D + 1 failures). Unsolved D + 1 heads have their clock in the map plane (off small) and read (readgain ~1): the free route existed in principle and was not open. P6 (1D). Rank D = 1 is not trapped (no direction to hide along); its failures, if any, show a large drift that the global route is still removing, and forbidding the drift only speeds it up. P7 (clock slot). Rank D plus a per-channel learned clock rate (a fixed extra latent coordinate that every token steps by 1; the map stays rank D) opens Prop. 1(a) without adding any map freedom. Prediction: it behaves like rank D + 1 (drift-free, well-conditioned frames), and on the T = 1024 torus solves like rank 3-4. An over-parameterisation (Burer-Monteiro) account, in which the extra rank helps by adding factorisation freedom to the map, predicts no rescue: this arm discriminates. P8 (escape). A stalled rank-2 model lifted to rank 4 can park only if its clock acquires an off-plane part; the lift adds new W_in rows at the default scale beside large trained rows, so the clock starts almost entirely in-plane. Lifted runs should solve in proportion to their initial off-plane share. P9 (D + k). P(clean) should rise with the expected off-plane share sqrt((r - D) / r) of a random clock: 0.71 (r 4, D 2) > 0.58 (r 3, D 2) > 0.50 (r 4, D 3); rank 5 in 3D should beat rank 4 on the wrap-heavy grid.

2.5 The alternatives (and where each stands)

- (T1 alone, end-state lemma) "a rank-D head is clean iff c = 0": correct but static; it does not say why training leaves c != 0 at rank D and removes it at D = 1. Wrap note: exactly, a rank-D head is drift-free iff beta in N Z^D (a whole number of laps per move), which is irrelevant in practice.
- (b) measure-zero / symmetry: the clean set at rank D is {c = 0} (codimension r in c), at rank D + 1 it also contains {F orthogonal to c, c outside range U}, which is

reachable through W_out alone. This is the static shadow of Prop. 1(a). On its own it fails the 1D test: at rank 1 the clean set is also {c = 0}, yet rank 1 solves 7/8.

- Burer-Monteiro / over-parameterisation (red team, `00_PLAN.md`): extra rank smooths the landscape of the map factorisation. Distinguished by P3 (what the extra direction holds) and decisively by P7 (clock slot).
- "Rank D is hard because of the exactly periodic code" (SCORE_RANK secondary, CORRECTED as not separated from long gaps): in this theory wrap-only revisits fail because they are long-gap revisits read through a hidden drift (R1); RANK_NOWRAP would separate them.

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3. Checks

All scripts: `docs/paper/expansion/scripts/rank/`, run from `/home/prashr` with `PYTHONPATH=/home/prashr`. Geometry readout `rank_geom.py` (verified: kappa and indep equal `analyze_score_rank.head_stats` to 4 decimals on rank_mi Vanilla s0, r4mi s0, r2ph s1, r4mibd s2). Final-checkpoint sweep `sweep_final.py` -> `sweep_final_out.txt` / `sweep_final.json` (25 sets, every stored matched-length rank checkpoint plus the escape arms and the T = 128 rank_sweep); summaries `summarise_final.py` -> `summarise_final_out.txt`. Note: `runs/rank_wrap/N10/D3` and `runs/rank_nd/D3` r4 (and N32/D2 vs rank_nd D2 r3) are bit-identical reruns (`RANK_WRAP_RESULTS.md`), counted once. SOLVED is the registered rule (final-5% training loss < 0.05, `stats_core.classify_run`).

C1. P1: drifting rank-D heads drift along their least-resolved axis -- PASS (ANALYSIS; not discriminating)

set (rank D, D >= 2)	drifting heads	median r_beta	share < 0.5
torus G64 shared r2 (rank_mi)	16	0.060	0.88
torus G64 per-head r2	9	0.194	0.89
torus G64 block-diagonal r4	8	0.106	0.75
torus G64 shared r2 (score_rank)	20	0.195	0.80
torus G64 shared r2, 1800 ep	12	0.214	0.67
ND D2 G32 r2	15	0.005	1.00
ND D3 G10 r3	14	0.088	0.86
escape C (r2 continued)	14	0.087	1.00
T = 128 rank_sweep r2	8	0.096	0.75
Pooled 116 heads: median 0.087, 85% below 0.5 (random direction given each head's own frame: 29%). Run level			
(`c1_runlevel.py`): 57/62 runs vs 28.7 expected, P < 5e-5 (20,000 simulations). Clean heads of rank > D sit at			
r_beta ~1 (0.93-1.05 medians). Caveat: this is what Prop. 3 predicts, but also what pruning of the visible drift after			
a collapse of any cause predicts (`03_formal.md` 1.3 CAUTION); it fixes the geometry, not the order of events.			

C2. P3: solved rank > D heads park the clock -- PASS (ANALYSIS)

set (solved heads)	r - D	leftover lam = abs(c)/abs(U)	off-plane share (random)	readgain
torus G64 per-head r3	1	0.004	0.978 (0.58)	0.15
torus G64 shared r4 (rank_mi)	2	0.004	0.996 (0.71)	0.11
torus G64 per-head r4	2	0.003	0.988 (0.71)	0.16
torus G64 shared r4 (score_rank)	2	0.003	0.998 (0.71)	0.09
ND D2 G32 r3	1	0.003	0.987 (0.58)	0.09
ND D3 G10 r4	1	0.008	0.964 (0.50)	0.23
ND D3 G18 r4	1	0.001	0.810 (0.50)	0.57
ring R2 (1D, rank 2), 300 ep	1	0.025	0.989 (0.71)	0.21
T = 128 rank_sweep r4, 300 ep	2	0.021	0.988 (0.71)	0.27
escape F (fresh r4 step map, 600 ep)	2	0.003-0.004	0.98 (0.71)	0.18
For comparison, solved rank-D heads: lam 0.000-0.001 (ring R1, torus r2 solved): there the clock is removed, because it				
cannot be parked. The parked remainder is ~6x larger after 300 epochs than after 600-900 (0.021-0.025 vs 0.003-0.004),				
of the order AdamW's decoupled decay alone gives an unread direction over a cosine schedule (factor exp(-wd * sum lr):				
0.44 at 300 epochs vs 0.09 at 900, ratio ~5). Consistent, not a test (different tasks).				

C3. P5: unsolved rank > D heads hold the clock in the map plane and read it -- PARTIAL PASS (ANALYSIS)

Torus per-head r3, 4 unsolved heads: lam 1.50, off-plane share 0.27 (random 0.58), readgain 0.61. Escape E unsolved, 10 heads: off-plane 0.08 (random 0.71), readgain 0.76. ND D3 G10 r4, 8 unsolved heads: off-plane 0.53 (random 0.50), readgain 0.78 -- here the clock is not especially in-plane, but it is read: the free route existed and was not taken (as `03_formal.md` 1.3 found, |c_perp| / |c| 0.16 by its metric). So "in-plane" holds in 2 of 3 sets, "read" in 3 of 3.

C4. P6: the 1D exemption -- consistent (ANALYSIS + ESTABLISHED)

In 1D r_beta = 1 identically and every channel sees the same phantom drift beta (Prop. 1c). Ring R1 (`RANK_TESTS`): 14 clean heads end at lam <= 0.001 (the clock removed, not parked); the 3 drifting heads (s20 h0, s22 h0, s26 h1) have the largest map steps |U| (5.0, 2.8, 3.25 vs 1.6-2.6 for clean heads): the only other global way to shrink beta = c / U. R1DF (drift forbidden) solves in 24 epochs vs 58 (ESTABLISHED), as Prop. 1c says the drift has to be removed through the latent in 1D. Not a test of the mechanism: the ring task also differs (T = 128) -- see G5.

C5. P8: the escape test re-read -- PASS (ANALYSIS, post hoc; `escape_start.py` -> `escape_start_out.txt`)

arm (600 ep, same 8 trained networks)	runs with a drift-free head at the end	heads drift-free	off-plane share at end
F: fresh rank-4 step map (start kappa 0.28-5.86)	8/8	16/16	0.91-1.00 (all)
E: stalled rank 2 lifted to rank 4	3/8	6/16	1.00 in the 3 solved; 0.01-0.31 in four unsolved runs, 0.66 in s4 (unsolved)
C: rank 2 continued	1/8	2/16	-- (rank 2)
F vs E on runs: Fisher p 0.026. Within E, the full-latent off-plane share of the clock at the START (the lift adds two			
W_in rows at default scale next to large trained rows): solved s1 0.09, s2 0.14, s3 0.11; unsolved s0 0.03, s4 0.06,			
s5 0.01, s6 0.01, s7 0.06 -- the three largest are the three solved (exact one-sided rank-sum p 0.018). Confound: the			
three solved sources were "descending" at 900 epochs (so was s4, unsolved). F's solves are only 2/8 (`RANK_TESTS`):			
parking removes the drift in every F run; learning the rest of the map within 600 epochs is a separate matter.			

C6. P2, P4, P7 dynamics -- PARTIAL (ANALYSIS, CPU pilot; `train_snap.py`, `analyze_snap.py`, `pilot_t128/`)

Paper torus T = 128 (grid 64), batch 32 (a quarter of RANK_SWEEP's), 40 epochs cosine, lr 1e-3, `--sdpa`; seeds 3000-3002 (outside every batch); arms shared r2, shared r4 (r2's base), R2Clock (= shared r2 at init plus a per-channel learned clock rate: a fixed extra latent coordinate every token steps by 1; `rank_geom.MapFormerWM_r2Clock`). Snapshots every 49 steps. | arm | seed | final loss | acc T=128 / 512 (eval mode) | heads: kappa, frame indep (end) | latent clock | |---|---|---|---|---|---| | r2 | 3000 | 0.005 | 1.000 / 0.937 | 0.004, 0.76 / 0.004, 0.67 | removed in lockstep with kappa (lam 0.63 -> 0.004) | | r2 | 3001 | **0.230** | 0.922 / 0.836 | 0.043, **0.00** / 0.039, **0.00** | kept, on the unresolved axis (r_beta 0.00) | | r2 | 3002 | 0.003 | 1.000 / 0.981 | 0.002, 0.68 / 0.001, 0.50 | removed in lockstep | | r4 | 3000 | 0.005 | 1.000 / 0.954 | 0.003, 0.38 / 0.004, 0.51 | parked: off-plane 0.62 -> 0.96 | | r4 | 3001 | 0.028 | 0.990 / 0.959 | 0.004, 0.57 / 0.005, 0.91 | parked: 0.55 -> 0.98 | | r4 | 3002 | 0.005 | 1.000 / 0.946 | 0.032, 0.48 / 0.037, 0.78 | parked: 0.81 -> 1.00, lam 0.09 vs kappa 0.03 | | R2Clock | 3000 | 0.003 | 1.000 / 0.949 | 0.020, 0.71 / 0.018, 0.48 | in the slot (off-plane 1.00) | | R2Clock | 3001 | **0.006** | **1.000** / 0.941 | 0.045, **0.44** / 0.049, **0.36** | in the slot | | R2Clock | 3002 | 0.004 | 1.000 / 0.959 | 0.025, 0.58 / 0.023, 0.53 | in the slot | - P4 (parking first): the visible drift fell 10x before the latent clock did in 4/6 rank-4 heads (the other 2 at the same snapshot) and 4/6 R2Clock heads; in rank 2 the two fall together (lam ~ kappa throughout). PASS (n = 3). - P2 (feedback creates the weak

axis): NOT SHOWN. The failing r2 seed 3001 started with a near-collinear frame (indep 0.03-0.04 at step 0) and its drift stayed on that axis (|cos(beta@200, beta_end)| 1.00); the anisotropy was inherited, not created. What it does show is Prop. 3's static half: drift on an unresolved axis is not removed. - P7 (clock slot): at the SAME initial weights as the failing r2 seed, R2Clock solved (loss 0.006 vs 0.230, held-out 1.000 vs 0.922 at T = 128) and repaired its frame (indep 0.03 -> 0.44 / 0.36). One matched pair: suggestive only. R2Clock does not drive its visible drift below 0.01 at T = 128 (0.02-0.05), where a drift of that size is harmless. - Budget and batch differ from RANK_SWEEP (40 vs 300 epochs, batch 32 vs 128): a dynamics testbed, not a reproduction.

C7. Initial conditions at T = 1024 -- MIXED (ANALYSIS; `init_frame.py` -> `init_frame_out.txt`)

Rebuilt initial weights (train_variant's seeding) for 36 rank-D runs and 8 rank-3 runs. Initial frame conditioning does not separate solved from unsolved rank-D runs (best head indep: solved 0.17-0.84 vs unsolved median 0.35; one-sided p 0.19), agreeing with `03_formal.md` 1.5. Initial drift (min over heads of kappa) is lower in the runs that solve: solved 0.28-1.25 (5 of 8 below 0.5) vs unsolved median 1.11 (n 28), p 0.012 pooled (seeds 0-7 share bases across sets: not independent); score_rank alone (independent seeds 10-21) 3 vs 9, p 0.073. At rank 3 there is no such relation (solved runs start at 0.28-1.45). Consistent with Prop. 3 (the rank-D plateau grows with the initial drift; rank D + 1 parks whatever the start); weak and post hoc.

C8. Prop. 2 at initialisation (readout gradient direction) -- NEGATIVE / uninformative (`grad_hiding.py`)

On the real task (T = 1024, 2 batches of 16, seeds 0-3, shared r2 and r4), a small readout step along -grad or -sign(grad) moves frame resolution toward or away from the drift axis with no consistent sign (hide index from -1.00 to +1.00 in both ranks) and changes kappa by < 0.4%. At initialisation the drift is 0.4-2.7 cells per move, so g |tau| >> 1 for almost every gap: assumption A1 (small drift) does not hold and the drift gradient is incoherent (R1). Prop. 2 is a statement about the small-drift phase (after the first ~200 steps in the pilot, where kappa has fallen to 0.01-0.2); it was not tested there.

C9. MapFormer's 5D failure under our recipe -- UNINFORMATIVE for rank (`train_nd_cpu.py`, `pilot_5d/`)

5D torus, grid 5, T = 16 (the paper's Table 6 setting; our directed walk and recipe, 100 epochs, batch 128, seeds 4000-4002): per-head rank 5 (= D), 6, and 5 + clock slot all SOLVED 3/3 (final loss 0.005-0.012; held-out 0.999-1.000 at T = 16, 0.996-0.998 at T = 32); index RoPE STALLED (0.95 / 0.84-0.88). The paper's 0.75 for MapWM does not reproduce here (recipe and walk differ; the paper's 5D recipe is not stated). Every MapWM head in every arm ends drifting (kappa 0.2-0.9) while solving: at T = 16 a drift is harmless (R1), and nothing removes it at any rank. So "SOLVED iff a clean head" is a long-walk statement, and the 5D short-walk task cannot test this theory.

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4. Which theory survives

candidate	what it says	verdict
(a) clock / free-route account (Props. 1-3), as stated in the Summary	rank > D lets each channel cancel the clock off the map plane; at rank D the clock can only leave through the latent, at a rate set by the resolution along the drift	**SURVIVES** every end-state check (C1, C2, C3 partly, C5) and the pilot dynamics' parking (C6 P4); one matched clock-slot rescue (C6 P7). THEORY, not established: no registered test
(a') its feedback half (hiding creates the weak axis, Prop. 2)	the readout rotates resolution off the drift axis because that is free to first order	**NOT SHOWN**: C6's failure inherited its weak axis; C8 uninformative (drift too large at init). Keep as a derivation with its assumptions, not as a finding
T1 end-state lemma (`03_formal.md`)	clean at rank D iff c = 0	true, kept as Lemma 0; explains nothing about training by itself
(b) measure-zero / symmetry of the rank-D solution set	the rank-D clean set is a thin target	**FAILS** the 1D test (rank 1's clean set is the same codimension, yet 7/8 solve) and does not predict C1, C2 or C5
(c) 1D exemption by "a scalar clock separates by sign"	--	not needed: Prop. 1(c) gives the exemption (no hiding direction; the readout cannot change the drift in cells)
Burer-Monteiro / over-parameterisation of the map factorisation	spare rank smooths the map's landscape	**NOT RULED OUT**, but C2 shows the spare direction ends up holding the clock, not the map (the map is rank D: RANK_PROJ's top-2 projection keeps 0.9955), and the n = 1 clock-slot pair favours (a). G1 decides
"the rank defect is the periodic (wrap) code" (SCORE_RANK secondary)	--	subsumed as a consequence (R1: wrap-only revisits are long-gap revisits read through a drift); not separated -- RANK_NOWRAP would
"skew is the mechanism" (WITHDRAWN), "training speed" (WITHDRAWN), packing (WITHDRAWN)	--	not revived. In this account skew (collinear axes, indep -> 0) is where a drift along an unresolved axis leaves a rank-D head: a symptom, as the withdrawal says

Status to carry into the paper: the account is a THEORY with ANALYSIS support on stored checkpoints (post hoc) and a CPU pilot (n = 3); the one causal prediction that would make it a finding (G1) is untested.

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5. Proposed paper text

5.1 Addition to Sec. 6.1 (after "Rank against the map's dimension"), ~260 words

Why the extra rank helps (theory and analysis). Every move adds to the latent a step common to all moves, a clock c, on top of the displacement U a. A channel stops seeing

the clock either through the latent (c shrinks, one coordinate shared by the head) or through its own readout. When the per-head rank exceeds D, part of c lies outside the plane the map occupies, and the readout can turn away from that part without changing the map; at rank D, c = U beta lies inside the map plane, so a channel can stop seeing it only by stopping to resolve the drift direction, and the latent removes it at a rate proportional to the head's resolution along that direction (App. M, Props. 1-3). In 1D there is no direction to turn into, so only the latent route exists, and rank 1 learns the ring (7/8). The stored checkpoints show the predicted signatures (analysis, post hoc). Solved heads of rank above D keep a small leftover clock almost entirely outside the map plane (off-plane share 0.96-1.00 in nine of ten sets, against 0.50-0.71 for a random direction) and read it at a tenth to a quarter of the map's gain; failing heads hold it in the plane. Drifting rank-D heads drift along their least-resolved axis (57 of 62 runs, against 29 expected). A fresh rank-4 step map on the stalled rank-2 networks removes the drift in 16 of 16 heads, each time by parking it off the plane, and of the stalled runs lifted to rank 4 the three that solve are the three whose clock started with the largest off-plane share (rank-sum p 0.018). Whether the extra rank helps by giving the clock somewhere to go, rather than by adding freedom to the map, is decided by a rank-2 model with a fixed clock coordinate (App. M; untested at T = 1024).

5.2 Appendix M (new): the clock and the rank

M1 Setup (Sec. 1 of this note, verbatim). M2 Lemma 0 (T1, with the wrap remark). M3 Proposition 1 with proof. M4 Proposition 2 with assumptions A1-A3 and the derivation; state plainly that it was not observed at initialisation (C8) and that the pilot failure inherited its weak axis (C6). M5 Proposition 3 and refinements R1-R2; the reduced plateau law marked as a cartoon. M6 Checks C1-C5 as tables (C1, C2, C5 tables above); C6 table with the caveat "CPU pilot, n = 3, quarter batch, 40 epochs"; C7 and C9 as one paragraph each. M7 Predictions P1-P9 with which were checked. Figure M1 (suggested, from `pilot_t128/*.json`): kappa and off-plane share against step for rank 2, rank 4 and R2Clock (one panel each, three seeds), showing lockstep removal at rank 2 and parking at rank 4.

5.3 Wording constraints

- Say "parks the clock outside the map plane" only with the off-plane share and its random baseline beside it.
- Do not say "rank fails because of drift" (causal); say "failing rank-D heads drift along their least-resolved axis; rank above D gives the drift a map-free exit" until G1 is run.
- Keep the per-head caveat: our shared latent vs the paper's per-head latent (CLAUDE.md invariants).

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6. GPU experiments that would decide it

Throughput from SCORE_RANK / RANK_TESTS: ~2,700 run-epochs per hour on the two 4090s at the T = 1024 recipe (RANK_MI: batch 16, lr 1e-3, cosine, 900 epochs, 1 layer, 2 heads, d 128). Each would need a pre-registration, a pilot on outside seeds and an independent code audit (rules 7, 29).

#	experiment	arms x seeds x epochs	run-epochs	wall	what decides it
G1	**Clock slot at T = 1024 (decisive).** Shared r2 (`Vanilla`), r2 + clock slot (`MapFormerWM_r2Clock`: identical to Vanilla at init, +64 parameters), shared r4 (`Vanilla_r4mi`), fresh seeds; snapshots every 10 epochs (geometry only)	3 x 8 x 900	21,600	~8 h	Primary SOLVED, slot vs r2 (Fisher). Account predicts slot >= 6/8 (like r3/r4); over-parameterisation predicts <= 2/8. Branches as RANK_TESTS. The snapshots test P2/P4 at full scale for free
G1+	add r2 with drift forbidden (odd action steps, zero observation steps) and its ActOnly control (observation steps 0, actions free)	+2 x 8 x 900	+14,400	+5.3 h	separates the clock from the observation leak (the R1DF / R1AO logic of RANK_TESTS)
G2	**Seeded escape.** The 8 stalled rank-2 RANK_MI runs lifted to rank 4 with new W_in rows set (still function-preserving, W_out new columns 0) so that the clock's off-plane share is ~0.7, vs the default lift (share 0.01-0.14)	2 x 8 x 600	9,600	~3.6 h	account predicts seeded >= 6/8 vs default ~3/8 (C5)
G3	rank D + 2 in 3D on the wrap-heavy grid (per-head r5, grid 10, vs r4 in the same batch)	2 x 8 x 900	14,400	~5.3 h	P9: expected off-plane share 0.63 vs 0.50 predicts r5 > r4's 4/8; a flat result says the 3D-G10 shortfall is the grid (memorisation / lattice), not the clock
G4	1D at long walks: ring rank 1 vs rank 2 at T = 1024 (ring 32 cells)	2 x 8 x 300	4,800	~2 h	Prop. 1(c) predicts rank 1 still learns the ring at long T (no hiding direction); if rank 1 then fails like rank 2 in 2D, the 1D exemption was the short task (RANK_TESTS caveat) and the account loses its D-dependence

Minimum to settle the account: G1 (8 h). With G2 (3.6 h) the mechanism and the escape reading are both registered.

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7. Reproduce (all CPU; `cd /home/prashr; export PYTHONPATH=/home/prashr`; D = `mapformer/docs/paper/expansion/scripts/rank`)

- `python3 $D/sweep_final.py` (~3 min) -> `sweep_final_out.txt`, `sweep_final.json`; `python3 $D/summarise_final.py`; `cd $D && python3 c1_runlevel.py` (C1-C4).
- `python3 $D/escape_start.py` (C5); `python3 $D/init_frame.py` (C7); `python3 $D/grad_hiding.py` (C8, ~10 min).
- C6: `python3 $D/train_snap.py --arm {Vanilla,Vanilla_r4mi,R2Clock} --seed {3000,3001,3002} --epochs 40 --batch-size 32 --sdpa --threads 2 --out OUT` (~85 min each on a loaded 32-core host), then `python3 $D/analyze_snap.py OUT`; stored outputs `pilot_t128/`.

- C9: `python3 $D/train_nd_cpu.py --arm {PH5,PH6,PH5C,RoPE} --seed {4000,4001,4002} --epochs 100 --out OUT` (~55 min each); stored outputs `pilot_5d/`.
- Pilot seeds (3000-3002, 4000-4002) lie outside every registered batch (rule 29). Checkpoints of the pilots were kept in the session scratchpad only (not in `runs/`).

Search theory for the two failure axes (sign and rank)

*2026-10-20. Expansion note for the main paper (`docs/paper/MAIN_DRAFT1.md` Sec. 6, 8, 9; theory appendix `docs/paper/APPENDIX_theory_v3.md`). CPU only; nothing here is registered and nothing changes a registered verdict. Every statement carries one label: ESTABLISHED (a registered result, cited), ANALYSIS (a post hoc or exploratory computation on stored checkpoints or a CPU run, script named), THEORY (proved here or in the appendix; proof given or pointed to), SPECULATION (an account or prediction not yet tested). Scripts and outputs: `docs/paper/expansion/scripts/sign/` (every `*.py` has a `*_out.txt` beside it). Notation follows the appendix: channel c = (head, block), angle a_c(x) = omega_c * step_c(x), step = W_out W_in e (pre-|.|), latent z_x = W_in e_x, readout row w_c.*

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0. Summary

The unifying statement (THEORY for the static part, ANALYSIS for the end states, SPECULATION for the dynamics). Both failure axes are failures to remove the *common mode* of the move code -- the part of a move's angle that an opposite move does not undo, i.e. a clock riding on the map. Write, for an inverse pair (x, y) in channel c, the clock k_c = a_c(x) + a_c(y) (the phase a back-and-forth adds at zero displacement) and the map m_c = a_c(y) (the phase per unit displacement). A map code needs k_c = 0 mod 2 pi in every channel attention reads. There are two ways to get there from a generic small start:

- **R1, the collective route**: a linear subspace of the move latents on which every channel's clock vanishes at once and the map part is untouched (u = z_x + z_y = 0). Along it every channel's clock shrinks in proportion, and in a quadratic surrogate the latent gradient flow converges to it from every start (Proposition S2).
- **R2, per-channel decoupling**: in each channel separately, a readout row with zero clock and full map rank, reachable without crossing a fold or losing a map dimension.

class	**R1**	**R2 (per channel; its success depends on how many channels start near their target)**	**ring (1D)**	**torus (2D)**
signed, per-head rank >= D + 1	yes (S2)	yes (S6)	8/8 (every batch)	8/8 (rank 3: 6/8)
signed, per-head rank = D	yes (S2)	**no**: a clock-free channel loses a map dimension (S6)	rank 1: 7/8	rank 2: 0-2/8; 3D rank 3: 1/8
signed, rank 4, move latents frozen (ACCOUNTS T2)	**no** (frozen)	yes, through zero per channel	default start (half the channels start opposite-signed): 7/8; |.|	not run

class	R1	R2 (per channel; its success depends on how many channels start near their target)	ring (1D)	torus (2D)
			start (none do): 1/8	
|.|, rank 4, small start	**no**: the latent route ends at a clock; no latent choice wraps > 2r channels at a generic readout (S3)	only to the wrap sheets: an unwrapped |.| channel has clock >= map (S1), its zero-clock points lie at |k| = 2 pi n, across |k| = pi (P3); every channel starts below pi	0-2/8 (4/8 at lr x3)	0-2/8
|.|, start past pi (MECHANISM, TORUS_START)	no	per-channel wrap targets near the start	8/8	0/8 (signed from the same start: 2/8)

Proved: the R1/R2 inventory of each class (S1, S2, S3, S6). R1 does not depend on the start (S2(c)); R2 does, and the measured counts follow that: R2 alone succeeds when many channels start near their per-channel target (signed frozen latents from the default start 7/8; |.| past pi on the ring 8/8) and fails when none do (1/8; 0/8). Measured (ANALYSIS, `sign_clock.py`, `rank_clock.py`): every failing arm, on both axes, ends with a usage-weighted clock as large as its map (clock/map 1.1-1.9 for |.| from small starts, 1.39 for frozen-latent signed, 0.5-1.9 for unsolved rank-2 heads), and every success from a small start ends clock-free (0.001-0.065), by R1 (latent clock 0.045-0.065 of the map at rank 3/4; u = z_x + z_y at 0.044-0.053 of the latent norm for signed ring models). Theory (S5): any code with clock rate k serves revisits with gap g < pi / k and loses the rest, which is the single failure signature of Sec. 6.3. **What the inventory alone does not explain**: why R1 alone suffices on the ring (rank 1: 7/8) but not on the 2D torus (rank 2: 0-2/8). The parallel rank note (`RANK_THEORY.md`, Props. 2-3) supplies it: at rank D >= 2 a free local move ("hiding" the drift by turning channels' resolution off its direction) starves the global route; in 1D there is no such move. So, in one frame (S7): **sign removes the global route; rank D >= 2 keeps it but lets hiding starve it; rank >= D + 1 adds a local move (parking) that costs no map and does not starve it.**

The sign axis in particular. 1. *Why* |.| *fails from small starts* (THEORY + ANALYSIS). An unwrapped |.| channel satisfies |k_c| >= |m_c| and |k_c| >= 2 |delta_c| (the cone; S1): it is at least as much clock as map, and its only zero-clock non-blind points are the wrap sheets k_c = 2 pi n. The latent move that cancels every signed channel (R1) produces a(y) = a(x) under |.|, a clock (T1 / Remark R), and at a generic readout no latent choice wraps more than 2r of the C channels (S3, proved by an omega-scaling argument: the wrap targets in step units, 2 pi n / omega_c, differ across the omega ladder). Measured: 34% (unweighted) / 56% (usage-weighted) of the change |.| needs at initialisation lies in the latent-reachable subspace, against exactly 100% for signed (`latent_reach.py`). Along straight paths with the content of a solved model held fixed, the |.| path to the wrapped solution rises above its running minimum by 0.6-1.9 nats in every seed, the signed path by 0.04-0.48 (complete separation, 8 vs 8); the signed collective latent route alone takes accuracy from 0.30 to 0.72, the same route under |.| leaves it at 0.30 (`path_profiles.py`). 2. *Why the wall* (THEORY + ANALYSIS). In the quadratic surrogate, the |.| latent flow that

removes the clock is a contraction to the blind origin (S4, the +1/-1 paragraph); at fixed clock the largest map a |.| channel can carry sits on the cone's boundary, where one move's step is zero (a "one-sided counter"). Trained |.| ring models put 0.30-0.36 of their sub-pi channels within 10% of that boundary (chance at initialisation 0.056) and end with 19-22 one-sided and 14-16 blind channels of 64 (signed: 2-4 and 8). The reanalysis's MIXED wall verdict is consistent with this: collapse is the boundary optimum of a clock-map trade-off, not only a sticky kink. 3. *Why* |.| "*fails*" *on +1/+2 although no zero crossing is needed* (ANALYSIS, `latent_geometry.py`, `lattice_p.py`). It does not fail in the step code. With moves +1/+2 the |.| class keeps a collective route (z_2 = +-2 z_1; two mirror copies, S4) and the |.| models find it: the relation a(+2) = 2 a(+1) holds in >= 0.98 of the used channels in 8/8 |.| runs (signed 7/8 at >= 0.99), the latent relation in 5/8 |.| runs (one through the mirror copy z_2 = -2 z_1, cos -0.98) against 7/8 signed. The runs that stay unsolved (|.| 6/8, signed 3/8) have the relation in place; what they miss is not visible in the step code. So GAUGE G1's UNMEASURED cell is not evidence of a sign cost without an inverse pair, and nothing here points to one; the +1/+2 task is harder for both classes for a reason outside the step map. 4. "*Exists, held, not found*" *as a statement about the landscape* (THEORY definitional + ANALYSIS, S8, Sec. 3.4). The three facts together say the obstruction is where the initialisation distribution sits relative to the basin of the solution set. With the content of a solved model frozen and only the step map trained (`step_basin.py`, 2000 steps), the start decides: signed from the default start reaches 0.83 accuracy (3/8 >= 0.9), while from the same-direction start signed (0.47), frozen-latent signed (0.40) and |.| (0.36) all stay near the start (0.29), and |.| from past pi reaches 0.64 (p 0.0007 against its small start); started halfway along the straight segment to the exact |.| solution, the |.| step map reaches 0.61 and returns to the solution in 0/8 runs (the solution itself is held). The signed escape from the same-direction start that full training shows (7/8) is NOT reproduced by the step map alone: the collective route needs the content to co-adapt, or more steps.

What does not survive. "The |.| failure needs an inverse pair" (no: the +1/+2 shortfall is not in the step code, and G1 is unmeasured); "R1 is the whole story for rank" (no: rank 2 has R1 and fails on the torus); the strong form of the requested unification ("the failing class lacks a linear route to cancellation") holds for sign and for frozen latents but is FALSE for rank D, which has the linear route.

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1. Setting

One layer, observation-step-free models (only the moves step) unless stated. Ring: moves +1 / -1 (inverse pair) or +1 / +2 (GAUGE). Torus: four moves, two inverse pairs. Per channel c and inverse pair (x, y):

- clock k_c = a_c(x) + a_c(y), map m_c = a_c(y), delta_c = (a_c(y) - a_c(x)) / 2.
- Over an interval with n_x moves x and n_y moves y, displacement d = n_y - n_x, the phase is phi_c = m_c d + k_c n_x (exact: a_y n_y + a_x n_x = a_y (d + n_x) + a_x n_x).
- A *map code* has k_c = 0 mod 2 pi in every used channel; a *cancelling* channel has k_c = 0 mod 2 pi with m_c != 0 mod 2 pi; a *blind* channel has m_c = 0 = a_c(x) mod 2 pi.

- Latents z_x = W_in e_x in R^r; signed step w_c . z_x; |.| step |w_c . z_x|. The relation latent u = z_x + z_y.
- Usage (attention's reading of a channel) = mean |q_c| over move tokens x mean |k_c| over observation tokens at layer 0, exact per token because layer 0's input is the embedding (`model_wrap_exist.usage_weights` up to token frequencies).

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2. Theory statements

S1. The clock floor of the |.| class (THEORY, proved)

Proposition S1. In the |.| class, for an inverse pair in channel c (a_c(x) = omega_c |w_c . z_x|, a_c(y) = omega_c |w_c . z_y|, one omega_c of either sign): (a) a_c(x) and a_c(y) have the same sign, so |k_c| = |a_c(x)| + |a_c(y)| >= max(|m_c|, 2 |delta_c|), with |k_c| = |m_c| iff a_c(x) = 0 and |k_c| = 2 |delta_c| iff one of the two angles is 0. (b) As real numbers, k_c = 0 iff the channel is blind. So the non-blind zero-clock points of the channel are exactly the wrap sheets |k_c| in 2 pi Z_{>=1}. (c) On a zero-displacement revisit with n_x = g / 2 backward moves (g the number of moves in the interval), the channel's phase is k_c g / 2, with |k_c| g / 2 >= |m_c| g / 2. If |k_c| < 2 pi (unwrapped), the phase is within eps of 0 mod 2 pi only for g <= 2 eps / |k_c| <= 2 eps / |m_c| (or g near a multiple of 4 pi / |k_c|).

Proof. (a) same sign of omega_c times non-negative numbers; triangle equality cases. (b) from (a). (c) phi = m d + k n_x with d = 0 and n_x = g / 2. []

Reading. An unwrapped |.| channel is at least as much clock as map, and the better it resolves neighbouring cells (|m_c| large), the shorter the backtracks it can read. The signed class has k_c = 0 with any m_c (through zero). The appendix's P3 (the cut circle) is the geometric form of (b): the near-side cancelling point is the blind origin, the non-blind ones lie at |k| = 2 pi, across |k| = pi.

Check (ANALYSIS, `sign_clock.py`). Usage-weighted clock/map ratio rms(wrap k) / rms(wrap m) at the end of training: |.| from small starts 1.10-1.68 per arm mean (ring MECH AO 1.68, AO_absmag 1.63, GAUGE AO_m 1.55; torus AO4 1.22), signed 0.001-0.018 (SO, SO_absmag, SO_m, SO4). The inequality itself holds with zero violations on raw angles (it is an identity: 0 violations in 4,059 sub-pi |.| channels over all arms; a first version of the script compared signed sums and counted channels with omega < 0 as violations, corrected before this run).

S2. The collective route of the signed class (THEORY, proved)

Proposition S2. Signed class, inverse pair, readout W in R^{C x r} of full column rank, omega_c != 0. (a) The clock vector is k = diag(omega) W u with u = z_x + z_y. The latent pairs at which every channel has zero clock form the r-dimensional linear subspace {u = 0}; the map part delta = (z_y - z_x)/2 is free on it, so it contains non-blind points (any z_y not in ker W). (b) Along z_y(tau) = (1 - tau) z_y - tau z_x (W, omega fixed), u(tau) = (1 - tau) u(0) and every channel's clock is (1 - tau) k_c(0): all channels shrink together, monotonically. (c) Quadratic surrogate R(z) = 1/2 sum_c q_c k_c^2 (q_c >= 0 usage weights). Gradient flow on (z_x, z_y) gives du/dt = -2 H u with H = W^T diag(omega) diag(q) diag(omega) W, positive definite when the used channels' rows span R^r, and d(delta)/dt = 0. So u(t) = exp(-2 H t) u(0) -> 0 from every start, the map part is

unchanged, and {u = 0} is the only stationary set. (d) Generic uniqueness. For Lebesgue-almost every (W, omega), {u = 0} is the only latent set at which every channel cancels mod 2 pi (no latent u != 0 makes diag(omega) W u a non-zero vector of 2 pi Z^C when C > r).

Proof. (a)-(c) linear algebra: k_c = omega_c (w_c . z_x + w_c . z_y); dR/dz_x = dR/dz_y = H u. (d) diag(omega) W u in 2 pi Z^C iff the vector (2 pi n_c / omega_c)_c lies in col(W); for fixed n != 0 the map omega -> (2 pi n_c / omega_c) is a diffeomorphism of (R \ 0)^C onto itself, so its preimage of the r-dimensional subspace col(W) is null when r < C; union over countably many n. []

ACCOUNTS T2 (ESTABLISHED) is the causal test of (b): freezing the two move latents blocks the escape from the |.| start (1/8 vs 7/8, Fisher p 0.010; 15% vs 82% opposite-signed channels). Trained signed ring models end on {u = 0}: |u| / (|z_x| + |z_y|) = 0.044-0.053, cos(z_x, z_y) = -0.994 to -0.995 (ANALYSIS, `latent_geometry.py`; MECH SO, SO_absmag, GAUGE SO_m); MapFormer's Fig. 3d / 9b describe the same antipodal end state.

S3. The |.| class has no collective route for an inverse pair (THEORY, proved)

Proposition S3. |.| class, inverse pair, C channels, rank r. (a) The latent route of S2(b) ends, under |.|, at a(y) = a(x) in every channel: |w . (-z_x)| = |w . z_x|. It is a clock. (b) For Lebesgue-almost every (W, omega), at that fixed (W, omega) no latent pair (z_x, z_y) makes more than r channels of each sign-coherence class (pre-|.| steps of equal or of opposite sign), hence more than 2r channels, cancel non-blindly. (c) Hence every |.| map code with more than 2r cancelling channels has its readout (W, omega) tuned channel by channel: the solution set is a countable union of sheets indexed by the wrap counts n in Z^{C'} of the C' used channels, and on each sheet the channel-c condition, in step units, is |w_c . z_x| + |w_c . z_y| = 2 pi n_c / |omega_c|, an offset that differs across the omega ladder (omega_max / omega_min = grid size: 32 on the ring, 64 on the torus).

Proof. (a) evenness of |.|. (b) Fix a set S of 2r + 1 channels, a sign pattern (e_c(x), e_c(y)) per channel and non-zero integers n_c (a non-blind |.| channel cancels only by wrapping, S1(b)). On the sign region the angle sum is k_c = omega_c (e_c(x) w_c . z_x + e_c(y) w_c . z_y), linear in zeta = (z_x, z_y) in R^{2r}: k = diag(omega) N zeta with N the (2r + 1) x 2r matrix of rows (e_c(x) w_c, e_c(y) w_c). The conditions k_c = 2 pi n_c for c in S hold for some zeta iff v(omega) = (2 pi n_c / omega_c)_{c in S} lies in col(N), a subspace of dimension <= 2r < 2r + 1 that does not depend on omega. As in S2(d), omega -> v(omega) is a diffeomorphism of (R \ 0)^S, so the set of omega_S for which it holds is null; by Fubini the set of (W, omega) is null. (Within one coherence class the sum depends on zeta only through z_x + z_y or z_x - z_y, an r-vector, so the same argument with r + 1 channels of one class gives the per-class bound.) Countable union over S, the sign patterns and n. (c) from (b): at any (W, omega) outside the null set, more than 2r cancelling channels requires moving (W, omega); the channel-c equation is the |.| sum omega_c (|w_c . z_x| + |w_c . z_y|) = 2 pi n_c written in step units. []

Reading. The signed class reaches every channel's zero clock by r numbers (u = 0), whatever the readout. The |.| class needs the readout itself to carry per-channel integer offsets. The constructed solution AO_bal is such a point: its latents are orthogonal (c e1, c

e2) and its cancellation sits entirely in W_out (W_out[c, 0] + W_out[c, 1] = 2 pi n_c / (omega'_c c)). This is the precise sense in which the wrapped code needs "coordinated, channel-specific" configurations.

How many latent directions? (ANALYSIS, `aobal_geometry.py`, the held AO_bal models after 300 epochs, seeds 600-607.) Both codes are carried by ONE latent combination: in the balanced construction every pre-|.| step is >= 0 (same-sign share 0.77-0.86 after training), so on that sign region the |.| clock is diag(omega) W u with u = z_x + z_y, exactly as in the signed class. The difference is the target: the signed code needs W u = 0, which every readout contains; the wrapped code needs diag(omega) W u = 2 pi n (here n_c = 1 on every cancelling channel), which a readout drawn at initialisation carries 0.3-11% of (share of the target in col(diag(omega) W_init); random-subspace baseline 4/64 = 6%). The held models cancel in only 9-13 channels, which carry 92-98% of the usage, so S3(b)'s bound of 2r = 8 latent-cancellable channels leaves a handful of channels to tune: S3 alone is a modest obstruction; the larger ones are S3(a) (the collective move makes a clock) and S1 + P3 (each channel's target lies at k = 2 pi, across pi, from a small start).

Check (THEORY check, `prop_checks.py` (1)). At the default readouts of seeds 800-807, least-squares latents for random channel sets, sign regions and wrap counts n in {1, 2, 3} (4,000 trials per size): 9 channels never cancel exactly (smallest max error 0.027 rad); 8 channels cancel exactly (1e-14) only when they split 4 / 4 between the coherence classes (median error 7.6 rad over random splits), as the per-class bound says.

Check (ANALYSIS, `latent_reach.py`). At the default initialisation of MECHANISM seeds 800-807, the share of the needed change of the angle sums that a latent move can produce: signed (to zero) 1.000 in every seed (by S2(a)); |.| (to the nearest wrap) 0.339 unweighted, 0.561 usage-weighted (random-subspace baseline 8/64 = 0.125); at the |.| (absmag) start in the balanced gauge 0.285 / 0.463 (that start's latent-reachable space has rank 2). A latent move does help |.| partly (it scales all sums up together), but the residual 44-66% must come from per-channel readout changes.

S4. Moves +1 / +2: mirror copies and frustration (THEORY in the surrogate; ANALYSIS)

Proposition S4. Relation n = (n_x, n_y) (the code needs n_x a(x) + n_y a(y) = 0 per channel: +1/-1 is n = (1, 1); +1/+2 is n = (-2, 1)). (a) Signed: the residual is omega_c w_c . u with u = n_x z_x + n_y z_y; one linear zero set {u = 0} (as S2). (b) |.|: on a sign region the residual is omega_c w_c . (n_x e_c(x) z_x + n_y e_c(y) z_y). If n_x n_y < 0 (no inverse pair) the class contains two collective zero sets, {n_x z_x + n_y z_y = 0} and its image under z_y -> -z_y (the |.| class is invariant under that reflection; S3(a)), and both contain non-blind points. If n_x n_y > 0 (an inverse pair) neither is non-blind (S3). (c) Frustration. In the quadratic surrogate R = 1/2 sum_c q_c (residual_c)^2 restricted to a sign region in which both coherence classes (e_c(x) e_c(y) = +1 and = -1) contain channels whose rows span R^r, the only latent stationary point is the blind origin zeta = 0, and R is a positive-definite quadratic in zeta there, so the latent flow contracts toward it until some channel's step reaches zero (the wall).

Proof. (a), (b) as S2, S3. (c) On the region, residual = A_+ (n_x z_x + n_y z_y) for the coherent channels and A_- (n_x z_x - n_y z_y) for the others (rows omega_c w_c, sign

absorbed); dR/dz_x = n_x (H_+ u_+ + H_- u_-), dR/dz_y = n_y (H_+ u_+ - H_- u_-), with H_+- positive definite; both vanish iff u_+ = u_- = 0 iff z_x = z_y = 0. []

Check (`prop_checks.py` (3)). +1/+2 surrogate R = 1/2 mean_c (omega_c (step_c(+2) - 2 step_c(+1)))^2 with z_1 fixed at unit norm, Adam on z_2 from random starts, real readouts of seeds 800-807: signed reaches the collective zero z_2 = 2 z_1 from 80/80 starts; |.| reaches one of its two collective zeros from 30/80 (16 of them the mirror copy z_2 = -2 z_1) and stalls elsewhere from 50/80. Frustration costs the |.| class the collective route from most random starts even with no inverse pair; in the trained transformers the per-channel route then supplies the relation (below).

For +1/-1 the same computation gives (b) of the summary: every channel's residual is the clock k_c >= 0, so on any region R is a positive-definite quadratic with minimum at the blind origin: **the |.| latent gradient that removes the clock is a contraction toward the wall**, while the signed one (S2(c)) projects onto u = 0 and leaves the map alone. This is the surrogate-level mechanism of the collapse the reanalysis measured. *Check (`prop_checks.py` (2)):* gradient descent on the clock surrogate from the real initial latents and readouts (4 seeds): signed ends at u = 0 exactly (|u| / |z| < 1e-3) with the map part unchanged (relative change 0.000) and mean |map angle| 0.12-0.50; |.| ends with both latents at 0 (norm ratio 0.000), map part gone, mean |map angle| 0.000 -- the blind origin. Trained models show the same collapse in part (`collapse_wall_out.md`: 24-42% of |.| move steps on the ring, 60-70% on the torus; signed 10-16%, 34-40%).

Check (ANALYSIS; GAUGE seeds 1300-1307, `latent_geometry_out.txt`, `lattice_p_out*.txt`). - At initialisation the pre-|.| sign coherence is 0.52-0.77 (mixed: both classes populated). - +1/+2, latent relation (min over the two mirror copies of |z_2 -+ 2 z_1| / (|z_2| + 2 |z_1|) < 0.1): |.| 5/8 (s1307 via the mirror copy: cos(z_1, z_2) = -0.980, ratio 1.99, same-sign share 0.02), signed 7/8. The 3 |.| runs without it stay mixed (coherence 0.53-0.67, residual 0.47-0.73): frustrated. - In the channels attention uses the |.| models satisfy the relation anyway (usage share with |wrap(a(+2) - 2 a(+1))| < 0.05: 0.98-1.00 in 8/8 |.| runs; signed 0.99-1.00 in 7/8): where the collective route is frustrated the per-channel route supplies it, because on +1/+2 the per-channel targets lie on the near side (no wall, no far wrap). - Unsolved runs of both classes have the relation in place (|.| s1300-s1304, s1306; signed s1304, s1306); the lap lattice 32 a(+1) = 0 mod 2 pi does not separate them either (usage share within 0.05 rad: solved |.| runs 0.15 and 0.38, unsolved |.| runs 0.00-0.99, four of them >= 0.75). What the unsolved +1/+2 runs miss is not visible in the step code (ANALYSIS; the usage proxy is per token, not per context; content and readout not examined). - Inverse pair: the same readout gives 1-collapsed / 2-collapsed channels (of 64) 19.4 / 13.9 (MECH AO), 22.2 / 16.0 (AO_absmag), 18.8 / 13.2 (GAUGE AO_m), against 2.1-4.1 / 8.1-8.9 signed and 15.6 / 5.0 at initialisation. Sub-pi |.| channels within 10% of the cone boundary (|delta| >= 0.9 k / 2): 0.30 (AO), 0.36 (AO_absmag), 0.25 (AO_m); 0.056 at initialisation; 0.00-0.02 for the large starts.

S5. A clock produces the failure signature (THEORY, proved; ANALYSIS for the measured clocks)

Proposition S5. Let a channel's code have clock k_c and map m_c (any class), and let a revisit at displacement 0 have n_x backward moves. Its phase is k_c n_x. For a population of channels with usage weights q_c (sum 1), the same-cell score's position factor F(n_x) = sum_c q_c cos(k_c n_x) equals 1 at n_x = 0; if a usage share s of the channels has kappa

<= |k_c| <= K (unwrapped), then F(n_x) <= 1 - s (1 - cos(kappa n_x)) for n_x <= pi / K. Index RoPE is the extreme case: every token, observations included, advances every channel, so the whole code is clock.

Proof. phi = m d + k n_x with d = 0. For n_x <= pi / K, kappa n_x <= |k_c| n_x <= pi, and cos is decreasing on [0, pi], so cos(k_c n_x) <= cos(kappa n_x) on the share s and <= 1 elsewhere. []

So any code with a non-zero clock serves short-gap revisits and loses long-gap ones; wrap-only revisits also need the lattice N m_c = 0 mod 2 pi, which a code with a large clock cannot read either (its long-gap revisits are already lost). This is the common failure signature of the main draft Sec. 6.3 (index RoPE 0.784 / 0.462 / 0.471, shared rank 2 0.950 / 0.671 / 0.557, |.| 0.870 / 0.630 / 0.515 for short / long / wrap-only; ESTABLISHED), derived rather than observed.

Measured clocks (ANALYSIS, `sign_clock.py`, `rank_clock.py`; usage-weighted rms clock / rms map).

arm	SOLVED	clock / map
ring signed (MECH SO, SO_absmag; GAUGE SO_m)	8/8, 7/8, 8/8	0.015-0.018
ring |.| from small starts (MECH AO, AO_absmag; GAUGE AO_m)	0/8, 0/8, 1/8	1.55-1.68
ring signed, move latents frozen, |.| start (T2 SO_absmag_fz)	1/8	1.39 (the one solved run 0.06)
ring |.| from past pi (MECH AO_high)	8/8	0.31 (wrapped; usage on cancelling channels 0.82)
torus signed SO4 / |.| AO4 (TORUS_START)	8/8 / 1/8	0.001 / 1.22 (the solved AO4 run 0.02)
torus rank 2 heads, unsolved (shared, per-head, block-diagonal)	0/8, 2/8, 2/8	0.51-1.91 (latent clock 1.6-3.2)
torus rank 3 / 4 heads, solved	6/8, 8/8, 8/8	0.043-0.051 (latent clock 0.045-0.065)

Every arm that fails from a small start ends with a clock about as large as its map; every arm that succeeds from a small start removes it, and at rank 3 / 4 it is removed in the latent (R1), not parked. (Rank 2 heads that solved keep a moderate clock, 0.26-0.40; large-start arms carry wrapped codes whose rms clock is not small although they cancel in the used channels: the ratio is a coarse summary there.)

S6. The rank analogue: clock and map share every channel at rank D (THEORY, proved; it restates the rank theory T1)

Proposition S6. One head, latent dimension r, move latents z_m in R^r; for a D-dimensional map write the common mode c (the sum of the latents of each inverse pair, stacked) and the map frame F = [d_1 ... d_D] in R^{r x D} (half-differences), rank F = D. Channel c's clock along pair j is omega_c w_c . c_j and its map wavevector is omega_c F^T w_c. (a) r = D: if c_j != 0 then c_j = F alpha_j with alpha_j != 0, and channel c's clock equals its wavevector's component along alpha_j: omega_c w_c . c_j = (omega_c F^T w_c) . alpha_j. A channel is clock-free iff its wavevector is orthogonal to every

alpha_j, i.e. it resolves at most D - (number of independent alpha_j) dimensions. So with a non-zero common mode, no channel is both clock-free and a full D-dimensional map (no R2), and the only way to a clean code is c = 0 (R1). (b) r >= D + 1: if the components of the clock vectors outside col(F) are linearly independent (possible only for q <= r - D independent clock directions), every channel can be clock-free with any wavevector (choose the part of w_c orthogonal to col(F) to cancel every w_c . c_j). R1 and R2 both exist. On the torus a single common clock (all four moves advancing alike, the index-like start) needs r >= 3; two independent pair clocks need r >= 4.

Proof. (a) c_j in R^D = col(F) when F is invertible. (b) w = F (F^T F)^{-1} kappa + w_perp with w_perp in col(F)^perp free; w . c_j = kappa . (F^T F)^{-1} F^T c_j + w_perp . c_j,perp for j = 1..q, a linear system in w_perp with linearly independent coefficient vectors c_j,perp, hence solvable. []

Reading. At rank D a clock is a uniform drift of the decoded position (alpha_j per move pair): the rank theory's T1 ("a rank-D head cannot place a clock off the map"). In route terms, rank D keeps R1 (u = 0 is linear at every rank) but loses R2. The measured rank-3 / 4 successes nevertheless end on R1 (latent clock 0.045-0.065 of the map), and the rank-2 failures end clock-coupled (latent clock 1.6-3.2): **the end states match the sign axis; the route that rank 2 lacks is R2, not R1**.

S7. The unification, stated once (THEORY: inventory; ANALYSIS: end states; SPECULATION: dynamics)

Proposition U (route inventory). For an inverse-pair map task, a class reaches the zero-clock set from a generic small start through R1 iff its clock is a linear function of one latent vector whose zero set contains non-blind points, and through R2 iff each channel has a clock-free, full-map readout reachable without crossing a fold. Then: signed r >= D + 1: R1 and R2 (S2, S6(b)); signed r = D: R1 only (S6(a)); signed with frozen move latents: R2 only (the latents cannot move; each channel can still cross zero through W_out); |.|: neither from a small start (S3(a, b): the latent route ends at a clock, and no latent choice wraps more than 2r channels; S1 + P3: per channel the zero-clock points are at |k| = 2 pi n, across |k| = pi).

Empirical claim (ANALYSIS + ESTABLISHED counts). R1 and R2 together: 6/8-8/8 in every batch measured (ring and torus). Neither (|.| from small starts): 0-2/8 in every ring and torus recipe but one (ROBUST lr x3 on the ring, 4/8, unmeasured against 8/8). R2 only depends on the start, as it should: signed with frozen move latents solves 7/8 from the default start (about half the channels start opposite-signed, i.e. near their through-zero target) and 1/8 from the |.| start (none do); |.|, whose per-channel targets are the wrap sheets, solves 8/8 on the ring from a start past pi and 0-2/8 from small starts. R1 only: 7/8 on the 1D ring (rank 1), 0-2/8 on the 2D torus (rank 2), 1/8 on a small 3D torus (rank 3). Every failure ends clock-coupled (S5 table). So: **from a small start whose channels are not already near their per-channel targets, learning needs the collective route; on the 2D and 3D torus it has needed per-channel decoupling as well**.

Why R1 alone fails in 2D: the rank note's answer (THEORY for the gradient statements, SPECULATION for the dynamics; `docs/paper/expansion/RANK_THEORY.md`, written in parallel, read-only here). RANK_THEORY Prop. 1 is the same two-route

split as S6 (its "global" route = R1, its "local" route = R2; at r > D the local route "parks" the clock off the map plane at zero map cost, at r = D it "hides" the drift by removing the channel's resolution along the drift direction beta). Its Props. 2-3 supply the missing dynamics: at r = D >= 2, hiding costs nothing to first order when the head's resolution is isotropic (rotating a channel's wavevector off beta leaves tr P unchanged), so the readout takes it at once; but the global route removes the drift at a rate proportional to the head's resolution along beta, which hiding has just lowered -- a positive feedback that leaves the drift in place (a plateau, or the CLOCK / COLLAPSE end states). At D = 1 there is no direction to hide along (hiding kills the channel and leaves the drift-to-map ratio unchanged), so the global route is not starved: rank 1 solves 7/8. This explains the one anomaly of Proposition U.

The sign axis in the same terms (THEORY). For |.| the local route to a zero clock *below pi* is the cone's: shrinking k_c shrinks the largest map the channel can carry (S1(a), |m_c| <= |k_c|), so it is "hiding" with no free direction, as at D = 1 -- the channel dies (blind) or becomes a one-sided counter -- and the global route does not exist (S3(a)). The two axes therefore fail for different reasons inside one frame: **sign removes the global route; rank D >= 2 keeps it but lets a free local move (hiding) starve it; per-head rank >= D + 1 adds a free local move (parking) that does not starve it**. The common end state (a clock as large as the map, S5 table) and the common signature (S5) follow.

Consistency with the measured end states (ANALYSIS, `rank_clock.py`). Unsolved rank-2 heads keep a latent clock 1.6-3.2x the map frame with the clock pattern inside the map span (in-span 0.91-0.98, wrapped); solved rank-3/4 heads have a latent clock of 0.045-0.065 of the frame (and a read clock 0.043-0.051), i.e. by the end the clock is mostly removed, not merely parked; whether it was parked first (RANK_THEORY P4) needs mid-training snapshots (Sec. 6, item 3). RANK_THEORY P3 (the leftover clock lies off-plane) is compatible with these numbers (a small leftover), not tested here.

Predictions shared by the two notes: (i) a rank-2 torus model whose move latents are tied to cancel (drift forbidden, as RANK_TESTS' R1DF) solves like rank 4; (ii) rank D plus a learned per-channel clock slot (RANK_THEORY P7) behaves like rank D + 1; (iii) on the sign axis, a model with a non-negative readout but a signed latent (global route restored, Sec. 6 item 4) loses the sign cost.

S8. "A solution exists, is held, and is not found" as a statement about basins (THEORY, definitional; ANALYSIS)

Let Phi_B be the stochastic training map for budget B, S_eps = {theta : L(theta) <= eps} the solution set of a class, and P_0 the initialisation distribution. Define the basin Bas = {theta : P(Phi_B(theta) in S_eps) >= 1/2}. Then: - *exists*: S_eps is non-empty in the class (Proposition 1 / P1(a), constructed); - *held*: the constructed points lie in Bas (ESTABLISHED: ring 8/8, torus re-found 8/8, rank-2 projection 7/8); - *not found*: P_0(Bas) is small (ESTABLISHED: SOLVED fractions 0-2/8 estimate it, with binomial error). Together they say the obstruction is the position of supp(P_0) relative to Bas, not the emptiness or instability of S_eps. Two cheap measurements of that position follow (Sec. 3.3, 3.4): the loss profile along straight paths from the start to a constructed solution (barriers), and the basin's extent along such a segment (train from (1 - tau) theta* + tau theta_0 and record success).

3. Analyses run (CPU; scripts in `docs/paper/expansion/scripts/sign/`)

3.1 Latent geometry and per-channel patterns (`latent_geometry.py` -> `latent_geometry_out.txt`, `_rows.json`)

Eval-only on GAUGE (+1/-1 and +1/+2, seeds 1300-1307, initialisation rebuilt exactly and checked bit-exact against MECHANISM's stored default angles: max difference 0.0) and MECHANISM (seeds 800-807). Results quoted in S2, S4.

3.2 Clock / map ratios (`sign_clock.py`, `rank_clock.py`) and the reachable share (`latent_reach.py`)

Results in S1, S3, S5, S6. `rank_clock.py` reads RANK_MI / RANK_SEP / RANK3 checkpoints (full models: observations step); the observation clock (mean observation step) is 0.015-0.018 of the map in solved rank-3/4 heads and 0.18-0.52 in unsolved rank-2 / rank-3 heads.

3.3 Loss along straight paths with the content of a solved model fixed (`path_profiles.py` -> `path_profiles_out.txt`)

Ring, MECHANISM SO seeds 800-807; content (everything but W_in, W_out, omega) from the solved signed model; balanced gauge, so a straight line in W_out is a straight line in angle space. 64 fixed walks, T = 128, eval mode. The start loss is high (8.8-10.4 nats) because the solved content is confident.

path	loss at tau = 0, 0.2, ..., 1 (mean of 8)	accuracy at tau = 0 -> 1	max rise above running minimum, per seed
S_lin: signed, |default| -> signed solution	9.46 9.18 8.65 7.50 6.41 0.00	0.30 -> 1.00	0.27 0.04 0.30 0.40 0.48 0.38 0.05 0.36
A_lin: |.|, |default| -> wrapped solution	9.46 8.23 7.87 7.96 7.44 0.00	0.30 -> 1.00	1.92 0.87 1.41 0.60 1.09 1.01 1.14 0.74
S_lat: signed, collective latent route only (W_out, omega at the start)	9.46 9.39 9.49 9.25 8.35 2.98	0.30 -> 0.72	0.03 0.00 0.00 1.36 2.15 1.44 0.21 0.02
A_lat: |.|, the same latent route	9.46 9.41 9.66 9.90 9.68 9.21	0.30 -> 0.30	0.17 1.11 0.18 2.40 2.23 1.68 1.33 0.51

Reading (ANALYSIS, n = 8, exploratory): the |.| straight path to its exact solution is rugged (every seed's largest rise exceeds every signed seed's: complete separation, exact two-sided permutation p = 2 / C(16, 8) = 1.6e-4 for "all 8 |.| rises above all 8 signed"), the signed one nearly monotone. Rotating one move latent toward the negative of the other, with nothing else changed, takes the signed model from 0.30 to 0.72 accuracy (monotone in 5/8 seeds) and leaves the |.| model at 0.30 (it ends at a clock, S3(a)). This is the transformer version of the review's toy `path_cut` (where the route started on a plateau); here it descends from the start in 5/8 seeds.

3.4 The step map's basin with the content frozen (`step_basin.py`, driver `run_basin.sh` -> `runs_basin/`)

Ring, MECHANISM seeds 800-807. Content (embeddings, attention, FFN, readout) from the solved signed model, FROZEN; only W_in, W_out, omega train (AdamW lr 1e-3, wd 0.05, cosine, 2000 steps of batch 32, eval mode: about 2% of the samples of a full 300-epoch run). Both classes have an exact solution with this content (signed_balanced / balanced construction). The start loss is 7-10 nats (confident content, wrong step map). No run reaches the house SOLVED loss (0.05) in this budget, so the readout is fresh-stream accuracy (100 walks, T = 128). `basin_summary_out.txt`; exploratory, n = 8.

start (balanced gauge)	class	accuracy start -> end (mean)	per seed (end)	acc >= 0.9
default angles	signed (S_def)	0.33 -> **0.83**	0.89 0.86 0.95 0.70 0.98 0.75 0.96 0.56	3/8
|default| (the |.| start)	signed (S_abs)	0.29 -> 0.47	0.37 0.94 0.34 0.75 0.28 0.37 0.35 0.32	1/8
|default|, move latents frozen	signed (S_abs_fz)	0.29 -> 0.40	0.36 0.32 0.33 0.76 0.33 0.35 0.37 0.36	0/8
|default|	|.| (A_abs)	0.29 -> 0.36	0.33 0.37 0.35 0.55 0.28 0.35 0.36 0.31	0/8
a(+1), a(-1) iid U(1.7, 4.5) (past pi)	|.| (A_high)	0.36 -> 0.64	0.81 0.50 0.74 0.56 0.92 0.53 0.55 0.53	1/8
the exact wrapped solution (pilot, seed 800 only, 300 steps)	|.| (A_sol)	1.00 -> 1.00	held	--

Contrasts (permutation p on accuracy): default vs |.| start, signed: +0.365 (p 0.005); past-pi vs small start, |.|: +0.280 (p 0.0007); signed vs |.| from the |.| start: +0.104 (p 0.32, not separated); signed free vs frozen latents from the |.| start: +0.069 (p 0.56, not separated).

Reading (ANALYSIS, exploratory). (i) With the content fixed, the START decides the step map's search: from the default start the signed step map recovers most of the map in 2000 steps; from the same-direction (|.|) start neither class does, and the past-pi start helps the |.| class (the MECHANISM direction, reproduced in the step map alone). (ii) **The signed escape from the |.| start (7/8 in full training, MECHANISM SO_absmag) is NOT reproduced when only the step map trains**: 2/8 signed runs move off the start (0.94, 0.75) against 1/8 frozen-latent and 1/8 |.| runs. So the collective route, although a descent direction along the straight latent path (Sec. 3.3: 0.30 -> 0.72 with nothing else changed), is not what gradient descent on the step map alone takes from that start within this budget; in full training the escape also involves the content weights (or more steps). This qualifies S2(c): the surrogate's guarantee is about the clock term alone. (iii) In the language of S8: the default signed start lies in (or near) the step map's basin of its solution, the same-direction start lies outside it for both classes, and the past-pi start is closer for |.|. The basin's extent along the segment from the |.| start (tau = 1) to the wrapped solution (tau = 0), angles (1 - tau) a* + tau |a_0|, |.| class:

tau	1 (A_abs)	0.75	0.5	0 (A_sol pilot)
accuracy at the start of training	0.29	0.32	0.32	1.00
accuracy after 2000 step-map steps (mean of 8)	0.36	0.51	0.61	1.00 (held, n = 1)

tau	1 (A_abs)	0.75	0.5	0 (A_sol pilot)
runs >= 0.9	0/8	0/8	0/8	--

(permutation p against tau = 1: 0.012 at 0.75, 0.0017 at 0.5; tau = 0.25 and the signed segment were dropped to save CPU time.) Halfway along the straight segment the |.| step map recovers part of the map but returns to the solution in no run, while the signed step map from its default start (a generic point, not on any segment) reaches 0.83 (p 0.013 against tau = 0.5). In S8's terms the wrapped solution's basin, for the step map with this content and budget, does not reach halfway from the |.| start; midpoints of the segment score like the start itself (0.32), consistent with the rugged straight path of Sec. 3.3.

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4. What survives

- THEORY, proved: S1 (clock floor of |.|), S2 (signed collective route; uniqueness), S3 (no collective route for |.| on an inverse pair; at most 2r latent-cancellable channels at a generic readout; channel-specific wrap offsets), S4 (mirror copies; frustration and contraction in the quadratic surrogate), S5 (a clock gives the short-gap-only signature), S6 (rank D: no clock-free full-map channel while the common mode is non-zero), U (the route inventory).
- ANALYSIS: every failing arm on both axes ends clock-coupled (clock/map 0.5-1.9), every small-start success clock-free (<= 0.065) by R1; |.| ends with one-sided and blind channels at the cone boundary (5-6x the initial rate); on +1/+2 the |.| step code is learned (relation in 8/8 runs) and the unsolved runs fail elsewhere; the signed collective route is a descent direction on the real model with solved content (0.30 -> 0.72), the |.| one is not; the |.| straight path to its solution is rugged, the signed one is not; with the content frozen the start decides the step map's search for both classes, and the signed escape from the same-direction start is not reproduced by the step map alone.
- Not surviving: the strong unification "the failing class has no linear route" (false for rank D); "|.| fails without an inverse pair" (no step-code evidence); the wall as the sole account (it is the boundary optimum of the cone trade-off plus a contraction, and the reanalysis's end-of-training gradients are MIXED).
- Open: whether hiding (RANK_THEORY Props. 2-3) is what starves R1 at rank 2 in training (its gradient statements are proved there; the dynamics and the predictions P2, P4, P7 are untested); what the unsolved +1/+2 runs miss; whether per-channel capture or a collective effect produces the past-pi rescue on the ring and why it fails on the torus.

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5. Proposed paper text

5.1 Main text: a theory subsection (~1 page; for Sec. 6, after 6.2, replacing the Limitations line "why the

non-negative parameterisations stall from small starts is not identified")

6.3 Why search fails: one common mode, two missing routes. A map code must cancel opposite moves: in each channel the clock k = a(x) + a(y), the phase a back-and-

forth adds at zero displacement, must vanish mod 2 pi while the map m = a(y) does not. Any non-zero clock makes a revisit's phase grow with the number of moves since the last visit, so the code reads short-gap revisits and loses long-gap and wrap-only ones (Proposition S5) -- the single failure signature of every failing model (Sec. 6.4). The question is how a class removes its clock from a small random start. There are two routes. In the *collective* route the clock vector is diag(omega) W_out (z_x + z_y), so the r-dimensional latent condition z_y = -z_x zeroes every channel at once and leaves the map untouched; in a quadratic surrogate the latent gradient converges to it from every start (Proposition S2). In the *per-channel* route each channel's readout is tuned to a clock-free row of full map rank. A signed step at rank D + 1 has both. A non-negative step has neither from a small start: under |.| the collective move produces a(y) = a(x), a clock, and at a generic readout no choice of latents makes more than 2r of the C channels wrap (Proposition S3: the wrap targets 2 pi n / omega_c differ across the frequency ladder); per channel, an unwrapped |.| angle pair has clock at least as large as its map (|k| >= |m|, Proposition S1), and its clock-free non-blind points lie at |k| = 2 pi, across |k| = pi. At rank D the collective route remains, but no channel can be clock-free and a full map while the common mode is non-zero (Proposition S6, the rank theory's drift). The trained models match this inventory: every model that fails from a small start, on either axis, ends with a usage-weighted clock as large as its map (ratio 0.5-1.9; non-negative, frozen-latent signed and rank-2 alike), and every success ends clock-free (<= 0.065) through the collective route (Table X). Freezing the move latents of a signed model removes the collective route and leaves it where the non-negative step ends (1/8; clock ratio 1.39 against 1.6). With the content of a solved model held fixed, rotating one move latent toward the negative of the other takes the signed model from 0.30 to 0.72 accuracy with nothing else changed, and leaves the non-negative one at 0.30 (gradient descent on the step map alone, with that content frozen, rarely takes this route from the same-direction start; in full training the content co-adapts). The two axes fail differently inside this frame. The non-negative step has no collective route. At rank D the collective route exists, but for D >= 2 a channel can stop seeing the drift at no first-order cost by turning its resolution away from the drift direction, and that lowers exactly the resolution that drives the collective removal -- a feedback that, we conjecture, keeps the drift in place (the gradient statements are proved; the dynamics are not tested); in 1D there is no direction to turn to, and rank 1 learns the ring (7/8). One more latent dimension gives a per-channel move that hides the clock off the map plane at no cost to the map.

5.2 Appendix material (for App. E, after P5 / Remark R)

- **E.x Clock and map.** Definitions of Sec. 1; Proposition S1 with proof; the cone-boundary readout (S4 check).
- **E.y The collective route.** Proposition S2 (a)-(d) with proofs; ACCOUNTS T2 as the causal test; latent end states.
- **E.z No collective route under |.|.** Proposition S3 with the omega-scaling proof; the reachable-share check; the remark that AO_bal is a point of a channel-tuned sheet.
- **E.w Same-direction moves.** Proposition S4; the GAUGE +1/+2 readouts (relation learned in 8/8 |.| runs; mirror copy in s1307; unsolved runs fail outside the step code). This replaces the main draft's "why |.| also struggles on +1/+2 is open" with "the |.| step code is learned on +1/+2; the remaining failures, shared with signed, lie elsewhere".

- **E.v Rank.** Proposition S6 (restating the rank theory's T1 as a route statement) and Proposition U, with the clock table of S5 and the explicit open problem.
- **E.u Basins.** The definitional statement of S8 and the measurements of Sec. 3.3-3.4.

Remark R of the appendix (the |.| side "open, fits loosely") can then say: the |.| failure from small starts is accounted for at the level of routes (S1, S3) and end states (S5 table); the per-channel dynamics remain underived (per-channel flow captures only near 2 pi, review `mech_capture_out.txt`).

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6. Decisive GPU experiments (when GPUs return)

Costs from the batch logs: ring observation-step-free run (T = 128, 300 epochs) ~10-12 min on one GPU slot (GAUGE: 40 runs in 1 h 40 min on two GPUs); torus T = 1024, 900 epochs ~45-60 min per run per slot (TORUS_START: 40 runs in 6 h 10 min on two GPUs).

1. **R1 alone vs R1 + R2 on the torus: drift-forbidden rank 2** (tests S7's central claim). Torus T = 1024, 900 epochs, observation-step-free, per-head rank 2 with the inverse-pair latents tied (z_W = -z_E, z_S = -z_N: the clock forbidden by construction, as R1DF), against per-head rank 2 untied and rank 4 untied; n = 8 each, 24 runs, ~6 h on two GPUs. Prediction: tied rank 2 solves like rank 4 (>= 6/8) if the rank-2 failure is a failure to remove the clock; if tied rank 2 still fails, the rank axis has a second obstacle (the S6 account is incomplete).
2. **R2 only on the torus: frozen move latents** (completes the 2 x 2). Signed rank 4, torus, move latents frozen at the default and at the |.| start, n = 8 each, 16 runs, ~4 h. Prediction: fails like |.| (<= 2/8) from the |.| start.
3. **Parking (S7 prediction ii), eval of snapshots.** Rerun 4 rank-4 and 4 rank-2 torus seeds with checkpoints every 50 epochs (8 runs, ~2 h incl. snapshots); eval-only readout of the clock's share outside span{map} over training. Prediction: rank 4 parks then removes; rank 2 cannot park and keeps it.
4. **A |.| class with a collective route** (tests S3 constructively). |.| applied after a learned sign-free reparameterisation that keeps an odd latent: e.g. steps |w . z_x| for x and the same with an inverse-pair tie z_y := -z_x (the clock is then omega (|w . z| + |w . z|) = 2 omega |w . z|, never zero): predicted to fail; vs a "signed-latent, |.|-readout" hybrid a(x) = omega |w| . z_x elementwise (readout non-negative, latent signed: collective route restored) predicted to succeed. Ring, n = 8 per arm, 16 runs, ~1.5 h. This separates "non-negative readout" from "no collective route".
5. **+1/+2 with the content examined.** GAUGE AO_p / SO_p at 900 epochs (16 runs, ~3 h) plus an eval-only read of what the unsolved runs miss (content: same-observation key retrieval; readout). Decides whether G1 is a budget effect.

Cheapest decisive item: 4 (1.5 h); most informative for the unification: 1 (6 h).

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7. Reproduce

From `docs/paper/expansion/scripts/sign/` (CPU; paths are absolute): `python3 latent_geometry.py > latent_geometry_out.txt` (~2 min); `python3 sign_clock.py > sign_clock_out.txt`; `python3 rank_clock.py > rank_clock_out.txt`; `python3 latent_reach.py > latent_reach_out.txt`; `python3 aobal_geometry.py > aobal_geometry_out.txt`; `python3 -u`

`prop_checks.py > prop_checks_out.txt` (~10 min); `python3 lattice_p.py > lattice_p_out.txt` and `python3 lattice_p.py 0.05 > lattice_p_out_tol005.txt`; `python3 path_profiles.py > path_profiles_out.txt` (~15 min, 4 threads); `./run_basin.sh` (56 step-only runs of 2000 steps, 10 at a time, ~3 h on a host at load ~80; the 16 runs of A_seg_t0.25 / S_seg_t0.75 were dropped by writing `{"skipped": true}` placeholders, which the summary skips; pilot in `runs_basin_pilot/`), then `python3 basin_summary.py > basin_summary_out.txt`.

The depth exchange rate: what index attention computes, layer by layer (expansion note, 2026-10-20)

Scope: main draft Section 5 (`docs/paper/MAIN_DRAFT1.md`), outline claims 5.1-5.4 (`docs/paper/OUTLINE_main_v2.md`). Every statement carries one label:

- **ESTABLISHED** -- a registered result in a results file (the results file is authoritative; quoted, not re-derived).
- **ANALYSIS** -- computed here, post hoc, eval-only on stored checkpoints (CPU), after the batches were read. Not pre-registered; descriptive. Each number names its script and output file.
- **THEORY** -- an argument (a construction, an upper bound, or a cited theorem applied to our setting). Not a measurement.
- **SPECULATION** -- a mechanism the data are consistent with but do not establish.

Scripts and outputs: `docs/paper/expansion/scripts/depth/` (outputs in `out/`). Every stored accuracy was reproduced before any new readout was taken (Section 4.0). No GPU was used; nothing was trained.

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1. Summary

1. **ESTABLISHED (the rates).** One path layer is worth about three index layers on Dyck-2 at matched depth and on the biased ring wherever a step can be undone; one index layer suffices when the walk never reverses (p_plus 1.0); on the torus at T = 1024 four index layers or a loop of four fall far short (0.860 / 0.864 vs 1.000), and depth helps only short-gap revisits.
2. **THEORY (why path integration needs one layer).** A revisit at move t must retrieve an observation at any earlier s with P_s = P_t (mod N), P the prefix sum of the moves. In the path model the score depends on omega (P_t - P_s) directly: the integrator supplies the prefix sum, exactly and for any gap, before attention. Index attention sees position only as t - s, so the predicate "the moves in (s, t] sum to zero" must be computed in the residual stream.
3. **THEORY (two routes for index attention).** (A) *Global count*: uniform causal attention averages the moves, giving P_t / n_t; rescaled and matched, this is a depth-2 solution in principle (Liu et al. 2023, Lemma 6; Yao et al. 2021's depth-2 Dyck construction), but it needs a normaliser and precision 1/n_t, its MLP width and weights grow with N x T in the known construction, and the target is highly sensitive (every one of the g moves since the last visit decides it; Hahn and Rofin 2024). Its cost grows with T, not with depth. (B) *Local return set*: a layer can realise rules whose return offset depends on a bounded number of recent direction runs (*segments*): undoing the last run (retrace), or a constant offset (a clock). Its cost grows with the number of segments a revisit spans, and depth extends that number.
4. **ANALYSIS (route B is what deep index attention learned on the torus).** Of the torus's short-gap revisits, 82% are one-level retraces; four index layers or the loop solve them (0.996 / 1.000), and on the rest accuracy falls with the number of segments spanned: 0.93 / 0.98 at 3-4 segments, 0.74 / 0.65 at 5-8, 0.56 / 0.50 at

9-16, floor beyond. At a fixed gap of 32-127 moves the same fall appears (0.87 / 1.00 at 2 segments to 0.49 / 0.50 at 17+), so segments, more than the gap in moves, limit the deep models (Section 4.4 finds a smaller residual gap effect). This accounts exactly for "just above retrace": 0.943 = (0.680 x 0.996 + 0.148 x 0.699) / 0.828. Their last layer puts 0.55-0.58 of its attention on the last visit's observation for retrace queries, 0.22 for other short gaps, and 0.00-0.01 for long-gap and wrap-only revisits (path model: 0.81 / 0.74 on long gaps). Linear probes find no global count in their residual stream (unwrapped displacement R^2 <= 0.10; normalised count P_t / t <= 0.21), and a displacement code over the recent window that degrades with the window (R^2 0.96 at 4 moves, 0.77 at 64, 0.37 at 256).

5. **ANALYSIS (route A is what three index layers learned on the ring).** On the ring (T = 128, fresh seeds 2-7), one index layer learns exactly two rules: retrace (0.68-0.80) and, when the walk is biased, the one-lap clock (0.987-1.000); every other revisit is at the floor (0.49-0.59). Two layers add segment composition that decays with segments (0.89 / 0.80 / 0.68 at 3-4 / 5-8 / 9-16 segments, p_plus 0.5). Three layers are flat in segments (0.997 / 0.993 / 0.972) and their residual stream carries a global count (P_t / t R^2 0.69, P_t 0.60, against 0.18 / 0.09 at one layer): the count route is found on the ring and not on the torus.
6. **ANALYSIS (the 0.75 dip, descriptive only).** One-layer index accuracy across p_plus is the share-weighted sum of those two rules and the floor: a1 = 0.709 / 0.678 at p_plus 0.5 / 0.75 is reproduced from class shares and class accuracies. The dip is the retrace share falling (0.72 -> 0.49) before the clock's share is large (0.18). The pre-registration attached no mechanism sentence to the dip; this is a post hoc decomposition, not a verdict.
7. **ANALYSIS (Dyck).** One index layer predicts a closer only when its opener is the previous bracket (0.998), and is near chance (0.61-0.64) once one balanced block intervenes; each further layer extends the number of blocks it skips (two blocks: 0.78 / 0.89 / 0.93 at 2 / 3 / 4 layers), with a residual dependence on distance. Dyck has a BOS token, so the count route's normaliser is available; the data are a mixture and do not separate the routes cleanly.
8. **Proposed paper text** (Section 5) and **five GPU experiments** that test the account (Section 6). The decisive one varies the walk's run length at fixed T: route B predicts accuracy is a function of segments spanned, route A a function of T and the gap. A CPU eval-only version on the stored models (Section 4.4) agrees: retraces are solved at any gap, and at a fixed gap of 16-63 moves 4L scores 0.98 / 0.87-0.93 / 0.72-0.80 / 0.64-0.70 at 2 / 3-4 / 5-8 / 9-16 segments, with a smaller residual dependence on the gap.

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2. What is being explained (ESTABLISHED)

task (trained and tested at the same length / depth)	path, 1 layer	index 1 / 2 / 3 / 4 layers	rate	file
Dyck-2, L32 D12, A2f (floor 0.594)	0.996	0.648 / 0.889 / 0.964 / 0.982 (RoPE)	~3	`DYCK_MDEPTH_RESULTS.md`
ring 32, T = 128, p_plus 0.5 (fresh seeds 2-7)	1.000	0.710 / 0.949 / 0.998	3	`CANCEL_RESULTS.md`, `docs/audits/2026-09-27/fresh_seeds.txt`

task (trained and tested at the same length / depth)	path, 1 layer	index 1 / 2 / 3 / 4 layers	rate	file
ring, p_plus 0.75 / 0.9	1.000	0.677 / 0.962 / 0.998; 0.824 / 0.987 / 0.999	3 (0.9 knife-edge)	same
ring, p_plus 1.0	1.000	1.000	1	same
torus 64x64, T = 1024 (n = 4)	1.000	0.719 / -- / -- / 0.860; loop x4 0.864	> 4	`FACTORIAL_RESULTS.md`

Torus strata (registered secondary; floors post hoc, `docs/paper/figs/fig_factorial_strata_out.txt`): index 1 layer / 4 layers / loop: short gap 0.769 / 0.943 / 0.938 (retrace floor 0.912), long gap 0.480 / 0.458 / 0.507, wrap-only 0.482 / 0.457 / 0.509 (constant floors 0.515 / 0.517). Every index run STALLED; budget-scoped (900 epochs torus, 300 ring, 1x Dyck ladder budget; at 3x budget four index layers reach Dyck's ceiling, 0.997-0.998).

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3. Theory

3.1 What a revisit prediction requires (THEORY, elementary)

Tokens interleave moves and observations, [a_1, o_1, a_2, o_2, ...]; o_t is predicted at the token a_t. Let P_t be the unwrapped displacement after move t (an integer in 1D, a pair in 2D). The map is redrawn per evaluation, so the only source of o_t is an earlier observation token o_s at the same cell: P_s = P_t (mod N per axis). Because the map is fixed within a walk, *any* earlier visit will do; no recency tie-break is needed.

A causal attention score between query t and key s in an index-RoPE layer is

```
score(t, s) = q(h_t)^T R((t - s) Omega) k(h_s),
```

so position enters only through the offset t - s, and the content h_t, h_s are the residual streams. The target predicate P_t - P_s = 0 (mod N) is a property of the *interval* (s, t]. There are two ways to make the score depend on it:

- **(A) absolute code**: write a function of P_t into every h_t, so that q(h_t)^T k(h_s) peaks at P_s = P_t (mod N). This requires the prefix sum P_t in the residual stream.
- **(B) relative return set**: write into h_t a description of the set of offsets d <= W at which the recent path returns to the current cell, and let the RoPE term select those offsets: score(t, t - d) = sum_c Re(alpha_c(t) e^{i omega_c d}) approximates the indicator of the return set on [1, W]. This needs only the last W moves, but the query must encode their return structure.

3.2 Path integration supplies the prefix sum (THEORY, exact)

MapWM rotates queries and keys by theta_t = omega * cumsum(Delta(x)). With observation steps near zero and move steps a fixed vector per move (MapFormer's learned solution), theta_t = omega P_t and score(t, s) = q^T R(omega (P_t - P_s)) k: route

(A) with the prefix sum computed outside attention, exactly (a float32 cumsum), for every gap and every T. In a channel whose frequency times the learned move step is a multiple of 2 pi / N, the score is N-periodic in P, and wrap-around costs nothing (the slowest default frequency has period N moves at unit step; Prop. 1 of the draft concerns which step tables represent this). One layer suffices; the stored path models show it (same-cell attention 0.81 / 0.74 on long-gap torus revisits, 0.66 / 0.58 on wrap-only; Section 4.2).

3.3 Route A for index attention: depth 2 in principle, precision in practice (THEORY)

Construction (upper bound on depth). Layer 1: one head with constant query and key attends uniformly to the prefix; its value carries the move vector on move tokens, so its output is P_t / n_t with n_t = 2t the prefix length. A second feature proportional to 1 / n_t is available in the same layer (for example, a head with a fixed self-attention bonus beta has self-weight e^beta / (e^beta + n_t - 1)). The MLP rescales to P_t and forms periodic features cos(2 pi k P_t / N), sin(2 pi k P_t / N). Layer 2 matches q(h_t)^T k(h_s) on those features. This is Liu et al.'s (2023) Lemma 6 (one layer simulates a mod-n counter: prefix average + MLP mod) followed by a match, and Yao et al.'s (2021) depth-2 Dyck construction (counter in layer 1, proximity-based depth matching in layer 2; as summarised in Liu et al., App. A.5). In RASP terms the task has depth 2 in 1D and in 2D alike (two counts are two value dimensions of one head) and with or without wrap (the mod is in the MLP). **So the measured rates (3, 3, > 4) are not representational depth bounds.**

What the construction costs. - A normaliser. Lemma 6 uses a padding token and an absolute position channel gamma_t = log(2T - t) to undo the 1/t of causal averaging. Our torus and ring sequences have neither (no BOS; RoPE is relative); Dyck has a BOS. The model must improvise the 1/n_t feature (it does: 1/t is linearly decodable at R^2 0.67-0.89 at the best pass of every multi-layer index model, Sections 4.3, 4.5). - *Precision.* Adjacent cells differ by 1/n_t in the averaged code: 1/256 at the end of a ring walk (T = 128), 1/2048 per axis at the end of a torus walk (T = 1024). - *Width and weights.* Lemma 6's mod gate uses MLP width 4 n T and weights up to 4 n T + 2 (width O(n) only with periodic activations). Read as an upper-bound construction: 32 x 256 for the ring, 64 x 2048 per axis for the torus (16x per axis, two axes); our MLP has width 512. - *Sensitivity.* For a revisit at gap g, flipping any one of the g moves in (s, t] moves the target cell, while flips before s do not; the prediction's sensitivity grows with g. Hahn and Rofin (2024, Cor. 5 and Thm. 6) show that representing a function whose average sensitivity grows faster than sqrt(n) log n needs parameter norms or layer-norm blow-up that grow with input length, and that such minima are sharp. Long-gap torus revisits (g in the hundreds against sqrt(n) ln n ~ 345 at n = 2048 tokens) approach that regime; short-gap revisits (g < 16 moves) are far from it. This cost does not by itself separate the ring from the torus (ring gaps reach 128 moves against sqrt(n) ln n ~ 89 at n = 256); precision, width and the second axis do. Their theorem is for Boolean inputs under the uniform distribution; we use it for direction only.

Prediction. Route A is found when T and the dimension are small and a normaliser is easy (ring at T = 128; Dyck with a BOS at length 32) and not when T is large (torus at T = 1024). Where it is found, accuracy does not depend on how many segments a revisit spans.

3.4 Route B: bounded return rules, extended by depth (THEORY, with an empirical reach)

One layer. Two rules need no prefix sum. - *Retrace.* While the current run of moves reverses the previous run, the walk returns along its own track: after j steps of reversal the cell is the one seen 2j moves (4j tokens) earlier. The query must encode j (the current run's length, a local average over the last <= 10 moves) and rotate its RoPE phases by the matching offset (4j tokens). This is a bounded-window rule (runs are at most 10 moves); its sensitivity is bounded by the two runs involved. - *Clock.* When the walk never reverses (p_plus 1.0) P_t = t, and the cell visited N moves earlier is the current cell: a constant offset, which RoPE represents in one head. Position equals index; there is nothing to integrate.

Depth. A return that spans k segments (maximal runs of one action) is a function of those k segments' directions and lengths. To select its offset the query needs a summary of k segments; segment boundaries sit at content-dependent offsets, so fetching the summary of the segment before the last is a pointer hop. With two heads a layer can make a bounded number of hops; stacking layers extends the reach. Two bounds bracket the reach: hops compose linearly if each layer adds one segment, and logarithmically if layers double their pointers (Liu et al.'s Theorem 1 gives O(log T) depth for any semiautomaton by such doubling; for an Abelian group like ours Lemma 6's depth-1 route exists, which is route A). We have only L = 1 and 4 on the torus, so the shape (linear or logarithmic) is open (experiment E2).

Prediction. At fixed depth, accuracy falls with the number of segments spanned, and depends on T or the gap only through them. The depth needed to solve a task is set by the segment counts of its revisits: on a 2D walk at T = 1024, long-gap revisits span 17 or more segments in 98.5% of cases (median 37; wrap-only: 99%, median 65; Section 4.1), far beyond what 4 layers reach.

3.5 What the two routes predict for each rate (THEORY, checked in Section 4)

task	route B covers	route A	predicted rate	checked
ring, p_plus 1.0	everything (clock: constant offset)	not needed	1	4.5: lap class 1.000 at 1 layer
ring, p_plus 0.5-0.9, T = 128	retrace + clock at 1 layer; 3-8 segments partly at 2	ring is small: count learnable	3 (count found at 2-3)	4.5: 3 layers flat in segments, count decodable
Dyck-2, L32 D12	adjacent opener at 1 layer; blocks to skip grow with depth	BOS available	~3-4	4.6: block dependence shrinks with depth
torus, T = 1024	retrace; ~4 segments at 4 layers	not found	> 4, growing with segments	4.1-4.4

3.6 Assumptions and limits of the theory

- Route A's depth-2 construction is an upper bound on the depth needed and an existence statement; it says nothing about what SGD finds. Route B's reach is

measured, not derived; the per-layer hop model is a SPECULATION about the mechanism, consistent with the attention and probe data.
- The two routes are not exhaustive (a model can mix them, as on the ring at 2 layers and on Dyck).
- The sensitivity argument transfers Hahn and Rofin's Boolean, uniform-input theorem to structured walks; it predicts a direction (long gaps harder), not a threshold.
- Linear probes bound what attention can read linearly from LN(h); a nonlinear code could exist undetected. The torus models' retrieval failure at long gaps (attention mass 0.00-0.01 on the cell) does not depend on the probes.

---

4. Checks run (ANALYSIS; post hoc, eval-only, CPU)

4.0 Reproduction (every check passed before any new readout)

- Torus (`torus_depth.py`): the FACTORIAL eval stream re-generated (env seed 10000, np seed 1234 + s, 100 walks); overall accuracy per run equals `FACTORIAL_{L1,L4,LOOP}_STRATA.json` "all" to 4 decimals for all 16 runs; the re-implemented forward (to expose attention and residuals) equals the model's logits (max |diff| 0.0).
- Ring (`ring_depth.py`): the CANCEL eval stream (train_cancel.evaluate: np seed 0, env seed 10000, 200 walks); overall accuracy equals each run's `eval.json` "128" for all 128 runs (asserted < 2e-3; printed values equal to 4 decimals). Only fresh seeds 2-7 are read below.
- Dyck (`dyck_distance.py`): A2f recomputed per run; arm means equal `DYCK_MDEPTH_RESULTS.md` (RoPE 0.648 / 0.889 / 0.964 / 0.982, PoPE 0.640 / 0.850 / 0.946 / 0.971, MapWM 0.996 / 1.000).

4.1 Torus: which revisits depth buys (`torus_depth.py`, `summarize_torus.py` -> `out/torus_depth_out.txt`)

Per revisit: s = the last visit to the cell; *retrace* = the current run reverses the previous one and has not overshot it (then s = t - 2 cur, the one-level retrace rule of the floors); *segments* = maximal single-action runs among moves s+1..t. Mean of per-seed accuracies (n = 4 seeds, range in brackets); share = fraction of all revisits.

revisit class	share	index 1L	index 4L	loop x4	path 1L
short gap, retrace	0.680	0.833 (0.79-0.86)	0.996 (0.99-1.00)	1.000	1.000
short gap, not retrace	0.148	0.475	0.699 (0.51-0.77)	0.655	1.000
... 3-4 segments	0.031	0.477	0.926 (0.74-0.99)	0.977	1.000
... 5-8 segments	0.056	0.476	0.735 (0.52-0.81)	0.645	1.000
... 9-16 segments	0.052	0.474	0.561 (0.38-0.64)	0.499	1.000
... 17+ segments	0.009	0.472	0.493	0.502	1.000
long gap (>= 128)	0.090	0.480	0.458	0.507	1.000
wrap-only	0.083	0.482	0.457	0.509	1.000

Gap in moves (plain revisits): index 4L / loop 1.000 / 1.000 at 2-7, 0.978 / 0.986 at 8-31, 0.614 / 0.551 at 32-127, 0.458 / 0.506 at 128-511. One layer: 0.935 at 2-7, 0.681 at 8-31, floor beyond.

Segments at a fixed gap (plain revisits, gap 32-127 moves; pooled over seeds):

segments spanned	n	index 1L	index 4L	loop x4
2	240	0.412	0.874	0.996
3-4	503	0.483	0.832	0.911
5-8	4565	0.464	0.688	0.575
9-16	6569	0.474	0.560	0.499
17+	1202	0.472	0.493	0.502

At gap 8-31 the same: 4L 0.993 / 0.940 / 0.818, loop 1.000 / 0.987 / 0.768 at 2 / 3-4 / 5-8 segments.

Accounting for "just above retrace". The short-gap retrace floor is (0.680 x 1 + 0.148 x ~0.5) / 0.828 = 0.911 (measured 0.912). Four index layers: (0.680 x 0.996 + 0.148 x 0.699) / 0.828 = 0.943 (measured 0.943). Depth adds the exact retrace rule plus partial returns over 3-8 segments; that is the whole of its short-gap margin over retrace.

Long-gap and wrap-only revisits span many segments: 98.5% of long-gap revisits span 17 or more (median 37, minimum 11), and 99% of wrap-only revisits (median 65). They are *below the constant floor* (0.458 / 0.457 vs 0.515 / 0.517) with high NLL (2.1-4.8 in `FACTORIAL_L4_STRATA.json`): the deep models predict a specific wrong object rather than abstaining to blank.

4.2 Torus: where index attention looks (same script)

Attention at the revisit query (token a_t), per pass and head (h0 / h1), mean over revisits and seeds. *same* = mass on earlier observation tokens of the same cell (uniform-attention chance 0.001-0.003); *last* = mass on the last visit's observation and arrival tokens.

model, final pass	retrace: same / last	short, not retrace: same	long gap: same	wrap-only: same
index 1L	0.05 / 0.10 (h0), 0.06 / 0.12 (h1)	0.00	0.00	0.00
index 4L, layer 4	0.55 / 0.56, 0.58 / 0.58	0.22 / 0.22	0.01 / 0.01	0.00 / 0.00
loop x4, pass 4	0.50 / 0.52, 0.49 / 0.50	0.20 / 0.20	0.00 / 0.00	0.00 / 0.00
path 1L	0.83 / 0.70, 0.76 / 0.65	0.82 / 0.76	0.81 / 0.74	0.66 / 0.58

In the deep index models retrieval happens in the last pass only (layers 1-2: same-cell mass <= 0.04 everywhere; layer 3 0.21-0.33 for retrace), and on retraces all of it is on the last visit (same = last): they find the mirror point, not the cell. Split by outcome, the final pass's max-over-heads same-cell mass is 0.75 on correct and 0.30 on wrong retrace predictions (4L), 0.45 vs 0.21 on other short gaps, and 0.02 vs 0.02 on long gaps: at long gaps there is no retrieval to get right or wrong.

4.3 Torus: no global count (`torus_probe.py` -> `out/torus_probe_out.txt`; `torus_depth.py` probes)

Ridge probes from the residual stream at move tokens (train walks 0-29, test 30-39, t >= 16; R^2 out of sample, mean of 4 seeds, x and y averaged). Best pass shown.

target	index 1L	index 4L	loop x4	(ring index 3L, p_plus 0.5, for comparison; 4.5)
unwrapped displacement P_t	-0.06	0.10	-0.01	0.60
normalised count P_t / (t+1)	-0.03	0.21	0.06	0.69
1 / (t+1)	-0.01	0.78	0.74	0.89
displacement over the last 4 moves	0.90	0.96	0.95	0.97
... last 16	0.47	0.93	0.86	0.86
... last 64	0.09	0.77	0.56	0.70
... last 256	-0.01	0.37	0.14	--
torus phase cos / sin (2 pi P / 64)	-0.02	-0.01	-0.01	(ring phase 0.04-0.10)

The deep torus models know elapsed time (1/t) but carry no normalised count; they carry a recent-displacement code whose error in cells grows with the window (rmse = sd x sqrt(1 - R^2) with the window sds of `out/walk_stats_out.txt`: 0.5 cells at 4 moves, 1.8 at 16, 7.1 at 64, 25 at 256 for 4L). That is route B's signature. (The path model's residual stream carries no position either, as expected: its position is in the rotation.)

4.4 Torus: run length moved at evaluation (`torus_runlen.py` -> `out/torus_runlen_{L1,L4,LOOP,PATH}_out.txt`)

The stored FACTORIAL models re-evaluated (no training) on walks whose run length is fixed at 2, 5 or 10 moves instead of drawn from 1-10 (within the training support of run lengths, but a shift in walk statistics; held-out map; 60 walks per seed for 1 layer, 30 for 4L / loop, 10 for path). Plain revisits, pooled over the 4 seeds; cells with >= 50 revisits. Rows: segments spanned; columns: gap in moves (2-15 / 16-63 / 64-127 / 128+). Index 4 layers (loop in brackets):

condition	2 segments (retrace)	3-4	5-8	9-16	17+
training walk (1-10)	0.999 / 0.978 (1.000 / 1.000)	0.978 / 0.925 (0.969 / 0.978)	0.875 / 0.759 / 0.527 (0.838 / 0.675 / 0.418)	-- / 0.635 / 0.539 (0.512 / 0.505)	0.440 / 0.459 (0.481 / 0.509)
runs of 2	0.999 (1.000)	0.934 / 0.898 (0.884 / 0.966)	0.836 / 0.783 (0.744 / 0.621)	0.702 (0.526)	0.580 / 0.524 / 0.461 (0.483 / 0.489 / 0.493)
runs of 5	0.997 / 0.979 (1.000 / 1.000)	0.997 / 0.912 (0.999 / 0.982)	0.803 / 0.819 (0.683 / 0.511)	0.701 / 0.654 (0.517 / 0.503)	0.551 / 0.465 (0.509 / 0.509)
runs of 10	1.000 / 0.978 (1.000 / 0.995)	0.867 / 0.662 (0.939 / 0.382)	0.715 / 0.605 (0.556 / 0.503)	0.557 / 0.449 (0.511 / 0.494)	0.453 (0.506)

(Each cell lists the populated gap columns left to right.) Overall plain accuracy of 4L: 0.901 on the training walk, 0.843 / 0.858 / 0.841 with runs of 2 / 5 / 10; path 1.000 in every cell of every condition.

- *Two-segment returns are gap-invariant for deep index models*: retraces at gaps of 16-63 moves (runs of 10) score 0.978 / 0.995. One index layer's retrace is window-limited instead: 0.83-0.93 at gaps of 2-15 moves and 0.44-0.46 at 16-63 in every condition (`torus_runlen_L1_out.txt`).
- *Segments set the level, the gap trims it*: down a column of fixed gap (16-63 moves) 4L falls from 0.98 (2 segments) to 0.87-0.93 (3-4), 0.72-0.80 (5-8), 0.64-0.70 (9-16) and 0.58 (17+, runs of 2); within a segment bin across gaps it moves by 0.02-0.2 (0.35 in one cell of 55 revisits). The loop is sharper: 0.88-1.00 up to 4 segments at gaps below 64, at or near the floor (0.50-0.53) from 9 segments at any gap.
- Shorter runs put more segments into each gap and lower overall accuracy (0.843 with runs of 2), as route B predicts. Runs of 10 also lower it (0.841), through 3-8 segment returns at gaps of 16-127 moves (4L 0.60-0.87), so the gap is not irrelevant.
- Reading: route B with a reach of about 4 segments at 4 layers (partial to 8), plus a weaker gap dependence. Because the conditions shift the walk distribution, this is suggestive; E1 (Section 6) trains on each condition.

4.5 Ring: rules, segments and a count (`ring_depth.py`, `summarize_ring.py` -> `out/ring_depth_out.txt`)

Classes: *retrace* as on the torus; *lap* = the displacement over the last 32 moves is 0 mod 32 (a fixed one-lap offset lands on the cell; the clock); *neither* = the rest, split by segments. Fresh seeds 2-7, mean (range).

p_plus 0.5:

class	share	index 1L	index 2L	index 3L	path 1L
retrace	0.716	0.797	0.987	1.000	1.000
lap, not retrace	0.035	0.497	0.977	1.000	1.000
neither, 3-4 segments	0.127	0.487	0.891 (0.72-0.94)	0.997	1.000
neither, 5-8	0.097	0.489	0.802 (0.55-0.87)	0.993	1.000
neither, 9-16	0.025	0.485	0.678 (0.48-0.75)	0.972	1.000
wrap-only	0.118	0.503	0.819	0.989	1.000

p_plus 0.75 / 0.9 (index 1L): retrace 0.692 / 0.677, lap 0.987 / 0.994, neither 0.487 / 0.592. p_plus 1.0: every revisit is a lap revisit, 1.000 at every depth.

- One index layer learns exactly retrace and (when it is common) the clock, and nothing else.
- Two layers compose segments, decaying with their number (route B).
- Three layers are nearly flat in segments, and the residual stream at layer 2 carries a global count: P_t R^2 0.60, P_t / t 0.69, 1/t 0.89 (one layer: 0.09 / 0.18 / 0.14; two: 0.40 / 0.60 / 0.67). At p_plus 0.5 there is no drift, so P_t is not a proxy for t here. Route A is found on the ring.

- Its last-layer attention reaches the cell on wrap-only revisits (same-cell mass 0.36 / 0.38) and other non-local revisits (0.17 / 0.18), where 1-2 layers do not (0.00-0.14). The probes find no linear phase code (cos / sin R^2 <= 0.10), so how the count is matched across the wrap is not identified.

The non-monotone a1(p) (descriptive). With per-class accuracies at each p, a1 = sum over classes of share x accuracy: p_plus 0.5: 0.716 x 0.797 + 0.035 x 0.497 + 0.249 x 0.487 = 0.709 (observed 0.710); 0.75: 0.493 x 0.692 + 0.179 x 0.987 + 0.329 x 0.487 = 0.678 (0.677); 0.9: 0.214 x 0.677 + 0.530 x 0.994 + 0.256 x 0.592 = 0.823 (0.824). The dip at 0.75 is retrace's share falling faster than the clock's share rises (an identity given the class accuracies; the explanatory content is that one layer implements these two rules and only these). Registered as "no mechanism sentence"; this is a post hoc decomposition and should be presented as such or not at all.

4.6 Dyck-2: blocks to skip (`dyck_distance.py`, `dyck_crosstab.py` -> `out/dyck_distance_out.txt`, `out/dyck_crosstab_out.txt`)

Closing accuracy (A2f's per-prefix renormalised legal-closer probability, closer-feasible prefixes of the registered L32 D12 cell) by the number of complete balanced blocks between the top opener and the prefix; mean over 8 seeds.

blocks to skip (n prefixes)	RoPE 1L	2L	3L	4L	MapWM 1L
0 (3682)	0.998	0.999	0.999	0.999	0.999
1 (5839)	0.640	0.925	0.984	0.991	0.997
2 (572)	0.641	0.778	0.885	0.934	0.993
3-4 (161)	0.613	0.760	0.860	0.925	0.993

Cross-tab (`dyck_crosstab_out.txt`): at a fixed distance of 16-31 tokens, 3 layers score 0.966 / 0.896 / 0.853 at 1 / 2 / 3-4 blocks; at a fixed 1 block, 2 layers score 0.970 / 0.948 / 0.835 at distance 2-7 / 8-15 / 16-31. Both blocks and distance matter for the index models; the path model is flat in both (0.987-0.998). Reading: a mixture of the routes, consistent with but not specific to the account. 93% of scored prefixes need at most one block, which is why three layers already reach 0.964 in aggregate. By stack depth, one layer falls from 0.88 at depth 1 to 0.60 at depth 9-10; two layers or more are nearly flat.

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5. Proposed paper text (Section 5)

Replace the paragraph "What depth buys index attention" and the closing positioning paragraph with the following (labels for the editor: the first two sentences restate registered numbers; the rest is post hoc analysis and theory and must be marked "(analysis)" in the paper as the draft already does for floors):

> **Why the rate differs by task.** A revisit asks for an earlier observation at the same cell, that is, for a key s whose moves since s sum to zero. Path integration puts the prefix sum into the rotation, so one layer compares P_t - P_s directly at any gap. Index attention sees only t - s, and must compute the predicate in its residual stream. It can do so in two ways. A global count (uniform attention over the moves, rescaled, then matched) needs only two layers in principle (Liu et al.

> 2023; Yao et al. 2021), but its precision must grow with the walk length and the target is sensitive to every move since the last visit (Hahn and Rofin 2024). A local rule needs no count but only reaches returns that span a few direction runs: undoing the last run (retrace) or, when the walk never reverses, a fixed one-lap offset (a clock). Each extra layer reaches returns across more runs. The stored models show both routes (analysis; App. F). One index layer learns exactly retrace and the clock: on the ring it scores 0.68-0.80 on retraces, 0.99-1.00 on one-lap returns once they are common, and the floor on everything else, so at p_plus 1.0, where every revisit is a one-lap return, it matches the path layer. Three index layers on the ring are flat in the number of runs a revisit spans and carry a global count in the residual stream (linear-probe R^2 0.69 for P_t / t, against 0.18 at one layer). On the torus at T = 1024 no index model carries a count (R^2 <= 0.21); four layers or the loop solve retraces (0.996 / 1.000) and decay with the number of runs spanned (0.93 / 0.98 at 3-4 runs, 0.74 / 0.65 at 5-8, 0.56 / 0.50 at 9-16), at a fixed gap as well as overall, and their last layer attends to the remembered cell for retraces (0.55-0.58 of its mass) but not at long gaps (0.00-0.01; path layer 0.81). Nearly all long-gap revisits span 17 or more runs (median 37). Depth therefore buys index attention a local reach measured in direction runs; the torus exchange rate is "more than four" because its long-range revisits lie beyond that reach and the global count is not found.

Figure suggestion (Fig. 2a companion or App. F): accuracy against runs spanned (2, 3-4, 5-8, 9-16, 17+) for index 1L, 4L, loop and path on the torus, with the ring's 1L / 2L / 3L beside it; caption numbers from Sections 4.1 and 4.5.

Two sentences for Limitations: "The explanation of the exchange rate is post hoc: the class splits, attention masses and probes were computed on stored checkpoints after the registered results were read, and the reach of route B was measured only at one and four layers on the torus. The routes are distinguished by probes and by accuracy against runs spanned, not by an intervention."

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6. Proposed GPU experiments (each with a registered prediction)

Costs from stored logs: torus T = 1024 at 900 epochs ~3.1 s/epoch per index layer (L1 50 min, L4 3.2 h per run, two jobs per GPU); ring T = 128 at 300 epochs 9 min (1 layer) to 25 min (3 layers).

E1. Run length at fixed T (decisive between routes; cheapest). Torus T = 1024, walks with run length fixed at 2, 5 or 10 moves (or drawn 1-3 / 8-10), index RoPE 4 layers and path 1 layer, trained and tested on each. Prediction (route B): the accuracy-against-runs-spanned curve of the index model is the same in every condition, so short runs (more segments per gap) lower overall accuracy and long runs raise it; route A predicts dependence on T and gap only. Path flat at 1.000. 3 conditions x 2 arms x 4 seeds: ~12 x 3.2 h + 12 x 0.8 h ~ 48 GPU-h; at T = 512 ~ 24 GPU-h. A CPU eval-only version on the stored models is Section 4.4 (distribution-shifted, so it can only suggest).

E2. Reach against depth. Torus T = 1024, index RoPE at 1, 2, 3, 4, 6, 8 layers, n = 4, all retrained in one batch (rule 12). Readout: the largest runs-spanned bin with accuracy >= 0.9, per depth. Linear reach predicts ~L + 1 runs at L layers; doubling predicts

~2^L. Decides whether any practical depth reaches the torus (long gaps need 17+). Cost: 24 layer-units per seed x 47 min x 4 seeds ~ 75 GPU-h; trimmed to {2, 4, 8} ~ 44 GPU-h.

E3. Give index attention a normaliser. Add a BOS token and Lemma 6's absolute channel log(2T - t) (or t / T) to index RoPE at 2 and 4 layers on the torus. If route A is blocked by the missing normaliser, two layers should then solve long-gap and wrap-only revisits, and probes should find P_t / t; if precision or sensitivity block it, nothing changes. 2 arms x 2 depths x 4 seeds ~ 16 runs ~ 35 GPU-h. A cheap first step: the ring at T = 128 with and without BOS at 2 layers (does the count appear at 2 instead of 3? ~3 GPU-h).

E4. The ring as a function of T. Ring p_plus 0.5, T in {128, 256, 512, 1024} at matched length, with the ring size scaled so the wrap-only share is fixed (N ~ sd of P_T), index 1-4 (6) layers, n = 4. Route A predicts the rate grows with T (precision 1/T, sensitivity ~ gap) and the count probe weakens; route B predicts the rate grows with runs spanned. With E1 this separates the two. ~60 GPU-h full; T in {128, 512} x L in {2, 3, 4, 6} ~ 15 GPU-h.

E5. Dyck at matched depth across D. Dyck at L32 / L64 with D in {4, 8, 12, 16}, index 1-4 layers: does the rate rise with D (blocks to skip) at fixed length? ~20 GPU-h. Lower priority: Dyck mixes the routes (Section 4.6).

Priority: E1 (cheap, decisive, ties to the paper's main exchange-rate claim), then E3's ring step (3 GPU-h), then E2. Each needs a pre-registration and a results-blind code check (rule 29) before launch; pilots on seeds outside the batch.

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7. Caveats

- Everything in Sections 4.1-4.6 is post hoc and eval-only; it describes stored models, all of whose index runs are STALLED at their budget. "Reach" is the reach of these budgets.
- n = 4 seeds on the torus (complete separations only in the registered contrasts); seed-to-seed spread in the 4-layer model is large on non-retrace classes (0.51-0.77).
- Segment and gap are correlated in natural walks; the fixed-gap tables and Section 4.4 reduce but do not remove that.
- Linear probes can miss a nonlinear code; the attention readout (no mass on the cell at long gaps) does not rely on them.
- The ring eval stream overlaps training batch 0 of seed-0 runs (`CANCEL_RESULTS.md` correction); only seeds 2-7 are read.
- Liu et al.'s and Yao et al.'s constructions use absolute positions or a padding token, dense state supervision (Liu), and hard or near-hard attention; our task supervises the state only through retrieval. They bound depth; they do not predict what is learned. Yao et al. (2021) is cited through Liu et al.'s App. A.5 (not in `papers/`).
- Hahn and Rofin's bounds are for Boolean functions under uniform inputs; we use them for direction only.

Recurrence: what looping a shared block does to a path-integrated model

Expansion note for `docs/paper/MAIN_DRAFT1.md` Secs. 5, 6.1 and 6.3 (2026-10-20). Labels: **ESTABLISHED** (registered, results file named; a CORRECTED / AUDIT block wins), **ANALYSIS** (computed here on stored checkpoints, CPU, inference only unless stated; script named; all post hoc), **THEORY** (statement, assumptions, the check that could break it), **SPECULATION**. Scripts and outputs: `docs/paper/expansion/scripts/recurrence/` (`out/` holds every JSON and text table quoted below). No GPU used, nothing committed, no existing file edited.

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0. Summary

Key insight. In MapFormer the phase theta is computed once, from the token embeddings, before any block runs (`model.py` `MapFormerWM.forward`, `model_looped.py` `MapFormerWM_Looped.forward`). A loop or extra layers therefore cannot change the position code's class: rank-2 plain (A), rank-2 looped x4 (L) and rank-2 at four real layers (L4) draw their code from exactly the same family (`token_emb`, `W_in`, `W_out`, `omega`). What extra passes change is the **reader**: in one layer the query at an action token depends only on the action's identity and the key only on the token's identity, so position reaches the score only through theta differences. From the second pass on, queries and keys carry context, and the model can read the same code in ways one pass cannot. The stored checkpoints show this is what happens (ANALYSIS):

1. **The loop and depth do not find a better code.** A one-pass, position-only reader fitted to each model's own theta (an idealised single head) scores 0.893 (L) and 0.897 (L4) against 0.866 for A (perm p 0.28 / 0.27) and 1.000 for rank 4 (p 0.0002). On long-gap revisits the code supports 0.61 / 0.65 / 0.56 / 1.00, on wrap-only revisits 0.41 / 0.45 / 0.45 / 1.00.
2. **They read it far better.** The models beat their own code's one-pass reader by +0.32 / +0.34 on long-gap and +0.34 / +0.44 on wrap-only revisits. For the one-pass models the margin is +0.11 / +0.11 (A) and -0.005 / -0.02 (rank 4) (gap vs A's gap: p 0.0025 / 0.0045 long, 0.0002 / 0.0002 wrap). A one-layer model is almost exactly its code (A +0.028 overall). The looped model is a 4-step computation, not an improved 1-layer model: run with one pass, the same weights score 0.381, below the 0.51 always-blank floor. Retrieval of the right cell appears only at passes 2-4: same-cell attention mass rises 0.03 -> 0.36 -> 0.46 -> 0.52 (head 0) while attention entropy falls 6.1 -> 2.4 nats.
3. **Rank 4's code is never found.** Classifying each step map by its omega-weighted geometry: a two-dimensional, cancelling ("2D map") code in 1/8 A, 0/8 L, 1/8 L4 and 8/8 rank-4 runs (aids pooled vs rank 4: Fisher p 1.2e-5). What the aids do reach, in 8/16 runs against 0/8 for A (Fisher p 0.022, post hoc), is a **cancelling but one-dimensional** code: the x and y phase steps are collinear (sigma_2/sigma_1 <= 0.002), so the code reads a single coordinate, x + kappa y (|kappa| 0.62-1.72; 1.38-1.42 in all four L runs). That is a clock-free code with a lost map dimension, a third end state beside the "drifting" (clock-coupled) code of failing rank-2 runs and the 2D code of rank 4.

How this strengthens the paper. The current text says rank 2 fails "by search, not capacity" (Sec. 6.1). The loop and depth results, read this way, sharpen that claim instead of muddying it. Four times the computation (L, same parameters and initialisation as A) or four times the parameters and computation (L4) lift accuracy a long way (registered: +0.079 and +0.096, perm p 0.0034 / 0.0012). The gain comes from computation downstream of the code, and the code itself stays a rank-2 failure. So the deficit is local to the content-to-angle map: nothing downstream repairs it, and spending rank is the only intervention that finds the 2D code. It also gives Sec. 5's exchange rate a second axis. Extra passes add accuracy exactly where the code carries information one pass cannot read: index code, short gaps only (+0.17; long and wrap unchanged); rank-2 path code, long gaps and wrap (+0.26-0.33, +0.19-0.33); rank-4 code, nothing left to add.

What to add. (a) One paragraph in Sec. 6.1 after "The rank-2 solution exists and is held" (text in Sec. 6.1 of this note). (b) One sentence in Sec. 5 tying the index loop to the path loop. (c) An appendix section with the reader, per-pass and geometry tables. (d) A registered in-batch rank x sign x passes factorial when GPUs return (Sec. 7, E1). The LOOP_RANK contrast should **not** be presented as "a search change at constant capacity": parameters and initialisation are identical, but the function class of the reader is not (Sec. 4).

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1. Established facts (results files are authoritative)

fact	numbers	status	file
Loop and depth partly recover rank 2 (torus, T = 1024, 900 ep, n = 8, one batch; A and C reused, A s0 reproduced bitwise)	SOLVED (final-5% loss < 0.05) and accuracy: A r2 1 layer 0/8, 0.894; **L r2 + loop x4 2/8, 0.973** (vs A: Fisher 0.47, perm **0.0034**, +0.079 CI [+0.026, +0.131]); **L4 r2 4 layers 5/8, 0.990** (Fisher **0.026**, perm **0.0012**); C r4 8/8, 0.998. L4 - L +0.017 (perm 0.0009). Holm 0.0034 / 0.0028 / 0.0028	registered, verdict **UNMEASURED** (needed L >= 4/8 SOLVED); cutoff sensitivity: at 0.08 L 6/8, L4 8/8	`LOOP_RANK_RESULTS.md`, `LOOP_RANK_ANALYSIS.txt`
Three loss regimes	A 0.05-0.68; L 0.045-0.127; L4 0.011-0.070; C 0.002-0.010: the aids leave A's regime, never enter C's	registered secondary	`LOOP_RANK_RESULTS.md`
L and A identical in parameters and initialisation	204,373 params, max weight difference 0.0 at every seed (gate); L4 799,189	registered gate	`LOOP_RANK_PREREG.md`
Strata (T = 1024)	short / long / wrap-only: A 0.950 / 0.671 / 0.557; L 1.000 / 0.927 / 0.746; L4 1.000 / 0.990 / 0.887; C 1.000 / 0.995 / 0.977	registered secondary	`LOOP_RANK_STRATA.json`, `RANK_MI_STRATA.json`

fact	numbers	status	file
Other lengths	T = 512: A 0.934, L 0.996, L4 0.998, C 1.000; T = 2048: 0.819 / 0.825 / 0.868 / 0.973	registered secondary	`LOOP_RANK_ANALYSIS.txt`
Twice the budget does not rescue plain rank 2	1800 ep: r2 0/8 (0.908) vs r4 7/8 (Fisher 0.0014); 5/8 r2 still descending; the L / L4 arms at 1800 ep were never run (part 2 stopped)	registered	`LOOP_RANK_E1800_P1_RESULTS.md`
Index attention: loop and depth help, do not catch up (torus T = 1024, n = 4, `--fast-attn`, seeds 500-503)	path 1L 1.000; index 1L 0.719, 4L 0.860, loop x4 0.864 (each 0/4, STALLED; complete separations p 0.029). Strata short / long / wrap: 1L 0.769 / 0.480 / 0.482; 4L 0.943 / 0.458 / 0.457; loop 0.938 / 0.507 / 0.509; floors 0.508 / 0.515 / 0.517	registered	`FACTORIAL_RESULTS.md`, `FACTORIAL_*_STRATA.json`
Short-walk torus (T = 128, 300 ep, n = 3)	loop x4 matches 4 real layers at a quarter of the parameters (index horizon 9-16 -> 17-32); on MapFormer L1 0.948 -> loop 0.994 at long intervals	pilot (n = 3)	`LOOPED_PILOT.md`
Match-Query: loop helps path integration	loop vs no loop **+0.346 unpaired** (t 3.75; loop pooled over two batches 0.803 +/- 0.200, 1/16 failures). Per-seed drift between batches 0.185; the paired +0.315 interaction and "never fails" are withdrawn; the pooled contrast is asymmetric (loop 16 runs over 2 batches, no-loop 8 runs over 1); loop matches 3 real layers (+0.032, t 0.30); refining theta per pass adds nothing (+0.090, t 1.56; learned gate	g	<= 0.14)
Match-Query rank x loop (one batch, n = 8)	r2 0.416, r4 0.421, r2 + loop 0.833, r4 + loop 0.986 (8/8 >= 0.941); interaction +0.149	unregistered, one batch	`MQ_RANK_2X2.md`
No per-position router can pay (Match-Query)	loop-count curve flat above k = 3; oracle router +0.007 vs seed sd 0.152	analysis	`LOOP_DEPTH_STRATA.md`

fact	numbers	status	file
Loop beats Level 1.5 under action noise; loop-count sampling makes one pass work	+0.138 / +0.205 at p = 0.10 / 0.25 vs Level15 +0.023 / +0.004 (n = 3, found in a control column); sampled counts {2..6}: count-vs-accuracy spread 0.178 -> 0.001, 0.998 at one pass; the loop's training-length win on the clean torus is convergence (loss-matched +0.006)	unregistered, n = 3-5; needs replication	`.claude-memory/project_loop_and_correction.md`, `L15_LOOP_2X2.md`, `MQ_NOISE_2X2*.md`
Loop x path integration on parity is additive, not super-additive	interaction +0.028 / +0.003 / +0.007 at L = 64 / 128 / 256 (inside MDE); the loop matches or beats 3x its parameters	unregistered	`.claude-memory/project_loop_and_correction.md` (ALGORITHMIC / FRONTIER)

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2. Design facts that decide how the contrasts can be read

- **The code is computed once.** `MapFormerWM.forward` computes `delta = action_to_lie(token_emb(x))`, `cos, sin = path_integrator(delta)` and hands the same `cos, sin` to every layer. `MapFormerWM_Looped.forward` does the same and applies `layers[0]` `n_loops` times. So A, L and L4 share the code family exactly. L4's three extra blocks never feed theta.
- **L vs A is clean in parameters, initialisation, data and recipe** (registered gate). It is **not** clean in the function class. L computes `B(B(B(B(x))))` with the same weights. It cannot, in general, express A's function (the residual block cannot be switched off for passes 2-4 without switching off pass 1), and it can express functions A cannot (four rounds of attention). "Same parameter count" is not "same capacity" for the reader. It is the same capacity for the code.
- **The loop applies at evaluation as a runtime knob** (`n_loops`), so the per-pass readouts below are exact: the readout after pass j of L *is* L run with `n_loops = j`. For L4, an intermediate readout is a logit lens (out_norm + out_proj on an intermediate residual; the head was trained only on layer 4).
- **Refining theta per pass was tested and declined by the optimiser** (`REFINE_RESULTS.md`; action-noise version in the memory file). Every positive loop result is therefore a result about re-reading a fixed code.

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3. Theory

T1. A one-pass model reads its code with a fixed, token-typed kernel (THEORY, exact for the code in `model.py`)

Statement. In a one-layer MapWM, the pre-softmax score between the query at position i and the key at position j, head h, is

s_h(i, j) = sum_c |q_c(x_i)| |k_c(x_j)| cos(theta_c(i) - theta_c(j) + psi_c(x_i, x_j)) / sqrt(d_head),

where q_c, k_c are 2-D blocks of W_q LN(e(x_i)), W_k LN(e(x_j)), and psi_c is the angle between them. Both depend only on the token **identities** x_i, x_j. Position enters only through theta(i) - theta(j). The output at i is a function of x_i and of the softmax-weighted mixture of values V(x_j), which also depend on identity alone. So for scored revisit queries (x_i is an action, j ranges over earlier observations), a one-layer model is a **position-only reader of its own code**: a channel-weighted cosine kernel with token-type-dependent weights and phase offsets, a soft vote over the observations it lands on, and an MLP on the vote.

Consequence. A one-layer model's revisit accuracy should be close to that of the best reader of this form fitted to its own theta. More heads and the MLP give a small margin. **Check (ANALYSIS, `phase_reader_soft.py`):** model minus reader overall is +0.028 for A and -0.002 for rank 4 (Sec. 5.3). The check could have failed; it did not.

T2. Extra passes enlarge the reader, not the code (THEORY)

Statement. From pass 2 on, q and k at a position depend on the residual stream, i.e. on context gathered by earlier passes, so both the per-channel weights |q_c||k_c| and the offsets psi_c become context-dependent. The values can carry content retrieved earlier (relay). The code theta is unchanged (Sec. 2). Extra passes can therefore extract from theta, together with context, information that no fixed token-typed kernel can extract. They cannot change which positions theta distinguishes, but they can resolve theta's ambiguities with content.

Predictions. (P1) The code of L and L4 is no better than A's when read by the T1 reader. (P2) L and L4 exceed their own reader by much more than A does. (P3) The gain from extra passes sits in the strata where the code carries information a fixed kernel cannot read. For an index code (no map information) that means short gaps only, where local content matching works; for a partial path code it means long gaps and wrap. (P4) Nothing is gained where one pass already reads a complete code (rank 4, ceiling). **Checks:** P1 holds (reader +0.027 / +0.031 over A, p 0.28 / 0.27; long +0.05 / +0.10, p 0.58 / 0.32; wrap -0.04 / +0.01). P2 holds (Sec. 5.3: p 0.0025 / 0.0045 long, 0.0002 wrap). P3 holds across batches (index loop: short +0.17, long +0.03, wrap +0.03, `FACTORIAL`; rank-2 loop: short +0.05, long +0.26, wrap +0.19, `LOOP_RANK`) and within single models (Sec. 5.7: the index loop's long-gap and wrap-only accuracy sits at 0.50-0.52 after every pass while short gaps climb 0.41 -> 0.94; the rank-2 loop's long gaps climb 0.36 -> 0.93). P4 is untested on the torus (no rank-4 loop arm; MQ_RANK has one on Match-Query: r4 + loop 0.986 vs r4 0.421, so on Match-Query rank 4's one-pass code is not complete).

T3. Why the aids leave A's regime but never enter rank 4's (THEORY with a partial check)

A rank-4 model reads a complete, wrap-consistent 2D code in one pass (reader 1.000 on every stratum), so its loss floor is set by optimisation alone (0.002-0.010). The aids' accuracy is the code's information plus what context can add. Their code does not resolve wrap-only revisits (reader cell top-1 0.04-0.12 on wrap, `phase_code.py`), and a revisit whose unwrapped position is new can be recognised only by content (the local observation

sequence matches one seen in another unwrapped copy of the map). That is unreliable for a reader of any depth. Prediction: the aids' residual loss concentrates in wrap-only and long-gap revisits. **Check (partial):** wrap-only accuracy is 0.747 (L) and 0.887 (L4) against 1.000 on short gaps; the aids' unsolved runs sit at losses 0.05-0.13 with short-gap accuracy 1.000. The loss split by stratum was not computed; see E2.

T4. Search on the code: what the loop does and does not do (THEORY + ANALYSIS + SPECULATION)

The search theory for the rank axis (`RANK_THEORY.md`, `SEARCH_THEORY.md` S6) says that at per-head rank D a clock-free channel cannot also be a full D-dimensional map unless the common mode is removed collectively (R1: z_x + z_-x = 0). The aids' geometry fits this inventory (ANALYSIS, `geometry.py`):

- **A (one pass)**: 7/8 runs end **drifting** (common mode 0.34-2.9 of the map step): the clock-coupled failure.
- **L, L4**: half end drifting (4/8, 3/8), half **cancelling but one-dimensional** (4/8, 4/8): the common mode is gone (0.01-0.05) but the x and y map steps are collinear (sigma_2/sigma_1 0.000-0.002; latent ratio 0.000-0.006). The code is x + kappa y with |kappa| 1.38-1.42 in 4 L runs and 1 L4 run, 0.62-1.72 in the other three.
- **Rank 4**: 8/8 cancelling and two-dimensional (sigma_2/sigma_1 0.72-0.97).

SPECULATION (to be tested, E2). Extra passes make a 1D clock-free code *good enough*. On an unwrapped plane a 1D projection with an irrational slope is injective, but nearly colliding cells exist: for kappa near sqrt 2 the displacements (3, -2), (7, -5), (17, -12) leave phase residuals of only 0.17, 0.07 and 0.03 x-steps. One fixed kernel would have to be very sharp to separate them; a reader that can also check content can reject them. So with a deep reader the collapsed code is a usable basin, while one pass cannot use it (A never lands there: 0/8). Read this way, the aids *change* which basin the step-map search ends in rather than helping it find rank 4's. This is a compensation basin: the reader absorbs the error signal that, at one pass, would keep pushing the code. The registered training curves fit the picture (ANALYSIS, `training_curves.py`). The aids leave the initial plateau early (first epoch below loss 0.5: L 42-67, L4 19-78, against A 183-508 or never, 3/8, and rank 4 83-294) and then creep down without the sharp drop rank 4 shows (rank 4: loss 0.5 -> 0.05 in 57-228 epochs, after a plateau). Not tested: whether the 1D code forms before or after the reader absorbs the short-gap revisits.

What the naive "more gradient paths" account predicts, and why it is not the mechanism (ANALYSIS, `grad_at_init.py`). At initialisation, with weights and batches identical, the 4-pass forward multiplies the step map's gradient norm by 2.9 (2.4-3.8) and the block's by 3.4 (3.0-3.9). Adam is invariant to that scale, and the step-map gradient's batch-to-batch coherence is unchanged (0.46 -> 0.41). The loop does not deliver a stronger or more consistent signal to the code at the start. Its effect on the code's search, if any, runs through the reader (above).

Summary of the theory

code \ reader	one pass	four passes (loop or depth)
none (index RoPE)	short gaps only (0.769 / 0.480 / 0.482)	short gaps better (0.938-0.943); long and wrap still at the floor (FACTORIAL)

code \ reader	one pass	four passes (loop or depth)
rank-2 path code (drifting or 1D)	what a fixed kernel reads (A 0.950 / 0.671 / 0.557; reader 0.940 / 0.558 / 0.448)	context-dependent reading: long 0.93-0.99, wrap 0.75-0.89; code unchanged
rank-4 path code (2D)	complete: 1.000 / 0.995 / 0.977 (reader 1.000)	nothing to add on the torus (untested)

4. Question (2): are loop and depth search aids? How clean is "same parameters, more solved"?

- **As a test of the code's capacity, yes, cleanly.** A, L and L4 have the same code family (Sec. 2). Neither aid changes what the code can represent, and both lift accuracy a long way (registered). The analyses add what the registered numbers could not: the lift is not a better code (P1) but better reading (P2). So the rank-2 deficit is in the code, and more downstream compute or parameters do not repair it. That is a stronger form of "not capacity" than the paper states: the content-to-angle map at rank 2 can represent a solving code (RANK_PROJ), it is not found from scratch, and giving the rest of the network four times the compute and four times the parameters still does not find it (2D code 1/16 vs rank 4 8/8).
- **As "a search change without a capacity change", no.** The pre-registration framed L vs A that way ("the cleanest contrast available"). L has the same parameters and initialisation but a larger reader function class. Its gain is a reader-capacity gain on a fixed code, not search success on the code. The text should say so; the UNMEASURED verdict stands as registered.
- **Strength.** The registered accuracy contrasts are strong (+0.079, p 0.0034; +0.096, p 0.0012; Holm <= 0.0034). The solved counts are knife-edge (L 2/8 at the registered 0.05 cutoff, 6/8 at 0.08; two spot runs at 0.0528 / 0.0538, `REPRO_AUDIT_nonsign.md` note 5). The reader and geometry readings are post hoc, n = 8 per arm, from one batch, and untested against fresh seeds (rule 6).
- **Depth beats the loop** (L4 - L +0.017, p 0.0009; 5/8 vs 2/8). Both have the same code; L4 has four distinct blocks. Under T2 that is a richer reader. The aids' code classes are the same (4/8 1D each).

5. Analyses run (CPU, stored checkpoints; post hoc)

All on the registered LOOP_RANK arms at T = 1024 (A `runs/rank_mi/p0/Vanilla_s*` -> `runs/rank_matched_e900`, L and L4 `runs/loop_rank/p0`, C `runs/rank_mi/p0/Vanilla_r4mi_s*`), seeds 0-7, held-out map (env seed 10000). Evaluation trajectories use np seed 1234 + s, the registered stream; the readers are fitted on np seed 777 + s.

5.1 Reproduction check (`summarize.py`)

The re-implemented forward (returns every pass's logits and attention) matches each model's own forward exactly on the checked trajectory. The trained-readout accuracy over 100 trajectories matches the stored registered evaluations to max |diff| 3.1e-4 (A), 2.8e-4 (L), 9.4e-5 (L4), 3.1e-5 (C).

5.2 Accuracy and attention per pass (`pass_strata.py`, `attn_offsets.py`)

Accuracy by pass (mean of 8 seeds; L: `n_loops = j`, passes 5-8 beyond training; L4: logit lens after layer j):

arm	stratum	1	2	3	4	5	6	7	8
L	all	0.381	0.887	0.964	**0.973**	0.971	0.954	0.899	0.847
L	short	0.382	0.951	0.999	1.000	1.000	0.987	0.932	0.881
L	long	0.363	0.612	0.875	0.927	0.921	0.871	0.799	0.731
L	wrap	0.393	0.519	0.700	0.747	0.730	0.699	0.666	0.623
L4	all	0.507	0.575	0.935	**0.990**				
L4	long	0.500	0.504	0.757	0.990				
L4	wrap	0.504	0.507	0.601	0.887				

The best eval-time pass count is the trained one (4) on 6/8 seeds and 5 on 2/8 (+0.001-0.003). Past it, accuracy falls (8 passes 0.847, two seeds 0.61-0.62). The loop is a fixed-depth computation, not an iteration to a fixed point. This is the free secondary the LOOP_RANK pre-registration listed and never read. One pass of L's trained weights is below the always-blank floor (0.381 vs 0.51): the looped model is not a better one-layer model.

Same-cell attention (scored revisit queries; mass on earlier observation tokens at the same wrapped cell, head 0 / head 1; "argmax" = the most-attended key is a same-cell token):

arm	pass	mass, all	argmax, all	mass, long	mass, wrap	argmax, wrap	entropy (nats)
A	1	0.47 / 0.46	0.69 / 0.70	0.19 / 0.23	0.02 / 0.13	0.04 / 0.19	1.90
L	1	0.03 / 0.01	0.19 / 0.21	0.01 / 0.00	0.00 / 0.00	0.01 / 0.00	6.06
L	2	0.36 / 0.16	0.60 / 0.42	0.12 / 0.04	0.04 / 0.01	0.15 / 0.05	4.13
L	3	0.46 / 0.32	0.75 / 0.66	0.39 / 0.22	0.24 / 0.16	0.36 / 0.27	2.70
L	4	0.52 / 0.38	0.87 / 0.75	0.52 / 0.36	0.32 / 0.20	0.44 / 0.29	2.43
L4	1	0.01 / 0.01	0.04 / 0.09	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00	6.46
L4	2	0.10 / 0.15	0.18 / 0.25	0.03 / 0.07	0.00 / 0.03	0.01 / 0.06	4.43
L4	3	0.62 / 0.35	0.80 / 0.56	0.41 / 0.18	0.17 / 0.04	0.25 / 0.04	2.49
L4	4	0.78 / 0.79	0.94 / 0.95	0.67 / 0.73	0.50 / 0.54	0.64 / 0.64	0.71
C	1	0.77 / 0.75	0.92 / 0.94	0.75 / 0.71	0.73 / 0.62	0.88 / 0.80	1.25

Pass 1 of both aids is near-uniform over the prefix (6.1-6.5 nats; uniform over ~1000 keys is ~6.5-6.9). It gathers a summary of the whole context, not the revisit. Retrieval forms over passes 2-4 and sharpens (L4's last layer is sharper than rank 4's single layer, 0.71 vs 1.25 nats). Mass on same-cell **action** tokens, which could hold a relayed observation only after an earlier pass, stays at 0.01-0.12 in every pass: no evidence of a relay through earlier arrivals. On wrap-only revisits L4's last layer finds the right cell 64% of the time, from a code whose own one-pass reader finds it 12% of the time.

5.3 Code quality, read without the block (`phase_code.py`, `phase_reader_soft.py`)

Two readers of each model's own theta, fitted on 24 training-seed walks and scored on the 50 evaluation walks: (i) a nonnegative channel-weighted cosine kernel with top-1 retrieval (`phase_code.py`); (ii) the T1 reader, which adds token-type phase offsets on query and key, a key bias and a soft vote over observations (`phase_reader_soft.py`). (i) is reported as cell retrieval, (ii) as observation accuracy, comparable to the model's.

arm	stratum	model	soft reader (ii)	model - reader	reader (i) cell top-1
A	all / short / long / wrap	0.894 / 0.950 / 0.671 / 0.558	0.866 / 0.940 / 0.558 / 0.448	+0.028 / +0.010 / +0.113 / +0.110	0.688 / 0.800 / 0.182 / 0.105
L	all / short / long / wrap	0.973 / 1.000 / 0.927 / 0.747	0.893 / 0.971 / 0.608 / 0.406	+0.080 / +0.029 / **+0.319** / **+0.341**	0.786 / 0.918 / 0.221 / 0.044
L4	all / short / long / wrap	0.990 / 1.000 / 0.990 / 0.887	0.897 / 0.967 / 0.653 / 0.453	+0.094 / +0.033 / **+0.337** / **+0.435**	0.801 / 0.923 / 0.297 / 0.115
C	all / short / long / wrap	0.998 / 1.000 / 0.995 / 0.977	1.000 / 1.000 / 1.000 / 0.997	-0.002 / 0.000 / -0.005 / -0.020	0.999 / 1.000 / 0.999 / 0.988

Permutation tests (8 vs 8): code quality (ii) L - A +0.027 (p 0.28), L4 - A +0.031 (p 0.27); long +0.050 (0.58) / +0.095 (0.32); wrap -0.042 (0.77) / +0.005 (0.93). C - L +0.107, C - L4 +0.103 (p 0.0002 each; long +0.39 / +0.35, p 0.0002). Model-minus-reader margin vs A's: L overall p 0.0036, long 0.0025, wrap 0.0002; L4 0.0031, 0.0045, 0.0002. Reader (i) gives the same verdicts (code L - A p 0.20, L4 - A p 0.15; C - aids p 0.0002).

Observation-token phase step (the leak, |omega * delta(obs)| / |omega * delta(action)|): A 0.385, L 0.168, L4 0.163, C 0.026. The aids reduce the leak in their 1D-code runs (0.004-0.051) but not in their drifting runs (0.17-0.41).

5.4 Step-map geometry (`geometry.py`)

Per seed: omega-weighted x- and y-step phase vectors (antisymmetric parts of E/W and N/S), dim2 = sigma_2 / sigma_1, cancel = common mode / map (not wrap-aware), kappa, and the stored tail loss. Classes (thresholds post hoc; dim2 is bimodal among cancelling runs, <= 0.002 vs >= 0.57, so the 1D call does not depend on the threshold):

arm	2D map (cancel < 0.1, dim2 > 0.3)	1D map (cancel < 0.1, dim2 <= 0.3)	drifting (cancel >= 0.1)
A	1/8 (s1, the run at 0.998)	0/8	7/8
L	0/8	4/8 (s1, s3, s6, s7; kappa -1.38, -1.40, +1.42, -1.42)	4/8
L4	1/8 (s1, the best L4 run, tail 0.011)	4/8 (s3, s4, s5, s7; kappa -1.60, -0.62, -1.39, -1.72)	3/8
C	8/8	0/8	0/8

Aids pooled vs A, 1D map: 8/16 vs 0/8 (Fisher p 0.022, post hoc). 2D map, aids vs rank 4: 1/16 vs 8/8 (p 1.2e-5). Within the aids, 1D-map and drifting runs reach similar accuracy (L 0.828-0.974 vs 0.872-0.992 on long gaps). Both kinds are compensated by the reader. Per-seed table: `out/GEOMETRY.txt`. Corroboration in the registered delta-space geometry

(LOOP_RANK_GEOMETRY.md): the same four L seeds have opposition 0.03-0.11 and |cos(N, E)| >= 0.9997.

5.5 Training curves (training_curves.py, stored per-epoch losses)

arm	first epoch loss < 0.5	first epoch < 0.15	first epoch < 0.05	shape
A	183-508; never on 3/8	515, 834; never on 6/8	773 (s1); never on 7/8	slow, from a high plateau
L	42-67	261-649	716-870 on 5/8	leaves the plateau at once, then creeps
L4	19-78	198-580	414-899 on 7/8	as L, faster
C	83-294	173-308	261-471	plateau, then a sharp drop (0.5 -> 0.05 in 57-228 epochs)

5.6 Gradient at initialisation (grad_at_init.py)

Same random init per seed, same six batches (B 16, T 1024), same dropout masks, 1 vs 4 passes: step-map gradient norm x2.9 (2.4-3.8), omega x3.0, block x3.4, embedding x1.1, head x1.2. Batch coherence of the step-map gradient 0.46 -> 0.41. No larger relative or more coherent signal reaches the code at the start (Adam removes the scale).

5.7 Index attention per pass (index_loop_passes.py, FACTORIAL checkpoints, seeds 500-503)

Same evaluation stream; trained-depth accuracy matches FACTORIAL_{LOOP,L4}.json to 1e-6 on all 8 models.

model	stratum	pass 1	2	3	4 (trained)	5	6
index loop x4	short	0.408	0.543	0.904	**0.938**	0.917	0.796
index loop x4	long	0.374	0.506	0.509	**0.507**	0.514	0.515
index loop x4	wrap	0.378	0.509	0.510	**0.509**	0.517	0.517
index 4 layers (lens)	short	0.473	0.471	0.620	**0.943**		
index 4 layers (lens)	long	0.510	0.499	0.480	**0.458**		
index 4 layers (lens)	wrap	0.512	0.502	0.476	**0.457**		

Floors (constant per stratum) 0.508 / 0.515 / 0.517. With an index code every pass works on short gaps; long-gap and wrap-only accuracy never leaves the floor (the 4-layer model drifts slightly below it). The rank-2 path loop, on the same readout, moves long gaps 0.363 -> 0.612 -> 0.875 -> 0.927 and wrap 0.393 -> 0.519 -> 0.700 -> 0.747 (Sec. 5.2). That is P3 inside single models: passes add only where the code has something to read.

Not done (and why)

- Match-Query loop checkpoints exist (runs/loop_headroom/PI_loop, PI_flat, PI_L3; runs/mq_rank), but the necessity / destruction runs do not (REPRO_AUDIT_nonsign.md note 8). The reader analysis there needs Match-Query's cell bookkeeping; it is a half-day port (E5).
- Loss by stratum (T3's check): cheap, not run.

---

6. Proposed paper text

6.1 Main text: Sec. 6.1, after "The rank-2 solution exists and is held" (about 190 words)

More computation reads a rank-2 code; it does not repair it. Looping the single block four times (identical parameters and initialisation) or using four real layers raises rank-2 accuracy from 0.894 to 0.973 and 0.990 (permutation p 0.0034 and 0.0012; SOLVED 2/8 and 5/8 against 0/8; rank 4 0.998, 8/8). In MapFormer the phase is computed once from the token embeddings, so neither change touches the code. They change how it is read. A one-layer model reads position through a kernel fixed by token identities. Fitting such a reader to each model's own phase recovers the one-layer models' accuracy (rank 2 0.866 against 0.894; rank 4 1.000 against 0.998), but falls 0.32-0.44 short of the looped and deep models on long-gap and wrap-only revisits. Their codes are no better than plain rank 2's (0.893 / 0.897 against 0.866, p 0.27-0.28) and far from rank 4's (p 0.0002). Run with one pass, the looped weights score 0.381; the right cell is found only at passes 2-4. None of the 16 runs finds rank 4's two-dimensional code: half end with x and y moving the phase along one direction (App. X). Downstream computation compensates for the deficit but does not remove it.

6.2 Main text: Sec. 5, end of "What depth buys index attention" (one sentence)

The same four passes on a rank-2 path code lift long-gap and wrap-only revisits instead (0.671 -> 0.927-0.990 and 0.557 -> 0.747-0.887; Sec. 6.1): passes add accuracy where the code carries position information that one pass cannot read, and none where it carries none.

6.3 Sec. 8 (Discussion), "What the two failure axes share" (one sentence to append)

Giving the rest of the network four times the computation, or four times the parameters, does not find the rank-2 solution either: the deficit is local to the content-to-angle map.

6.4 Appendix X: "Loops and depth on a rank-2 code" (tables from Sec. 5)

- X.1 Registered arms and three loss regimes (table of Sec. 1, rows 1-4).
- X.2 Per-pass accuracy and same-cell attention (Sec. 5.2 tables), with the note that L at `n_loops = j` is exact and L4's intermediate readouts are a logit lens.
- X.3 The one-pass reader: definition (T1), why it bounds a one-layer model up to heads and MLP, and the model vs reader table (Sec. 5.3). State that it is post hoc and fitted on separate walks.
- X.4 Step-map geometry classes (Sec. 5.4).
- X.5 Training curves (Sec. 5.5).
- Caveat paragraph: post hoc; one batch; n = 8; registered verdict UNMEASURED; solved counts sensitive to the cutoff; the reader is an idealised single head, so "the code supports" means "a single fixed-kernel pass can read".

Figure suggestion (`fig_loop_reader`): per stratum (short / long / wrap), bars for A, L, L4, C, each split into "read by one pass" (soft reader) and "added by later passes" (model -

reader). Index 1L / loop from FACTORIAL as a fifth and sixth group with a "cross-batch" label. Data in `out/SUMMARY.txt`; floors from `FACTORIAL_*_STRATA.json` and `RANK_MI_STRATA.json`.

6.5 Limitations (amend the "No recurrent or SSM baselines" bullet)

"Looped (weight-shared) blocks are tested as readers of a fixed code; a loop that recomputes the phase each pass was declined by training on Match-Query and under action noise, and is not tested on the torus."

---

7. Proposed GPU experiments (ranked by value / cost)

Costs from `LOOP_RANK_PREREG.md`: at T = 1024, B 16, one pass ~2.0 s/epoch, loop x4 ~7.4, four layers ~7; 900 epochs: one-pass run ~0.5 h, loop or 4-layer run ~1.8 h, at 2 jobs/GPU.

E1. Registered rank x sign x passes factorial, in one batch (highest value). Torus T = 1024, 900 ep, n = 8, cells {shared r2, shared r4} x {signed, |.|} x {1 pass, loop x4}; 64 runs, 32 x 0.5 h + 32 x 1.8 h = ~74 GPU-h (~19 h on 2 GPUs x 2 jobs). Trimmed: drop r4 |.| (48 runs, ~55 GPU-h). Primary readouts (registered): (a) geometry class per run (2D map / 1D / drifting) and the soft-reader code quality; (b) accuracy. Predictions from T2: the loop does not change the 2D-map count in any cell (r4 signed 8/8 both; r2 <= 2/8 both; |.| <= 2/8 both); the loop raises accuracy in r2 and |.| cells by raising model-minus-reader, not reader; no loop gain in r4 signed (ceiling). Branches: "loop changes the code" if 2D-map counts differ by Fisher $p < 0.05$ in any cell; "reader only" if no count differs and the margin rises. Value: tests the paper's search claim on both axes with one control. A positive "reader only" result is a second axis-general statement; a loop that changes the code would be a direct search-aid result.

E2. Trajectory snapshots of the existing LOOP_RANK arms (medium-high value, cheap if deterministic). Re-run A, L, L4, C for 4 seeds with checkpoints every 50 epochs (A s0 reproduced bitwise, so snapshots should match the stored runs; verify at epoch 900). ~4 x (0.5 + 1.8 + 1.8 + 0.5) = ~18 GPU-h. Readouts on CPU: soft-reader code quality, geometry class and loss by stratum over training. Tests T4's compensation-basin speculation (does the 1D code form after the reader absorbs short-gap revisits?) and T3 (residual loss in long and wrap). Snapshots must not change the RNG stream: check `train_variant.py` before launch (rule 13).

E3. Pass count and code at rank 2 (medium). L at loop x8 and with sampled counts {2..6} (n = 8 each, one batch with L x4): ~2 x 8 x (2.5-3.7 h) = ~50 GPU-h; trimmed to x8 only, ~30 GPU-h. Prediction: more passes raise long-gap and wrap accuracy further and leave the code class unchanged. If wrap stays below rank 4 at any depth, T3 is supported.

E4. Loop vs Level 1.5 under action noise, registered replication (medium value, cheap). The n = 3 finding (+0.205 vs +0.004 at p 0.25, T = 128) fits T2: a noisy code is compensated by a deep reader. n = 8, 3 arms x 2 noise levels at T = 128 (minutes per run): ~4 GPU-h. Readout: accuracy and the soft reader on the noisy code.

E5. Match-Query loop, registered, fresh seeds (medium; fixes a weak citation). The +0.346 is cross-batch and asymmetric, and the necessity runs are gone

(`REPRO_AUDIT_nonsign.md`). One batch: path 1L, path loop x4, path 3L, index 1L, index loop x4, n = 8, fresh seeds, with checkpoints kept: ~5-8 GPU-h. Then port the reader to Match-Query on CPU.

E6. (done on CPU, Sec. 5.7.) Index loop per pass: long and wrap at the floor after every pass.

---

8. What does not survive, and caveats

- "The loop is a search aid at constant capacity" (the H1 framing): constant parameters, larger reader. The registered verdict stays UNMEASURED; the reading in this note is post hoc.
- "More gradient paths": at initialisation the loop scales every gradient about 3x (Adam-invariant) and does not make the step-map gradient more coherent.
- "Iterated attention chains partial matches through earlier arrivals" (relay): mass on same-cell action tokens stays <= 0.12 at every pass; not supported as the main route.
- The reader is an idealised single head with nonnegative channel weights. It is not a proven upper bound on a one-layer model (two heads plus an MLP), but A's margin over it is small (+0.028), so it is close.
- Geometry thresholds are post hoc; "cancel" is not wrap-aware (rule 8). The reader columns are wrap-aware by construction and agree.
- The kappa ~ sqrt 2 clustering (4 of 4 L 1D runs) is an observation, not a claim.
- One batch, n = 8 per arm, no fresh-seed replication of any analysis (rule 6).

9. Reproduce (CPU, from `/home/prashr`)

```
D=mapformer/docs/paper/expansion/scripts/recurrence
for a in A L L4 C; do python3 $D/pass_strata.py $a 100; done      # ~3 min/arm; L runs 8
passes
for a in A L L4 C; do python3 $D/phase_code.py $a; done           # ~30 s/arm
for a in A L L4 C; do python3 $D/phase_reader_soft.py $a; done    # ~4 min/arm
python3 $D/attn_offsets.py; python3 $D/training_curves.py; python3 $D/grad_at_init.py   #
~15 / <1 / ~45 min
python3 -u $D/index_loop_passes.py 100 > $D/out/INDEX_LOOP_PASSES.txt                    #
~3 min at 4 threads
python3 $D/summarize.py > $D/out/SUMMARY.txt; python3 $D/geometry.py > $D/out/GEOMETRY.txt
```

Language side and the integrator framing: expansion for Section 7 and the Discussion (2026-10-20)

Scope: what a one-layer path model learns from words, when a learned phase becomes a map and when a clock, the integrator (Howard) framing, and whether what/where and the leak belong in the main paper. For `docs/paper/MAIN_DRAFT1.md` Sections 2.1, 7 and 8. No training was run; every new number is a post hoc, eval-only CPU analysis of stored checkpoints (scripts and outputs in `docs/paper/expansion/scripts/language/`, section 8). Results files are authoritative; where this note and a results file differ, the results file wins.

Labels. **ESTABLISHED** = a registered verdict or a declared secondary in a results file (with its file). **ANALYSIS** = post hoc, on stored checkpoints, not pre-registered (new here unless a file is named). **THEORY** = an argument, with the check that tests it. **SPECULATION** = untested. Terms: *step* = the per-token increment Delta_w = W_out W_in e_w of a word w (one per word type; the model's step is context-free); *move* = one step of the walk (one movement clause in the text world); *tick* tau_c = the phase a channel gains per move whatever its direction; *channel* = one rotary 2D block of one head (64 per model); *used channel* = weighted by how much retrieval attention uses it (mean ||q_c|| x mean ||k_c|| over the query words and the object words); *re-scored* = read with the x1.111 attention-dropout scale (the registered correction, `docs/audits/2026-10-04/DROPOUT_RESCORE.md`).

---

0. Summary: the key additions

1. **A complete step account of the text-world models (ANALYSIS, 32 models).** One path layer learns: direction words as signed steps whose differentials cancel (opposition 0.001-0.013 on all 24 learned-step models, wrap-aware); optional words (adverbs, fillers) with steps of 0.004-0.03 of a move on every model; aside sentences whose words have sizeable steps that cancel within the sentence, leaving a net offset of 0.04-0.16 of a move (usage-weighted); and, on 10 of 24 models, a per-move tick in 26-38 channels.
2. **The aside offset is causal, both ways (ANALYSIS).** Putting each aside's words at its clause's object-slot phase, with the phase after the aside unchanged, drops the 19 learned-step models at ceiling in eval mode from 0.989-1.000 to 0.952-0.975, and 71-100% of their errors then name an object told in an aside at that cell: the direction-words-only oracle's ceiling and error signature (`TW_NORMSTEP_RESULTS.md`). Conversely, offsetting the oracle's aside words by half a move (phase after the aside unchanged) lifts it from 0.962-0.981 (100% aside errors, 8/8 seeds) to 0.999-1.000 on 8/8 seeds.
3. **The text-world clock counts moves, so it is path length, Howard's Case II (ANALYSIS).** An aside sentence adds at most 0.06 of a tick, a filler or adverb word at most 0.007: the clock is neither a word nor a sentence counter. On the clock seeds one phase runs Case III (position) in some channels and Case II (path length) in others.

4. **The clock is idle (ANALYSIS).** The per-move tick alone predicts which channels drift between visits (agreement 0.98-1.00 per model, 32 models). 79-99% of retrieval usage sits in channels whose tick is 0 mod pi (pi: on the even 64-torus every return takes an even number of moves). Removing the 26-38 clock channels from the score costs <= 0.009 on all 7 clock models at ceiling in eval mode, while the 3 below ceiling still need them (cost 0.09-0.26); removing as many map channels collapses accuracy (0.02-0.65).
5. **A usage-weighted drift/content ratio separates map from clock where the growth exponent cannot (THEORY + ANALYSIS).** The pooled growth exponent alpha reads 0.98-1.02 on the text-world clock seeds (which carry a perfect map) and on TinyStories alike. The usage-weighted median of rho_c = |mu_c| L / range(content_c, L) (clock part over content part, per channel, at the training length) reads: torus 0.064-0.079, text world 0.06-0.23 (converged seeds), code 4.5-6.6, TinyStories 9.9-12.3, recency 21.8-22.1. Same architecture, same signed step: the data decide whether the used channels are a map or a clock.
6. **The text world is a short-walk task, and the paper must say so (ANALYSIS).** Its walks are ~132 moves, and the paper's own retrace floor scores **0.934** on its registered eval stream (0.937 on TW_NORMSTEP's), not the 0.597 of the one-step reversal-copy rule the results file reports. Path still clears it, and on the revisits retrace gets wrong (5.9%; best constant 0.112) path models score 0.944-1.000 on the six seeds at ceiling (0.961 on the fresh seeds 2-7) against 0.10-0.17 for two index layers and 0.00-0.01 for one (section 2.6).
7. **Integrator paragraph** (section 4): a positive statement of what is ours (the three-way identification for rotary attention; that in the rotary slot the Case II / III distinction is one of learnability, not expressivity; and the measurement that a learned rate realises Case I, II or III per channel as the data demand), with precise prior work.
8. **What/where and the leak** (section 5): one paragraph in Section 7, as a single principle -- a token's phase offset from its place must depend on its role, not its identity -- with the leak registered result (and its failed accuracy replication) and the what/where checks in the appendix.

---

1. Established facts (from the results files)

fact	numbers	label	file
One path layer learns navigation told in English	fresh seeds (n = 6): 0.979 +/- 0.048, 6/6 SOLVED vs RoPE 1 layer 0.505, 0/6; +0.474, permutation and Fisher p 0.0022; RoPE 2 layers 0.793 (0/6). All 8 seeds: 0.969 (eval mode), **0.993 re-scored** vs 0.506 / 0.772	ESTABLISHED (A; the re-score is registered in DROPOUT_RESCORE)	`TEXTWORLD_RESULTS.md`
Floors on that eval stream	best constant 0.505, one-step reversal-copy 0.597	ESTABLISHED	`TEXTWORLD_SECONDARY.txt`
Opposite directions	after removing the direction words' common	ESTABLISHED (declared secondaries; registered	`TEXTWORLD_RESULTS.md`

fact	numbers	label	file
cancel on every seed	component: 0.002-0.012 (8/8); synonyms one step (cosine 1.000, vs 0.36-0.61 across directions); zeroing direction steps -> 0.39-0.51	B branch did not fire, 4/8)	
Half the seeds also count	4/8 seeds: common component parallel to the verbs' step (cos +0.999 to +1.000); 31-38 of 64 channels drift between visits (2-4 in map seeds)	ESTABLISHED (secondary + CORRECTED block)	`TEXTWORLD_RESULTS.md`
Learned steps beat "only direction words move"	DirOnly 0.970-0.974 on 8/8 seeds; on 2 seeds 100% of its errors name an aside object told at that cell; learned arms make none. Re-scored DirOnly - MapWM -0.024 (p 0.0002), a post hoc correction (eval mode: no difference)	ESTABLISHED (post hoc in the file)	`TW_NORMSTEP_RESULTS.md`
NormStep on words	accuracy +0.006, UNMEASURED below 0.096; the registered "per-word clock" is the aside offset (0.126 of 0.133 rad)	ESTABLISHED	`TW_NORMSTEP_RESULTS.md`
State sentences	MapWM 0.994 vs RoPE 0.516 (sanity check); C = location + recency among nouns at a place, NOT verb semantics (a verb-agnostic rule scores 1.000 on T2drop); state sentences displace the phase 0.028 moves, like asides; DirOnly worse only under the re-score	ESTABLISHED (CORRECTED 2026-10-11)	`TW_STATECHANGE_RESULTS.md`
What/where, causal	converged path models keep 0.974-0.989 with object identity removed from the score, 0.988-1.000 with a shared kernel x content gain; index models fall below the floor	ANALYSIS (post hoc)	`docs/WHAT_WHERE_CHECKS.md`
The what-to-where leak	in distribution, removing it (ActOnly / NormStep) gains +0.0107 (p 0.0002, 8/8), equal to MapWM's leak; both remedies SOLVED 16/16 vs MapWM descending. Fresh seeds: SOLVED replicates (8/8 vs 0/8), accuracy does not (+0.0045, p 0.46)	ESTABLISHED (LEAK registered; GAIN_PHASE D1r)	`LEAK_RESULTS.md`, `GAIN_PHASE_RESULTS.md`

fact	numbers	label	file
TinyStories	MapWM 1.5053 vs RoPE 1.5050 nats/token (n = 3); clock share 0.79-0.87; `mr`/`mrs` cancel '.'; a uniform clock costs 0.07-0.09 nats	PILOT (CORRECTED 2026-10-11)	`TINYSTORIES_PILOT_RESULTS.md`
Growth exponent on text and code	alpha 0.999-1.006 (TinyStories), 0.990-1.016 (code); R^2(S ~ t) 0.994-0.999	ANALYSIS	`docs/audits/2026-10-12/alpha_language_out.txt`
Code at matched length	position main +0.0056 bpc (3/3, t p 0.024): path integration costs on code	ESTABLISHED (n = 3)	`CODE_FULLVAL_RESULTS.md`

Not to be revived (CLAUDE.md, withdrawn): "the model tracks state changes"; TinyStories "nothing cancels", "spatial 0%", "20x the MapWM - RoPE difference"; "two defects, two fixes".

---

2. What a one-layer path model learns from words (question 1)

Models: the 8 registered text-world path runs (`runs/textworld/p0`, seeds 0-7) and the 8 MapWM, 8 NormStep and 8 DirOnly runs of TW_NORMSTEP (`runs/tw_normstep/p0`, seeds 10-17). Eval: held-out map, np seed 424242 (used by no registered script), 40 walks x 1024 words, unless stated. The manual forward used for manipulations reproduces every model's logits exactly (max |diff| 0.0) and the role-annotated generator reproduces `TextWorld.generate_trajectory` token for token (`tw_lib.py` checks).

2.1 The step table, by word class (ANALYSIS; `tw_step_account_out.txt`)

Steps omega-scaled (radians), relative to the clause-level direction differential |D_N| (one move):

word class	map-type models (14 learned-step)	clock-type models (10)	what it means
direction words (differential)	1.00 by normalisation; opposites cancel 0.001-0.013	same	signed, cancelling steps
adverbs, fillers	0.004-0.03	0.004-0.025	words whose count per move varies get no step
`and` / `a` / `she` (shared by fillers, seeing phrases, asides)	0.02-0.07	0.02-0.08	near zero
verbs, seeing words, object words, `.`	0.02-0.14	0.24-1.49	one per move: free to carry a per-move tick
aside-only words (`thought`, `about`, `remembered`, `story`)	0.07-0.16 per word	0.41-1.33 per word	large, but cancel within the sentence
aside sentence, net (usage-weighted)	0.05-0.12 of a move	0.04-0.16 of a move	a small offset off the place
DirOnly (oracle): every non-direction word	0 by construction	--	asides sit at the place's phase

The clause tick tau (what every move adds whatever its direction) is 0.03-0.23 of a move (map type) and 3.3-6.8 moves (clock type, raw norm, all channels), parallel to the verbs' step (cos +0.999 to +1.000 on all 10 clock models).

2.2 The tick predicts the drift; the clock counts moves (ANALYSIS)

- **The tick alone predicts which channels drift.** Predicted drift per channel = mean |wrap(lag x tau_c)| over the consecutive-visit pairs (median lag 8 moves, 1,221 pairs); observed = mean |wrap(dtheta_c)| at those pairs. Classified at 1 rad, they agree on 0.98-1.00 of channels on every one of the 32 models (r 0.96-1.00 on the 24 learned-step models). The clock seeds drift in 26-38 channels (tw s0/s2/s3/s4: 26/29/35/36; MapWM s11/13/14/15: 38/27/28/30; NormStep s11/16: 27/28), map seeds in 0-3. (The CORRECTED block's 31-38 counted drift from the FIRST visit; consecutive visits give the numbers here.) `tw_step_account_out.txt`.
- **It counts moves, not words or sentences.** Regressing each move's net phase on direction, aside present, filler count and adverb present, in the drifting channels: an aside sentence adds 0.009-0.062 of a tick, a filler word 0.001-0.007, an adverb word 0.001-0.007 (10 clock models; `tw_clock_unit_out.txt`). A word counter would give fillers and adverbs ~1/8 of a tick each; a sentence counter would give asides ~1 tick. Every move is one cell, so the move count is the path length: **these channels integrate Howard's Case II (distance travelled) while the others integrate Case III (position)**.
- **Pi ticks are map features.** On all 10 clock models 2-6 channels tick by exactly pi per move (e.g. MapWM s11: -3.141, -3.140 rad). On the even 64-torus every return to a cell takes an even number of moves, so a pi tick is invisible to revisits: it encodes the checkerboard parity of the cell. These channels carry up to 0.24 of retrieval usage (MapWM s11).
- **Why clock seeds lean on non-direction words.** The registered secondary found that zeroing every non-direction step costs the clock seeds 0.33-0.69 and the map seeds 0.05-0.30. In the channels retrieval uses (tick 0 mod pi), the direction words' common component is 0.06-0.34 of a move on the clock seeds and 0.01-0.03 on the map seeds (registered text-world batch), and the other words cancel it within the move. Zeroing them leaves it as a tick in exactly the used channels: across the 8 seeds the cost tracks that component (Spearman +0.905; `tw_common_used_out.txt`).

2.3 The clock is idle: usage and channel ablation (ANALYSIS)

- **Usage.** 79-99% of retrieval usage sits in channels whose tick is 0 mod pi (lowest: tw s0 0.79, unconverged in eval mode; NormStep s16 0.82); channels where the clock dominates the phase (rho_c > 1 and more than half a turn per window, section 3) carry 0.13-0.30 of usage on clock models and 0.01-0.15 on map models (`usage_tick_out.txt`).
- **Ablation.** Removing the drifting channels' q and k components from the score (re-scored accuracy):

model	intact	clock channels removed	same number of map channels removed (5 draws)	only clock channels kept
	0.973	**0.717**	0.43 [0.24, 0.58]	0.07

model	intact	clock channels removed	same number of map channels removed (5 draws)	only clock channels kept
tw s0 (eval mode 0.875, unconverged)				
tw s2	0.999	0.990	0.25 [0.03, 0.63]	0.03
tw s3	0.998	0.993	0.06 (all 29 map channels)	0.06
tw s4	0.999	0.999	0.03 (all 28)	0.03
MapWM s11	0.989	0.985	0.48 (all 26)	0.48
MapWM s13	0.997	0.996	0.65 [0.50, 0.84]	0.48
MapWM s14 (eval mode 0.960)	0.995	**0.902**	0.25 [0.03, 0.72]	0.03
MapWM s15	0.995	0.994	0.07 [0.02, 0.23]	0.02
NormStep s11	0.998	0.996	0.46 [0.20, 0.60]	0.44
NormStep s16 (eval mode 0.835)	0.982	**0.809**	0.22 [0.04, 0.48]	0.04

On the 7 clock models at ceiling in eval mode the clock costs at most 0.009 to remove; the 3 that need it (tw s0, MapWM s14, NormStep s16: 0.09-0.26) are exactly the ones below ceiling in eval mode, i.e. not converged to the clean solution. A clock in the channels retrieval does not use is how half the seeds and a perfect map coexist. That unconverged models still lean on it suggests the clock is used early and abandoned later (SPECULATION; E1 tests it). Why training builds it at all is not identified.

2.4 Asides: the offset is what beats the oracle (ANALYSIS; `tw_attention_asides_out.txt`)

`TW_NORMSTEP_RESULTS.md` found that DirOnly stops at 0.970-0.974 because an aside's noun sits at the place's phase, and that learned-step models make none of these errors. Two manipulations at eval, each keeping the phase after the aside unchanged (so only the within-sentence offset changes):

- **Collapse** (learned models): every aside word is put at its clause's object-slot phase. The 19 learned-step models at ceiling in eval mode (>= 0.98) fall from 0.989-1.000 to **0.952-0.975**, and **71-100%** of their errors then name an object told in an aside at that cell (tw s1-s6 0.94-1.00; MapWM s10, s11, s13, s15-s17 0.71-1.00; NormStep s10-s15, s17 0.86-1.00); before the manipulation they make 0-13 errors, none of that kind. That is DirOnly's ceiling and its error signature, produced in the models that had escaped it. (The 5 models below ceiling in eval mode respond partly: 0.966-0.994, 0-85% aside errors.)
- **Lift** (DirOnly, `tw_attention_asides_dironly_out.txt`): intact, DirOnly scores 0.962-0.981 and **100%** of its errors are aside errors on **8/8** seeds (TW_NORMSTEP checked 2). Offsetting its aside words by half a move in-plane, with the phase after the aside unchanged, gives **0.999-1.000 on 8/8**; by one move off the map plane (a random direction orthogonal to the direction steps) 1.000 on 7/8 (s16 0.972); by a quarter move off-plane 0.963-0.996 (the kernel is not sharp at that distance). The oracle's ceiling is exactly the missing aside offset.

So the learned non-direction steps do two jobs: they cancel within every move (so the place's phase is the same whatever the wording) and they leave a small net offset inside aside sentences (so a mentioned-but-absent object is not filed at the place).

2.5 Attention at revisits (ANALYSIS)

At each revisit query (head mean), the 19 learned-step models at ceiling put 0.55-0.76 of their attention on the object slot of the most recent visit to that cell, 0.07-0.14 on older visits' object slots, 0.004-0.035 (mean 0.015) on aside nouns told at the cell, 0.07-0.30 on other tokens of earlier clauses at the cell, and 0.01-0.06 elsewhere. Retrieval is by place: 0.94-0.99 of the attention lands on tokens told at the current cell. DirOnly puts 0.041-0.077 (mean 0.047, about 3x) on aside nouns told at the cell and 0.10-0.23 elsewhere.

2.6 The text world is a short-walk task: the retrace floor (ANALYSIS; `tw_retrace_floor_out.txt`, `tw_strata_out.txt`)

The text world renders ~132 moves per 1024 words, the walk length at which the main paper's Section 4 shows a map-free retrace rule scoring 0.934 on the torus. On the text world's own registered eval stream the same rule (port of `docs/audits/2026-10-16/sign_floors.py`) scores **0.9337** (TW_NORMSTEP stream: 0.9365), while the constant and the one-step reversal-copy rule reproduce the registered 0.5054 / 0.5965 exactly. Consequences:

- The text world's accuracy margin over the best map-free rule is small: path 0.993 re-scored (0.969 eval mode) vs retrace 0.934; DirOnly's 0.972 is barely above it. Index RoPE (0.505 / 0.772) does not reach retrace.
- The map claim must rest on the revisits retrace gets wrong and on the mechanism (cancelling steps; direction steps zeroed -> floor). On the first 100 walks of the registered stream, 179 of 3,055 revisit targets (5.9%) are ones retrace gets wrong (best constant there 0.112: 'cat'; 'nothing' never):

model (registered batch, n = 8)	all	RETRACE-OK	NEEDS-MAP (n 179)
path, 1 layer, re-scored	0.994	0.998	**0.920** (seeds 1-6: 0.944-1.000; s0 0.603, s7 0.832, the two below ceiling in eval mode); fresh seeds 2-7: 0.961
path, 1 layer, eval mode	0.970	0.975	0.885
RoPE, 2 layers	0.779	0.819	**0.138** (0.10-0.17)
RoPE, 1 layer	0.507	0.538	0.002

Two-layer index RoPE learns part of retrace (two seeds reach 0.93, just under the 0.934 floor; RETRACE-OK 0.98) and stays at the constant floor where a map is needed. Path solves those revisits. This is the text-world analogue of Section 4's strata and the strongest single piece of evidence that the words are integrated into a map. - Section 7 of the draft should report retrace beside the constant floor. This does not change any registered verdict.

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3. Theory: when a learned phase is a map and when it is a clock (question 2)

3.1 Setting and decomposition (THEORY)

Channel c's score between query t and key t' depends on position only through omega_c (Phi_c(t) - Phi_c(t')), the summed step over the interval. With token frequencies f_w, split every step into a common part and a content part: Delta_c(w) = mu_c + d_c(w), mu_c = sum_w f_w Delta_c(w), so sum_w f_w d_c(w) = 0. Over an interval of L tokens

```
Phi_c(t) - Phi_c(t') = mu_c L + sum of d_c over the interval.
```

The first term is a clock (Howard Case I if mu is carried by every token; Case II if only by selected tokens, as in the text-world move counter); the second is content-driven and can be a map (Case III) only if it cancels over closed loops. "The common component tracks the frequency-weighted mean step" is true by definition of mu; the substance is in which channels mu_c survives and why.

3.2 T1: the tick rule for a retrieval channel (THEORY, checked)

If the target at t is the key at the last visit to the same latent state z (a cell), matched pairs come at lags spread over a distribution, and for a channel to score matched pairs consistently its phase difference over them must be constant mod 2 pi. With the map part cancelling over the closed loop, the remainder is lag x tau_c. Hence: **a channel can serve latent-keyed retrieval only if its tick is 0 mod 2 pi / g**, g the gcd of the possible return lags (g = 2 on a bipartite lattice such as the even torus), and otherwise it decorrelates once |wrap(tau_c)| x sd(lag) exceeds ~1 rad.

Checks (section 2.2-2.3): the tick alone predicts each channel's revisit drift (agreement 0.98-1.00, 32 models); used channels have tick 0 mod pi (79-99% of usage), including exact-pi parity channels; channels that violate T1 are not used, and removing them costs <= 0.01 on converged models.

3.3 T2: the readout decides, channel by channel (THEORY + ANALYSIS)

T1 constrains mu_c only where the readout looks. Elsewhere a tick is free, and the text world shows one surviving in 26-38 channels on 10 of 24 models without cost. Conversely, when targets are keyed to a count back (k-back recency) or to recent context (language), a nonzero mu_c in the used channels is what the readout needs. So the prediction is about **used** channels: their drift/content ratio rho_c = |mu_c| L / range(content_c, L) at the training length L should be << 1 when targets are keyed to a latent state reached at variable lags, and >> 1 when they are keyed to a token or event count.

Check (`crossover_usage_out.txt`, `usage_tick_out.txt`; usage-weighted median rho over channels, all-token usage for the multi-layer language models):

data (all MapWM, signed step)	used-channel rho (median)	usage in clock-dominated channels	seeds
2D torus, 1 layer, matched length (sign_matched Signed_r4)	0.064-0.079	0.000 (0/64 channels)	8

data (all MapWM, signed step)	used-channel rho (median)	usage in clock-dominated channels	seeds
text world, 1 layer (24 learned-step models; tw s0 unconverged 2.5)	0.06-0.23	0.01-0.30	24
code, bytes, 9 layers (trained at 2048)	4.5-6.6	0.50-0.55	3
TinyStories, words, 6 layers (trained at 512)	9.9-12.3	0.42-0.43	3
k-back recency with fillers, 1 layer (recency Signed_r4)	21.8-22.1	0.93-1.00	8

The same architecture and parameterisation (the torus and recency rows are the identical `Signed_r4` class) put its used channels at opposite ends; language and code sit on the recency side. (Clock-dominated: rho_c > 1 and more than half a turn per training window. On language part of the usage is in slow channels with rho > 1 that turn less than half a turn per window -- slow recency kernels -- which is why the share is 0.42-0.55 while the median rho is 4.5-12.)

3.4 What the growth exponent can and cannot show (ANALYSIS; `tw_alpha_drift_out.txt`)

The pooled exponent alpha (range of the raw summed step ~ T^alpha, the readout of `alpha_language.py`) weights channels by their raw range, so a large tick in channels attention ignores dominates it:

models	alpha	pooled rho(1024)	used-channel rho	accuracy
text-world map seeds (14)	0.61-0.74 (the true walk: 0.668)	0.12-0.86	0.06-0.13	0.985-1.000 (re-scored)
text-world clock seeds (10)	0.98-1.02	9.3-13.9	0.07-0.23 (tw s0, unconverged: 2.5)	0.989-0.999 at ceiling (7)
DirOnly (8)	0.60-0.65	0.05-0.07	--	0.962-0.981 (this stream)
TinyStories (3)	1.00-1.01	11.9-12.5	9.9-12.3	MapWM = RoPE
code (3)	0.99-1.01	2.9-5.0	4.5-6.6	path costs +0.0056 bpc

The clock seeds and TinyStories are indistinguishable by alpha and pooled rho, yet one carries a perfect map. So the draft's "it grows linearly with token index (R^2 0.994-0.999)" does not by itself show the absence of a map. The used-channel rho does separate them (0.07-0.23 vs 9.9-12.3): on TinyStories the channels attention uses are dominated by the clock part. The two-part model accounts for alpha: the exponent of |mu| T + range(content, T) is within 0.01-0.07 of the measured alpha on every model (it over-predicts by 0.01-0.03 on map-type models and under-predicts by 0.03-0.07 on clock-type models, TinyStories and code, since the range of a sum is not the sum of the ranges).

3.5 Howard's cases in one model (THEORY + ANALYSIS)

A signed, content-dependent rate can realise all three of Howard et al.'s cases: Case I (mu carried by every token), Case II (a non-negative tick on selected tokens: the text-world move counter; the TinyStories punctuation and boundary ticks read as an event

counter of the same kind, an interpretation), Case III (signed steps that cancel). Which one a channel ends in is set by what the readout keys on, not by the parameterisation: torus and text world put their used channels in Case III, recency and language in Case I/II. The sign result of Section 6.2 is the complement: a non-negative step can represent Case III in the rotary slot (Proposition 1) but does not learn it from a small start.

SPECULATION, not tested: language would put used channels in Case III wherever prediction pays for a latent state reached by variable routes (dialogue about places, procedural text). The text world says a one-layer model finds it when the words that move are context-free; TW_AMBIG (section 7, E2) would test the context-dependent case.

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4. The integrator framing (question 3)

4.1 What is prior art (from `docs/lit/LIT_HOWARD_TRANSFORMER.md, LIT_CLOCK_MAP.md`)

- Howard et al. 2014 (J Neurosci 34:4692): one integrator bank driven by a rate alpha(t) codes time (alpha constant, Case I), distance travelled (alpha = speed, Case II) or position (alpha = signed velocity, Case III: "like a bead on a wire"). The dichotomy, with the collision property, is theirs.
- Velocity-modulated integrators in trained networks: Alipour et al. 2025 (J Comput Neurosci 53:131; Cases I and III, alpha given, real s); Mochizuki-Freeman et al. 2023 (IJCNN; alpha = dz/dt from the environment). The Howard and Tiganj labs keep s real and explicitly reject complex s for stability (Mochizuki-Freeman 2023 fn. 1; Kabir et al. 2025, AAAI; Dickson et al. 2025).
- RoPE as a velocity-controlled oscillator: GridPE (Li et al., arXiv:2406.07049): Phi(t) = omega_s t + beta omega^T x(t), reducing to RoPE in 1D; coordinates given, not learned from content.
- Laplace memory and SSMs: Sarkar and Howard 2026 (arXiv:2601.02618) relate SITH's diagonal recurrence to discretised SSMs (decay side, not the step Delta_t, not phase).
- Content-gated phase rate motivated by Howard: RoMem (Li et al. 2026, EMNLP; arXiv:2604.11544), monotone (Case II-like), for temporal knowledge graphs.
- Delta_t as a learned stand-in for time or arc length: TIDES (Soydan et al. 2026, arXiv:2605.09742), no signed case.
- Index position as "linear structure" and path-integrated position as a learned map in a transformer: TEM-t (Whittington et al. 2022). Sign and state tracking: Sarrof et al. 2024, Grazzi et al. 2025 (Thm 3 for rotations), Mamba-3 Sec. 3.2.

4.2 What is ours

1. The explicit identification, for rotary attention: index RoPE, monotone content-dependent phase (CoPE, CARoPE, softplus or |.| steps) and signed content-dependent phase (MapFormer, Selective RoPE, Mamba-3's data-dependent RoPE) are Howard's Cases I, II and III with an imaginary rate, and Howard's alpha(t) is the selective-SSM step Delta_t and MapFormer's content-dependent angle step. Not found stated (LIT_HOWARD_TRANSFORMER sec. 1); one sentence once the pieces are side by side, so present it as an organising remark, not a result.

2. The rotary slot is where the signed rate is bounded (a rotation), the move the Howard lab declined for real s; and in that slot Cases II and III coincide in what they can represent (Proposition 1, extending Grazzi Thm 3 / Mamba-3 to rank-limited step maps). The distinction becomes one of learnability (Section 6.2).
3. Measured: which case a learned signed rate realises is decided per channel by the data (section 3.3), and one model can run Case III and Case II in different channels at once (section 2.2).

---

5. What/where and the leak (question 4)

Recommendation: keep both in the main paper only as one paragraph closing Section 7, stated as a single principle with the text world, and move the detail to App. K.

- **The principle (THEORY, light).** Retrieval by place needs every token that should be retrieved from a place to sit at the same phase offset from that place, and every token that should not be retrieved there to sit elsewhere. So a token's offset must depend on its role, not its identity. The leak violates the first half (an object's step depends on its identity, so objects at one place spread over identity-dependent offsets: LEAK, registered); DirOnly violates the second (an aside's noun sits at the place's phase); learned steps satisfy both in the text world (asides offset 0.04-0.16 of a move, observed objects at a fixed offset).
- **The leak, as registered** (`LEAK_RESULTS.md`): +0.0107 in distribution (p 0.0002) for removing it, equal to MapWM's own leak, and the remedies converge where MapWM is still descending. Must be said with its replication: on fresh seeds (GAIN_PHASE D1r) SOLVED replicates (8/8 vs 0/8) but accuracy does not (+0.0045, p 0.46). Effect size ~1%, one layer, partly convergence speed (r(loss, acc) -0.94). That is a supporting result, not a contribution line.
- **What/where checks** (post hoc, `docs/WHAT_WHERE_CHECKS.md`): already in the draft's Discussion with the right status; keep it there and add that separation was shown in embeddings by MapFormer (Fig. 9), so our addition is that the trained models rely on it.
- Do not include: GAIN_PHASE's gain score (no difference, D2), NormStep on words (unmeasured), the state-change C as anything beyond location + recency.

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6. Proposed paper text

6.1 Section 7, replacement (about 0.9 page plus figure)

> **7. Maps from words**
>
> The text world asks whether the path model can find its actions when nobody marks them. Each move of the torus walk is written as a sentence: a direction word chosen from three synonyms, verbs that do not move the walker, optional adverbs and fillers, and, after some moves, an aside that names an object without visiting it ("she thought about a cat ."). Nothing in the input says which words move the walker (vocabulary and step tables: App. J). The walks are about 132

moves long, so the map-free retrace rule of Section 4 already scores 0.934 here; we report it with the constant (0.505) and reversal-copy (0.597) floors.

One path layer learns navigation told in English. On fresh seeds (n = 6) one path layer scores 0.979 +/- 0.048 (6/6 SOLVED) against 0.505 for one-layer index RoPE (0/6), +0.474 (permutation and Fisher p 0.0022); two index layers reach 0.793 (0/6). Both index models stay below retrace on every seed; path clears it (0.993 on all eight seeds with the attention-dropout re-score, App. C). On the revisits retrace gets wrong -- places reached by a route other than undoing the last run, 5.9% of targets, best constant 0.112 -- path scores 0.961 on the fresh seeds (0.944-1.000 on every seed at ceiling) and two index layers 0.10-0.17 (analysis).

What the step table holds. On every seed, opposite direction words cancel once the component common to the four directions is removed (0.002-0.012), synonyms collapse to one step, and zeroing the direction steps drops accuracy to the constant floor (declared secondaries). The rest of the vocabulary is organised by how often a word occurs per move (analysis, 32 models, Fig. 7d): adverbs and fillers, whose count varies, get steps of 0.004-0.03 of a move; words that occur once per move (verb, seeing words, object, full stop) may carry a common tick; aside sentences get sizeable word steps that cancel within the sentence, leaving the aside's noun 0.04-0.16 of a move off the place.

The offset is why learned steps beat being told. A model whose steps are restricted to the twelve direction words stops at 0.970-0.974 on every seed, and its errors name objects told in an aside at that place: with no other word stepping, the aside's noun sits at the place's phase (Sec. 6 of `TW_NORMSTEP`). Moving each aside back onto its place's phase at evaluation, with the phase after it unchanged, drops the learned-step models from 0.989-1.000 to 0.952-0.975, with 71-100% of their errors now of the same kind; offsetting the oracle's aside words by half a move lifts it to 0.999-1.000 on every seed (analysis). "Only actions move" is the wrong separation in language: some words that are not actions need steps, so that what is mentioned is not filed where it is not.

Half the models also count moves, in channels they do not use. On 10 of 24 learned-step models a tick per move rides on every move, parallel to the verbs' step. It counts moves -- an aside sentence adds at most 0.06 of a tick, a filler word at most 0.007 -- so in Howard et al.'s terms these channels integrate distance travelled (Case II) beside the others' position (Case III). The tick alone predicts which channels drift between visits (agreement 0.98-1.00), and those channels are idle: 79-99% of retrieval attention uses channels whose tick is zero modulo pi (on the even torus a return always takes an even number of moves), and removing the 26-38 clock channels from the score costs at most 0.009 on every clock model at ceiling (7), while removing as many map channels collapses accuracy (analysis).

What must and must not move. The same rule covers the what-to-where leak on a task with new objects at every episode: a step that depends on an object's identity spreads objects at one place over identity-dependent offsets, and removing it gains +0.011 in distribution (registered, p 0.0002; on fresh seeds the

> convergence gain replicated and the accuracy gain did not, App. K). A token's offset from its place should depend on its role, not on its identity.

Figure 7: keep panels (a)-(c); add (d) the step table by word class (bars: direction differential, once-per-move words, optional words, aside words, aside net) for map and clock seeds, and (e) the aside collapse/lift test (accuracy and aside-error share). Caption must say "eval mode" for (c) as now and "re-scored" for (e).

6.2 Background 2.1, replacement of the Howard sentences

> **2.1 Rotary position as an integrator.** [first two sentences unchanged] Howard et al. (2014) drive one bank of integrators by a rate alpha(t): a constant rate codes elapsed time (Case I), the speed codes distance travelled (Case II), and the signed velocity codes position, the same for every visit to a place "like a bead on a wire" (Case III). The rotary families are these three cases with an imaginary rate: index RoPE adds a constant angle per token; CoPE, CARoPE and non-negative content steps advance by a non-negative, content-dependent amount; MapFormer, Selective RoPE and Mamba-3's data-dependent RoPE add a signed one, and alpha(t) is the selective-SSM step Delta_t. Velocity-modulated integrators have been trained in networks with a real Laplace variable (Alipour et al. 2025), and RoPE has been derived as a velocity-controlled oscillator on given coordinates (GridPE, Li et al. 2024); the machine-learning models of the Howard and Tiganj labs keep the Laplace variable real and set the complex (oscillatory) case aside as unstable (Mochizuki-Freeman et al. 2023; Kabir et al. 2025). In the rotary slot a signed rate is a bounded rotation, and the three cases differ not in what a channel can represent (Sec. 2.3) but in what the data and the optimiser make of it. TEM and TEM-t (Whittington et al. 2020, 2022) frame the Case III code as a "where" learned apart from the "what".

6.3 Discussion, replacement for "The integrator view" and "Text"

> **The integrator view.** A learned, signed, content-dependent rate can realise all three of Howard et al.'s cases, and which one a channel realises is set by what the readout keys on. Split each step into the frequency-weighted mean (a clock) and a content part. For targets keyed to a place reached at variable lags, a channel is usable only if its tick is zero modulo the lattice's return period, and the models we train put their used channels there (torus: clock part / content part 0.06-0.08; text world 0.06-0.23). For targets keyed to a count back, the used channels are clocks (k-back recency, the same one-layer architecture: 22). One text-world model can do both at once, with distance travelled in idle channels and position in used ones. In the rotary slot the cases coincide in what they can represent (Proposition 1), so the sign result of Section 6.2 is about learnability: a non-negative step could hold a Case III code and does not find one.
>
> **Text.** On ordinary text at small scale (analysis, n = 3), the channels attention uses are clock-dominated (clock part / content part 9.9-12.3 on TinyStories, 4.5-6.6 on code, against 0.06-0.23 in the text world), MapWM matches index RoPE on TinyStories (1.5053 against 1.5050 nats per token), and at matched length path integration costs on code (+0.0056 bits per byte, 3/3 seeds, t-test p 0.024). The phase grows linearly with the token index (R^2 0.994-0.999), but that alone does

> not exclude a map: text-world models that carry a perfect map show the same growth, carried by channels attention does not use. Where a map would pay in natural language, and whether a step that reads context would find it, is open.

6.4 Limitations, add

> - The text world is scripted (58 words, one movement clause per move, context-free direction words) and its walks are short enough that retrace scores 0.934; the map evidence there is the mechanism and the revisits retrace gets wrong.

6.5 Corrections to the current draft (Sections 7-8)

- Section 7 reports only the constant and reversal-copy floors; add retrace (0.934).
- "Half the seeds add a per-step clock alongside the map, in 31-38 of 64 channels, and solve as well": add that the clock counts moves (not words) and sits in channels retrieval does not use (section 2.2-2.3), else a reader takes it as a failure mode.
- "zeroing the non-direction words' steps costs it more (0.33-0.69 against 0.05-0.30)": the reason is the direction words' common component in the used channels, which the other words cancel (Spearman +0.905, section 2.2), not that the clock is used.
- Discussion "Text": "grows linearly with token index ... consistent with there being no map to build" -- replace by 6.3 (the linear growth is not diagnostic).
- References to add: Alipour et al. 2025; GridPE (Li et al., arXiv:2406.07049); Mochizuki-Freeman et al. 2023; Kabir et al. 2025 (arXiv:2412.15292); optionally Sarkar and Howard 2026 (arXiv:2601.02618), RoMem (arXiv:2604.11544).

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7. Proposed GPU experiments (with cost)

Costs from TW_STATECHANGE (32 text-world runs, 900 epochs, T = 1024, 1 layer, on one GPU in 5.7 h at 2 jobs/GPU, about 0.18 GPU-h per run) and TW_AMBIG's own estimate (64 runs at 1800 epochs, ~15 h). Each needs the rule-29 audit and a pre-registration before launch; pilots on outside seeds.

id	question	design	cost	priority
E1	Why do half the seeds build an idle move counter? Hypothesis: it appears early (when recency -- reversal copying, floor 0.597 -- is the cheapest cue) and attention later leaves its channels rather than unlearning it	8 fresh MapWM text-world seeds, 900 epochs, step table and per-channel usage saved every 25 epochs (no change to training). Registered readouts: epoch at which the tick exceeds 1 move and the map's opposition falls below 0.05; usage of later-clock channels over time	8 runs, ~1.5 GPU-h	medium (explains a figure panel; not a claim)
E2	Does a context-dependent step learn which uses of a word move (TW_AMBIG,	direction words also used as non-moving content; MapWM vs	64 runs at 1800 epochs, ~15 h	low for this paper (a separate question;

id	question	design	cost	priority
	prepared, STOPPED 2026-10-11)	oracle vs HSR (hidden-state step), cue near/far	(prereg's estimate)	mechanisms published, LIT_CONTEXT_STEPS)
E3	Does the used-channel ratio move with the readout inside one stream? Double dissociation	text world where a fraction of targets ask "what did she see k moves ago" (count-keyed) and the rest are revisits (place-keyed); 8 seeds. Prediction: used channels split into rho << 1 and rho >> 1, and ablating each set removes only its query type	16 runs (with a revisit-only control), ~3 GPU-h	high if the theory paragraph is kept: it turns section 3.3's across-task comparison into a within-model test
E4	Is the text-world advantage over retrace a map at long walks?	text world at 4096 words (~530 moves), where retrace should fall as on the torus (0.843 at 1024 moves); path vs RoPE 1-2 layers, 8 seeds	24 runs at 4x tokens per run (attention cost grows faster than 4x), ~20-35 GPU-h; --fast-attn valid (no MapEM)	medium: removes the short-walk caveat of 2.6
E5	Retrace floor and strata re-scored on TW_NORMSTEP / TW_STATECHANGE batches	CPU, no training	0 GPU, ~1 h CPU	do before any text-world number goes in the paper

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8. Analyses run (scripts and outputs)

All in `docs/paper/expansion/scripts/language/`; run from `/home/prashr` with `PYTHONPATH=/home/prashr python3

The paper as a whole: verdict, framing, additions, experiments, venues (2026-10-20)

Scope: `docs/paper/MAIN_DRAFT1.md` (+ `MAIN_APPENDIX1.md`, `APPENDIX_theory_v3.md`) read as a reviewer for TMLR, a strong ICLR/NeurIPS workshop and CCN. Sources read: the outline v2, `DRAFT6_workshop.md`, `review_draft5/` (mock review, synthesis, prior art, repro audits), CLAUDE.md's citable table and rank section, `docs/WHERE_THINGS_STAND.md`, `VENUES_2026-10-15.md`, `docs/lit/LIT_REPRESENTABILITY_2026-10-14.md`, and for the EM question `EM_WM_STATE.md`, `SEARCH_RESULTS.md`, `PAIRSPLIT_RESULTS.md`, `RECENCY_EM_RESULTS.md`, `WARM_RESULTS.md`, `NOLEAK_RESULTS.md`, `SPREAD_RESULTS.md`. CPU only; no file other than this one was written; nothing committed. Every number below is copied from the draft or the file named beside it. GPU costs are estimates scaled from measured timings (Sec. 4, "calibration").

0. Top three recommendations

1. **Make one claim and organise the paper around it.** The claim: *in content-dependent rotary models, per-head rank and the sign parameterisation of the step decide whether the map is found, not whether it can be represented.* The advantage results (floors, exchange rate, words) then show why finding the map matters: index depth does not buy it. Use revisit strata (short-gap / long-gap / wrap-only) as the measuring tool from Sec. 3 onward, not as a late Sec. 6.3. New title, abstract and contributions in Sec. 2.
2. **Run the sign test outside attention.** Use a one-layer diagonal complex SSM (Mamba-3-style input-dependent rotation) on Z_m tracking with inverse moves, with signed, |.|, softplus and exp (LRU-style) phase steps, plus an exp arm in the attention ring test. About 6-10 GPU-h and 1-2 days of code and checks. This is the one experiment that turns a MapFormer study into a claim about content-dependent phases, in the literature (Grazzi, Mamba-3, Shakerinava) that so far asks only what such phases can represent.
3. **Close the torus learning-rate gap.** On the torus, signed vs |.| has been run at one learning rate only (and rank too). Fix: each arm at its best of three rates, n = 8, about 15 h wall (sign only) or about 25 h (sign + rank). Every reviewer raised the learning rate. The ring answer (2/8 vs 8/8 at best rates) does not yet cover the task that carries the accuracy evidence.

EM's recency deficit (question 2) is **not** a clean third case of "representable but not found". It fits the pattern only after changing the model, and a larger budget mostly closes it (Sec. 3.2). Use it as a contrast in the Discussion and an appendix, not as a third axis.

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1. Verdict on Draft 1

Overall, as a reviewer. TMLR: accept with minor revisions. The claims match the evidence, the controls are unusually careful, and TMLR weighs correctness over significance. Strong ICLR/NeurIPS workshop (Sci4DL type): accept, but only as the 4-page sign paper; Draft 1 is three times that length. CCN 8-page: borderline. The cognitive-map content (words, what/where, rank D+1) is there, but the core of the paper

is ML learnability, and CCN reviewers will ask about egocentric actions and neural data. ICML/NeurIPS main track: weak reject as is. The reasons would be one model family at 1-4 layers, d 128-256 and 2 heads, all tasks synthetic, the sign result only in attention, and no recurrent baseline. Each of these has a costed fix (Sec. 4).

The single most compelling story. Path-integrated attention does something depth does not: one path layer holds a 1024-move torus map that four index layers or a looped block do not (0.860 / 0.864 vs 1.000). Each extra index layer buys only short-gap revisits (to 0.94); long-gap and wrap-only revisits stay at the constant floor. So the map has to come from the step map, and whether training finds it depends on two choices that do not limit what the model can represent: per-head rank and the sign of the step. In both, the solution exists, training keeps or re-finds it when it is installed, and from scratch it is not found. Every failing model, index or path, ends with the same code: local revisits served, long-gap and wrap-only revisits at the floor. No other paper has this story. MapFormer owns "path integration helps"; Grazzi and Mamba-3 own representability; Shakerinava owns "optimisation, not expressivity" for non-Abelian SSMs (`PRIOR_ART.md`: the learnability cost of a non-negative rotary step and Lemma 1 were not found anywhere).

Is the structure delivering it? Partly.

- *Two papers in one.* Title and abstract close on "optimisation, not expressivity", but Secs. 4, 5 and 7 (about 1,350 words body) never refer to it. The connection is implicit: depth does not substitute, so the map must be found. The introduction should say this in one sentence, and Sec. 5 should end on it.
- *The best tool is introduced too late.* Strata (short-gap / long-gap / wrap-only) are the paper's clearest way to tell a map from a local rule. They appear in Sec. 4, return in Sec. 5 and are unified only in Sec. 6.3 (129 words). Define them once in Setup, and make Fig. 2 the figure the paper keeps returning to. Sec. 6.3 can then be cut to one sentence closing Sec. 6.
- *The two failure axes are presented as parallel, but their evidence is not.* For rank, the solution is held (7/8) and 2x budget leaves 0/8, but there is no learning-rate test. For sign, the solution is held on the ring, perturbed and re-found on the torus, and the best-rate test is ring only. Neither has a budget test on the torus with an in-batch control for both arms. A reviewer will build this table; build it first (Sec. 2.4) and let the experiments in Sec. 4 fill its gaps.
- *Sec. 6.2 dominates.* It has 779 words of body, 684 words of captions and tables, and three of seven figures. Fig. 6 (the mechanism) can move to the appendix in an 8-page version, with one sentence left.
- *Background (678 words) is long for what it carries.* The Howard paragraph is used once in the Discussion (mock review W8). Sec. 2.4's list of orientation-fixing parameterisations belongs in Related Work at the end, or in the Discussion.
- *The protocol sentence over-generalises.* "Every verdict was pre-registered" sits beside the ring exchange rate, which is a declared secondary (the registered CANCEL primary landed in no branch; `CANCEL_RESULTS.md`), and beside "(analysis)" floors. Change it to "every contrast in Tables 1-3 and Figs. 1, 3-5 was pre-registered; the rest is marked". Check the figure list against App. A.
- *The what/where paragraph uses the wrong regime.* The checks ran on the T = 128 paper-torus checkpoints (`docs/WHAT_WHERE_CHECKS.md`), where the paper's own Sec. 4

argues a map is not needed (retrace 0.934). Re-run them on the stored T = 1024 checkpoints (eval only, CPU; Sec. 3.1), or move the paragraph to the appendix.
- *Positioning against MapFormer.* Mostly right. Say once, early, that MapFormer already shows path integration helps, and that this paper's contributions are the floors, the matched-depth rate, the learnability axes and the words.

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2. Improved title, abstract, contributions

2.1 Title

Recommended: **Representable, kept, but not found: what decides whether path-integrated attention learns a map**

- "Kept" names the leg that separates this work from a plain optimisation failure (installed solutions survive training). "What decides" names the two variables. It does not undersell Secs. 4-5, because the abstract opens with them.

Alternatives: - *Depth does not buy the map, and training must find it: path-integrated rotary attention.* This puts the advantage first. Use it only if the torus depth series is extended (Sec. 4, E8); "does not buy" is otherwise measured only up to four layers. - *Rank and sign are learnability variables: path-integrated rotary attention learns maps that index depth does not.* For an ML main track once the SSM generalisation (E1) has landed; then retitle to "content-dependent phases".

2.2 Abstract (about 215 words; for the current evidence, no new experiments)

> Content-dependent rotary position encodings let each token add a learned step to a running phase; when the steps are driven by actions, attention compares positions on a map. We study what path-integrated attention (MapFormer) learns and what decides whether training finds the map. Against measured map-free floors at matched length, one path-integrated layer solves 1024-move torus navigation (0.998, 8 of 8 seeds), while index RoPE stays below a rule that retraces the last run of moves (0.731 against 0.843), and four index layers or a looped block lift only short-gap revisits (0.860 and 0.864). At matched depth one path layer is worth about three index layers on Dyck-2 and on a biased 1D walk, and the map is learned from navigation told in English (0.979 against 0.505), with opposite direction words cancelling. Two choices in the content-to-angle map decide whether it is found: its per-head rank (rank 2: at most 2 of 8 seeds; rank 4: 8 of 8) and the sign of its step (non-negative |.|: 0 of 8; signed: 8 of 8). In both, the constrained model represents the map exactly, and training keeps or re-finds an installed solution; from scratch the gap persists at twice the budget (rank) and at each parameterisation's best learning rate (sign). Every failing model, index or path, ends with the same code: short-gap revisits served, long-gap and wrap-only revisits at the floor. Rank and sign are learnability variables, not expressivity variables.

If E1 lands positive, replace the last sentence: "The same cost of a non-negative phase step appears in a diagonal complex SSM tracking Z_m, so it is a property of content-dependent phases, not of attention."

2.3 Contributions

1. *Measuring map use.* On MapFormer's 128-move walks a map-free retrace rule scores 0.934. We measure at 1024 moves (retrace 0.843) with revisits split by stratum (short-gap / long-gap / wrap-only); only long-gap and wrap-only revisits show a map. There one path layer scores 0.998 and index RoPE 0.731 (8/8 vs 0/8 SOLVED, p 0.0002).
2. *Depth does not substitute on a 2D map.* At matched depth, one path layer is worth about three index layers on Dyck-2 (+0.353 / +0.130 / +0.045 / +0.024 at 1-4 layers) and on a biased 1D walk. On the torus it is worth more than four: index depth and a looped block lift short-gap revisits to 0.94 and leave long-gap and wrap-only revisits at the floor (0.46-0.51).
3. *Two learnability variables.* - Rank: with per-head rank 2 the torus is solved on 0-2 of 8 seeds (rank 3: 6/8; rank 4: 8/8; Fisher p 0.0070 at matched sharing). A rank-2 solution exists (0.9955) and training keeps it (7/8). - Sign: a non-negative step represents the map exactly (Proposition 1) and fixes only relative signs (Lemma 1). Training keeps an installed solution (ring 8/8) or re-finds it (torus 8/8), yet from scratch |.| solves 0/8 against 8/8 for signed steps. The gap holds at each parameterisation's best of four learning rates (2/8 vs 8/8, p 0.007). - Signed steps reach the map collectively, through zero: freezing the move latents gives 1/8 vs 7/8.
4. *Maps from words.* One path layer learns the torus told in English (0.979 vs 0.505, 6/6 vs 0/6 fresh seeds, p 0.0022), learns which words move it, and cancels opposite direction words on every seed.

The CoPE/CARoPE results (CoPE can represent the ring map yet fails it; CARoPE provably cannot cancel) go inside contribution 3, as "published monotone phases fail too, for a learnability (CoPE) or a representational (CARoPE) reason". They are not a separate contribution.

2.4 The evidence table the paper should carry (new; CPU, from existing files)

One row per axis and task, one column per leg. Cells give the status from the results files; "--" means not run.

axis, task	exists (frozen)	kept / re-found when installed	from scratch, 1x	2x-3x budget	best LR
rank 2, torus	0.9955 (`RANK_PROJ`)	kept 7/8	0-2/8 vs 8/8	0/8 vs 7/8 at 1800 ep (5/8 still descending)	-- (E2)
|.|, ring	exact (Prop. 1; `WRAP_EXISTENCE`)	kept 8/8	0-2/8 vs 8/8	3/8 at 900 vs 3/8 at 300 ep, no in-batch signed arm (`RING_BUDGET`)	2/8 vs 8/8 (`LRSWEEP`)
|.|, torus	exact	re-found 8/8 (`GENERALITY` E2)	0/8 vs 8/8	--	-- (E2)
	exact (Prop. 1f)	--	1/8 vs 8/8	--	--

axis, task	exists (frozen)	kept / re-found when installed	from scratch, 1x	2x-3x budget	best LR
softplus, torus					

The empty cells are what reviewers will ask about. E2 fills the LR column; E6 fills the budget column at scale.

2.5 Structure and page budget (8-page version; TMLR adds the bracketed items)

section	pages	change from Draft 1
1 Introduction	0.9	one sentence joining the halves: depth does not buy the map, so the step map must find it
2 Background	0.5	keep 2.1-2.3 and 2.5; move 2.4 to a closing Related Work (TMLR) or cut to two citations
3 Setup, floors and strata	0.7	define strata here; protocol sentence scoped
4 Measuring map use (Table 1)	0.7	as now, shorter
5 Depth does not substitute (Figs. 1, 2a)	0.9	end on the bridge sentence
6 Learnability: rank, sign (Table 2, evidence table, Figs. 3, 4, 5)	2.4	Fig. 6 to appendix [TMLR: keep]; 6.3 cut to one closing sentence; add SCORE_RANK and LOOP_RANK sentences (Sec. 3.1) [+ E1 SSM paragraph 0.3]
7 Maps from words (Fig. 7)	0.5	clock-seed detail to App. J
8 Discussion and limitations	0.6	EM contrast (2 sentences); egocentric-action boundary (2 sentences); TinyStories moved to limitations as a pilot
total	7.2 + refs	leaves 0.8 for figure slack

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3. Existing results: add, keep out

3.1 Add

result	where	why it strengthens	status / caveat
SCORE_RANK: PoPE's score leaves rank 2 at 3/12 SOLVED vs MapWM 3/12 at T = 1024 (rank 4 8/8 in both)	Sec. 6.1, one sentence; App. L	Shows the rank failure lives in the step map, not the score rule. Rules out the obvious alternative account	registered, NO RESCUE
LOOP_RANK search aids: rank 2 + loop x4 2/8 (0.973), 4 real layers 5/8 (0.990), plain 0/8 (0.894), rank 4 8/8	Sec. 6.1, one sentence; App. G	More search (depth) partly recovers rank 2, as a search account predicts; three loss regimes	registered, verdict UNMEASURED; depth adds 4x params, so not "search at constant capacity". Say so
What/where checks re-run on the T = 1024 checkpoints	Discussion + App. K	Moves the TEM "where apart from what" claim into the regime where a map is needed.	new eval-only CPU job (~1-2 h); the current checks are T = 128 only,

result	where	why it strengthens	status / caveat
(SIGN_MATCHED, RANK_SEP: path, index, |.|, rank 2)		Tests whether failing models also lose separation	where retrace scores 0.934
Rotation actions and allocentric recoding (+0.050 -> +0.488, 8/8)	Discussion, as a boundary (2 sentences) + appendix	The scope of an Abelian (cumsum) phase: egocentric turn/forward actions need a heading-dependent step; recoding to allocentric moves restores it. This is the head-direction / grid-cell link a CCN audience expects	16-epoch recipe, index arm on the floor, not registered. Cite as a boundary only, or rerun (E5)
EM recency, as a contrast case	Discussion (2 sentences) + App. M (one table)	Sharpens what "search failure" means; see 3.2	registered pieces; the 4x-budget contrast is not registered
MapFormer v4 Table 2 reproduction	already App. A4	keep	--

3.2 EM's recency deficit: a third case of "representable but not found"? Not a clean one

The three legs, from the files:

- **Exists: yes.** Single-p0 MapEM with the rewind installed and frozen scores 1.000 on 8/8 at T = 1024 and 2048 (`WARM_RESULTS.md` W1, registered).
- **Kept: only after changing the model and the install scale.**
- Installed at the 1/64 scale, the trainable model loses it (0.642).
- At 8x the scale it reaches 0.941 (4/8 >= 0.95).
- It is fully kept (1.000, 8/8) only when `w_in`'s content columns are held at zero, a restricted model (`NOLEAK_RESULTS.md`). Rank and sign were kept in the same class, within about 5x of trained scale.
- **Not found from scratch: partly.**
- From scratch P0 scores 0.600, 0/8 to loss < 0.5 (`RECENCY_EM_RESULTS.md`).
- But `SEARCH_RESULTS.md` shows EM does find the rewind, per query token and wrapped, for about half the offsets k.
- With one k it finds a 63-symbol rewind on 7/8 seeds (0.985), faster than WM.
- The obstacle is queries per token (`SPREAD2`). Four times the budget moves the 64-offset task from 0.578 to 0.928 (6/8 near 1.000; `SPREAD_RESULTS.md`, an arm of a registered batch, but the e1200-vs-e300 contrast was not registered).
- **Other differences from the main paper.**
- The failing model is single-p0 MapEM, an ablation. Paper-faithful MapEM scores 0.837.
- The task is k-back recency, a clock task, not a map.
- The line carries a withdrawal ("0/40 never found", withdrawn by SEARCH).

Verdict. It belongs to the same family: the solution is representable, and training from scratch is the problem. But it is a different kind of search failure. More data or budget mostly closes it (exposure per query token). Rank, by contrast, stays at 0/8 at twice the budget, and |.| on the ring stays at 3/8 at three times. As a third main-text axis it would

add a model, a task and a mechanism, and weaken "kept". Use it as a **contrast** that defines the main claim more sharply:

> "Not every search failure is of this kind: single-origin MapEM represents k-back recency exactly and finds it per query token, and four times the budget closes most of its deficit (0.578 -> 0.928), whereas rank 2 stays at 0/8 with twice the budget."

Caveat for that sentence: at 1800 epochs 5/8 rank-2 runs were still descending, so "stays" is budget-scoped. Appendix M: one table with the WARM, NOLEAK, SEARCH and SPREAD rows.

3.3 Keep out (dilutes, or is not citable as needed)

result	reason
What-to-where leak and NormStep (+0.0107)	Effect about 1%, partly training speed (r(loss, acc) -0.94); the accuracy gain did not replicate on fresh seeds (+0.0045, p 0.46); off-thesis
Loop on Match-Query (+0.346)	Cross-batch and asymmetric; per-seed drift 0.185 across batches; checkpoints missing (`REPRO_AUDIT_nonsign.md` note 8). The torus factorial already carries the loop
Match-Query "necessary for in-context maps" (0.730 vs 0.154; destruction 0.918 -> 0.074)	Not registered; checkpoints and the destruction script are missing, so it cannot be reproduced. Cite only after a registered rerun (E7)
Map-extent threshold (-0.010 / +0.015 / +0.305)	Not registered, n = 5, measured noise floor 0.150, index arms still descending. Its message (small maps need no map) is already carried, better, by the retrace floor. At most one appendix line
CSCG stitching control (+0.131)	n = 3, a probe, not a capability result
Parallel-scan timing (2.6-3.3x)	Engineering, and MapFormer's claim; one Setup clause at most
MapPoPE pair / gain grain / gain phase	Paper torus T = 128 at ceiling, or about 1% effects; App. L already has what the Dyck pairing needs
TW_STATECHANGE	Its strongest reading (state semantics) is withdrawn; what remains (location + recency) adds little to Sec. 7
TinyStories "token clock"	Pilot, n = 3, with withdrawn sub-claims; one limitations sentence, labelled pilot
Clock/map crossover (-0.280 / -0.004)	Out of distribution on both tasks
Code / Bach / Indirect replications	Off-thesis

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4. Experiments that would raise the paper's level, ranked

Calibration. - 1-layer torus at T = 1024 (2,048 tokens): about 2 s/epoch alone, and about 10 s/epoch at 4 jobs per GPU under load (`RANK_NOWRAP_PREREG.md` pilot timing). A 900-epoch run is therefore about 2.5 slot-hours, with 8 slots on the two 4090s. - 2-layer d 256 torus: 3.6 s/epoch, GPU-bound at one run per GPU (`ROBUST_PREREG.md`). - 4-layer index torus: 7.4 s/epoch alone (`LOOP_RANK_E1800_PREREG.md`). - Ring (32 cells, T = 128, 300 epochs): 64 runs take about 5.3 GPU-h, about 2.7 h wall on two idle GPUs.

Every batch below needs, per rules 7, 11, 12 and 29, a pre-registration, a CPU task gate, pilots on seeds outside the batch and a results-blind verification agent. Add about half a day of CPU work per batch. The user is cost-sensitive and shares the GPUs: check `nvidia-smi` owners before launching.

rank	experiment	design	cost	claim it u
1	**Sign beyond attention: diagonal complex SSM**	One-layer diagonal complex recurrence h_t = exp(i omega Delta(x_t)) h_{t-1} + B x_t, readout from Re/Im. Task: Z_m position tracking with +1 / -1 / stay tokens (m = 8 and 32; T = 128-512; label = current position). Steps: signed, |.|, softplus, exp. 8 seeds. Existence construction per Prop. 1 (exp and softplus: annihilated tokens step exp(0) = 1 or ln 2, made a full turn by omega). Prediction from attention: non-negative steps rarely learn from a small start; a past-pi start rescues	about 64-96 tiny runs, about 5-8 GPU-h, about 3-4 h wall; 1-2 days of code (new model; `model_baselines_extra.MambaLikeBaseline` exists from the withdrawn lm200 era, but has no input-dependent phase)	The learna cost is a pr of content- dependent parameter shown in b attention a SSMs, on t state-track benchmark Grazzi and Mamba-3 p only representa Moves the from "a MapForme to the SSM literature; answers m review W5 one uses |. since exp i LRU parameter
2	**Torus learning-rate sweep (sign, + rank)**	Observation-step-free torus, signed vs |.| at lr 1e-3 / 3e-3 / 1e-2, n = 8 fresh seeds, selection-aware best-rate test copied from `LRSWEEP`. Optional rank arms: shared r2 vs r4 at 3e-3 and 1e-2	sign only: 48 runs, about 120 slot-h, **about 15 h wall**; + rank: 80 runs, about 25 h wall; trimmed (signed 2 rates, |.| 3): 40 runs, about 12.5 h	"The cost h each arm's rate" on bo tasks, for s (optionally Removes th limitation e reviewer ra fills the evi table's LR
3	**Exp (and sigmoid-rate) steps in attention**	Ring, observation-step-free; signed, |.|, softplus, exp at each arm's best of 3 rates, n = 8. Can share a batch, code gate and verification agent with #1	about 72-96 ring runs, about 6-8 GPU-h, about 3-4 h wall	"Every non negative parameter we tried", including t used in the literature (exp)

rank	experiment	design	cost	claim it u
4	**Recurrent baseline on the torus**	LSTM/GRU, 1 and 2 layers, d 128 / 256, T = 1024, n = 8. `model_baselines_extra.LSTMBaseline` exists but was used only in the withdrawn lm200 era: re-verify it	about 16-24 runs, about 6-10 GPU-h (cuDNN LSTM is fast; untimed at 2,048 tokens, so pilot first)	Answers "v attention?" Expected: recurrent r can integra must store map (16 ob over hundr cells) in its so long-gap revisits fai memory. T separates p integration map storag TEM "whe "what" in architectur
5	**Egocentric actions, registered, at T = 1024**	Torus T = 1024; path MapWM with {translate, turn/forward commanded, turn/forward allocentric} + index RoPE (turn/forward); n = 8	32 runs, about 80 slot-h, about 10 h wall	The scope Abelian ph measured map is nee commande break it, allocentric recoding (v head-direc cells suppl restores it. headline; a limitation t into a resu
6	**Scale for sign and rank**	(a) sign, 2 layers d 256, 1800 epochs, n = 8 per arm: does the quarter-size gap (6/8 |.| still descending) close? (b) per-head rank 2 / 3 / 4 at 8 heads d 256	(a) 16 runs at about 1.8 h each GPU-bound, about 15 h wall; (b) 24 runs, about 12 h wall; together about 27 h	"The varial decide at v 256 / 8 hea budget", or honest "at the sign co speed". Fil evidence ta budget col answers th layer d 128 objection
7	Registered Match-Query rerun with destruction ablation	path vs index, n = 8, + context destruction; time a pilot (cost unmeasured; estimate 5-10 h)	5-10 h (estimate)	Restores "t is built in c with a reproducib record; a s 2D task be torus

rank	experiment	design	cost	claim it u
8	Torus index depth to 8 layers, n = 8	index 6 and 8 layers + loop x8, and L1 / L4 / path at n = 8	about 16 deep runs at about 3.7 h each GPU-bound, about 60 GPU-h + about 8 h: **about 35-40 h wall**	Turns "mo four" into a number or than eight' licenses "d does not bu map"
9	Real navigation data	--	unscoped	--

Cheapest set that changes the venue tier: #1 + #3 in one batch (about 8-12 GPU-h) and #2 sign-only (about 15 h wall), about two days of GPU in all. Add #4 (about 8 GPU-h) for any ML main-track submission. Add #5 for CCN. Then

6, if a main-track reviewer is expected to press on scale.

CPU-only additions (no GPU). - (a) The what/where checks on the T = 1024 checkpoints (Sec. 3.1). - (b) The evidence table (Sec. 2.4). - (c) Strata for the rank-2 + loop / depth arms (LOOP_RANK), to show that extra search lifts the long-gap stratum and not only short gaps. - (d) An existence construction for exp steps (as for softplus), needed by #1 and #3 anyway.

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5. Venue plan

All dates are from `VENUES_2026-10-15.md` unless marked. ICML / NeurIPS 2027 dates are **UNVERIFIED** (from past editions); check them before planning.

track	paper	venue, deadline	pages	condition
A (recommended)	full paper, Secs. 1-8 + appendix	**TMLR**, rolling; submit about mid-Dec 2026 after #2 (and ideally #1 + #3)	no limit; aim for 12-14 pages main (all 7 figures, mechanism kept, EM contrast, what/where at T = 1024) + appendix A-M	TMLR forbids concurrent archival submission; non-archival workshops are fine

track	paper	venue, deadline	pages	condition
A, parallel	sign short paper (Draft 6, cut to Secs. 4.1-4.3 + Lemma 1 + loss curves + existence figure)	**ICLR 2027 workshop** (Sci4DL-type; list 29 Nov 2026; papers about 1 Feb 2027)	4 pages + unlimited appendix	Non-archival. Sci4DL asks that previously accepted work add new results; the SSM arm (#1) would serve
A, parallel	cognitive-map abstract: depth does not buy the map, maps from words, rank D+1, what/ where (+ #5 if run)	**CCN 2027 2-page** extended abstract, 1 Apr 2027	2 pages	Non-archival, lightly reviewed; collides with nothing
A, parallel (optional)	same as TMLR	CPAL 2027 Recent Spotlight, 18 Jan 2027	250-word abstract + manuscript	Non-archival; concurrent submissions allowed
B (reach)	full paper reframed per Sec. 2.1, alternative 3 ("content-dependent phases")	**ICML 2027 main**, expected late Jan 2027 (**UNVERIFIED**)	8 pages + unlimited appendix; use the 8-page budget in Sec. 2.5	Only if #1, #2 and #4 land by early January. Fallback: NeurIPS 2027 (expected mid-May, UNVERIFIED) with #6 added, or TMLR
not recommended	full paper	CCN 2027 8-page proceedings (4 Feb 2027)	8	Archival (DOI in 2026), so it blocks TMLR/ ICML for the same content. The ML learnability core is not CCN's centre. Use the 2-page track instead

Order of work. 1. CPU now: reframe the text (Secs. 1-2 here), build the evidence table, re-run what/where at T = 1024, and add the SCORE_RANK / LOOP_RANK sentences. 2. GPU next: #1 + #3 in one batch, then #2 sign-only. 3. Decide A vs B in the first week of January, on #1's outcome. If the SSM shows the cost, B is worth trying. If not, the SSM result is a scope paragraph and A is the plan.

Page budgets by venue, in brief. - TMLR: 12-14 pages main. - ICML: 8 pages, per the table in Sec. 2.5. - Workshop: 4 pages, sign only. - CCN: 2 pages, three claims and one figure (the exchange rate or the text world).

Rotary phase steps parameterised to be non-negative rarely learn a map, even when they can represent it (one-layer models, synthetic maps)

Draft 6, 2026-10-16. Changes and word counts: `docs/paper/DRAFT6_changes.md`. *Appendices A-D and F-K:* `docs/paper/APPENDIX_draft6.md`; *appendix E (theory):* `docs/paper/APPENDIX_theory_v3.md`. *Delete this line before submission.*

Abstract

Content-dependent rotary encodings accumulate a phase theta_t = omega * sum_{u <= t} Delta_u. A non-negative step *value* costs nothing there (Delta mod 2 pi/|omega| gives the same angles), but a non-negative *parameterisation*, |s| or softplus(s), puts a barrier at zero that gradient steps cannot cross. In one-layer models trained and tested at equal length on synthetic maps (a 2D torus, a 1D ring), barrier steps solve the map on 0-4 of 8 seeds per batch against 8 of 8 for signed steps at the default recipe (6 of 8 at a third of the learning rate); on the ring |.| reaches 0.95-0.97 accuracy but rarely an exact map. Both barrier classes can represent the map exactly; built |.| solutions are kept (ring) or re-found (torus) by training, yet from scratch the class solves 0-1 of 8. The barrier removes the through-zero route by which signed steps cancel an inverse pair, but that does not explain the failure: with moves +1/+2, where no step need cross zero, |.| still solves 2 of 8 (signed 5 of 8; unmeasured). The cost holds with each step at its own best of four learning rates (2 of 8 vs 8 of 8), without weight decay and in a 2-layer d 256 torus model, where its accuracy gap shrinks. Negatives are scoped to our budgets.

1. Introduction

A rotary position code is an integrator. Index RoPE adds a constant per token; content-dependent variants let each token set its own step, non-negative by construction (CARoPE's rate in (0, 1), Veisi et al.; in a different slot, CoPE's gated counts, Golovneva et al.) or of either sign (MapFormer's path-integrated phase, Rambaud et al.; Selective RoPE, Movahedi et al.; Mamba-3, Lahoti et al.).

Sign decides what real-valued recurrences can represent: real non-negative eigenvalues cannot track parity (Sarrof et al. 2024, Thm. 2; Grazzi et al. 2025, Thm. 1), and real eigenvalues cannot count modulo m for m not a power of two (Grazzi et al., Thm. 2). Rotations escape this: a rotation by theta has eigenvalues e^{+-i theta}, real and non-negative only at multiples of 2 pi, and rotations by non-negative angles build cyclic groups (Grazzi et al., Thm. 3; Mamba-3, Sec. 3.2). A non-negative step *value* is harmless there; a non-negative *parameterisation* may not be: |s| folds and softplus(s) flattens the step at zero, so training cannot move a step through zero. Trained signed models cancel opposite moves with antipodal latents (Rambaud et al., Figs. 3d, 9b), a through-zero code the barrier excludes; signed steps reach it collectively, but losing this route does not explain the barrier's failure (Sec. 4.5).

Contributions. 1. *A learnability cost at matched length, within budget*: on a torus and a ring, barrier steps rarely reach an exact map (Sec. 4.1). 2. *Existence*. Both barrier classes represent the map exactly; built |.| solutions are held (ring) or re-found (torus) by

training, yet from scratch the class solves 0-1/8 (Secs. 3, 4.2). 3. *Lemma 1*: the barrier fixes only relative signs within a channel; for an inverse pair it removes the through-zero route, not the function. Its learnability prediction was not confirmed (Sec. 4.5). 4. *The route signed steps use* (ring): through zero, collectively via their latents; |.| learns reliably only from a start whose opposite angles sum past pi, which fails on the torus within 900 epochs (Secs. 4.3-4.4). 5. *Published monotone methods* (faithful CARoPE, CoPE) do not learn these maps within budget, CoPE although it can represent the ring map (Sec. 4.6).

2. Setup and terms

Model. MapFormer-WM (MapWM; Rambaud et al.): one layer, 2 heads, d = 128. Query and key are rotated per rotary channel c (a head and a 2D block) by theta_{t,c} = omega_c * sum_{u <= t} Delta_{c,u}, with Delta_u = W_out W_in x_u through a rank-4 latent shared across heads (MapFormer's is per head). Theta_{c,x} = omega_c Delta_{c,x} is token x's angle in channel c. **Step parameterisations.** Signed: Delta = s = W_out W_in x. |.|: Delta = |s|. Softplus: Delta = softplus(s), which starts every token at the same step (softplus(0) = 0.69), an initialisation confound |.| does not have. |.| and softplus are *barrier parameterisations*: no gradient step carries Delta through zero. *Observation-step-free* versions give steps to the move tokens only. **Tasks.** A 2D torus walk (64 x 64, T = 1024 moves interleaved with observations, 900 epochs): predict the observation at revisited cells, scored on a held-out map. A 1D ring (32 cells, T = 128, 300 epochs; maps redrawn for every walk) with forward-step probability p_plus (0.5 is balanced). A line: a 256-cell ring the walk never wraps. **Terms.** The +1 and -1 moves form an *inverse pair*. A channel *cancels* it if Theta_{c,+} + Theta_{c,-} = 0 mod 2 pi (Theta_{c,+} != 0 mod 2 pi): *through zero* if the sum is 0, *by wrapping* if a non-zero multiple of 2 pi. *SOLVED*: training loss over the final 5% of training < 0.05. **Protocol.** Every verdict was pre-registered with its decision rule (post hoc readouts are marked); every model is trained and tested at the same length; n = 8 seeds per arm unless stated. Every main-text batch but the torus sign batch (Sec. 4.1; authors' pre-launch gate, audit after the run) was checked before launch by an independent code-verification agent, an automated reviewer blind to results whose findings were amended into the pre-registration. Pilots: outside seeds. "Unmeasured": below the registered threshold; "does not learn": within the stated budget.

3. Theory

Representability. Per channel a non-negative step reaches any angle mod 2 pi (-1 = m - 1 in Z_m; Grazzi et al., Thm. 3), but our step is a shared rank-r map. *P1(a) (proved).* If at most r tokens step, with embeddings independent of the others', a |.| rank-r model reproduces a signed angle table exactly mod 2 pi. Annihilating the observation embeddings in W_in leaves only the moves stepping (2 on the ring, 4 on the torus; r = 4), so every |.| arm can represent any code in which only the moves step (built; appendix D). *P1(f) (proved).* Softplus cannot set a step to zero but need not: with the observations annihilated each steps softplus(0) = ln 2, a full turn at omega' = 2 pi / ln 2 (built from a solved ring model, logits equal to 2.8e-11; appendix E). Its existence under training is untested.

Lemma 1 (relative-sign gauge). Negate every token's step in a set of channels (equivalently omega_c; in the signed class also row c of W_out) and the second

coordinate of each such channel's query and key projections (weights and biases, every layer). Every attention score is unchanged. *Proof:* with J = diag(1, -1), J R(phi) J = R(-phi) (checked bitwise on CPU; appendix E). (i) Without wrapping, a |.| model realises a signed table with the same latent iff every row of the angle table is sign-coherent (all stepping tokens in a channel step the same way): |.| fixes only *relative* signs within a channel. (ii) It excludes no map *function* that has a sign-coherent code, which needs no wrapping: one stepping token (parity, mod-m counting) or moves stepping the same way (+1/+2: a(+2) = 2 a(+1)); that learning is then cheap is a prediction (Sec. 4.5). (iii) Through-zero cancellation of an inverse pair needs opposite signs and is excluded; the function remains as the wrapped code (a, 2 pi - a) (P1): the barrier removes a route to cancellation, not cancellation.

One channel's geometry (proved; appendix E). Let sigma = Theta_{c,+} + Theta_{c,-}. *P3 (cut circle):* from a start in the open non-negative quadrant with sigma < pi, the nearest cancelling point (in angle coordinates) lies through zero, out of |.|'s reach, and every |.| path to cancellation crosses |sigma| = pi; from pi <= sigma < 2 pi both classes are equally near, at the wrap. *P4 (sticky wall):* with a = omega |s|, dL/ds = omega sign(s) dL/da, so where the loss at the wall increases with |s|, s = 0 is a strict local minimum along s, other parameters fixed (cf. Amid and Warmuth 2020, Prop. 2). Softplus has no fold, but its slope vanishes toward zero.

The collective route (hypothesis). P3-P4 are geometry, not dynamics (appendices C, E). For signed steps, the latents at which every channel cancels through zero, {z(-1) = -z(+1)} with z = W_in x (W_out of full column rank), form a linear subspace: one latent move cancels every channel, while under |.| it gives a(-1) = a(+1), a clock. Sec. 4.3 tests whether signed models take it.

4. Results

4.1 Barrier parameterisations rarely learn the map within budget

On the torus at the training length, |.| costs -0.177 in accuracy (0.821 vs 0.998; permutation p 0.0002) and solves 0/8 against 8/8, with 3/8 |.| runs still descending at 900 epochs (Fig. 1); softplus costs -0.218 (1/8; initialisation confound). Post hoc floors (appendix B) on the same walks: blank 0.506, best n-gram 0.575, and 0.843 for "retrace", a map-free rule that copies the observation 2j steps back while a run of moves reverses the previous one; the |.| mean is below retrace (3/8 runs), as are softplus (0.781; 4/8) and index RoPE (every run, 0.72-0.74). On the ring, |.| solves 2/8 and 1/8 at p_plus 0.5 and 0.75 against 8/8 (Fisher p 0.007, 0.0014; at 0.9 unmeasured, 4/8 vs 8/8), and 3/8 with three times the training. Ring |.| models reach 0.95-0.97, above the floors (n-gram 0.554; retrace 0.860, post hoc) and the one-layer index model (0.730 at p_plus 0.5): they rarely reach an exact map, and the ring evidence is the SOLVED criterion. Post hoc, at a loss threshold of 0.1 the p_plus 0.5 contrast no longer fires (4/8 vs 8/8, Fisher p 0.077); every other contrast in Table C10 holds at 0.02, 0.1 and accuracy >= 0.99.

Barrier codes do not cancel (Fig. 2, post hoc): signed models cancel opposite moves mod 2 pi in 64-98% of channels, |.| and softplus models in 3-11%, solved runs included (unweighted; weighted by use 0.70-0.93 vs 0.01-0.10). Observation-step-free versions fail

too: ring 0/8 vs 8/8 (two batches), torus 0/8 vs 8/8 (-0.120) and 1/8 vs 8/8 (-0.104) (unlike the torus sign batch: fused attention kernel, observation-step-free classes; appendix A3).

Recipe and scale (registered, fresh seeds; appendix C13). On the observation-step-free ring the cost holds with weight decay 0 (0/8 vs 8/8, -0.127), on SOLVED at a third of the learning rate (0/8 vs 6/8, Fisher p 0.007; accuracy p 0.089), and at the default (1/8 vs 8/8); at three times the rate it was unmeasured (4/8 vs 8/8). A learning-rate sweep with each step at its own best rate (1e-3 to 3e-2, fresh seeds, selection-aware) keeps it: |.| peaks at 3e-3 with 2/8, signed solves 8/8 at 1e-3 (p 0.007; 3/8 |.| runs still descending). A 2-layer d 256 torus model (8x the parameters) keeps it (2/8 vs 8/8, Fisher p 0.007; 0.967 vs 0.999, p 0.0005), with a quarter of the one-layer accuracy gap (out of batch), 6/8 |.| runs still descending and no SOLVED contrast at loss 0.1 (5/8 vs 8/8, p 0.20; a declared secondary): at scale part of the cost may be speed. In every cell |.| shows little wrapped code, even at lr x3 with 4/8 solved (unweighted arm share 0.04-0.13 vs 0.43-1.00 signed; a declared secondary).

4.2 The |.| solution exists: kept on the ring, re-found on the torus

We convert each solved signed observation-step-free model exactly into |.| form (P1(a); appendix D): angles unchanged mod 2 pi (error <= 1.4e-6 rad), frozen accuracy equal to the source's on 8/8 seeds, weights within ~5x of the signed model's. A naive construction keeping the signed latents is also exact but ill-conditioned, and Adam loses it within ~10 epochs: the test must be posed near a trained scale.

On the ring, the constructed model trained 300 more epochs stays solved on 8/8 seeds and never leaves the solved region; weighted by how strongly attention reads each channel, it keeps its cancelling code (99% of used channels at construction, 98% at the end; a registered readout of ACCOUNTS T1, whose retrained runs are bit-identical to these: determinism, not replication). On the torus it is knocked out early (peak loss 0.11-0.21 vs <= 0.017 for the signed control; 6/8 leave the solved region) and re-finds the solution (8/8 SOLVED, 1.000) at lr 1e-4, fixed before launch (at 1e-3, descriptive, 6/8). From scratch the class solves 0/8 on both tasks.

4.3 The mechanism on the ring: the route signed steps take

We vary only the initial move angles of observation-step-free ring models, every other weight identical at a seed (Table 1).

initial move angles	signed	|.|
default (small, signs independent)	8/8	0/8
|default|: both moves rotate the same way (the |.| start)	**7/8**	0/8
large, summing to less than pi	--	2/8
large, summing to more than pi	8/8	**8/8**

Table 1. SOLVED, ring p_plus 0.5, 300 epochs, n = 8 per cell; threshold checks in Table C10.

Four registered contrasts read Table 1. (i) From the same start the sign matters: 0/8 vs 7/8 (Fisher p 0.0014; accuracy -0.092 [-0.135, -0.045]). (ii) A start past pi rescues |.|: 8/8 vs 0/8 (+0.109 [+0.085, +0.135], p 0.0002). (iii) Size alone does not: a large start with

sums below pi solves 2/8 (Fisher p 0.007 vs 8/8; sum and size are only partly separated, appendix C12). (iv) From past pi the sign makes no difference (8/8 vs 8/8). Post hoc (Fig. 3): signed models from the small starts end with 98-99% of channels opposite-signed (from 0% at the same-direction start), no cancelling channel wrapping; from past pi both signs cancel by wrapping (88% and 100% of cancelling channels).

Signed steps escape collectively (registered; fresh seeds). From the same-direction start, freezing the two move latents (W_out and omega free per channel) blocks the escape: 1/8 vs 7/8 SOLVED (Fisher p 0.010; -0.075 [-0.130, -0.020]); 15% of channels end opposite-signed against 82% with free latents. From the default start freezing has no detected effect (7/8 vs 8/8). MapFormer described the end state; this is the causal test.

Loose ends (post hoc; appendix C12). Rescued |.| models cancel in only 23% of channels, not those the one-channel flow predicts (kappa +0.09); the final gradient points into the wall for only about half of the collapsed small-start |.| steps (mixed, by a rule fixed in advance); a transformer-free toy does not reproduce the signed escape (appendix I). Why |.| fails from a small start (a wrap past |sigma| = pi, P3; the wall, P4; both; neither, Sec. 4.5) is open.

4.4 The start lever fails on the torus within 900 epochs

On the torus (observation-step-free, 4 moves) the past-pi start does not rescue |.|: 0/8 vs 1/8 from the default start (+0.005 [-0.060, +0.070]; no effect detected), and it slows the signed model to 2/8 against 8/8 (secondary); 5/8 runs per large-start arm are still descending. The default-start cost replicates (1/8 vs 8/8, Fisher p 0.0014).

4.5 Does the cost follow the relative sign? (registered)

By Lemma 1(ii) a |.| step represents, without wrapping or crossing zero, a map whose moves step the same way. Does the cost then disappear? Ring models as in Sec. 4.3, fresh seeds; moves +1/-1 (an inverse pair) vs +1/+2 (map code a(+2) = 2 a(+1), sign-coherent).

moves	signed	|.|	difference (95% CI)	label
+1/-1	1.000, 8/8	0.931, 1/8	-0.069 [-0.095, -0.045], Fisher p 0.0014	signed learns better
+1/+2	0.947, 5/8	0.883, 2/8	-0.064 [-0.200, +0.075], Fisher p 0.31	unmeasured

Table 2. Accuracy (T = 128) and SOLVED, n = 8 per cell.

The cost replicates on the inverse pair. +1/+2 needs integration (index RoPE 0.649, 0/8; floor ~0.50), and signed models learn it with a sign-coherent code (0.92 of channels end with both move angles of one sign, 0.01 on +1/-1; a declared secondary), as Lemma 1 allows. The registered prediction is not confirmed: |.| solves +1/+2 on only 2/8, no more than +1/-1 (1/8; Fisher p 1.00), though no zero crossing is needed; the comparison is unmeasured because +1/+2 (all revisits long-gap laps) is harder for signed steps too (5/8). On laps, the stratum both tasks share, |.| scores 0.893 and 0.884 (post hoc reading of a declared secondary). So losing the through-zero route does not account for |.|'s difficulty here; Lemma 1 is a statement about representation only.

4.6 Published monotone methods (within budget)

Faithful CARoPE (a per-head rate in (0, 1)) cannot represent cancellation (*P6, proved*: the shortest return advances each channel by (0, 4] < 2 pi); it scores 0.667 on the torus, below index RoPE's 0.722 (p 0.040), and 0.741 vs 0.730 on the ring (unmeasured). MapFormer reports CoPE failing 1D and 2D navigation (their Table 2) and attributes it to the gate's sign. But CoPE can represent ring cancellation (constructed; appendix H), so its obstacle is not representational; it solves 0/8 (0.887 vs signed 1.000, p 0.0002; 1/32 runs at the paper's initialisation over three ring batches, pooled post hoc). On the line CoPE solves 1/16 (ring 0/16), signed steps 8/8, and the index fails (0.772). For CoPE the loss criterion is fragile: 5/16 line runs score >= 0.9997 held out yet miss the (dropout-on) loss cut, and at held-out accuracy >= 0.999 the line reads 5/16 vs 2/16 (post hoc), so whether the wrap adds to CoPE's difficulty is unmeasured. The precision its count needs is a plausible, untested account (appendix H).

5. Related work

Integrators and position. Per channel the rotary position factor is the propagator of Howard et al.'s (2014) rate-driven integrators with an imaginary rate constant (appendix E; Mamba-3, Prop. 3): constant, non-negative and signed rates give elapsed time (index RoPE), a clock and a map. TEM-t (Whittington et al. 2022) and MapFormer put action-driven position codes into transformers. GRAPE's (Zhang et al.) non-negative content phase is a real-slot bias, as in FoX (Lin et al.).

Sign, state tracking and learnability. In diagonal SSMs, parity needs input dependence and negative eigenvalues in the same layer (Khavari et al. 2025, Thm. 4.4); none of these theorems restricts an input-dependent rotation by a non-negative angle. Shakerinava et al. (2026) find a learnability gap for input-dependent rotations in unit-modulus SSMs: non-Abelian solutions exist but are not reached from scratch, and near-solution starts recover them; ours is a gap created by one parameterisation choice, on single-layer Abelian tasks that their signed-angle models learn. We give Sarrof et al.'s and Grazzi et al.'s advice, allow a sign, for a different reason.

Sign-constrained parameterisations. A reparameterised weight cannot change sign under gradient flow (Woodworth et al. 2020; Amid and Warmuth 2020, Prop. 2 at tau = 0: w = |u| is gradient descent on the non-negative orthant). In orthogonal RNNs the +/-1 diagonal of the scaled Cayley parameterisation is fixed during training, its count of -1s a per-task hyperparameter (Helfrich et al. 2018). The LRU (Orvieto et al. 2023) needs its positive phases initialised near zero to learn PathX; they are input-independent, so by Lemma 1(ii) their sign is a convention. Dale's-law networks ask the same question, constraint or start (Cornford et al. 2021; Li et al. 2023). With fixed frequencies the parameter symmetries of rotary attention are per-plane complex scalars, with no reflection (Wang and Wang 2025; Tran et al. 2026); to our knowledge the reflection a learnable step adds (Lemma 1), the loss of the through-zero route for inverse pairs, and the learnability cost of a non-negative rotary step have not been stated. We claim their conjunction and the measurements, not any single ingredient.

6. Limitations

"Does not learn" is scoped to our budgets: 3x the ring budget did not help; learning rates were swept on the ring only (each step at its best of four); in a larger torus model the accuracy gap shrinks and |.| runs were mostly still descending; no other optimisers or longer budgets at scale. Small models (one layer, 2 heads, d = 128, a rank-4 latent shared across heads), synthetic tasks, n = 8, no recurrent, SSM or exp(.)-type (LRU) baselines. On the ring the evidence is mainly a loss criterion, which at p_plus 0.5 does not survive a looser threshold (0.1; post hoc). On the torus the barrier class has less latent slack (the wrapped code also needs the 2 pi k / omega offsets), and per-head rank, known to decide torus learnability for signed steps, is uncontrolled. Softplus carries an initialisation confound; its representability is proved, its existence under training untested. The route is shown on the ring only and does not explain the |.| failure. CARoPE and CoPE were tested at one layer (CoPE on ring and line). Natural text is not studied.

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Titles and author lists checked against arXiv, OpenReview, Crossref, ACL Anthology or PMLR metadata on 2026-10-15/16; venues marked * were not re-checked. Liu et al., Hahn and Rofin, and Yang et al. are cited in appendix K only.

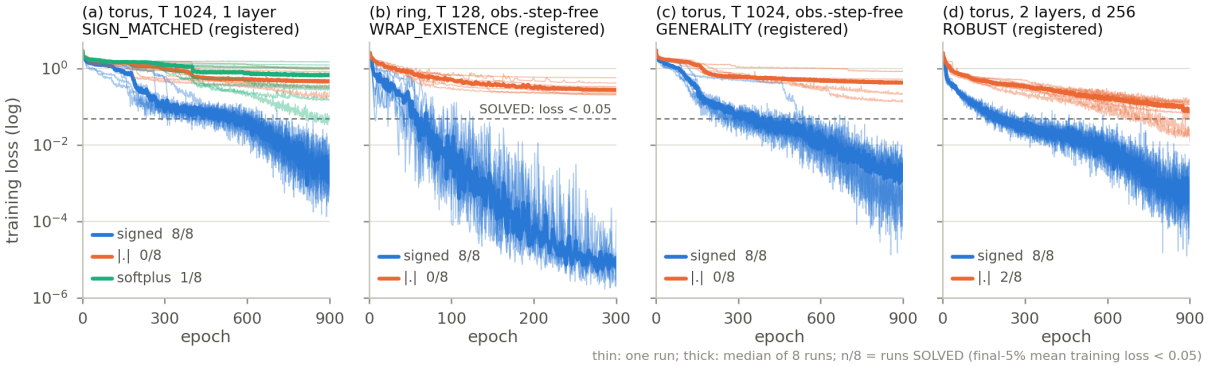
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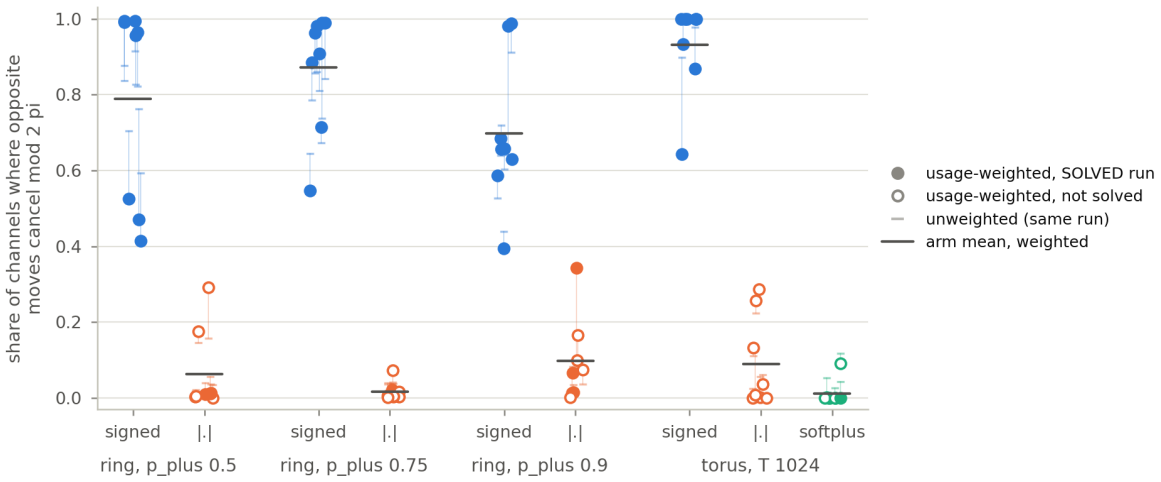
Figures and tables

- Table 1 (Sec. 4.3): start x step parameterisation. Table 2 (Sec. 4.5): moves +1/-1 vs +1/+2. Appendix tables C1-C13 (and C9b) are in `APPENDIX_draft6.md`.
- **Fig. 1** (`docs/paper/figs/fig_loss_curves.png`; Sec. 4.1). Training loss per epoch for the signed step (blue) and the non-negative steps |.| (orange) and softplus (green), every run of four registered batches (thin lines; thick: median of 8 runs; log scale; training loss with dropout on). Dashed line: the registered SOLVED threshold (final-5% mean training loss < 0.05); n/8 = runs SOLVED. (a) Torus, T = 1024, one layer (SIGN_MATCHED): signed 8/8, |.| 0/8, softplus 1/8. (b) Ring of 32 cells, observation-step-free (WRAP_EXISTENCE): 8/8 vs 0/8. (c) Torus, observation-step-free (GENERALITY): 8/8 vs 0/8. (d) As (c) at two layers and d_model 256 (ROBUST): 8/8 vs 2/8. The signed median crosses the threshold at epoch 480 / 57 / 309 / 183 of

900 / 300 / 900 / 900 and keeps descending; the |.| median never does, and most |.| runs flatten at 0.3-0.5 (STALLED on the ring 8/8; 5-6/8 on the 1-layer torus). The 2-layer |.| runs are still descending at 900 epochs (6/8), so there part of the cost may be speed (budget-scoped).

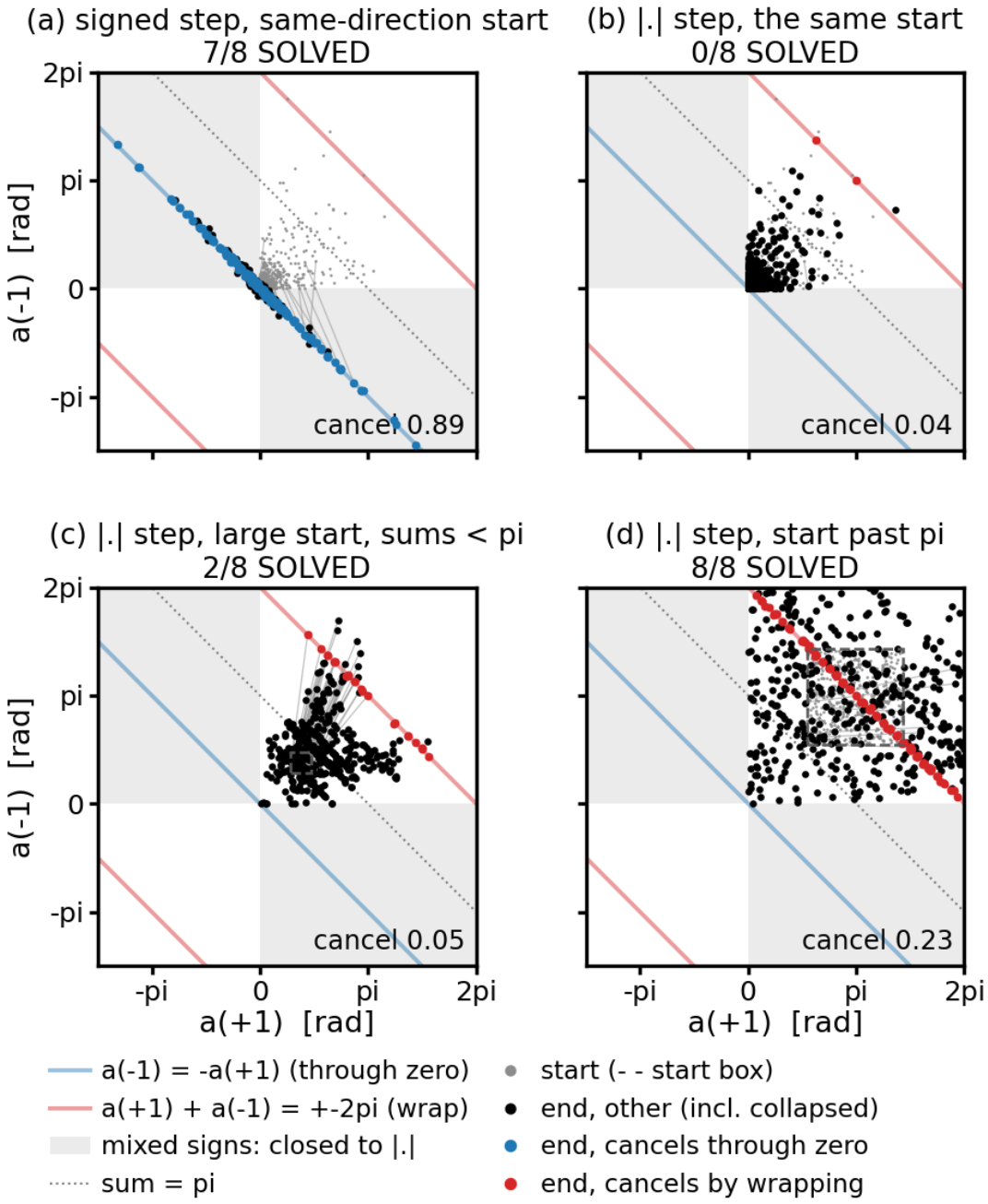


- **Fig. 2** (`docs/paper/figs/fig_wrap_share.png`; Sec. 4.1). Share of rotary channels in which opposite moves cancel mod 2 pi, per run, weighted by how much attention uses each channel (dots; filled = SOLVED) and unweighted (faint dashes, joined to the same run). A channel cancels when |wrap(a(move) + a(opposite move))| < 0.1 rad; channels whose move angle has collapsed to within 0.05 rad of 0 mod 2 pi are excluded from the denominator. Usage weight: mean query norm of the channel at move tokens times mean key norm at observation tokens (eval walks). Ring of 32 cells at three move biases p_plus (RING_DIAL, n = 8 each) and the torus at T = 1024 (SIGN_MATCHED, n = 8). Bars: arm means of the weighted share. Signed steps: 0.70-0.93; |.| steps: 0.02-0.10; softplus: 0.01. Seven of the eight non-negative runs that solve stay at or below 0.07.

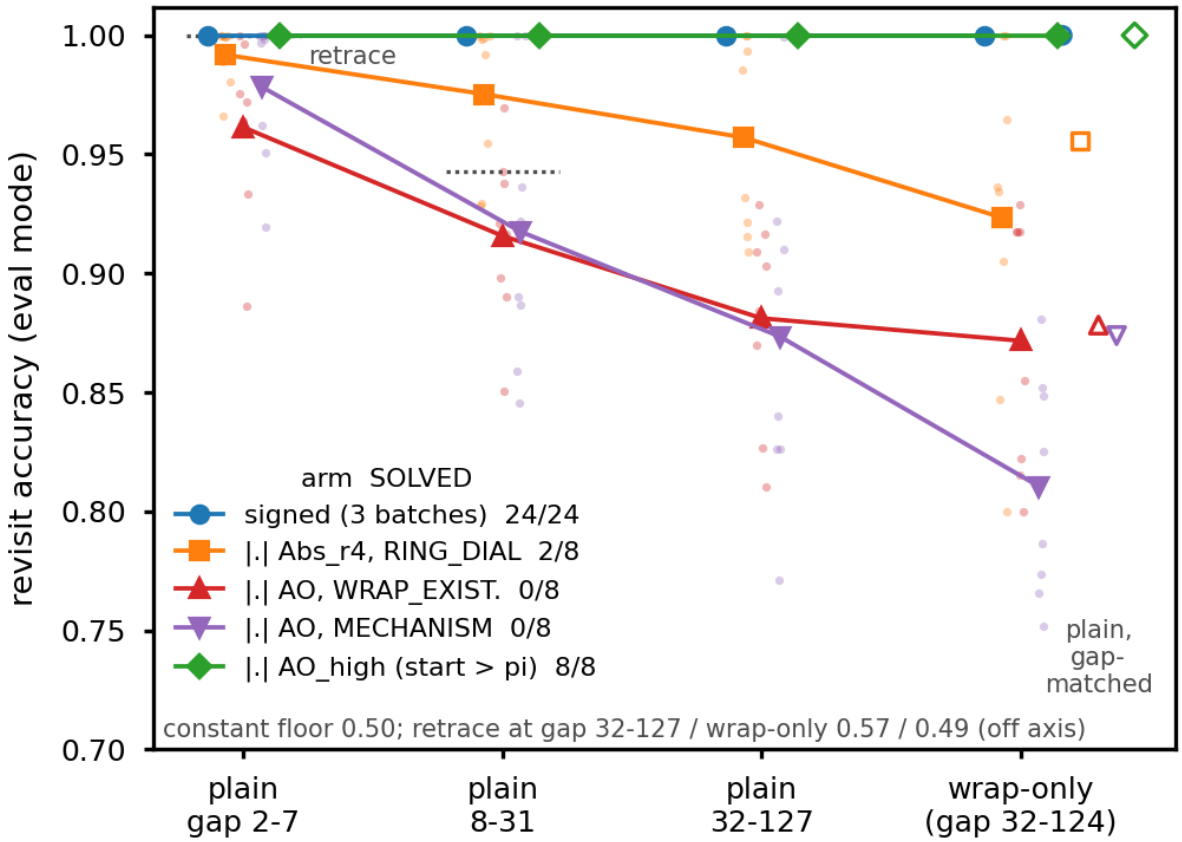


- **Fig. 3** (`docs/paper/figs/fig_mechanism.png`; Sec. 4.3; post hoc). Move angles (a(+1), a(-1)) of every rotary channel, at the start (grey) and after training (dots). There are 8 runs per panel, on the 1D ring. A channel cancels opposite moves when a(+1) + a(-1) = 0 mod 2 pi, on the blue line (through zero; reached only with opposite signs) or the red line (by wrapping). Grey shading marks the two mixed-sign quadrants, the only region the |.| step cannot reach (omega's sign is free, so both angles may be negative); the dotted line is a(+1) + a(-1) = pi; dashed boxes are the U(lo, hi) start boxes. The panels: - Signed step from the same-sign start (SO_absmag): 7/8 runs solve. One move's angle crosses zero: 6% of channels cancel at the start, 89% at the end (88% with opposite signs, none by wrapping). - The |.| step from the same

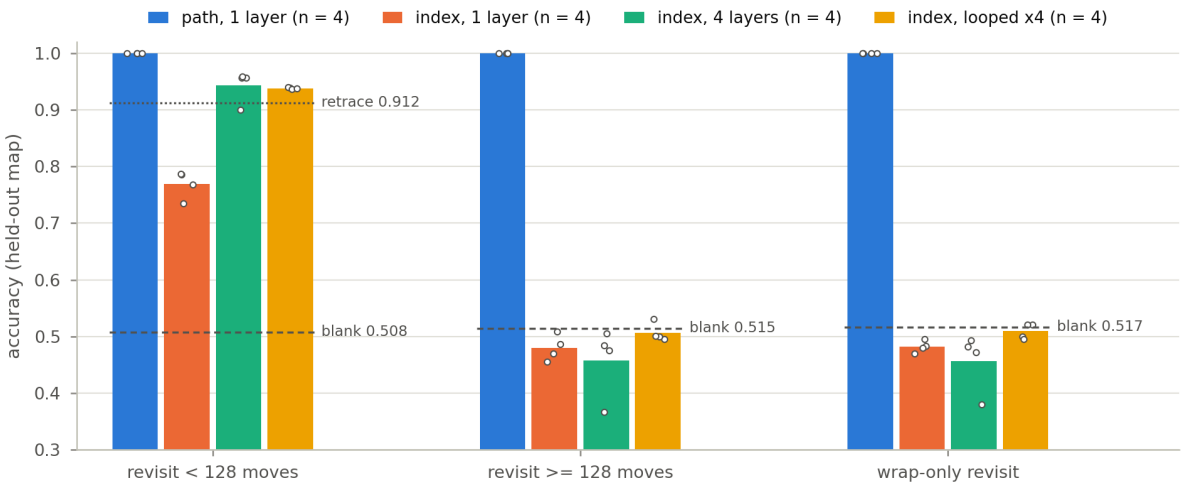
start (AO_absmag): 0/8 runs solve, with 4% cancelling. - The |.| step from a large start whose sums stay below pi (AO_mid): 2/8 runs solve, with 5% cancelling, all by wrapping. - The |.| step from a large start whose sums exceed pi (AO_high): 8/8 runs solve, and cancelling rises from 7% to 23%, all by wrapping. The cancelling share is the registered wrap share: |wrap(a(+1) + a(-1))| < 0.1 rad among channels with |wrap(a(+1))| > 0.05.



- **Fig. C1** (appendix C; `docs/paper/figs/fig_ring_strata.png`). Ring revisit accuracy by gap and wrap-only stratum (post hoc).



- **Fig. K1** (appendix K; `docs/paper/figs/fig_factorial_strata.png`). Revisit accuracy by stratum on the torus at T = 1024 (FACTORIAL; trained and tested at T = 1024; bars: means of n = 4 runs, dots: runs). Strata: plain revisits with a gap of fewer than 128 moves since the last visit, plain revisits with longer gaps, and wrap-only revisits (the cell was seen before only from a different unwrapped position). Dashed: always-blank (the best constant) within each stratum, 0.508 / 0.515 / 0.517; dotted: the retrace rule within the short-gap stratum, 0.912 (elsewhere it equals the constant). Floors are computed on this batch's own evaluation walks. Path integration with one layer is at 1.000 in every stratum. Four index layers or a loop of four lift short-gap revisits to 0.94, just above retrace; at long gaps and wrap-only revisits every index model stays at or below the constant (0.46-0.51).



Appendices A-D and F-K (for draft 6)

Draft 6, 2026-10-16. Main text: `docs/paper/DRAFT6_workshop.md`. *Appendix E (theory) is the separate file* `docs/paper/APPENDIX_theory_v3.md`; *its statements are cited here by their labels there (A0, B0, T1 = Lemma 1, P1-P7, Remark R). Every number below is copied from the file named beside it; a CORRECTED or AUDIT block at the top of a results file supersedes its body, and was read first. Status words: registered (a pre-registered decision or declared secondary), post hoc (computed after the registered analysis; never changes a verdict), descriptive (a registered arm with no decision attached), exploratory (not registered). Nothing here is new computation: the tables are copies of committed analysis outputs, reformatted.*

Contents. A: models and recipes. B: floors and gates. C: per-seed tables and post hoc re-analyses. D: constructions. (E: theory, separate file.) F: Dyck-2 at matched depth. G: navigation in words. H: faithful CARoPE and CoPE. I: the toy. J: the clock side. K: depth.

A. Models and recipes

A1. Architecture (every navigation batch)

Source: `model.py` (`MapFormerWM`, `WMTransformerLayer`, `ActionToLieAlgebra`, `PathIntegrator`), `model_sign.py`, `model_wrap_exist.py`, `model_mono_baselines.py`, `model_baseline_rope.py`, and the trainers named in A5.

component	setting
layers	1 (appendix K also 4 real layers and one block looped x4; appendix F 1-4 layers; ROBUST torus 2 layers, d_model 256, C13)
heads, width	2 heads, d_model 128, d_head 64; FFN 4 x d_model with GELU; LayerNorm before attention and before the FFN; dropout 0.1 on attention weights, attention output and FFN output
rotary channels	32 two-dimensional blocks per head, 64 channels; every query and key coordinate is rotated
score	channel c contributes q_c^T R(theta_{t',c} - theta_{t,c}) k_c (MapWM: content Q, K rotated by the path angle; not additive)
step map	Delta = W_out W_in x, W_in: 128 -> r, W_out: r -> 64, no bias, r = 4 (_r4), one latent shared across the heads (MapFormer's is per head)
frequencies	omega_c = 2 pi (1 / Delta_max)^(i / 31) for block i = 0..31, the same ladder in each head; Delta_max = the grid size (torus 64, ring 32, line 256); omega is a free parameter (its sign is not constrained)
vocabulary	torus: 4 moves + 17 observation tokens (16 types + blank); ring and line: 2 moves + 17
parameters	~204k for the one-layer torus MapWM (`FACTORIAL_RESULTS.md`)

A2. Step parameterisations and arm names

name in the files	main-text name	step Delta (per channel, after W_out)	source
`Signed_r4` (= `Vanilla_r4`)	signed	W_out W_in x	`model.py`

name in the files	main-text name	step Delta (per channel, after W_out)	source
`Abs_r4`	|.|	|W_out W_in x|	`model_sign.py` mode "abs"
`Pos_r4`	softplus	softplus(W_out W_in x); softplus(0) = 0.693, so every token starts at about the same step	`model_sign.py` mode "pos"
`SignedAO_r4`, `AbsAO_r4` (ring); `SignedAO4_r4`, `AbsAO4_r4` (torus)	observation-step-free (AO) signed / |.|	as above for the move tokens; every other token's step is 0 by construction	`model_wrap_exist.py`
`RoPE`	index RoPE	canonical ladder base^(-2c / d_head), base 10000	`model_baseline_rope.py`
`CARoPEF`	CARoPE	appendix H	`model_mono_baselines.py`
`CoPEF` (and `_Hi`, `_Lo`, `_LoS` table starts)	CoPE	appendix H	`model_mono_baselines.py`

The |.| constraint is applied after W_out, on each of the 64 channel steps, so every channel is non-negative; applying it on the r-dimensional latent would leave W_out free to mix signs (`model_sign.py` docstring). It adds no parameters.

Initial move angles (Table 1 arms; `train_wrap_mech.py`). default: the seed's initialisation. *Same-direction start* (`_absmag`): |default angles| in the balanced gauge (appendix D2) -- the start |.| imposes. *Large, sums below pi* (`AO_mid`): a(+1), a(-1) iid U(0.9, 1.5) rad per channel. *Large, sums above pi* (`AO_high`, `SO_high`): iid U(1.7, 4.5) rad. Every other weight is the seed's default initialisation, identical across arms at a seed. Torus large starts (`train_torus_start.py`): the same distributions, iid per channel and move, balanced gauge (`TORUS_START_PREREG.md`). *Frozen move latents* (`_fz`, ACCOUNTS T2): W_in fixed and the move embeddings detached in the step path (they still train through the content path), so only W_out and omega move the angles.

A3. Optimisation (all from `train.train`, the trainers' argparse defaults and the drivers)

	torus (Secs. 4.1, 4.2, 4.4, 4.6; K)	ring and line (Secs. 4.1-4.6)
optimiser	AdamW, weight decay 0.05 (`train.train` default; ACCOUNTS T1 also 0), default betas	same
peak learning rate	1e-3	1e-3
schedule	per step: 5% linear warm-up, then cosine to 10% of the peak	same
batch	16 walks of T = 1024 moves (2,048 tokens, interleaved move / observation)	128 walks of T = 128 moves
batches per epoch, epochs	98, 900	98, 300 (RING_BUDGET also 900)
loss	cross-entropy on revisited observation positions only	same
data	3 parallel data workers (`--data-workers 3`)	same

	torus (Secs. 4.1, 4.2, 4.4, 4.6; K)	**ring and line (Secs. 4.1-4.6)**
attention kernel	SIGN_MATCHED: standard; GENERALITY, TORUS_START, FACTORIAL, ROBUST: `--fast-attn` (SDPA, TF32 off)	standard
recipe variants	ROBUST torus: 2 layers, d_model 256 (C13)	ROBUST ring: lr 3e-3 and 3.3e-4 (wd 0.05), and wd 0 (lr 1e-3) (C13)
training map	one per run (GridWorld seed = run seed); walks are runs of 1-10 repeated moves	redrawn for every walk (`environment_cancel`)

Cross-batch torus comparisons. Sec. 4.1 sets SIGN_MATCHED (full `Signed_r4` / `Abs_r4`, standard attention) beside GENERALITY and TORUS_START (observation-step-free `SignedAO4_r4` / `AbsAO4_r4`, `--fast-attn`). These differ in attention kernel and model class: `--fast-attn` agrees with the standard path to ~1e-6 in logits but is not bit-identical in training (dropout RNG). Only within-batch contrasts carry verdicts. ROBUST's torus cell shares GENERALITY's kernel and classes and differs in depth, width and seeds, so "a quarter of the one-layer gap" (Sec. 4.1) is out of batch but not across kernels.

Continuations (appendix D5): ring 300 epochs at lr 1e-3, schedule re-warmed, fresh optimiser state, walk stream offset 10^6 (`train_wrap_exist.py`); torus 300 epochs at lr 1e-4 (primary) or 1e-3 (descriptive), `--data-seed-offset 1000000` (`run_generality.sh`).

A4. Evaluation

- **Torus** (`eval_noise_refine.evaluate`): one held-out map (GridWorld seed 10000), 100 walks per run with numpy seed 1234 + run seed, revisit accuracy at T = 1024 (registered) and 2048 (out of distribution, no verdict), eval mode (no dropout). Revisit strata from `eval_rank_strata` (lag < 128, lag >= 128, wrap-only).
- **Ring and line** (`train_cancel.evaluate`): maps redrawn per walk (env seed 10000), 400 walks, numpy seed 10^6 for every run, accuracy at T = 128 (registered) and 4T (out of distribution).
- **Dropout re-score**: attention scaled by 1/(1 - p) at eval. A declared secondary in RING_DIAL, RING_BUDGET, WRAP_EXISTENCE, GENERALITY, MECHANISM and WRAP_COPE (no label changed; each results file states it), and applied to SIGN_MATCHED and TEXTWORLD by the 2026-10-04 audit (`docs/audits/2026-10-04/DROPOUT_RESCORE.md`; no verdict changed). ACCOUNTS, TORUS_START, GAUGE and ROBUST were not re-scored (ROBUST: the correction is invalid at 2 layers); FACTORIAL declared it for the 1-layer cells and does not report it.
- **Run classes** (`stats_core.classify_run`, per-epoch training loss): *SOLVED* if the mean of the final 5% of epochs is < 0.05; otherwise *STALLED* if the final-10% mean is within 5% of the preceding 10% mean, else *DESCENDING*.
- **Cancelling share** (Fig. 2, "wrap share"; draft 5's Fig. 1): among channels with |wrap(a(+1))| > 0.05 rad, the share with |wrap(a(+1) + a(-1))| < 0.1 rad (wrap-invariant). *Usage-weighted* (ACCOUNTS T1): channels weighted by query x key norm. *Opposite-sign share*: channels whose raw move angles have opposite signs. *Collapsed*: |a| <= 0.05 rad. The unweighted share excludes collapsed channels from its denominator, and unused channels dominate it (ACCOUNTS T1: 0.87 -> 0.18 unweighted vs 0.99 -> 0.98 weighted on the same runs); Fig. 2 is redrawn usage-weighted for draft 6.

A5. Batches

batch (results file)	main text	task	arms	seeds	trainer / driver	independent check before results
SIGN_MATCHED	4.1	torus	Signed_r4, Abs_r4, Pos_r4, RoPE	0-7	`train_variant` / `run_sign_matched.sh`	no: authors' pre-launch gate (`docs/audits/2026-09-27/gate_sign_matched.py`) audit after the run (2026-09-30), which reproduced its probe (no file records it as independent)
RING_DIAL	4.1	ring p_plus 0.5 / 0.75 / 0.9	Signed_r4, Abs_r4	30-37	`train_ring_dial` / `run_ring_dial.sh`	yes (Amendment 1)
RING_BUDGET	4.1	ring 0.5	Abs_r4 at 300 and 900 epochs	40-47	`train_ring_dial` / `run_ring_budget.sh`	yes
WRAP_EXISTENCE	4.1, 4.2	ring 0.5	SO, AO, constructed arms	600-607	`train_wrap_exist` / `run_wrap_existence.sh`	yes
GENERALITY	4.1, 4.2, 4.6	torus; ring 0.5	SO4, AO4, constructed; Signed_r4, CARoPEF, CoPEF, RoPE	700-707	`train_variant`, `train_ring_variant` / `run_generality.sh`	yes
MECHANISM	4.1, 4.3	ring 0.5	Table 1 arms	800-807	`train_wrap_mech` / `run_wrap_mech.sh`	yes
WRAP_COPE	4.6, H	ring 0.5	CoPEF table starts	900-907	`train_ring_variant` / `run_wrap_cope.sh`	yes
ACCOUNTS	4.2, 4.3, 4.6	ring 0.5; line	T1 continuations (600-607), T2 frozen latents (1000-1007), T4 line (1100-1115)	see arms	`train_wrap_exist`, `train_wrap_mech`, `train_ring_variant` / `run_accounts.sh`	yes
TORUS_START	4.4	torus	SO4, AO4, AO4_high, SO4_high, AO4_mid	1200-1207	`train_torus_start` / `run_torus_start.sh`	yes
GAUGE	4.5	ring 0.5, moves +1/-1 or +1/+2	SO, AO, RoPE	1300-1307	`train_gauge` / `run_gauge.sh`	yes

batch (results file)	main text	task	arms	seeds	trainer / driver	independent check before results
ROBUST	4.1, C13	ring 0.5 (+1/-1; four recipes); torus 2 layers d 256	SignedAO / AbsAO (ring), SignedAO4 / AbsAO4 (torus)	1400-1407	`train_gauge`, `train_variant` / `run_robust.sh`	yes (Amendment 1, `docs/audits/2026-10-16/robust_audit.md`)
CANCEL	B (index gate only)	ring p_plus 0.5 / 0.75 / 0.9 / 1.0	path 1 layer; index 1-3 layers	0-7	`train_cancel` / --	no
FACTORIAL	K	torus	path 1 layer; index 1, 4 layers, looped x4	500-503 (n = 4)	`train_variant` / `run_factorial.sh`	yes
TEXTWORLD	G, K	torus in words	path 1 layer; index 1, 2 layers	0-7	`train_textworld` / `run_textworld.sh`	yes (Amendment 1), bu seeds 0-1 are byte-identical to the pilot (CORRECTED block)
DYCK_MDEPTH	F, K	Dyck-2	see F	0-7	`train_dyck` / `run_dyck_mdepth.sh`	no: predates the independent checks
CLOCK_SIDE, CLOCK_SPEED, CLOCK_SPEED_REP	J	k-back recency	Signed_r4, Abs_r4, CARoPE_r4 (CLOCK_SIDE also Pos_r4, RoPE)	200-207; 210-225; 230-245	`run_clock_side.sh`, `run_clock_speed*.sh`	yes

The independent check is an automated code-verification agent: a separately launched, read-only AI code-review session given the pre-registration and the code but no results (none exist at that point), which may run smoke tests on seeds outside the batch and whose findings are written into the pre-registration as an amendment before launch (example: `docs/audits/2026-10-16/robust_audit.md`). It is independent of the session that wrote the code, not of the project. The main text uses CANCEL only through GENERALITY's checked index value (0.730); CANCEL's own numbers appear in B, labelled.

B. Floors and gates

A floor is the accuracy of the best map-free predictor on the same kind of evaluation walk. Every value is from the file named; rows marked post hoc were computed for this draft (2026-10-16); the rest are copied.

task	predictor	value	computed on	source
torus T = 1024	always predict blank		same generator and	`docs/audits/2026-10-05/score_rank_floors_out.txt`

task	predictor	value	computed on	source
		0.507 (per stream seed 0.497-0.511)	protocol as A4 (held-out map 10000, numpy seed 1234 + s, 100 walks) for s = 10-25	
torus T = 1024	best token n-gram, orders 0-5, fit on the training maps	0.576 (0.568-0.582)	same	same
torus T = 1024	retrace (copy the observation 2j steps back while a run reverses the previous run)	0.843 (0.833-0.848)	same	same; `SCORE_RANK_PREREG.md`
torus T = 1024	blank / n-gram / retrace on the SIGN_MATCHED and GENERALITY walks (post hoc)	0.506 / 0.575 / 0.843 and 0.508 / 0.578 / 0.843	each batch's own eval stream	`docs/audits/2026-10-16/sign_floors_out.txt`, `generality_floors_out.txt`
torus T = 1024, 2 layers d 256 (ROBUST)	blank / n-gram / retrace (post hoc)	0.509 / 0.578 / 0.844; every run of both arms is above retrace (lowest |.| 0.910)	ROBUST's own eval stream, seeds 1400-1407	`docs/audits/2026-10-16/robust_floors_out.txt`
torus T = 128 (MapFormer's length; PAPER2X2 eval walks)	blank / n-gram / retrace (post hoc)	0.507 / 0.598 / 0.934 (retrace per seed 0.9277-0.9455; no wrap-only revisits at this length); index RoPE (0.805) is below retrace, path r = 2 (0.971) just above	PAPER2X2's eval stream, seeds 0-7	`docs/audits/2026-10-16/paper2x2_T128_floors_out.txt` (`sign_floors.py` with `FLOOR_T=128`; n-gram still fit on T = 1024 walks of maps 10-25)
ring p_plus 0.5 / 0.75 / 0.9	retrace (post hoc)	0.860 / 0.758 / 0.612	2000 walks, numpy seed 4242	`docs/audits/2026-10-16/ring_floors_out.txt`
torus T = 1024	best constant, on each batch's own evaluation walks (identical across arms at a seed)	SIGN_MATCHED 0.501-0.512 per seed (mean 0.506); GENERALITY 0.504-0.518 (0.508);	the batch's registered eval stream	`SIGN_MATCHED_STRATA.json`, `GENERALITY_TORUS_SO4_STRATA.json`, `FACTORIAL_L1_STRATA.json` ('all' stratum, field `floor`)

task	predictor	value	computed on	source
		FACTORIAL 0.505-0.515 (0.509)		
ring p_plus 0.5 / 0.75 / 0.9	better of best constant and n-gram (orders 1-5) over preceding tokens	0.554 / 0.544 / 0.520	1000 held-out walks (env seed 10000, numpy seed 1); n-gram fit on 3000 walks of map seed 0	`CANCEL_PREREG.md`; `docs/audits/2026-09-27/gate_cancel.py`
ring p_plus 0.5 / 0.75 / 0.9	fixed-lag copy (exactly one lap back)	0.284 / 0.375 / 0.626	same	same
ring p_plus 0.5	constant predictor, per revisit stratum	0.49-0.50	the registered ring eval stream (400 walks, numpy seed 10^6)	`docs/paper/figs/FIGS_NOTES.md`
line (256 cells)	predict blank	0.503	ACCOUNTS T4 eval stream	`ACCOUNTS_RESULTS.md`
ring, moves +1/+2	predict blank	~0.50	GAUGE eval stream	`GAUGE_RESULTS.md`
torus in words	best constant / reversal-copy rule	0.505 / 0.597	the batch's eval set	`TEXTWORLD_RESULTS.md`
Dyck-2 L32 D12	best A2f floor (n-grams of order 1-6 fit on the D12 training distribution; stack-free)	0.594	gate G4	`DYCK_MDEPTH_PREREG.md`, `DYCK_MDEPTH_RESULTS.md`

Own-stream torus floors (post hoc, 2026-10-16). The n-gram and retrace floors were first computed on stream seeds 10-25 only. Recomputed on each batch's own evaluation walks with the same script (`docs/audits/2026-10-16/sign_floors.py`, a copy of `score_rank_floors.py` with the stream seeds as a parameter; n-gram still fit on the 16 training maps 10-25): SIGN_MATCHED (seeds 0-7) blank 0.506, n-gram 0.575, retrace 0.843 (per seed 0.837-0.849); GENERALITY (700-707) blank 0.508, n-gram 0.578, retrace 0.843 (`sign_floors_out.txt`, `generality_floors_out.txt` there). Against its own seed's retrace floor, SIGN_MATCHED |.| falls below on 3/8 runs (0.637, 0.690, 0.717), softplus on 4/8, index RoPE on 8/8, signed on 0/8; GENERALITY's observation-step-free |.| (AO4, mean 0.880) is above the mean retrace floor. So the unsolved barrier models' torus accuracy is, on average, at or below what a map-free local rule reaches; the gap to the signed model (0.998) and SOLVED are the evidence that the map was not found.

Ring retrace floor (post hoc, 2026-10-16). The same retrace rule on the ring (`docs/audits/2026-10-16/ring_floors.py`, 2000 walks per p_plus, scored on revisits as `train_cancel.evaluate`): 0.860 / 0.758 / 0.612 at p_plus 0.5 / 0.75 / 0.9 (blank 0.497-0.499; share of revisits inside a retrace 0.72 / 0.52 / 0.23). It is a stronger map-free floor than the registered 0.554 / 0.544 / 0.520 at p_plus 0.5 and 0.75. The ring |.| models (0.95-0.97 at p_plus 0.5) are above it. For moves +1/+2 (GAUGE) no move reverses another and retrace reduces to predicting blank (~0.50).

Torus at T = 128 (post hoc, 2026-10-16). The draft cites no T = 128 torus number; the row is for readers who know MapFormer's T = 128 results. At that length almost every revisit is local, and the map-free retrace rule scores 0.934, so a T = 128 torus accuracy below ~0.93 is not evidence of a map (PAPER2X2's converged index RoPE, 0.805, is below it; the path model's 0.971 is above it by 0.04).

Gates. Each task was gated before any GPU run (rule 11): the floors above call the task code. Whether a task needs a map at all is read from the in-batch one-layer index RoPE model, which cannot integrate content-dependent steps: torus 0.722-0.731 (GENERALITY, SIGN_MATCHED), ring p_plus 0.5 / 0.75 / 0.9 0.710 / 0.677 / 0.824 on CANCEL's six fresh seeds (0.717 / 0.676 / 0.825 with its two pilot seeds; CANCEL had no independent pre-launch check; `CANCEL_RESULTS.md` CORRECTED block) and 0.730 (GENERALITY, the value the main text uses), line 0.772 (ACCOUNTS T4), moves +1/+2 0.649 (GAUGE). Each is above its blank and n-gram floors and below the signed path model (1.000 or 0.998); on the torus and on the ring at p_plus 0.5 / 0.75 it is below the retrace floor.

C. Per-seed tables and post hoc re-analyses

Accuracy is at the training length (torus T = 1024 on the held-out map; ring T = 128), eval mode. S / SOLV = SOLVED, STAL = STALLED, DESC = DESCENDING; a number in brackets is the final-5% training loss.

C1. Torus sign batch (Sec. 4.1; `SIGN_MATCHED.json`, `SIGN_MATCHED_ANALYSIS.txt`)

seed	signed	|.|	softplus	index RoPE
0	1.000 SOLV (0.003)	0.931 DESC (0.201)	0.590 STAL (1.250)	0.730 STAL (0.983)
1	0.997 SOLV (0.003)	0.917 DESC (0.314)	0.527 STAL (1.510)	0.729 STAL (0.939)
2	1.000 SOLV (0.005)	0.868 STAL (0.492)	0.630 STAL (1.084)	0.720 STAL (0.931)
3	1.000 SOLV (0.008)	0.637 STAL (1.043)	0.972 DESC (0.158)	0.739 STAL (0.910)
4	1.000 SOLV (0.004)	0.690 DESC (0.962)	0.674 STAL (1.009)	0.737 STAL (0.928)
5	0.990 SOLV (0.003)	0.888 STAL (0.446)	0.935 DESC (0.295)	0.744 STAL (0.885)
6	1.000 SOLV (0.004)	0.922 STAL (0.349)	0.922 DESC (0.345)	0.722 STAL (0.956)
7	1.000 SOLV (0.004)	0.717 STAL (0.834)	0.995 SOLV (0.049)	0.728 STAL (0.976)
mean	0.998	0.821	0.781	0.731

Table C1. Registered: |.| - signed -0.177 (MDE ~0.14, permutation p 0.0002; SOLVED 0/8 vs 8/8, Fisher p 0.0002); softplus - signed -0.218 (p 0.0003; 1/8 vs 8/8, p 0.0014). r(final

loss, accuracy) -0.992 over 32 runs. Seed 5's signed run (0.9896) is the one below 0.99 under the accuracy criterion of C10.

C2. Ring dial (Sec. 4.1; `RING_DIAL_VERDICTS.json`)

seed	0.5 signed	0.5 |.|	0.75 signed	0.75 |.|	0.9 signed	0.9 |.|
30	1.000 S	1.000 S	1.000 S	0.994 S	1.000 S	1.000 S
31	1.000 S	0.977	1.000 S	0.958	1.000 S	0.999 S
32	1.000 S	0.987	1.000 S	0.957	1.000 S	0.885
33	1.000 S	0.964	1.000 S	0.954	1.000 S	1.000 S
34	1.000 S	1.000 S	1.000 S	0.964	1.000 S	1.000 S
35	1.000 S	0.944	1.000 S	0.967	1.000 S	0.825
36	1.000 S	0.977	1.000 S	0.993	1.000 S	0.960
37	1.000 S	0.931	1.000 S	0.958	1.000 S	0.964

Table C2. Columns are p_plus and arm. Registered (`RING_DIAL_ANALYSIS.txt`): p_plus 0.5 d -0.0275 (perm p 0.0070, 95% CI [-0.046, -0.009]), SOLVED 2/8 vs 8/8 (Fisher p 0.0070); 0.75 d -0.0320 (p 0.0002), 1/8 vs 8/8 (p 0.0014); 0.9 UNMEASURED, d -0.0459 (p 0.0256, below the 0.05 materiality floor), 4/8 vs 8/8 (p 0.0769).

C3. Three times the budget (Sec. 4.1; `RING_BUDGET_ANALYSIS.txt`; |.|, ring p_plus 0.5)

seed	900 epochs	300 epochs
40	1.000 SOLV (0.002)	0.957 STAL (0.150)
41	0.986 STAL (0.070)	0.982 STAL (0.098)
42	0.969 DESC (0.129)	0.946 STAL (0.252)
43	1.000 SOLV (0.002)	1.000 SOLV (0.003)
44	0.951 DESC (0.215)	0.895 DESC (0.301)
45	0.944 STAL (0.195)	0.998 SOLV (0.012)
46	1.000 SOLV (0.000)	1.000 SOLV (0.008)
47	0.989 DESC (0.056)	0.960 DESC (0.197)

Table C3. Registered PARTIAL: 3/8 vs 3/8 (Fisher p 1.00); the solved sets differ. No in-batch signed arm (the signed comparison, 8/8 at 300 epochs, is across batches and descriptive).

C4. Observation-step-free ring, from scratch and constructed (Secs. 4.1, 4.2; `WRAP_EXISTENCE_ANALYSIS.txt`)

seed	SO (signed)	AO (|.|)	AO_bal: frozen -> trained (max continuation loss)	SO_bal: frozen -> trained (max loss)
600	1.000 S	0.905 STAL	1.000 -> 1.000 S (0.041)	1.000 -> 1.000 S (0.019)
601	1.000 S	0.949 STAL	1.000 -> 1.000 S (0.030)	1.000 -> 1.000 S (0.042)

seed	SO (signed)	AO (|.|)	AO_bal: frozen -> trained (max continuation loss)	SO_bal: frozen -> trained (max loss)
602	1.000 S	0.853 STAL	1.000 -> 1.000 S (0.154)	1.000 -> 1.000 S (0.072)
603	1.000 S	0.951 STAL	1.000 -> 1.000 S (0.046)	1.000 -> 1.000 S (0.017)
604	1.000 S	0.907 STAL	1.000 -> 1.000 S (0.024)	1.000 -> 1.000 S (0.039)
605	1.000 S	0.930 STAL	1.000 -> 1.000 S (0.032)	1.000 -> 1.000 S (0.028)
606	1.000 S	0.950 STAL	1.000 -> 1.000 S (0.031)	1.000 -> 1.000 S (0.112)
607	1.000 S	0.922 STAL	1.000 -> 1.000 S (0.036)	1.000 -> 1.000 S (0.015)

Table C4. Registered: E3 SIGNED LEARNS BETTER (0.921 vs 1.000, d -0.079, perm p 0.0002; 0/8 vs 8/8, Fisher p 0.0002); E2 HELD (AO_bal 8/8 SOLVED, 0/8 LOST). The ill-conditioned constructions are in D3.

C5. The start manipulation (Sec. 4.3, Table 1; `MECHANISM_ANALYSIS.txt`)

seed	SO	AO	SO_same	SO_absmag	AO_absmag	AO_high	SO_high	AO_mid
800	1.000 S	0.907 STAL	1.000 S	1.000 S	0.865 STAL	1.000 S	1.000 S	0.968 S
801	1.000 S	0.894 STAL	1.000 S	1.000 S	0.922 STAL	1.000 S	1.000 S	1.000 S
802	1.000 S	0.910 STAL	1.000 S	1.000 S	0.861 STAL	1.000 S	1.000 S	0.917 STAL
803	1.000 S	0.966 STAL	1.000 S	1.000 S	0.860 STAL	1.000 S	1.000 S	0.921 STAL
804	1.000 S	0.987 STAL	1.000 S	0.866 STAL	0.933 STAL	1.000 S	1.000 S	0.976 DESC
805	1.000 S	0.888 STAL	1.000 S	1.000 S	0.857 STAL	1.000 S	1.000 S	0.890 STAL
806	1.000 S	0.880 STAL	1.000 S	1.000 S	0.889 STAL	1.000 S	1.000 S	0.928 STAL
807	1.000 S	0.925 STAL	1.000 S	1.000 S	0.945 STAL	1.000 S	1.000 S	0.951 STAL
mean	1.000	0.919	1.000	0.983	0.892	1.000	1.000	0.944

Table C5. SO_same = the default angles re-set in the balanced gauge (gauge control); SO_absmag / AO_absmag = the same-direction start; AO_mid = large start with sums below pi; AO_high / SO_high = large start with sums above pi. Registered (all n = 8): M5 AO_absmag vs SO_absmag SIGN COST AT THE |.| START (d -0.092 [-0.135, -0.045], perm p 0.0030; Fisher p 0.0014); M2 AO_high vs AO_absmag RESCUES (+0.109 [+0.085, +0.135], p 0.0002); M2b AO_high vs AO_mid (+0.056 [+0.030, +0.085], p 0.0014; Fisher p 0.0070); M6 AO_high vs SO_high NO EFFECT DETECTED; M1 SO_absmag vs SO_same UNMEASURED (-0.017); M3 gauge controls NO DIFFERENCE DETECTED; M4 AO vs SO SIGNED LEARNS BETTER (d -0.081, p 0.0002). Declared secondaries, end-of-training cancelling share: SO 0.86, SO_same 0.96, SO_absmag 0.89, AO 0.07, AO_absmag 0.04, AO_mid 0.05, AO_high 0.23, SO_high 0.34.

C6. Frozen move latents (Sec. 4.3; ACCOUNTS T2, `ACCOUNTS_ANALYSIS.txt`, seeds 1000-1007)

arm	start	move latents	mean accuracy	SOLVED	opposite-sign channels at the end, per seed
SO_same	default (balanced gauge)	free	1.000	8/8	1.00 1.00 0.98 0.98 1.00 1.00 1.00 1.00
SO_same_fz	same	frozen	0.991	7/8	0.51 0.73 0.22 0.38 0.45 0.77 0.68 0.54
SO_absmag	same-direction	free	0.985	7/8	1.00 1.00 1.00 0.41 1.00 1.00 0.19 1.00
SO_absmag_fz	same-direction	frozen	0.910	1/8	0.21 0.15 0.09 0.18 0.14 0.16 0.04 0.21

Table C6. Registered: F1 WORSE (1/8 vs 7/8, Fisher p 0.010; -0.075 [-0.130, -0.020], perm p 0.009); F2 NO EFFECT DETECTED; reading SUPPORTED. Per-seed accuracies are not printed in the analysis output; the run-level SOLVED counts are in C10.

C7. Torus, observation-step-free, from scratch and constructed (Secs. 4.1, 4.2; `GENERALITY_ANALYSIS.txt`)

seed	SO4	AO4	SO_bal (lr 1e-4)	AO_bal (lr 1e-4): peak loss, epoch of peak / of re-entry	AO_bal_hi (lr 1e-3)
700	1.000 S	0.874 STAL	1.000 S	1.000 S; 0.168, 29 / 153	1.000 S
701	1.000 S	0.707 STAL	1.000 S	1.000 S; 0.110, 40 / 121	1.000 S
702	1.000 S	0.946 DESC	1.000 S	1.000 S; 0.173, 19 / 204	0.999 S
703	1.000 S	0.879 STAL	1.000 S	1.000 S; 0.143, 52 / 149	0.950 STAL
704	1.000 S	0.892 STAL	1.000 S	1.000 S; 0.131, 13 / 119	0.960 DESC
705	1.000 S	0.907 STAL	1.000 S	1.000 S; 0.155, 13 / 148	0.992 S
706	1.000 S	0.877 STAL	1.000 S	1.000 S; 0.208, 36 / 164	1.000 S
707	1.000 S	0.960 DESC	1.000 S	1.000 S; 0.117, 23 / 139	1.000 S

Table C7. SO_bal's peak continuation losses 0.004-0.017. Registered: E3 SIGNED LEARNS BETTER (0.880 vs 1.000, d -0.120, perm p 0.0002; 0/8 vs 8/8); E2 NOT HELD by the LOST rule (6/8 LOST vs 0/8, Fisher on not-LOST p 0.0070), read by the rule fixed before launch as "perturbed and re-found" (8/8 SOLVED at the end). AO_bal_hi (descriptive): 6/8 SOLVED, 0.988.

C8. Torus start manipulation (Sec. 4.4; `TORUS_START_ANALYSIS.txt`)

seed	SO4	AO4	AO4_high	SO4_high	AO4_mid
1200	1.000 S	0.801 STAL	0.934 DESC	0.922 DESC	0.937 DESC
1201	1.000 S	0.838 STAL	0.794 STAL	1.000 S	0.908 STAL
1202	1.000 S	0.998 S	0.975 DESC	0.926 DESC	0.824 STAL
1203	1.000 S	0.899 STAL	0.931 DESC	0.905 DESC	0.906 DESC
1204	1.000 S	0.948 STAL	0.951 DESC	0.745 STAL	0.840 STAL
1205	1.000 S	0.875 STAL	0.876 STAL	0.927 DESC	0.949 DESC
1206	1.000 S	0.882 STAL	0.915 DESC	1.000 S	0.939 DESC
1207	1.000 S	0.927 STAL	0.828 STAL	0.928 DESC	0.990 DESC
mean	1.000	0.896	0.900	0.919	0.912

Table C8. Registered: TS1 NO EFFECT DETECTED (+0.005 [-0.060, +0.070], p 0.89); TS2 UNMEASURED; TS3 UNREADABLE (SO4_high 2/8 < 4); TS4 SIGNED LEARNS BETTER (1/8 vs 8/8, Fisher p 0.0014; -0.104 [-0.150, -0.060]). Strata (lag < 128 / >= 128 / wrap-only): SO4 1.000 / 1.000 / 1.000; AO4 0.939 / 0.730 / 0.632; AO4_high 0.949 / 0.777 / 0.534; SO4_high 0.970 / 0.716 / 0.627; AO4_mid 0.950 / 0.809 / 0.621. An untested account of why the past-pi start does not transfer (moved from the main text of draft 5): with 64 cells per axis and two inverse pairs per channel, a random large-angle code is far from any map. The start lever is a ring result, not a design rule.

C9. Moves +1/-1 vs +1/+2 (Sec. 4.5; `GAUGE_RESULTS.md`)

seed	SO_m	AO_m	SO_p	AO_p	RoPE_p
1300	1.000 S	0.904 STAL	0.987 STAL	0.987 DESC	0.642 STAL
1301	1.000 S	0.953 DESC	1.000 S	0.910 STAL	0.663 STAL
1302	1.000 S	0.957 STAL	1.000 S	0.904 STAL	0.655 STAL
1303	1.000 S	1.000 S	1.000 S	0.590 STAL	0.661 STAL
1304	1.000 S	0.922 STAL	0.935 STAL	0.768 STAL	0.654 STAL
1305	1.000 S	0.928 STAL	1.000 S	1.000 S	0.602 STAL
1306	1.000 S	0.880 STAL	0.654 DESC	0.908 STAL	0.660 STAL
1307	1.000 S	0.903 STAL	1.000 S	1.000 S	0.660 STAL

Table C9. _m = moves +1/-1, _p = +1/+2. Declared secondaries (revisit strata): on +1/-1, 89% of revisits are backtracks (138,800 of 155,456 on the eval stream; 87.5% in the pre-launch audit's sample; `GAUGE_RESULTS.md` carries a CORRECTED banner); on +1/+2 every revisit is a lap with gap >= 16.

cell	mean accuracy	SOLVED	accuracy 4T	loss AUC	backtrack	lap	gap < 16	gap >= 16
SO_m	1.0000	8/8	0.997	0.090	1.000	1.000	1.000	1.000
AO_m	0.9309	1/8	0.871	0.537	0.936	0.893	0.952	0.902
SO_p	0.9470	5/8	0.913	0.515	--	0.947	--	0.947
AO_p	0.8834	2/8	0.863	0.661	--	0.884	--	0.884

cell	mean accuracy	SOLVED	accuracy 4T	loss AUC	backtrack	lap	gap < 16	gap >= 16
RoPE_p	0.6494	0/8	0.468	1.341	--	0.651	--	0.651

Table C9b. The laps comparison quoted in Sec. 4.5 (|.| 0.893 vs 0.884) is a post hoc reading of these declared secondaries. Signed models on +1/+2 end with 0.92 of channels having both (wrapped) move angles of one sign (per seed 0.57-1.00), against 0.01 on +1/-1.

C10. SOLVED under four criteria (post hoc; `docs/audits/2026-10-15/reanalyses/solved_thresholds_out.md`)

Predictions and decision rules were written before the run (`PREDICTIONS.md` there); all 47 arms reproduce their registered SOLVED counts at the registered 0.05.

batch	contrast	loss < 0.02	loss < 0.05 (registered)	loss < 0.1	accuracy >= 0.99
SIGN_MATCHED torus	|.| vs signed	0/8 vs 8/8 (p 0.0002)	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	0/8 vs 7/8 (0.0014)
SIGN_MATCHED torus	softplus vs signed	0/8 vs 8/8 (0.0002)	1/8 vs 8/8 (0.0014)	1/8 vs 8/8 (0.0014)	1/8 vs 7/8 (0.0101)
RING_DIAL p 0.5	|.| vs signed	2/8 vs 8/8 (0.0070)	2/8 vs 8/8 (0.0070)	**4/8 vs 8/8 (0.0769)**	2/8 vs 8/8 (0.0070)
RING_DIAL p 0.75	|.| vs signed	1/8 vs 8/8 (0.0014)	1/8 vs 8/8 (0.0014)	2/8 vs 8/8 (0.0070)	2/8 vs 8/8 (0.0070)
RING_DIAL p 0.9	|.| vs signed	4/8 vs 8/8 (0.0769)	4/8 vs 8/8 (0.0769)	4/8 vs 8/8 (0.0769)	4/8 vs 8/8 (0.0769)
WRAP_EXISTENCE ring	AO vs SO	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)
MECHANISM ring	AO vs SO	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	1/8 vs 8/8 (0.0014)	0/8 vs 8/8 (0.0002)
MECHANISM ring	AO_absmag vs SO_absmag	0/8 vs 7/8 (0.0014)	0/8 vs 7/8 (0.0014)	0/8 vs 7/8 (0.0014)	0/8 vs 7/8 (0.0014)
MECHANISM ring	AO_high vs AO_absmag	8/8 vs 0/8 (0.0002)	8/8 vs 0/8 (0.0002)	8/8 vs 0/8 (0.0002)	8/8 vs 0/8 (0.0002)
MECHANISM ring	AO_high vs AO_mid	8/8 vs 1/8 (0.0014)	8/8 vs 2/8 (0.0070)	8/8 vs 3/8 (0.0256)	8/8 vs 1/8 (0.0014)
MECHANISM ring	AO_high vs SO_high	8/8 vs 8/8 (1.00)	8/8 vs 8/8 (1.00)	8/8 vs 8/8 (1.00)	8/8 vs 8/8 (1.00)
ACCOUNTS T2 ring	SO_absmag_fz vs SO_absmag	1/8 vs 7/8 (0.0101)	1/8 vs 7/8 (0.0101)	1/8 vs 7/8 (0.0101)	1/8 vs 7/8 (0.0101)
ACCOUNTS T2 ring	SO_same_fz vs SO_same	7/8 vs 8/8 (1.00)	7/8 vs 8/8 (1.00)	7/8 vs 8/8 (1.00)	7/8 vs 8/8 (1.00)
GENERALITY torus	AO4 vs SO4	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)
GENERALITY ring	CoPE vs signed	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	1/8 vs 8/8 (0.0014)	0/8 vs 8/8 (0.0002)
TORUS_START torus	AO4 vs SO4				

batch	contrast	loss < 0.02	loss < 0.05 (registered)	loss < 0.1	accuracy >= 0.99
		0/8 vs 8/8 (0.0002)	1/8 vs 8/8 (0.0014)	1/8 vs 8/8 (0.0014)	1/8 vs 8/8 (0.0014)
TORUS_START torus	AO4_high vs AO4	0/8 vs 0/8 (1.00)	0/8 vs 1/8 (1.00)	0/8 vs 1/8 (1.00)	0/8 vs 1/8 (1.00)
ACCOUNTS T4	CoPE line vs signed line	0/16 vs 8/8 (< 1e-4)	1/16 vs 8/8 (< 1e-4)	8/16 vs 8/8 (0.0222)	7/16 vs 8/8 (0.0095)
ACCOUNTS T4	CoPE line vs CoPE ring	0/16 vs 0/16 (1.00)	1/16 vs 0/16 (1.00)	8/16 vs 2/16 (0.0538)	7/16 vs 2/16 (0.1134)

Table C10. Two-sided Fisher p in brackets. Within this table only the RING_DIAL p_plus 0.5 contrast changes its reading (at loss < 0.1). Outside it, ROBUST's torus SOLVED contrast does not survive loss < 0.1 (5/8 vs 8/8, Fisher p 0.20; a declared secondary of that batch, C13).

Where loss-based SOLVED and held-out accuracy disagree (reproduction audit, 2026-10-16). SOLVED is the train-mode (dropout on) final-5% loss. In ACCOUNTS T4, 5/16 CoPE line runs score 0.9997-1.000 held out (NLL <= 8e-4) with loss tails 0.062-0.089, so they count as unsolved, while the one SOLVED line run (s1104) scores 0.988; the CoPE ring arm has 2/16 at

> = 0.9999 and 0/16 SOLVED. At a held-out accuracy cut of 0.999 the line reads 5/16 vs 2/16 instead of 1/16 vs 0/16 (post hoc; no test run). Single runs show the same pattern: WRAP_COPE CoPEF_Hi s904 (0.9997, tail 0.055), CoPEF_LoS s902 (0.9992, 0.059), ROBUST ring lr x3 AO s1400 (1.000, 0.138), WRAP_EXISTENCE AO_wrap s604 (0.9998, 0.162); conversely MECHANISM AO_mid s800 is SOLVED at 0.968. SIGN_MATCHED Pos_r4 s7 is the softplus arm's only SOLVED run at tail 0.049 (knife-edge); that contrast fires on accuracy regardless.

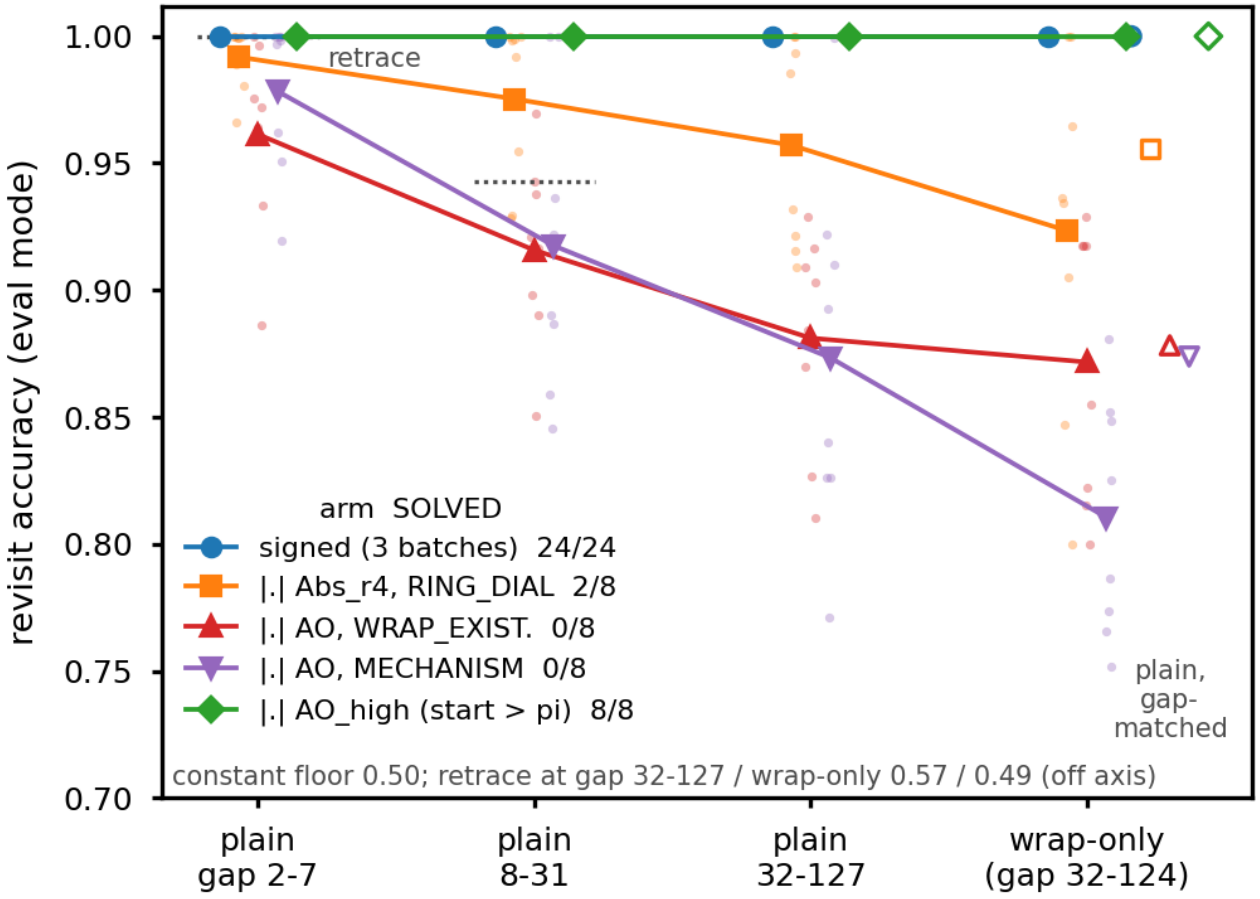
C11. Ring revisit strata and Fig. C1 (post hoc; `docs/paper/figs/ring_strata_out.txt`, `FIGS_NOTES.md`)

A revisit's *gap* is the number of moves since the last visit to the cell; a *wrap-only* revisit has every earlier visit at unwrapped displacement +-32k with k != 0. Stream: the registered ring eval stream (400 walks); overall accuracy reproduces every run's registered value (max |diff| 5e-5).

batch	arm	SOLVED	plain gap 2-7	plain 8-31	plain 32-127	wrap-only	plain, wrap-gap-matched	all
(stream)	n targets		12183	17420	5110	4134		38847
RING_DIAL	Signed_r4	8/8	1.000	1.000	1.000	1.000	1.000	1.000
RING_DIAL	Abs_r4	2/8	0.992 (0.97-1.00)	0.975 (0.93-1.00)	0.957 (0.91-1.00)	0.923 (0.80-1.00)	0.956	0.973
WRAP_EXISTENCE	SO	8/8	1.000	1.000	1.000	1.000	1.000	1.000
WRAP_EXISTENCE	AO	0/8	0.961 (0.89-1.00)	0.916 (0.85-0.97)	0.881 (0.81-0.93)	0.872 (0.80-0.93)	0.878	0.921

batch	arm	SOLVED	plain gap 2-7	plain 8-31	plain 32-127	wrap-only	plain, wrap-gap-matched	all
MECHANISM	SO	8/8	1.000	1.000	1.000	1.000	1.000	1.000
MECHANISM	AO	0/8	0.978 (0.92-1.00)	0.918 (0.85-1.00)	0.873 (0.77-1.00)	0.811 (0.75-0.88)	0.874	0.919
MECHANISM	AO_high	8/8	1.000	1.000	1.000	1.000	1.000	1.000

Table C11 (Fig. C1, `docs/paper/figs/fig_ring_strata.png`). Means over runs, range in brackets. |.| models lose accuracy with the gap in every batch (gap 2-7 to 32-127: -0.035 for Abs_r4, sign-flip p 0.031; -0.080 and -0.105 for the two AO batches, 8/8 runs lower, p 0.0078 each). The wrap-only deficit beyond gap-matched plain revisits varies between batches: -0.063 (MECHANISM AO, p 0.031), -0.007 (WRAP_EXISTENCE AO, p 0.88), -0.032 (Abs_r4, p 0.31). Signed models are at 1.000 in every stratum. Uncorrected for 3 batches x 2 contrasts; accuracy tracks loss, so the strata locate errors, not a mechanism. The constant predictor scores 0.49-0.50 in every stratum.



C12. Loose ends of the ring mechanism (post hoc; moved from the main text of draft 4)

Rules for the wall check and the channel-by-channel flow were fixed before the run (`docs/audits/2026-10-15/reanalyses/PREDICTIONS.md`).

- **The |.| start is small in most channels, not all.** Per-channel initial sums span 0.01-6.58 rad; 5% of channels start past pi (`docs/paper/figs/fig_mechanism_out.txt`; `docs/audits/2026-10-14/theory_review/mech_channels_out.txt`). Of the channels whose step does not collapse to zero, the 20 (over 8 runs) that start with sums between pi and 3 pi / 2 end with none cancelling (`mech_bins_out.txt`): being past pi is not sufficient per channel.
- **Rescued models use the wrap sparingly.** From past pi, |.| models solve with 23% of channels cancelling (signed 34%; 7% at the start by chance; `MECHANISM_RESULTS.md` secondaries). Capture falls gently with distance from 2 pi: 0.34 within 0.25 rad, 0.08 at 2-3 rad (`mech_capture_out.txt`).
- **The one-channel flow does not predict which channels cancel.** Gradient flow on the exact one-channel retrieval potential at a fresh model's attention scale (appendix E, Remark R) reproduces neither the signed escape (12% of channels cross zero) nor a sharp past-pi rescue (only channels within ~0.5 rad of 2 pi are captured: 28% of a U(1.7, 4.5) start). Run from each AO_high channel's own start (505 channels over 8 runs; `capture_by_channel_out.txt`) it predicts 0.329 cancelling against 0.230 observed; agreement with the trained models is barely above chance (Cohen's kappa +0.086; per run -0.26 to +0.51); it captures 95% of the channels starting within 0.5 rad of 2 pi, where the models cancel 28%; 59% of the models' cancelling channels are ones it does not capture. For SO_high, kappa +0.198. Whether the rescue is collective is untested; what the other 77% of channels do is unknown.
- **The wall.** 44% of channels in |.| models from the same-direction start end with the forward-move angle within 0.05 rad of zero, against 17% for signed models from that start and 0.2% and 1.4% for |.| from the large starts (`fig_mechanism_out.txt`). Per move step (`collapse_wall_out.md`): collapse is common to every |.| arm from a small start (ring 0.235-0.424, torus 0.601-0.695, softplus 0.624 included) and smaller in signed controls from the default start (ring 0.104-0.157, torus 0.343-0.402). It is partly inherited: of the |.| steps that start within 0.05 rad of zero, 77-81% end there (signed 29-35%). Attention reads the collapsed |.| channels (usage ratio collapsed / not 0.39-1.10). At the end of training the loss gradient points into the wall for 51-57% of the collapsed steps in the AO arms (eval mode, ring eval stream), which the pre-stated rule reads as MIXED, not as P4. Torus gradients were not computed.

batch	arm	collapsed move steps, mean (run range)
SIGN_MATCHED torus	Signed_r4 / Abs_r4 / Pos_r4	0.343 (0.28-0.41) / 0.695 (0.62-0.88) / 0.624 (0.54-0.71)
RING_DIAL p 0.5 / 0.75 / 0.9	Signed_r4	0.104 / 0.107 / 0.156
RING_DIAL p 0.5 / 0.75 / 0.9	Abs_r4	0.302 / 0.270 / 0.235
	Abs_r4	0.366

batch	arm	collapsed move steps, mean (run range)
RING_BUDGET (900 epochs)		
WRAP_EXISTENCE	SO / AO	0.157 / 0.337
MECHANISM	SO / SO_absmag / SO_high	0.145 / 0.171 / 0.001
MECHANISM	AO / AO_absmag / AO_mid / AO_high	0.368 / 0.424 / 0.010 / 0.001
GENERALITY torus	SO4 / AO4	0.357 / 0.660
TORUS_START torus	SO4 / AO4 / SO4_high / AO4_high / AO4_mid	0.402 / 0.601 / 0.062 / 0.061 / 0.127

Table C12 (`collapse_wall_out.md (a)`). Collapsed = |a| <= 0.05 rad, over (move, channel) entries.

- **Start size.** A start nearer the wrap also moves the angles 13-22x faster per Adam step; the sub-pi control (AO_mid, ~5x) separates sum from size only partly (`MECHANISM_RESULTS.md`, M2b).

C13. Recipe and scale (Sec. 4.1; `ROBUST_RESULTS.md`, `ROBUST_ANALYSIS.txt`; registered, seeds 1400-1407)

Observation-step-free SignedAO / AbsAO. Ring: `train_gauge.py --deltas=1,-1`, 32 cells, T = 128, 300 epochs, four recipes. Torus: GENERALITY's recipe at 2 layers, d_model 256 (1.59M parameters), T = 1024, 900 epochs. Reading: PARTLY UNMEASURED.

Registered decisions (AO vs SO; Fisher on SOLVED, permutation on accuracy) | cell | recipe | AO acc, SOLVED | SO acc, SOLVED | d [95% CI] | perm p | Fisher p | label | |---|---|---|---|---|---|---|---| | ring def | lr 1e-3, wd 0.05 | 0.911, 1/8 | 1.000, 8/8 | -0.089 [-0.125, -0.055] | 0.0002 | 0.0014 | **SIGNED LEARNS BETTER** | | ring wd0 | lr 1e-3, wd 0 | 0.873, 0/8 | 1.000, 8/8 | -0.127 [-0.200, -0.075] | 0.0002 | 0.0002 | **SIGNED LEARNS BETTER** | | ring lrx3 | lr 3e-3, wd 0.05 | 0.987, 4/8 | 1.000, 8/8 | -0.013 [-0.025, -0.005] | 0.026 | 0.077 | **UNMEASURED** | | ring lrd3 | lr 3.3e-4, wd 0.05 | 0.841, 0/8 | 0.950, 6/8 | -0.109 [-0.215, +0.010] | 0.089 | 0.0070 | **SIGNED LEARNS BETTER** | | torus d256L2 | 2 layers, d 256 | 0.967, 2/8 | 0.999, 8/8 | -0.032 [-0.055, -0.015] | 0.0005 | 0.0070 | **SIGNED LEARNS BETTER** |

Gates passed in every cell (SO >= 4/8). lrx3 is UNMEASURED, not NO SIGN COST: SOLVED differs by 4 (Fisher p 0.077) and the accuracy gap (perm p 0.026) is smaller than the registered 0.05 materiality. lrd3's accuracy contrast is not detected (perm p 0.089, CI [-0.215, +0.010]); its label rests on SOLVED. Across recipes at matched seeds (descriptive) |.| solves 0/8 (lr /3), 1/8 (default), 4/8 (lr x3): the comparison is at the signed arm's default rate, and the lr cells also change the effective decay (lr x wd) and the cosine floor (ROBUST Amendment 1). Floors (post hoc): torus on this batch's walks blank 0.509, n-gram 0.578, retrace 0.844, every run of both arms above retrace (lowest |.| 0.910); torus strata (lag < 128 /

> = 128 / wrap-only) SO 1.000 / 0.999 / 0.992, AO 0.989 / 0.911 / 0.799. The SOLVED-at-loss-0.1 column below is a declared secondary (not post hoc).

Declared secondaries:

cell	arm	acc 4T (ring 512) / 2T (torus 2048)	loss AUC	SOLVED at loss 0.1	wrap-cancel	opposite-sign
ring def	SO / AO	0.996 / 0.773	0.117 / 0.607	8/8 / 1/8	0.85 / 0.07	0.99 / 0.04
ring wd0	SO / AO	0.978 / 0.758	0.150 / 0.796	8/8 / 1/8	0.43 / 0.11	0.86 / 0.05
ring lrx3	SO / AO	1.000 / 0.934	0.077 / 0.453	8/8 / 4/8	1.00 / 0.13	1.00 / 0.07
ring lrd3	SO / AO	0.911 / 0.747	0.294 / 0.825	7/8 / 0/8	0.69 / 0.04	0.79 / 0.03
torus	SO / AO	0.983 / 0.888	0.070 / 0.319	8/8 / 5/8	1.00 / 0.11	1.00 / 0.06

Per seed (accuracy at the training length, class):

```
ring  seed | def:SO | def:AO | wd0:SO | wd0:AO | lrx3:SO | lrx3:AO | lrd3:SO | lrd3:AO
1400 | 1.0000 SOLV | 0.8230 STAL | 1.0000 SOLV | 0.9078 DESC | 1.0000 SOLV | 1.0000 DESC |
1.0000 SOLV | 0.8893 STAL
1401 | 1.0000 SOLV | 0.9155 STAL | 1.0000 SOLV | 0.6239 STAL | 1.0000 SOLV | 0.9596 STAL |
0.6194 STAL | 0.6221 STAL
1402 | 1.0000 SOLV | 0.9237 STAL | 1.0000 SOLV | 0.8878 STAL | 1.0000 SOLV | 0.9702 STAL |
1.0000 SOLV | 0.8319 STAL
1403 | 1.0000 SOLV | 0.9136 STAL | 1.0000 SOLV | 0.8800 STAL | 1.0000 SOLV | 0.9995 SOLV |
1.0000 SOLV | 0.9167 STAL
1404 | 1.0000 SOLV | 0.9118 STAL | 1.0000 SOLV | 0.9119 STAL | 1.0000 SOLV | 1.0000 SOLV |
1.0000 SOLV | 0.8313 STAL
1405 | 1.0000 SOLV | 0.8791 STAL | 1.0000 SOLV | 0.8858 STAL | 1.0000 SOLV | 0.9650 STAL |
1.0000 SOLV | 0.8526 STAL
1406 | 1.0000 SOLV | 0.9211 DESC | 1.0000 SOLV | 0.9110 STAL | 1.0000 SOLV | 1.0000 SOLV |
0.9794 STAL | 0.9062 STAL
1407 | 1.0000 SOLV | 0.9997 SOLV | 1.0000 SOLV | 0.9735 DESC | 1.0000 SOLV | 0.9998 SOLV |
1.0000 SOLV | 0.8801 STAL
torus 1400 SO 1.0000 SOLV | AO 0.9941 DESC     1404 SO 1.0000 SOLV | AO 0.9935 SOLV
      1401 SO 1.0000 SOLV | AO 0.9981 SOLV     1405 SO 1.0000 SOLV | AO 0.9706 DESC
      1402 SO 0.9941 SOLV | AO 0.9705 DESC     1406 SO 1.0000 SOLV | AO 0.9098 DESC
      1403 SO 1.0000 SOLV | AO 0.9454 DESC     1407 SO 1.0000 SOLV | AO 0.9544 DESC
```

D. Constructions

D1. Per channel (B0, P1.0)

For a signed step Delta and omega != 0, Delta' = Delta mod (2 pi / |omega|) lies in [0, 2 pi / |omega|) and omega Delta' = omega Delta (mod 2 pi) for either sign of omega. The wrap is taken in step space: the earlier form remainder(omega Delta, 2 pi) / omega is wrong for omega < 0, and trained omega does cross zero (pilot -0.012; RING_DIAL seeds 31-33) (`WRAP_EXISTENCE_PREREG.md` A2).

D2. The balanced construction (P1(a); the registered primary)

Source: `model_wrap_exist.wrap_construct(mode="balanced")`, `WRAP_EXISTENCE_PREREG.md` Amendment 1 (A1), on the torus `build_torus_exist.py`. Given a solved signed observation-

step-free model with m stepping move tokens (m = 2 ring, 4 torus; r = 4): 1. Re-solve W_in so that the move latents are c e_1, ..., c e_m (c = the source's mean move-latent norm). W_in acts only on the move embeddings (every other step is 0), so this is a free gauge. 2. Per channel set omega' = max(|omega|, 1) and the steps to remainder(omega Delta, 2 pi) / omega' (angles in [0, 2 pi) for either sign of omega; the code's form). 3. Set W_out[:, j] = step_j / c for the m move columns and 0 elsewhere. Every other weight is copied.

The signed control SO_bal is built by the same function at the same W_in and omega' with the *unwrapped* signed angle (steps omega Delta / omega'), so the two differ only in the wrap (same RNG consumption, same data stream). On the torus m = 4 = r, exactly at P1(a)'s limit.

Exactness (E1, registered). Ring: frozen accuracy = source on 8/8 seeds; max angle error 1.2e-6 rad (AO_bal), 6.1e-7 (SO_bal); max logit difference 5.1e-4 at T = 128, 1.1e-2 at T = 512 (`WRAP_EXISTENCE_ANALYSIS.txt`). Torus: frozen = source on 8/8; max angle error 1.4e-6 rad (AO_bal), 5.8e-7 (SO_bal) (`GENERALITY_ANALYSIS.txt`).

Weight scale. Ring: max |W_out| 0.31-0.38 in the signed sources, 1.07-1.70 built (AO_bal); |W_in| 0.15-0.21 -> 0.19-0.33; max step 6.28 (one turn) (`WRAP_EXISTENCE_ANALYSIS.txt`): within ~5x. Torus: max |W_out| 0.48-0.58 source -> 1.23-1.55 built (AO_bal), max |omega| 0.68-0.96 -> 1.00 (`GENERALITY_ANALYSIS.txt`).

D3. Why the naive construction fails as a test

Keeping the source's W_in and solving W_out (min-norm, `AO_wrap`) is exact (frozen accuracy 1.000 on 8/8, angle error 1.2e-5 rad) but ill-conditioned: in a solved signed ring model the two move latents are nearly antiparallel (cosine -0.987 to -0.998 over seeds 600-607), so W_out must be huge -- max |W_out| 653-16,320 against 0.31-0.38 signed, steps up to 9,333 (`WRAP_EXISTENCE_ANALYSIS.txt`). Under Adam's per-entry steps (rule 15: a parameterisation change is an optimiser change) the continuation loses the solution in the warm-up: LOST on 8/8 (max loss 0.6-2.0; Fisher on not-LOST p 0.0002 against SO_cont) and then re-finds it (7/8 SOLVED). A per-channel gauge version (`AO_wrapg`, max |W_out| 4.62 at seed 600) also loses it (8/8 LOST, 8/8 re-solved). By the reading fixed before launch, NOT HELD under an ill-conditioned construction says nothing about the class. The pilot reading of this as an intrinsic "path-length fragility" was confounded and is withdrawn (`WRAP_EXISTENCE_RESULTS.md`).

D4. The full |.| arms (observation steps present; ring audit M1)

To show that the *task* (not a given signed table) is representable by `Abs_r4`, the ring-dial audit chose W_in to project onto two directions orthogonal to all 17 observation embeddings and solved W_out per channel (`docs/audits/2026-10-12/ring_dial_audit.md` M1; `wrap_construction_t2.py`). In the real modules: |.| steps >= 0 (min 5e-11), observation-token |Delta| <= 1.7e-6, cos / sin equal to the signed code's to 6.3e-4 over a 512-step walk, circular mismatch <= 4.6e-4 on 2,000 same-cell revisit pairs. **Softplus (P1(f), appendix E; new in draft 6).** Softplus cannot reach 0, but observation steps need only be 0 mod 2 pi: with W_in annihilating every observation embedding, each observation steps softplus(0) = ln 2 in every channel, and omega'_c = 2 pi / ln 2 makes that exactly one turn; the move columns are set to softplus^{-1}((remainder(Theta, 2 pi) + 2 pi) /

omega'). Built in the real `MapFormerWM_Pos_r4` from the solved MECHANISM ring model SO_s800 (`docs/paper/review_draft5/theory_check_extra.py` (2), CPU): min step 0.693 > 0, angle error mod 2 pi 8.4e-15 rad over all 19 tokens, logits equal to the source's to 2.8e-11 (scale 21.6). On the torus m = 4 = r, so the same construction applies (not built). Existence under training (frozen and trainable continuation) was not tested.

D5. Continuations and the LOST rule

A constructed model is trained further on a fresh walk stream and compared with its signed control at the same seed. **LOST** (registered; `WRAP_EXISTENCE_PREREG.md` A4, `GENERALITY_PREREG.md`): the run's max per-epoch training loss during the continuation exceeds 0.3, or exceeds 10x its control's max at that seed and 0.05. **HELD** = SOLVED on >= n - 1 seeds AND mean accuracy >= control - 0.01 AND LOST on <= 1 seed. **NOT HELD** = Fisher (SOLVED) p < .05 with the control ahead, OR sign-flip p < .05 with d <= -0.05, OR Fisher (not LOST) p < .05 with more LOST in the constructed arm. Otherwise PARTIAL.

- **Ring** (lr 1e-3, 300 epochs, re-warmed schedule, fresh optimiser state): AO_bal HELD (8/8 SOLVED, 0/8 LOST; max per-epoch loss 0.024-0.154 vs the control's 0.015-0.112; Table C4).
- **Torus** (lr 1e-4 primary): at lr 1e-4 the control's losses are small, so LOST reduces to "left the solved region at some epoch"; the reading fixed before launch was that NOT HELD by LOST with SOLVED >= n - 1 at the end means "perturbed and re-found, not held in place" (`GENERALITY_PREREG.md` Amendment 1 A3). Why lr 1e-4: in the pilot (seed 690), re-warming to lr 1e-3 lifted the *signed* control itself from loss ~0 to ~0.1, so at 1e-3 the continuation cannot separate the arms; 1e-4 is about the learning rate at which the source solution was reached (`GENERALITY_PREREG.md`). At lr 1e-3 (descriptive): AO_bal_hi 6/8 SOLVED (0.988), max loss 0.46-0.63, SO_bal_hi 8/8 (max loss 0.15-0.21) (Table C7).
- Ring and torus E2 labels are not directly comparable (continuation lr 1e-3 vs 1e-4).

D6. What the held model keeps (ACCOUNTS T1, registered)

Weighted by usage (query x key norm per channel), the held ring AO_bal keeps 0.99 -> 0.98 of its cancelling code; the unweighted share falls 0.87 -> 0.18 because unused channels leave the code (`ACCOUNTS_RESULTS.md`). T1's wd 0.05 runs are bit-identical retrains of WRAP_EXISTENCE's AO_bal / SO_bal (reproduction audit 2026-10-16, integrity note 2), so T1 is a second readout of the same runs, not a replication of E2 HELD; its wd 0 arms are new runs. Weight decay 0 vs 0.05 changes nothing detected (D1 -0.003, NO EFFECT DETECTED). WRAP_EXISTENCE's secondary "87% -> 18% drift" and the reading "not an attractor" are withdrawn. On the torus the unweighted share goes 1.00 at construction -> 0.69 after training (GENERALITY secondary; not usage-weighted). The frozen perturbation probe (1% multiplicative noise on omega drops the wrapped ring solution to 0.802, the signed one stays 0.998) is a property of frozen models at a matched gauge, not of training (appendix E, P2).

F. Dyck-2 at matched depth

Source: `DYCK_MDEPTH_RESULTS.md`, `DYCK_MDEPTH_PREREG.md`, driver `run_dyck_mdepth.sh`, trainer `train_dyck.py`.

Task and metric. Dyck-2 (MapFormer v4 App. B.4), trained and scored at length L = 32 and nesting depth D = 12. Primary metric A2f: Hewitt's distance-averaged closing accuracy over prefixes where the sampler can emit a closer (the CE-optimal predictor scores 1.000; chance 0.500); best map-free floor 0.594 (appendix B).

Models and recipe. 2 heads, d_model 128; 1-4 layers; path-integrated arms MapWM and MapPoPE at rank 2 (`--rank 2`), index arms RoPE and PoPE at base 10000 (and base 32 in the 4-layer 3x cell); omega base 32 (the training context). AdamW lr 1e-4, weight decay 0.01, batch 128, 5% linear warm-up then cosine to 10% of the peak, dropout 0.1, next-token cross-entropy on every position; 560,000 training sequences (1x) or 1,680,000 (3x). n = 8 seeds (0-7), paired.

Results (registered). Position main effect = ((MapWM - RoPE) + (MapPoPE - PoPE)) / 2, A2f at L32 D12: - Depth ladder at 1x budget (declared secondary S1): +0.353 / +0.130 / +0.045 / +0.024 at 1 / 2 / 3 / 4 layers (8/8 seeds positive each, sign-flip p 0.0078; MDE 0.003 / 0.025 / 0.009 / 0.007). - Primary, 4 layers at 3x budget: every arm at ceiling (index 0.997-0.998, path 1.000); effect +0.002: CLOSES. The earlier ladder's +0.168 at 4 layers was depth extrapolation (trained at D = 4). - Mixture training (D drawn from 4..12, 4 layers): +0.110 at L32 D12 (8/8). - Index base 32 vs 10000 at 4 layers: within MDE (+0.001 / -0.001). The 1x budget limits the index arms (3x - 1x +0.021).

So one layer of path integration is worth about three layers of index attention on Dyck-2 at matched depth (at 3 layers the gap is +0.045; at 4, +0.024).

Protocol. The batch (2026-09-26) predates the independent code checks. Disclosed before registration: four n = 1 smoke runs with non-default flags at seed 0, whose training losses were seen (no A2 / A2f computed); the batch retrained seed 0 from scratch and nothing from the smoke runs enters it. The in-batch reproduction of two stored ladder runs was weight-exact.

G. Navigation in words

Source: `TEXTWORLD_RESULTS.md` (its AUDIT and CORRECTED blocks first), driver `run_textworld.sh`, trainer `train_textworld.py`.

Task. The 64 x 64 torus walk rendered as English: 58 words, three synonyms per direction, movement-free verbs, fillers and asides; T = 1024 words (~132 steps); held-out map; predict the observation word at revisits. Floors: best constant 0.505, reversal-copy rule 0.597 (appendix B).

Recipe. As the torus (A3): AdamW lr 1e-3, batch 16, 98 batches per epoch, 900 epochs, cosine; path MapWM r = 4 at one layer vs index RoPE at one and two layers; 200 evaluation walks.

Results. Registered (8 seeds per arm): path 0.969 +/- 0.054 (7/8 SOLVED) vs index one layer 0.505 (0/8) and two layers 0.772 (0/8); path - index one layer +0.464 (exact permutation p 0.0002; Fisher p 0.0014). **Seeds 0 and 1 are the pilot** (byte-identical to `runs/textworld_pilot`), so the numbers to quote are the six fresh seeds: path 0.979 +/- 0.048 (6/6 SOLVED) vs index 0.505 (0/6), +0.474, permutation and Fisher p 0.0022; index two layers 0.793. Dropout-scale re-score (AUDIT 2026-10-04, all 24 runs): path 0.969 -> 0.993, index 0.505 -> 0.506 and 0.772 -> 0.772; the verdict only strengthens. Declared secondaries: with the common component of the four direction steps removed, north + south and west + east cancel to 0.002-0.012 on all 8 seeds; 4/8 seeds also run a per-step clock (31-38 of 64 phase channels drift between visits to the same cell; CORRECTED block). The registered step-table branch (B) did not fire (4/8).

Scope. Scripted grammar; a direction word never appears outside a movement clause (context-free steps).

H. Faithful CARoPE and CoPE

Sources: `model_mono_baselines.py` (docstring), `GENERALITY_RESULTS.md`, `WRAP_COPE_RESULTS.md`, `ACCOUNTS_RESULTS.md` T4, `docs/audits/2026-10-14/cope_diagnosis/COPE_DIAGNOSIS.md` (post hoc), appendix E P6 and P7.

CARoPE (`CARoPEF`). Per head h and token t a scalar rate f_h(x_t) = 1 / (softplus(x_t W)_h + 1) in (0, 1) replaces RoPE's base frequency, so channel i advances by f_h(x_t)^i per token. Initialised as index RoPE (W = 0, a per-head bias with 1 / (softplus(b) + 1) = 10000^(-2 / d_head)); the bias is ours (without it the start is RoPE with base ~2e7). Every token, observations included, advances the phase; the layer is MapWM's, so only the phase differs. *P6 (proved, appendix E):* the shortest return advances each channel by a value in (0, 4] < 2 pi, so no channel reads every return at phase 0; a channel that reads the short returns is blind over the excursion's range. Registered (GENERALITY): torus 0.667 (0/8), WORSE THAN INDEX (0.722, -0.055, p 0.040); ring 0.741 vs index 0.730, UNMEASURED (+0.011, p 0.39).

CoPE (`CoPEF`). Golovneva et al.'s Listing 1, verbatim in torch: per layer one module shared by the heads; gates g_ij = sigmoid(q_i . k_j / sqrt(d_h)); count p_ij = sum_{k=j}^{i} g_ik, clamped to npos_max - 1 = 2047; position logit by linear interpolation of q_i . e[floor p] and q_i . e[ceil p], table e zeros at initialisation; added to the attention logits; no rotary position. *Representability (constructed, GENERALITY audit):* gates 17/16 and 15/16 per (move, observation) pair make p_ij mod 2 a function of net displacement, read by a period-2 table (1.000 revisit accuracy). *P7 (appendix E, proved under its setup):* the exact solutions form a lattice (period P = 2 or 3; alpha an odd multiple of P/32, i.e. move-gate difference an odd multiple of P/16).

Prior measurement. MapFormer v4 already reports CoPE failing its 1D and 2D navigation tasks in distribution (Table 2: 1 layer 0.63 +/- 0.21 in 1D and 0.65 in 2D, 4 layers 0.88 and 0.76, MapWM-r2 1.00) and attributes it to the gate's sign ("cannot represent an action and its inverse", their App.). So "CoPE fails these maps" is a replication; what is new here is that CoPE can represent ring cancellation and still fails, the line control, and the CARoPE results.

Results. | batch | task | CoPE | SOLVED | comparison | |---|---|---|---|---| | GENERALITY (700-707) | ring | 0.887 | 0/8 | signed 1.000, 8/8 (-0.113, p 0.0002); index 0.730 (+0.157, p 0.0002) | | WRAP_COPE (900-907) | ring, table at zeros (the paper's) / sinusoid past pi / below pi / slope-matched | 0.840 / 0.913 / 0.867 / 0.943 | 1/8, 2/8, 1/8, 0/8 | all contrasts UNMEASURED (bimodal seeds) | | ACCOUNTS T4 (1100-1115) | line (256 cells) / ring | 0.914 / 0.873 | 1/16 / 0/16 | line: signed 1.000 8/8, index 0.772 0/8; L1 line vs ring NO EFFECT DETECTED; L2 SIGNED LEARNS BETTER |

Per seed (ACCOUNTS T4): CoPE line 0.836 0.868 0.867 0.824 0.988 1.000 0.848 0.620 1.000 0.886 0.999 1.000 0.894 1.000 0.998 1.000; CoPE ring 0.850 0.872 0.799 0.874 0.834 1.000 0.797 0.666 0.915 0.861 0.957 0.896 0.823 0.974 0.848 1.000. Several CoPE seeds reach 0.99-1.00 accuracy without meeting the loss criterion; at loss < 0.1 half the line runs pass (8/16 vs 2/16 on the ring, Fisher p 0.054; post hoc, Table C10), and at a held-out accuracy cut of 0.999 the line reads 5/16 vs 2/16 (post hoc; C10 note). The loss-based SOLVED count is fragile for CoPE, so the line-vs-ring comparison is unmeasured, not null. The "1 of 32 runs at the paper's initialisation" of Sec. 4.6 pools GENERALITY (0/8), WRAP_COPE zeros (1/8) and ACCOUNTS T4 ring (0/16), post hoc.

Diagnosis (post hoc, eval-only, 40 ring runs; predictions written before any checkpoint was opened). Runs fall into three regimes: NONE (20/40 runs have a query type with no locked count; identity-dependent observation gates), OFF (11 runs: the count locks, local revisits are solved, but the move-gate difference is off the lattice, so wrap-only revisits sit near the floor, mean 0.55), LATTICE (9 runs, all 4 SOLVED runs among them). In the OFF runs an eval-only snap of the move gates to the nearest lattice point raises wrap-only accuracy 0.63 -> 0.90 and the straight path to the lattice rises in loss before it falls (11/11). The registered line test then showed the wrap is not CoPE's main obstacle (T4). The precision a count needs (on the line, a margin of at most P / (D_max + 1) ~ 0.016; appendix E, P7) is a plausible account, not a tested one; the near-wrap table start does not measurably help (WRAP_COPE, unmeasured).

I. The toy (exploratory, not registered)

Source: `MECHANISM_PREREG.md` (exploratory section and Amendment 1, A1), `ACCOUNTS_RESULTS.md` T3, `runs/toy_wrap/summary.txt`, `runs/accounts/t3/summary.txt`. A 64-channel retrieval model with no transformer and no observation tokens: two move angles per channel (rank-4 latents, signed or |.|, MapWM's frequency ladder), softmax retrieval of earlier moves at the same ring cell by sum_c w_c cos(theta_c(t) - theta_c(t')). SOLVED = retrieval accuracy >= 0.99.

start (move angles)	signed	|.|
default	6/12 (median 0.946)	0/12 (0.516)
|default| (the |.| start)	0/8 (0.406)	0/8 (0.385)
large, sums below pi, U(0.9, 1.5)	0/8	0/8
large, sums above pi, U(1.7, 4.5)	8/8	11/11

It reproduces the |.| failure without a transformer, but its signed model fails often from the default start and never escapes the |.| start (0/8), unlike the transformer (7/8). An observation-retrieval target (ACCOUNTS T3, seeds 10-17) does not open the escape

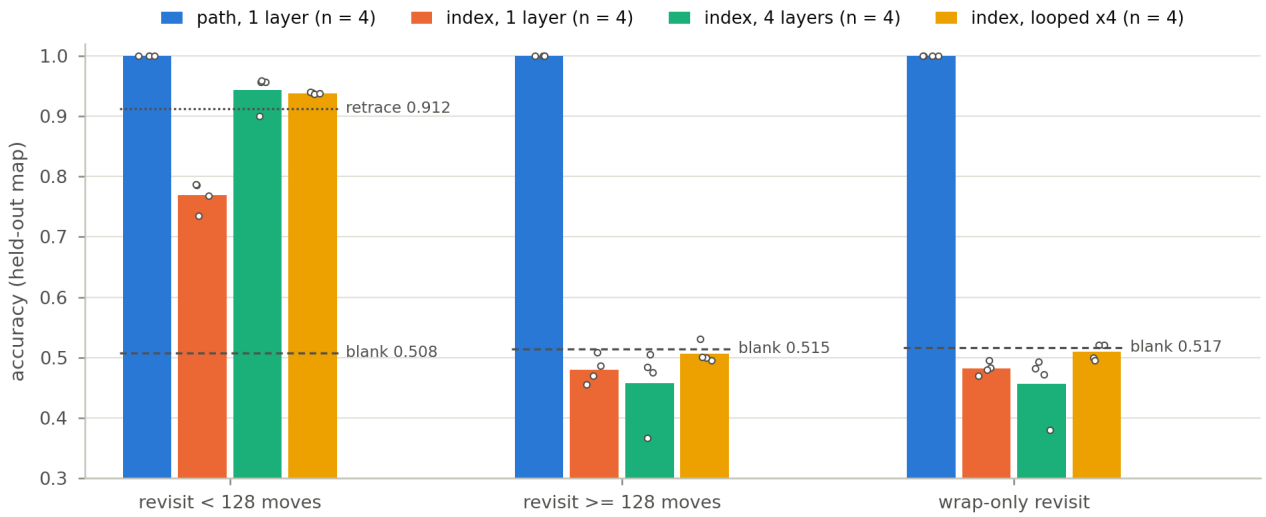
either (signed from the |.| start 0/8 MAP SOLVED; opposite-sign channels 0.27 vs 0.23 with the cell target). The toy and the transformer differ for an unidentified reason; the toy's stronger "the cost is the start, not the constraint" did not transfer and is not claimed.

J. The clock side is not established

A barrier step might learn a content *clock* faster than a signed one. On k-back recency (k_max 128, T = 1024, 300 epochs, one layer) it trends that way and is unmeasured twice (registered): loss AUC (mean training loss) |.| - signed -0.197 (exact sign-flip p 0.053, faster on 12/16 seeds; `CLOCK_SPEED_RESULTS.md`), then -0.106 on 16 fresh seeds (p 0.19, 10/16; `CLOCK_SPEED_REP_RESULTS.md`). Pooled post hoc -0.151 (p 0.018), pre-committed not to upgrade the verdict. On accuracy (`CLOCK_SIDE_RESULTS.md`) every barrier-vs-signed contrast is UNMEASURED (|.| +0.085, CARoPE +0.169, softplus +0.084; perm p 0.16-0.53) and no run converged (0/40 SOLVED); index RoPE fails the task (-0.450, p 0.0011). We do not claim a double dissociation. A consistent but untested account: signed steps contain every non-negative path, so a clock costs them a delay (moving minority signs through zero), while a map costs |.| a block.

K. Where a map is needed, one path layer beats index attention at the depths we tried

On the torus at T = 1024 (FACTORIAL, n = 4 per cell, seeds 500-503), one signed path layer scores 1.000 (4/4 SOLVED) against index RoPE at one layer 0.719, four layers 0.860 and a block looped four times 0.864 (0/4 each, all stalled within 900 epochs; complete separations, p 0.029; floors: blank 0.507, n-gram 0.576, computed on another stream of the same generator, appendix B). Depth and looping lift index accuracy only on revisits within 128 moves (0.74-0.79 -> 0.90-0.96); long-gap (>= 128 moves) and wrap-only revisits stay at 0.37-0.53, near or below the blank floor (Fig. K1, `docs/paper/figs/fig_factorial_strata.png`; its dashed line is the mean of the per-stratum best-constant floors on FACTORIAL's own stream, 0.513, overall 0.509; the 0.507 / 0.576 above are another stream of the same generator). Told in English words (appendix G; pilot seeds excluded): path 0.979 vs index 0.505, +0.474 (6/6 vs 0/6 SOLVED, p 0.0022); the eight-seed registered numbers are 0.969 vs 0.505 (p 0.0002), and with the dropout-scale re-score the path arm reads 0.993.



This is a learnability measurement, not a capacity bound. The torus walk is the group Z_64 x Z_64; Liu et al. (2023) show that one attention layer with an MLP can simulate such a group (their Lemmas 6 and 8), using absolute positions and width and weights growing with T, so their construction does not settle whether our 128-wide RoPE models can represent the task. Their experiments are mixed for our purpose: at T = 100 with every state labelled, the median of 20 runs stays below 90% for C_4 at 1-15 layers (and C_2, C_6 are unreliable), while C_5, C_7 and C_8 reach >= 98.5% from 3 layers. Our labels are sparse and our groups larger. On Dyck-2 at matched depth one path layer is worth about three index layers (appendix F).

Related work on depth (moved from the main text of draft 5). Shallow transformers learn automaton shortcuts unreliably (Liu et al. 2023; Hahn and Rofin 2024), while Selective RoPE (Movahedi et al.) and PaTH (Yang et al. 2025) solve parity or Flip-Flop in one layer.

Figure notes (2026-10-14; updated 2026-10-16; post hoc, CPU, eval-only)

Scripts, outputs and figures in this directory. Run from `/home/prashr` with `CUDA_VISIBLE_DEVICES="" PYTHONPATH=/home/prashr python3 mapformer/docs/paper/figs/<script>`. Every `fig_<name>.py` writes `fig_<name>.{pdf,png}` and `fig_<name>_out.txt` (the numbers drawn). Shared style: `figstyle.py` (dataviz reference palette, light mode: slot 1 blue `#2a78d6`, slot 2 orange `#eb6834`, slot 3 aqua `#1baf7a`; these three validate all-pairs for colour-vision deficiency; a fourth series (yellow `#eda100`) appears only where markers / legends also carry identity). Within a figure, signed / path integration is blue and |.| / index RoPE is orange. `fig_mechanism` and `fig_ring_strata` keep their older matplotlib default colours (only their content was fixed). The palette validator (`validate_palette.js`) was not run (no node on this machine); the slots used are the reference palette's documented passing order.

Status labels. "Registered" = the batch and its verdict / class were pre-registered; the figure only re-plots them. "Post hoc" = the quantity was computed after the results were read (floors, usage weights, strata re-cuts, plots of per-epoch losses). Each figure is a post hoc DRAWING; the label in each caption says what the drawn numbers are.

Index (2026-10-16)

figure	outline	shows	status of the numbers
`fig_loss_curves`	F4 (headline)	training loss per epoch, signed vs	.
`fig_loss_curves_rank`	F4	training loss per epoch, rank 2 / 3 / 4	registered batches and classes; plot post hoc
`fig_wrap_share` (redrawn)	F7 / draft-5 Fig. 1	cancelling share, usage-weighted (primary) + unweighted	post hoc
`fig_mechanism` (shading fixed)	F8 / draft-5 Fig. 2	move-angle planes, ring	registered batch; view post hoc
`fig_factorial_strata` (floor fixed)	F2 part / Fig. K1	torus strata, index depth / loop	registered secondary; floors post hoc
`fig_strata_modes`	F2	torus strata for index / rank / sign failures	registered strata; floors post hoc
`fig_exchange_rate`	F1	accuracy vs index layers, path 1 layer, Dyck / ring / torus	registered; floors post hoc
`fig_rank_solved`	F3	SOLVED by per-head rank; rank = D vs D + 1	registered
`fig_existence`	F5	from scratch vs installed (frozen / trained)	registered
`fig_textworld`	F6	text world: cancellation, clock, ablation	registered batch, declared secondaries
`fig_ring_strata` (floor added)	F9	ring strata	post hoc

Floor helpers (post hoc, CPU, no model): `torus_strata_floors.py` -> `torus_strata_floors_s500-503.json` (FACTORIAL's stream) and `torus_strata_floors_s0-7.json`

(the stream of SIGN_MATCHED / RANK_MI / RANK3 / RANK_SEP, seeds 0-7): per seed and stratum the best constant (asserted equal, n and value, to every `*_STRATA.json` 'floor' of those seeds), blank and retrace. Its overall retrace reproduces `docs/audits/2026-10-16/sign_floors_out.txt` per seed (0.8442, 0.8403, ...). `ring_strata_floors.py` -> `ring_strata_floors.{json,_out.txt}`: the same per stratum on the ring strata stream.

Not drawn anywhere: the torus T = 128 retrace floor 0.934 (`paper2x2_T128_floors_out.txt`) -- no figure draws T = 128 torus accuracies (PAPER2X2 is a table). Where floors are drawn, the caption names which.

Headline: training-loss curves, signed vs |.| (`fig_loss_curves.py`; outline F4)

Data: the `losses` array (per-epoch mean training loss, dropout on) of every stored checkpoint: (a) SIGN_MATCHED `runs/sign_matched/p0/{Signed,Abs,Pos}_r4_s0-7` (torus T = 1024, 900 epochs, 1 layer d 128, lr 1e-3 cosine); (b) WRAP_EXISTENCE `runs/wrap_existence/p0/{SO,AO}_s600-607` (ring 32 cells, T = 128, 300 epochs, observation-step-free, from scratch); (c) GENERALITY `runs/generality/torus/{SO4,AO4}/p0/*_s700-707` (torus T = 1024, observation-step-free); (d) ROBUST `runs/robust/torus/{SO,AO}/p0/*_s1400-1407` (as (c), 2 layers, d 256). Classes by `stats_core.classify_run` (= the registered ones; counts match each results file).

Numbers (`fig_loss_curves_out.txt`): SOLVED and class counts, median loss at the last epoch.

panel	signed	|.|	softplus
(a) SIGN_MATCHED	8/8 SOLVED; median 2.8e-3	0/8 (5 STALLED, 3 DESCENDING); median 0.47	1/8 (s7, final-5% 0.049; 4 STALLED, 3 DESC.); 0.67
(b) WRAP_EXISTENCE ring	8/8; ~1e-5	0/8 (8 STALLED); 0.28	--
(c) GENERALITY torus	8/8; 2.9e-3	0/8 (6 STALLED, 2 DESC.); 0.43	--
(d) ROBUST torus 2L d256	8/8; 6.3e-4	2/8 (s1401, s1404; 6 DESCENDING); 0.081	--

Epoch at which the median first falls below 0.05: signed 480 / 57 / 309 / 183 (a-d); |.| and softplus never. Median loss at 10% / 25% / 50% / 75% of training: (a) signed 1.21 / 0.23 / 0.075 / 0.025, |.| 1.54 / 1.16 / 0.56 / 0.50; (b) signed 0.37 / 0.019 / 8e-4 / 2e-5, |.| 1.04 / 0.60 / 0.37 / 0.30; (c) signed 0.89 / 0.10 / 0.023 / 0.006, |.| 1.56 / 0.66 / 0.55 / 0.47; (d) signed 0.15 / 0.047 / 0.013 / 0.003, |.| 0.65 / 0.39 / 0.24 / 0.15.

Caption (ready to paste; registered batches and classes, post hoc plot). Training loss per epoch for the signed step (blue) and the non-negative steps |.| (orange) and softplus (green), every run of four registered batches (thin lines; thick: median of 8 runs; log scale; training loss with dropout on). Dashed line: the registered SOLVED threshold (final-5% mean training loss < 0.05); n/8 = runs SOLVED. (a) Torus, T = 1024, one layer (SIGN_MATCHED): signed 8/8, |.| 0/8, softplus 1/8. (b) Ring of 32 cells, observation-step-free (WRAP_EXISTENCE): 8/8 vs 0/8. (c) Torus, observation-step-free (GENERALITY): 8/8 vs 0/8. (d) As (c) at two layers and d_model 256 (ROBUST): 8/8 vs 2/8. The signed median crosses the threshold at epoch 480 / 57 / 309 / 183 of 900 / 300 / 900 / 900 and keeps descending; the |.| median never does, and most |.| runs flatten at 0.3-0.5 (STALLED on the ring 8/8; 5-6/8 on the 1-layer torus). The 2-layer |.| runs are still descending at 900 epochs (6/8), so there part of the cost may be speed (budget-scoped).

Caveats: training loss is dropout-on (REPRO note 7: a run can score ~1.000 held out while missing the loss cut). Held-out T = 1024 accuracy of the unsolved non-negative runs (stored eval JSONs): SIGN_MATCHED |.| 0.64-0.93, softplus 0.53-0.97; GENERALITY |.| 0.71-0.96; ROBUST |.| 0.91-0.994 (s1400 0.994 is unsolved by loss but near ceiling held out -- at 2 layers the loss curve and the accuracy disagree most). The softplus SOLVED run (s7, 0.995 held out) sits at 0.049, on the threshold.

Training-loss curves by rank (`fig_loss_curves_rank.py`; outline F4, rank half)

Data: `runs/rank_mi/p0/{Vanilla,Vanilla_r4mi,Vanilla_r2ph}_s0-7`, `runs/rank3/p0/Vanilla_r3ph_s0-7`, `runs/rank_sep/p0/Vanilla_r4ph_s0-7` (torus T = 1024, 900 epochs; every arm built from our r = 2's initial weights at each seed). Numbers (`fig_loss_curves_rank_out.txt`): (a) shared r 2 0/8 SOLVED (4 STALLED, 4 DESCENDING; median final 0.35) vs shared r 4 8/8 (2.8e-3); (b) per-head r 2 2/8 (median 0.22), r 3 6/8 (3.5e-3), r 4 8/8 (3.1e-3).

Caption (registered batches and classes, post hoc plot). Training loss per epoch on the torus (T = 1024) by the per-head rank of the content-to-angle map; thin lines are runs, thick lines medians of 8; dashed: the SOLVED threshold. (a) One latent shared by the heads: rank 2 solves 0/8, rank 4 8/8 (RANK_MI). (b) Per-head latents: rank 2 2/8, rank 3 6/8, rank 4 8/8 (RANK_MI, RANK3, RANK_SEP). The rank-4 and rank-3 medians cross the threshold at epochs 478-532; the rank-2 medians never do, and most rank-2 runs flatten at 0.2-0.5.

Fig. 1 of draft 5, redrawn: cancelling share, usage-weighted (`fig_wrap_share.py`; outline F7)

Redrawn 2026-10-16 for MOCK_REVIEW_draft5 W4 / SYNTHESIS item 9. Same runs as before: RING_DIAL (`runs/ring_dial/p0`, ring 32, T = 128, 300 epochs, seeds 30-37, p_plus 0.5 / 0.75 / 0.9) and SIGN_MATCHED (torus T = 1024, seeds 0-7).

The measure. Entries are (move, channel) pairs, a = omega_c * step_c(move). Selected: |wrap(a)| > 0.05 rad. **The denominator excludes channels whose move angle has collapsed to ~0 mod 2 pi** (the old caption's "share of rotary channels" was wrong; the share is over non-collapsed channels). Cancelling: |wrap(a(move) + a(opposite))| < 0.1 rad; torus pairs (0,1), (2,3). Primary = usage-weighted: each entry weighted by mean |q_c| at move positions x mean |k_c| at observation positions (`model_wrap_exist.usage_weights`, ACCOUNTS Amendment 1; inlined for VARIANT_MAP models), on 32 eval walks (ring: GridWorldCancel(32, seed 10000, p_plus as trained), np seed 10**6, T = 128; torus: GridWorld(64, seed 10000), np seed 1234, T = 1024). Secondary (faint dash) = unweighted. The unweighted values reproduce the old figure exactly (0.79 / 0.06 / 0.78 / 0.03 / 0.64 / 0.06 / 0.98 / 0.11 / 0.03).

cell	arm	SOLVED	weighted mean (range)	unweighted mean (range)	excluded (collapsed) entries
ring p_plus 0.5	signed	8/8	0.79 (0.41-0.99)	0.79 (0.59-0.92)	0.11

cell	arm	SOLVED	weighted mean (range)	unweighted mean (range)	excluded (collapsed) entries
ring p_plus 0.5		.		2/8	0.06 (0.00-0.29)
ring p_plus 0.75	signed	8/8	0.87 (0.55-0.99)	0.78 (0.64-0.86)	0.11
ring p_plus 0.75		.		1/8	0.02 (0.00-0.07)
ring p_plus 0.9	signed	8/8	0.70 (0.39-0.99)	0.64 (0.44-0.91)	0.16
ring p_plus 0.9		.		4/8	0.10 (0.00-0.34)
torus T 1024	signed	8/8	0.93 (0.64-1.00)	0.98 (0.90-1.00)	0.35
torus T 1024		.		0/8	0.09 (0.00-0.29)
torus T 1024	softplus	1/8	0.01 (0.00-0.09)	0.03 (0.00-0.12)	0.62

Solved non-negative runs, weighted: ring 0.01, 0.01 (p 0.5); 0.02 (p 0.75); 0.34, 0.01, 0.07, 0.01 (p 0.9); torus softplus 0.00. Per-run values in `fig_wrap_share_out.txt`.

Reading. Weighting by use does not change the picture: signed means 0.70-0.93, |.| 0.02-0.10, softplus 0.01. It widens the signed spread (some signed runs read more of their non-cancelling channels: 0.39-0.64 minima), and 7 of the 8 solved non-negative runs stay at <= 0.07 weighted, so solved |.| runs do not use the wrapped code in the channels attention reads either. (ACCOUNTS T1's 0.87 -> 0.18 vs 0.99 -> 0.98 gap was for a held CONSTRUCTED solution; on these from-scratch runs the two measures agree to within ~0.1 on the means.)

Caption (ready to paste; post hoc). Share of rotary channels in which opposite moves cancel mod 2 pi, per run, weighted by how much attention uses each channel (dots; filled = SOLVED) and unweighted (faint dashes, joined to the same run). A channel cancels when |wrap(a(move) + a(opposite move))| < 0.1 rad; channels whose move angle has collapsed to within 0.05 rad of 0 mod 2 pi are excluded from the denominator. Usage weight: mean query norm of the channel at move tokens times mean key norm at observation tokens (eval walks). Ring of 32 cells at three move biases p_plus (RING_DIAL, n = 8 each) and the torus at T = 1024 (SIGN_MATCHED, n = 8). Bars: arm means of the weighted share. Signed steps: 0.70-0.93; |.| steps: 0.02-0.10; softplus: 0.01. Seven of the eight non-negative runs that solve stay at or below 0.07.

Fig. 2 of draft 4 (Figure 3 of the outline): the mechanism (`fig_mechanism.py` -> `fig_mechanism.{png,pdf}`, numbers `fig_mechanism_out.txt`)

Shading fixed 2026-10-16 (MOCK_REVIEW_draft5 W9): omega is free in sign, so the |.| class (omega * |s|) contains both the all-positive and the all-negative quadrant (appendix E P3); only the two MIXED-sign quadrants (II and IV) are closed to |.|. The figure now

shades only those, labelled "mixed signs: closed to |.|". No data moved; `fig_mechanism_out.txt` is byte-identical (checked with cmp). No |.| point lies in quadrant III (raw ranges >= 0), so the old figure's data were right and only the shading misstated the class.

Relabelled 2026-10-15: panel titles use draft 4's terms ((a) signed step, same-direction start = SO_absmag; (b) |.| step, the same start = AO_absmag; (c) |.| step, large start, sums < pi = AO_mid; (d) |.| step, start past pi = AO_high); the shading is labelled as the barrier side (closed to |.|). Numbers unchanged (`fig_mechanism_out.txt` byte-identical).

Data: MECHANISM batch (`MECHANISM_RESULTS.md`; ring 32, p_plus 0.5, T = 128, 300 epochs, seeds 800-807, n = 8 per arm). Start = `eval.json` `init_angles`; end = `model_wrap_exist.action_angles` of the checkpoint (omega * step). 64 channels x 8 seeds per panel. Init-to-end segments are drawn for seed 800 only.

View. Each angle has its whole turns removed toward zero: rep(a) = sign(a) (|a| mod 2 pi). Adding 2 pi to one move's angle does not change the model's function, so rep(a) is a valid representative, and it keeps the raw sign, which is what the non-negative constraint acts on. Neither plain wrapped view works: in (-pi, pi]^2 and in [0, 2 pi)^2 every cancelling pair falls on ONE line, so the two routes cannot be told apart. In this view the cancelling pairs lie on two lines: a(-1) = -a(+1) through the origin (needs opposite raw signs, so the |.| step cannot reach it) and a(+1) + a(-1) = 2 pi (or -2 pi). Signed raw angles are all within (-2 pi, 2 pi), so rep is the identity for them. AO_high's raw angles reach 17.7 rad (up to 3 turns), and its segments are skipped where a turn was removed. The through-zero / by-wrap split describes where the parameters sit. It is not a property of the function (rule 8). The cancelling share itself is wrap-invariant.

Caption (draft). Move angles (a(+1), a(-1)) of every rotary channel, at the start (grey) and after training (dots). There are 8 runs per panel, on the 1D ring. A channel cancels opposite moves when a(+1) + a(-1) = 0 mod 2 pi, on the blue line (through zero; reached only with opposite signs) or the red line (by wrapping). Grey shading marks the two mixed-sign quadrants, the only region the |.| step cannot reach (omega's sign is free, so both angles may be negative); the dotted line is a(+1) + a(-1) = pi; dashed boxes are the U(lo, hi) start boxes. The panels: - Signed step from the same-sign start (SO_absmag): 7/8 runs solve. One move's angle crosses zero: 6% of channels cancel at the start, 89% at the end (88% with opposite signs, none by wrapping). - The |.| step from the same start (AO_absmag): 0/8 runs solve, with 4% cancelling. - The |.| step from a large start whose sums stay below pi (AO_mid): 2/8 runs solve, with 5% cancelling, all by wrapping. - The |.| step from a large start whose sums exceed pi (AO_high): 8/8 runs solve, and cancelling rises from 7% to 23%, all by wrapping.

The cancelling share is the registered wrap share: |wrap(a(+1) + a(-1))| < 0.1 rad among channels with |wrap(a(+1))| > 0.05.

Notes: - AO_absmag's registered 0.04 is mostly not cancellation. 0.035 of it is same-sign channels near the origin (both angles < 0.1 rad), and 0.007 is by wrapping. Also, 44% of AO_absmag's channels end with |a(+1)| <= 0.05: their step collapses to ~0. The figure colours the near-origin channels black. - The |.| start is not uniformly "sums below pi". |default| angles reach 5.5 rad, and initial sums run 0.01-6.58. AO_high's start already has 7% cancelling channels by chance, since iid U(1.7, 4.5) sums pass near 2 pi.

Ring strata (plan item 7) (`ring_strata.py` -> `ring_strata.json`, `ring_strata_out.txt`; `fig_ring_strata.py` -> `fig_ring_strata.{png,pdf}`)

Stream. `GridWorldCancel(32, seed 10000, p_plus 0.5)`, `np.random.seed(10**6)`, maps redrawn per trajectory, T = 128 moves, eval mode with no dropout re-score. It uses **400** trajectories, not the 200 asked for: the 400 are the registered eval stream, the first 200 are identical to a 200-trajectory eval, and using 400 makes the check below exact.

Strata. Every stratum is a revisit, and the gap is the number of moves since the last visit to the cell. A revisit is "wrap-only" when every earlier visit is at unwrapped displacement +-32k with k != 0. Unwrapped position = cumsum of the move tokens.

Verification. - Constructed cases pass (`--test`): a full lap, the return, back-and-forth moves, laps in both directions, and runs of laps. - The revisit flag equals `environment_cancel`'s mask on 50 random trajectories plus all 400 eval trajectories, and the unwrapped walk matches the environment's cells. - Overall revisit accuracy reproduces every run's `eval.json` acc(128) (max |diff| 5e-5).

Floor. The constant predictor scores 0.49-0.50 in every stratum, so the strata do not differ in target composition.

batch	arm	SOLVED	plain gap 2-7	plain 8-31	plain 32-127	wrap-only	plain, wrap-gap-matched	all
(stream)	n targets		12183	17420	5110	4134		38847
RING_DIAL	Signed_r4	8/8	1.000	1.000	1.000	1.000	1.000	1.000
RING_DIAL	Abs_r4	2/8	0.992 (0.97-1.00)	0.975 (0.93-1.00)	0.957 (0.91-1.00)	0.923 (0.80-1.00)	0.956	0.973
WRAP_EXISTENCE	SO	8/8	1.000	1.000	1.000	1.000	1.000	1.000
WRAP_EXISTENCE	AO	0/8	0.961 (0.89-1.00)	0.916 (0.85-0.97)	0.881 (0.81-0.93)	0.872 (0.80-0.93)	0.878	0.921
MECHANISM	SO	8/8	1.000	1.000	1.000	1.000	1.000	1.000
MECHANISM	AO	0/8	0.978 (0.92-1.00)	0.918 (0.85-1.00)	0.873 (0.77-1.00)	0.811 (0.75-0.88)	0.874	0.919
MECHANISM	AO_high	8/8	1.000	1.000	1.000	1.000	1.000	1.000

Entries are means over runs, with the range in brackets. "Plain, wrap-gap-matched" re-weights plain revisits to the wrap-only gap distribution (8-move bins over gaps 32-127).

Answer. - The non-negative models fail mainly with distance. Accuracy falls with the gap since the last visit in every |.| batch: from gap 2-7 to gap 32-127 it drops by -0.035 for Abs_r4 (6/8 runs lower; the 2 solved runs sit at 1.000; sign-flip p 0.031), -0.080 for WRAP_EXISTENCE AO (8/8 lower, p 0.0078, the minimum for n = 8) and -0.105 for MECHANISM AO (8/8, p 0.0078). Short-gap revisits are near ceiling but not at it (0.96-0.99). This fits a code that works locally and drifts with distance. - Wrap-only revisits add a deficit beyond the gap that varies between batches. Against gap-matched plain revisits it is -0.063 for MECHANISM AO (6/8 runs, p 0.031), -0.007 for WRAP_EXISTENCE AO (2/8, p 0.88) and -0.032 for Abs_r4 (4/8, p 0.31). - The two AO

batches are the same arm on different seeds, and they disagree. One MECHANISM AO run (s804) is a clean case of "local clock, no wrap": 1.000 / 0.999 at short / long plain gaps, 0.881 on wrap-only revisits. - So the post hoc reading is: |.| models lose accuracy with gap in every batch, and the wrap-specific failure is real in some runs but not consistent. Signed models are at 1.000 in every stratum. - AO_high is at 1.000 in every stratum (8/8 SOLVED). It "fixes" the long-gap and wrap-only strata, but at ceiling this is the same fact as SOLVED, not separate evidence.

Caveats. - Post hoc and descriptive. The sign-flip p values are across runs within an arm, uncorrected for the 3 batches x 2 contrasts. - Accuracy tracks loss (rule 2), so the strata describe where the unsolved models' errors fall; they do not establish a mechanism. - Gap and wrap-only are confounded with each other by construction: wrap-only gaps are >= 32. That is why the gap-matched column exists.

Caption (draft). Revisit accuracy on the 1D ring by stratum (eval mode; 400 fresh trajectories, T = 128; maps redrawn). "Plain" revisits have an earlier visit at the same unwrapped position. "Wrap-only" revisits were seen only a lap away (+-32k). Lines show arm means; dots show runs; hollow markers show plain revisits re-weighted to the wrap-only gap distribution. Signed steps (24 runs over three batches) and |.| steps started past pi (AO_high) are at 1.000 everywhere. |.| steps from the default start lose accuracy with the gap (0.96-0.99 -> 0.87-0.96). Their extra wrap-only deficit varies between batches (-0.063, -0.007, -0.032). The constant-predictor floor is 0.50.

Floor added 2026-10-16 (post hoc; `ring_strata_floors.py`, the same 400-walk stream). Per-stratum retrace (one-level rule of `docs/audits/2026-10-16/ring_floors.py`): gap 2-7 **0.9998**, gap 8-31 **0.943**, gap 32-127 0.570, wrap-only 0.493 (blank 0.49-0.50 everywhere; overall retrace 0.864 on this stream vs 0.860 on ring_floors' stream). Drawn as dotted dashes where on axis; the off-axis values are printed in the panel. **Consequence:** in the short-gap strata a map-free rule is at or above the unsolved |.| arms (gap 2-7: retrace 1.000 vs |.| 0.96-0.99; gap 8-31: 0.943 vs WRAP_EXISTENCE AO 0.916 and MECHANISM AO 0.918; RING_DIAL Abs_r4 0.975 is above). The |.| models' advantage over map-free rules is in the long-gap and wrap-only strata (0.81-0.96 vs retrace 0.57 / 0.49). "Works locally" in the answer above therefore means "about as well as undoing the last run", not "better than a map-free rule".

Caption addition (post hoc). Dotted dashes: the retrace floor within each stratum (copy the observation seen when the walk last undid its most recent run of moves): 1.000 at gaps 2-7 and 0.943 at gaps 8-31; 0.57 and 0.49 in the long-gap and wrap-only strata (off axis).

Fig. K1 fixed: torus strata, index depth and loop (`fig_factorial_strata.py`)

Fixed 2026-10-16 (NUMBERS_AUDIT item 1). The old dashed line was the mean of the per-stratum constant floors over all arms and runs (recomputed: 0.513), while the caption said "overall 0.507" (a number from another stream, RANK_MI seeds 10-25). Now each stratum carries its own floors, computed on FACTORIAL's own eval stream (seeds 500-503; `torus_strata_floors.py`, asserted equal in n and value to the JSON 'floor' of every FACTORIAL run): best constant (= blank) 0.508 / 0.515 / 0.517 (short gap / long gap /

wrap-only), and retrace 0.912 in the short-gap stratum (retrace equals the constant in the other two: the walk's runs are at most 10 moves, so a retrace never reaches a gap >= 128 or a wrap-only revisit). Overall on this stream: constant 0.509 (0.505-0.515), retrace 0.844 (0.840-0.848).

Numbers (`fig_factorial_strata_out.txt`; mean (range), n = 4): path 1L 1.000 / 1.000 / 1.000; index 1L 0.769 (0.735-0.787) / 0.480 (0.456-0.508) / 0.482 (0.470-0.495); index 4L 0.943 (0.900-0.958) / 0.458 (0.367-0.506) / 0.457 (0.381-0.494); looped x4 0.938 (0.937-0.940) / 0.507 (0.496-0.531) / 0.509 (0.496-0.521).

Caption (ready to paste; registered strata secondary; floors post hoc). Revisit accuracy by stratum on the torus at T = 1024 (FACTORIAL; trained and tested at T = 1024; bars: means of n = 4 runs, dots: runs). Strata: plain revisits with a gap of fewer than 128 moves since the last visit, plain revisits with longer gaps, and wrap-only revisits (the cell was seen before only from a different unwrapped position). Dashed: always-blank (the best constant) within each stratum, 0.508 / 0.515 / 0.517; dotted: the retrace rule within the short-gap stratum, 0.912 (elsewhere it equals the constant). Floors are computed on this batch's own evaluation walks. Path integration with one layer is at 1.000 in every stratum. Four index layers or a loop of four lift short-gap revisits to 0.94, just above retrace; at long gaps and wrap-only revisits every index model stays at or below the constant (0.46-0.51).

F2: strata for every failure mode (`fig_strata_modes.py`)

Data: (a) `FACTORIAL_{L1,L4,LOOP}_STRATA.json` (seeds 500-503); (b) `RANK_MI_STRATA.json` (Vanilla = shared r 2, Vanilla_r2ph, Vanilla_r4mi) and `RANK3_STRATA.json` (Vanilla_r3ph), seeds 0-7; (c) `SIGN_MATCHED_STRATA.json` (Signed_r4, Abs_r4, Pos_r4, RoPE), seeds 0-7. All eval mode at T = 1024, held-out map. Floors per stratum from `torus_strata_floors_s500-503.json` (a) and `torus_strata_floors_s0-7.json` (b, c).

Numbers (`fig_strata_modes_out.txt`; short gap / long gap / wrap-only means): - (a) as Fig. K1. - (b) r 4 shared 1.000 / 0.995 / 0.977; r 3 per-head 1.000 / 0.967 / 0.875; r 2 per-head 0.919 / 0.784 / 0.644; r 2 shared 0.950 / 0.671 / 0.557. - (c) signed 1.000 / 0.995 / 0.982; |.| 0.870 / 0.630 / 0.515; softplus 0.805 / 0.729 / 0.574; index RoPE 1L 0.784 / 0.462 / 0.471. - Floors, seeds 0-7: constant 0.506 / 0.500 / 0.507; retrace 0.911 (short gap only).

Caption (registered strata; floors post hoc). Revisit accuracy by stratum on the torus at T = 1024 for the three ways a model fails to learn the map. Lines: arm means; dots: runs. (a) Index attention (FACTORIAL, n = 4): depth and a loop help short-gap revisits only. (b) Per-head rank of the content-to-angle map (RANK_MI, RANK3; n = 8, same seeds and initial weights): rank 4 is near 1.000 in every stratum; rank 3 loses wrap-only revisits on two runs (0.49, 0.63); rank 2 loses with gap and most at wrap-only revisits. (c) Sign of the step (SIGN_MATCHED, n = 8): signed near 1.000 everywhere; |.| and softplus fall from short gaps to wrap-only revisits; one-layer index RoPE for reference. Dashed: always-blank within each stratum; dotted: retrace within the short-gap stratum (0.91). Floors on each batch's own walks.

Note: in the short-gap stratum the mean |.| (0.870), softplus (0.805) and one-layer index (0.784, 0.769) arms sit BELOW retrace 0.911-0.912, and the rank-2 arms only just above

it (0.919, 0.950); their advantage over map-free rules is in the long-gap and wrap-only strata (where retrace = the constant, ~0.50-0.52).

F1: the exchange rate (`fig_exchange_rate.py`)

Data: (a) `DYCK_MDEPTH_RESULTS.json` cells `T12|{RoPE,MapWM}|{1-4}L|L32D12|A2f_close_acc_dist_feasible` (registered; trained and scored at L32 D12, 1x budget, n = 8); floor 0.5935 = `floors_A2f['T12']['L32D12']`. (b) `runs/cancel/p0/*/ eval.json` acc at T = 128, **fresh seeds 2-7 only** (n = 6; seeds 0-1 were the pilot; CANCEL_RESULTS CORRECTED block, and SYNTHESIS item 4: the main text cites fresh-seed values); retrace floors 0.860 / 0.758 / 0.612 from `docs/audits/2026-10-16/ring_floors_out.txt` (same task, another stream). (c) `FACTORIAL_{L1,L4,LOOP}.json` acc at 1024 (n = 4; --fast-attn); floors on its own stream (`torus_strata_floors_s500-503.json`).

Numbers (`fig_exchange_rate_out.txt`): - (a) Dyck A2f: path MapWM 1L 0.996; index RoPE 1L 0.648, 2L 0.889, 3L 0.964, 4L 0.982; floor 0.594. (MapWM 2-4L 1.000.) - (b) ring, fresh seeds: path 1L 1.000 at every p_plus; index 1 / 2 / 3 layers: p 0.5 0.710 / 0.949 / 0.998; p 0.75 0.678 / 0.962 / 0.998; p 0.9 0.824 / 0.987 / 0.999 (p 1.0: everything 1.000, not drawn). - (c) torus: path 1L 1.000; index 1L 0.719 (0.688-0.739), 4L 0.860 (0.814-0.881), looped x4 0.864 (0.860-0.868); floors blank 0.509, retrace 0.844.

Caption (ready to paste; registered results, floors post hoc). Accuracy of index-RoPE attention against its number of layers (orange; dots: runs, line: means) beside one layer of path integration (blue line), each trained and tested at the same length and depth. (a) Dyck-2 at length 32 and depth 12 (Hewitt closing accuracy A2f; n = 8): index attention needs about three layers to approach one path layer (0.964 at 3, 0.982 at 4, vs 0.996). (b) Ring of 32 cells at T = 128 for three move biases (n = 6 fresh seeds): three index layers reach one path layer (0.998-0.999 vs 1.000). (c) Torus at T = 1024 (n = 4): four index layers (0.860) or one layer looped four times (0.864) stay far below one path layer (1.000), just above the retrace floor (0.844). Dashed lines: map-free floors (Dyck: best A2f floor; ring: retrace per bias, on another stream of the same task; torus: always-blank and retrace on this batch's own walks).

F3: SOLVED by per-head rank (`fig_rank_solved.py`)

Data (parsed from the registered analysis outputs): `RANK_SEP_ANALYSIS.txt` (A, B, C_bd, C, D), `RANK3_ANALYSIS.txt` (E), `RANK_ND_ANALYSIS.txt` (A2, B2, A3, B3), `RANK_WRAP_ANALYSIS.txt` (3L), `RANK_TESTS_ANALYSIS.txt` (R1, R2). The exact lines drawn are copied into `fig_rank_solved_out.txt`.

Numbers: (a) torus T = 1024, n = 8: per-head rank 2 -- shared r 2 0/8 (acc 0.894), per-head r 2 2/8 (0.885), block-diagonal r 4 2/8 (0.948); rank 3 -- per-head r 3 6/8 (0.987); rank 4 -- shared r 4 8/8 (0.998), per-head r 4 8/8 (0.999). (b) rank = D vs D + 1: 1D ring 7/8 vs 8/8; 2D torus (32/side) 1/8 vs 8/8; 3D torus (10/side) 1/8 vs 4/8, and rank 4 on an 18/side 3D torus 8/8.

Caption (ready to paste; registered). Runs (of 8) that solve the map, by the per-head rank of the content-to-angle map. (a) Torus, T = 1024, six bottlenecks built from the same initial weights: every arm whose heads each read a rank-2 map solves 0-2/8, rank 3 solves

6/8 and rank 4 8/8, whether the latent is shared or per head (RANK_SEP, RANK3). (b) Rank equal to the task's dimension D against D + 1 (separate batches and recipes): on the 1D ring rank 1 is not hard (7/8 vs 8/8; RANK_TESTS); on a 2D torus rank 2 is (1/8 vs 8/8; RANK_ND); on a small 3D torus (10 per side) both rank 3 and rank 4 are (1/8, 4/8), and rank 4 solves 8/8 on an 18-per-side torus with few wrap-only revisits (hatched; RANK_WRAP). Budget-scoped (900 epochs on the torus; 2x the budget does not rescue rank 2: LOOP_RANK_E1800_P1).

F5: existence before mechanism (`fig_existence.py`)

Data (parsed from registered outputs): `RANK_SEP_ANALYSIS.txt` (A, C from scratch), `RANK_PROJ_ANALYSIS.txt` (FROZEN mean; TRAINABLE r2-proj, CONTROL r4-cont, REFERENCE r2-cont class counts; lr restart 1e-3), `WRAP_EXISTENCE_ANALYSIS.txt` (E2 PRIMARY, E3), `GENERALITY_ANALYSIS.txt` (E2 PRIMARY at lr 1e-4, E3).

Numbers (`fig_existence_out.txt`): (a) rank 2 torus: from scratch 0/8; r 2 continued 1/8; projected r 2 installed (frozen 0.9955) and trained 7/8; r 4 from scratch 8/8; r 4 continued 7/8. (b) ring |.|: from scratch 0/8 (acc 0.921); installed (frozen = source, 1.000) and trained 8/8, lost 0/8 (HELD); signed 8/8 / 8/8, lost 0/8. (c) torus |.|: from scratch 0/8 (0.880); installed and trained 8/8 but lost 6/8 (peak loss 0.11-0.21, re-found by epoch 119-204: NOT HELD, re-found); signed 8/8 / 8/8, lost 0/8.

Caption (ready to paste; registered). A solution exists in each constrained class and training keeps or re-finds it, but training from scratch does not find it. Bars: runs SOLVED of 8 (hatched: the solution was installed and then trained). (a) Rank 2 on the torus: a rank-2 projection of a solved rank-4 model scores 0.9955 frozen and is held when trained (7/8), while rank 2 from scratch solves 0/8, and 1/8 when the stalled runs are trained further (RANK_SEP, RANK_PROJ). (b) The |.| step on the ring: an installed exact non-negative solution is held under training (8/8 SOLVED, lost on 0/8), while |.| from scratch solves 0/8 against signed 8/8 (WRAP_EXISTENCE). (c) The |.| step on the torus: the installed solution is perturbed (loss above 0.05 on 6/8) and re-found (8/8 SOLVED); from scratch 0/8 vs 8/8 (GENERALITY). "Lost" (registered definition): the max per-epoch training loss after the start exceeds 0.3, or exceeds 10x the signed control's max at that seed and 0.05.

F6: navigation told in words (`fig_textworld.py`)

Data: `TEXTWORLD_SECONDARY.json` only (declared secondaries S1, S3, S5 of the registered TEXTWORLD batch; path arm Vanilla_r4, 1 layer, T = 1024, seeds 0-7, seeds 0-1 were the pilot). "Clock" seeds: |c| / |north| > 0.3 (s0, s2, s3, s4; values 0.48-1.14 vs 0.04-0.06; cos(c, verb step) +0.999 to +1.000). Ablation accuracies are eval mode without the 1/(1-p) attention re-scale (so s0 and s7 read 0.88; re-scaled path mean 0.993, TW_NORMSTEP CORRECTED block).

Numbers (`fig_textworld_out.txt`): opposition raw 0.080-0.120 (map seeds) and 1.43-1.77 (clock seeds); after removing the common component 0.002-0.012 on all 8 seeds; ablation: all steps 0.88-1.000; non-direction steps zeroed 0.59-0.95 (map seeds) and 0.19-0.67 (clock seeds); direction steps zeroed 0.39-0.51; floors constant 0.505, reversal-copy 0.597.

Caption (ready to paste; registered batch, declared secondaries). What a one-layer path model learns from a torus walk told in English (TEXTWORLD, n = 8 seeds). (a) Opposition of the learned direction-word steps, |A + B| / |A| averaged over north/south and west/east (0 = opposites cancel): after removing the component common to all four directions, opposites cancel to 0.002-0.012 on every seed. (b) That common component is small on four seeds (blue, 4-6% of a direction step) and large on the other four (orange, 0.48-1.14), where it points along the verbs' step: a per-step clock layered on the map (31-38 of 64 phase channels drift between visits to the same cell). (c) Held-out accuracy with every step, with the steps of all non-direction words zeroed, and with the direction words' steps zeroed: removing the direction steps drops every seed to the constant floor (0.505, dashed) or below; the clock seeds also depend on the other words. Eval mode without the dropout re-scale. Seeds 0-1 were the pilot.

Skipped / not made (2026-10-16)

- Nothing in outline section 4 needed a GPU; every listed CPU figure (F1-F9) now exists. T1-T4 are tables (text), not made here.
- F4's ring half uses WRAP_EXISTENCE only (one ring batch, as the review asked); RING_DIAL / MECHANISM / ROBUST-ring loss curves exist in the same checkpoints if wanted.
- `fig_mechanism` and `fig_ring_strata` were fixed in place, not restyled to the shared palette.